

Dictionary of Networking

You'll never have doubts about it again.

Share, don't be selfish, don't think only of yourself.

<https://www.linkedin.com/in/felipe-c-01a8b1280/>

Introduction

Networks are currently one of the fastest growing and most important developments in the computer industry. Not only are more and more PCs becoming parts of networks, but networked PCs are being incorporated into larger enterprise-wide applications so that everyone in a company can access and share data.

With the expanding technology of networking comes the terminology to describe it. This *Dictionary of Networking* provides definitions for all the terms you will encounter when dealing with networks of any type.

Who Should Use This Book?

This book is designed to meet the needs of people who work with networks, communications, and mobile computing systems. Whether you are networking previously unconnected computers or downsizing from a mainframe, this book is for you. And if you are studying for one of the network certification exams, you will find this book to be an essential reference.

Network users of all levels are barraged with an almost bewildering array of terms, abbreviations, and acronyms in books, magazine and newspaper articles, advertisements, and their day-to-day conversations. Jargon is a useful shorthand, but it can easily become incomprehensible and unmanageable, even to the most seasoned network administrator.

What You'll Find in This Book

Along with clear explanations of the jargon and slang associated with networking, you'll find definitions of more than 3,000 networking technical terms, abbreviations, and acronyms. The list that follows gives you a brief overview of the topics that this book covers:

- Acronyms and abbreviations
- Active Directory
- ActiveX
- Application software
- Cables, cards, and connectors
- Certification schemes
- Chips, memory, and adapters
- Communications

Introduction

- Connectivity tools, products, and equipment
- Disks and storage media
- E-mail
- Hardware
- File systems
- Industry standards
- Internet organizations
- Internet terms and abbreviations
- Intranet terms and abbreviations
- Java
- Leading hardware and software companies
- Linux, Free Software Foundation, GNU
- Microprocessors
- Microsoft Windows
- Microsoft Windows NT and NT Server
- Microsoft Windows 2000
- Mobile computing
- Networking theory and concepts
- Novell NetWare
- Novell Directory Services
- Operating systems and environments
- The OSI Reference Model
- Popular networking products
- Protocols and interfaces
- Security and network administration
- System architectures
- Trade associations

- Unix shells, tools, and utilities
- The World Wide Web
- Workstations

How This Book Is Organized

This book is organized for easy reference. Entries are arranged in letter-by-letter alphabetic order, ignoring punctuation and spaces, with terms that begin with an uppercase letter (or are all in uppercase) before those in all lowercase letters. So *Internet* comes before *internet*, and *link level* comes before *link-state routing algorithm*. Numbers and symbols are listed at the beginning of the book in ascending numeric order. If an entry begins with a letter or letters, but contains a number, it is listed alphabetically according to the letter, and then according to numerical order, so *V.42 bis* comes between *V.42* and *V.54*.

The information within each entry is always presented in the following order:

- Entry name
- Abbreviation or acronym
- Pronunciation, if it isn't obvious
- Definition, written in clear standard English
- URL pointing to further resources available on the Internet.
- Cross-references to other entries that provide additional or related information on the topic; more on the cross references in a moment.

If an entry has multiple definitions, each one is numbered to separate it from the next, and some of the entries also contain illustrations.

Extensive Cross-Referencing

The *Dictionary of Networking* is the most extensively cross-referenced dictionary of computing and networking terms available today. It contains two kinds of cross references:

- A *See* reference points to another entry that contains the information you are looking for. Thus, you can start with an abbreviation, such as *PPTP*, or with the complete term, such as *Point-to-Point Tunneling Protocol*, and be sure that you will arrive at the correct definition. You don't have to know what an abbreviation stands for to be able to look up a term. Some terms or concepts can be referred to by more than one name, such as *dialback modem* and *callback modem*; you will find both here, so you can always find your way to the appropriate definition.

Introduction

- A *See also* reference points to one or more entries that contain additional information about a topic. This kind of cross-reference allows you to follow through a related set of entries, broadening your search as you move from entry to entry.

We have also added an extra element in this dictionary to help you find information, and that is the pronunciation of an acronym or abbreviation that is pronounced differently from the way it is spelled. For example, if you are reading a magazine article and come across the abbreviation *SCSI*, you can look up the abbreviation, which will point to the main entry term, *Small Computer System Interface*. But if you are discussing hard disk interfaces with a colleague and hear the term *scuzzy*, you can look that up too, and you will also find your way to the main entry, *Small Computer System Interface*.

The Appendices

This book contains four appendices to provide additional reference material:

Appendix A: Internet Resources Collects together URLs under a wide variety of headings to cut down on the amount of time you have to spend with your favorite search engine. Using this appendix, you can go straight to the right Web site every time.

Appendix B: Certification Resources Provides a guide through the complex and confusing world of computer and networking certification programs.

Appendix C: ASCII Charts Contains both the standard and the extended American Standard Code for Information Interchange charts.

Appendix D: EBCDIC Chart Contains the most widely accepted Extended Binary Coded Decimal Interchange Code chart.

A Note about the URLs in This Book

Nothing is more annoying than a dead URL, and link rot is all pervasive. (If you don't know what link rot is, go look it up.) All the URLs in this book have been individually checked by our Technical Editor; and at the time of writing, they are all active, they all work, and they all contain the information that I say they contain. But that is not to say that some of them won't have changed by the time you try them out.

The better-organized sites will simply post a link to the new site if they make substantive changes, and you can use that new link to go right to the new or reorganized site. Other sites, such as the Microsoft Web site, reorganize themselves periodically as a part of their housekeeping; the information you want is still available, but you have to look in another place to find it, or use the site's built-in search engine to find it.

Some of the sites that contain the most advanced technical information belong to the .edu domain and are usually computer science departments at the major universities. I have tried to keep the number of such sites to a minimum in this book. Although they can be extremely useful, they usually have a lifespan that closely resembles that of the average graduate student. Once the student maintaining the information graduates, the site becomes neglected and is usually removed soon after. Another dead URL.

To be consistent with current usage, I have not specified the protocol used to access each Web site; unless a different protocol is specified, you can simply assume that HTTP will work in all cases. Just add `http://` to the beginning of each Web address in your browser when you access a site.

And finally, we have tried very carefully not to break a URL across a line; you should be able to type the characters you see without having to worry about whether to type that hyphen. If a URL has to break, the break is before a period (.) or after a slash (/).

About the Companion CD

The companion CD contains an electronic version of this entire book. You can use it to find entries quickly and follow cross-references without a great deal of page flipping.

And Finally...

Through more than 25 years of hands-on involvement in practical computer applications, including the management of minicomputer systems, PC-based networks, large-scale data communications systems, software development, and technical support, I have become intimately familiar with computer and networking terminology. The *Dictionary of Networking* is a direct result of that experience, and it represents a practical and down-to-earth approach to computers and computing.

Everyone who has worked on this dictionary has tried to make sure that it is as complete and accurate as possible. But if you think that we have missed a word or two that should be included in a future edition, or if you think that an entry doesn't contain enough information, please write to the following address:

Dictionary of Networking
c/o SYBEX Inc.
1151 Marina Village Parkway
Alameda, CA 94501-1044
USA

Symbols and Numbers

! *See* bang.

& *See* ampersand.

***** *See* asterisk.

. *See* star-dot-star.

. *See* period and double-period directories.

.. *See* period and double-period directories.

/ *See* slash.

// *See* double-slash.

: *See* colon.

<> *See* angle brackets.

> *See* right angle bracket.

? *See* question mark.

@ *See* at symbol.

**** *See* backslash.

| *See* vertical bar.

1/4-inch cartridge *See* quarter-inch cartridge.

2B+D Common abbreviation for Basic Rate ISDN, which has two B, or bearer, channels and one D, or data, channel.

See also 23B+D; Basic Rate ISDN.

3+ A network operating system, originally developed by 3Com, that implemented Xerox Network System (XNS) transport protocols and Microsoft MS-Net file sharing.

3Com Corporation One of the largest manufacturers of network hardware in the world, particularly known for LAN and WAN products, including remote access products, hubs, network interface cards, Gigabit Ethernet, and multimedia over networks. The company's PalmPilot handheld computer has proved to be extremely popular, with more than one million units sold to date. In 1997, 3Com merged with U.S. Robotics in a deal worth \$6.6 billion.

For more information on 3Com, see www.3com.com.

4B/5B encoding A data-translation scheme used to precede signal encoding in FDDI (Fiber Distributed Data Interface) networks. In 4B/5B, each group of four bits is represented as a five-bit symbol, which is then associated with a bit pattern, which in turn is encoded using a standard method, often NRZI (non-return to zero inverted).

See also Manchester encoding.

4.4BSD Lite A version of the 4.4 Berkeley Software Distribution (BSD) Unix from which all the AT&T code has been removed in an attempt to avoid licensing conflicts. It is not possible to compile and then run 4.4BSD Lite without a preexisting system because several important utilities and other files from the operating system are missing.

The 4.4BSD Lite version has served as the basis for several other important Unix implementations, including FreeBSD and NetBSD.

5B/6B encoding

See also Berkeley Software Distribution Unix; FreeBSD; NetBSD; Unix.

5B/6B encoding A data-translation scheme used to precede signal encoding in 100BaseVG networks. In 5B/6B, each group of five bits is represented as a six-bit symbol, which is then associated with a bit pattern, which in turn is encoded using a standard method, often NRZI (non-return to zero inverted).

See also Manchester encoding.

9-track tape A tape storage format that uses nine parallel tracks on 1/2-inch, reel-to-reel magnetic tape. Eight tracks are used for data, and one track is used for parity information. These tapes are often used as backup systems on minicomputer and mainframe systems; digital audio tapes (DATs) are more common on networks.

See also quarter-inch cartridge.

10/100 A term used to indicate that a device can support both Ethernet (at a data transfer rate of 10Mbps) and Fast Ethernet (at a data transfer rate of 100Mbps).

See also Ethernet.

10Base2 An implementation of the 802.3 Ethernet standard on thin Ethernet (RG-58) coaxial cable; sometimes called thinnet or cheapernet wire. The 10Base2 specification has a data-transfer rate of 10Mbps and a maximum cable-segment length of 185 meters (610.5 feet). A T-connector attaches the thin coaxial cable to the BNC connector on the Ethernet adapter.

10Base5 An implementation of the 802.3 Ethernet standard on thick Ethernet

coaxial cable; sometimes called thicknet. The 10Base5 specification has a data-transfer rate of 10Mbps and a maximum cable-segment length of 500 meters (1650 feet), over a bus topology. The cable attaches to the Ethernet adapter with a vampire, or piercing, connector and a transceiver.

See also bus network.

10BaseF An implementation of the 802.3 Ethernet standard over fiber-optic cable. This specification allows throughput of a maximum of 10Mbps and is separated into these three categories:

- **10BaseFP (fiber passive)** Used for desktop connections
- **10BaseFL (fiber link)** For intermediate hubs and workgroups
- **10BaseFB (fiber backbone)** Used for central facility lines between buildings

10BaseT An implementation of the 802.3 Ethernet standard over UTP wiring—the same wiring and RJ-45 connectors used with telephone systems. The standard is based on a star topology, with each node connected to a central wiring center and a maximum cable-segment length of 100 meters (330 feet).

See also star network.

23B+D Common abbreviation for Primary Rate ISDN, which has 23 B, or bearer, channels and 1 D, or data, channel.

See also 2B+D; Primary Rate ISDN.

24/7 Abbreviation for round-the-clock availability, implying that the service is available 24 hours a day, 7 days a week.

56K modem standard *See* V.90.

100Base-FX A specification for Fast Ethernet over fiber-optic cable.

See also Fast Ethernet

100Base-T4 A specification for Fast Ethernet over four pairs of Category 3, 4, or 5 untwisted-pair wiring.

See also Fast Ethernet

100Base-TX A specification for Fast Ethernet over two pairs of Category 5 unshielded twisted-pair or Category 1 shielded twisted-pair cable. The 100Base-TX specification requires two pairs of wires.

See also Fast Ethernet

100VG-AnyLAN A term applied to the IEEE 802.12 standard, originally developed by Hewlett-Packard and supported by Novell, Microsoft, AT&T, and many others.

The 100VG-AnyLAN specification modifies the existing Ethernet standard to allow speeds of 10 or 100Mbps and uses the demand priority access method rather than Carrier Sense Multiple Access/Collision Detection (CSMA/CD). A speed of 100Mbps transmission is supported over Category 3 four-pair unshielded twisted-pair cabling.

See also demand priority; Fast Ethernet.

680x0 A family of 32-bit microprocessors from Motorola, used in Macintosh computers and many advanced workstations. The 680x0 is popular with programmers, because it uses a linear-addressing mode to access memory, rather than the

segmented-addressing scheme used by Intel coprocessors.

Several models have been developed:

- **68000** The first microprocessor in this family, which used a 32-bit data word with a 16-bit data bus and could address 16MB of memory. It was used in the first Apple Macintosh computers as well as in Hewlett-Packard's LaserJet printers.
- **68020** A 32-bit microprocessor that runs at 16MHz, 20MHz, 25MHz, or 33MHz and is capable of addressing as much as 4GB of memory. The 68020 was used in the Macintosh II computer, but has been replaced by the 68030.
- **68030** Has a paged memory management unit built-in and, therefore, does not need external hardware to handle this function. The 68030 is used in the Macintosh II and SE computers.
- **68040** Incorporates a built-in floating-point processor and memory management unit, along with independent 4KB data and instruction caches. It can perform parallel execution by using multiple, independent instruction queues. The 68040 is used in the Macintosh Quadra line of computers.

See also PowerPC.

802.x A set of communications standards defining the physical and electrical connections in LANs, originally defined by the Institute of Electrical and Electronics Engineers (IEEE).

A number followed by an uppercase letter indicates a stand-alone standard; a number followed by a lowercase letter indicates

either a supplement to a standard or a part of a multiple-number standard.

Many of the IEEE standards have also been adopted by the International Organization for Standardization (ISO), whose standards are accepted all over the world; IEEE standards 802.1 to 802.11 are now also known as ISO 8802.1 to 8802.11. You will see both designations in networking books and magazines; it will take a while for everyone to get used to these numbers.

See also IEEE standards.

802.1 An IEEE standard that specifies the media-access-control level for bridges linking 802.3, 802.4, and 802.5 networks. It includes a spanning-tree algorithm for Ethernet media-access-control layer bridges and the heterogeneous LAN management specification for Ethernet and token-ring hubs.

See also IEEE standards.

802.2 An IEEE standard that specifies the logical link sublayer of the data-link layer in the OSI protocol stack. The data-link layer in the OSI protocol stack is divided into the logical link layer and the media-access-control layer. The logical link layer provides framing, addressing, and error-control functions.

See also IEEE standards.

802.3 An IEEE standard for CSMA/CD (Carrier Sense Multiple Access/Collision Detection) LANs, including both baseband and broadband networks. The baseband version is based on the Ethernet network, originally developed by Xerox Corporation.

The 802.3 standard includes the following:

- **10Base2** An implementation of the Ethernet standard on thin Ethernet cable, with a data-transfer rate of 10Mbps and a maximum cable-segment length of 185 meters (600 feet).
- **10Base5** An 802.3 Ethernet standard on thick Ethernet cable, with a 10Mbps data-transfer rate and a cable-segment length of a maximum of 500 meters (1650 feet), over a bus topology.
- **10BaseT** Establishes a standard for Ethernet over UTP wiring, the same wiring and RJ-45 connectors used with telephone systems. The standard is based on a star topology. Each node is connected to a wiring center, with a cable-length limitation of 100 meters (330 feet).
- **1Base5** A 1Mbps network standard with twisted-pair cable based on AT&T's StarLAN.
- **10Broad36** Defines a long-distance Ethernet with a 10Mbps data-transfer rate and a maximum cable-segment length of 3600 meters (11,880 feet).
- **10BaseF** Explicitly specifies fiber-optic cable in three settings; 10Base-FP (fiber passive) for desktops, 10Base-FL (fiber link) for intermediate hubs and workgroups, and 10Base-FB (fiber backbone) for central facility lines between buildings.
- **100BaseVG** A 100Mbps Ethernet network developed by Hewlett-Packard and AT&T Microelectrics.
- **100BaseT** A 100Mbps Ethernet developed by Grand Junction Networks.

See also 100VG-AnyLAN; Ethernet; Fast Ethernet; Gigabit Ethernet; IEEE standards.

802.4 An IEEE standard for bus topology networks that use token passing to control access and network traffic, running at 10Mbps a second. Token-bus networks are sometimes used in manufacturing settings, but they are not often found in office networks.

See also ARCnet; IEEE standards; Technical and Office Protocol.

802.5 An IEEE standard that defines ring networks using token passing to control access and network traffic, running at 4 or 16Mbps. It is used by IBM's Token Ring network over STP, UTP, or fiber-optic cabling. Also known as ANSI 802.1-1985.

See also IEEE standards.

802.6 An IEEE standard for metropolitan-area networks (MANs). It describes a DQDB (Distributed Queue Dual Bus) used for transmitting voice, video, and data over two parallel fiber-optic cables, with signaling rates in excess of 100Mbps per second.

See also IEEE standards; Switched Multi-megabit Data Services.

802.7 An IEEE Technical Advisory Group (TAG) report on broadband networks carrying voice, data, and video traffic.

See also IEEE standards.

802.8 An IEEE Technical Advisory Group (TAG) report on the use of fiber optics as alternatives to copper-based cable in LANs.

See also IEEE standards.

802.9 An IEEE advisory committee on integrated data, voice, and video networking.

The specification has been called IVD (Integrated Voice and Data), but is now more commonly referred to as Iso-Ethernet.

See also IEEE standards; Iso-Ethernet.

802.10 An IEEE Technical Advisory Group (TAG) working on the definition of a standard security model for use over a variety of network types that incorporates authentication and encryption methods.

See also IEEE standards.

802.11 A proposed IEEE standard that will define wireless LANs, including spread-spectrum radio, narrowband radio, infrared transmission, and transmission over power lines.

See also IEEE standards.

802.12 An IEEE working group defining the 100Mbps Ethernet 100VG-AnyLAN originally developed by Hewlett-Packard and several other vendors.

See also 100VG-AnyLAN; IEEE standards.

802.14 An IEEE working group defining standards for data transmission over traditional cable TV networks using cable modems.

See also IEEE standards.

1394 An IEEE standard for a digital plug-and-play bus, originally conceived by Apple Computer in 1986. The 1394 standard supports a maximum of 63 nodes per bus and a maximum of 1023 buses.

Three speeds for device connections are available:

- 100Mbps

- 200Mbps
- 400Mbps

All devices are hot pluggable, and both self-powered and bus-powered devices can be attached to the same bus. Also known as FireWire, 1394 uses six-pair shielded twisted-pair cable and is intended for high-end applications such as digitized video.

See also Universal Serial Bus.

3270 A general description for the family of products from IBM that includes terminals, printers, and terminal cluster controllers. These products all communicate with a mainframe computer using the SNA (Systems Network Architecture) protocol.

80286 Also called the 286. A 16-bit microprocessor from Intel, first released in February 1982, used by IBM in the IBM PC/AT computer in 1984. Since then, it has been used in many other IBM-compatible computers.

The 80286 uses a 16-bit data word and a 16-bit data bus, with 24 bits to address memory. It has two modes:

- Real mode effectively limits performance to that of an 8086 microprocessor and can address 1MB of memory.
- Protected mode prevents an application from stopping the operating system be-

cause of an error and can address 16MB of memory.

80386 Also called the 386DX and the 386. A full 32-bit microprocessor introduced by Intel in October 1985 and used in many IBM and IBM-compatible computers. The 80386 has a 32-bit data word, can transfer information 32 bits at a time over the data bus, and can use 32 bits in addressing memory. It has the following modes:

- Real mode effectively limits performance to that of an 8086 microprocessor and can address 1MB of memory.
- Protected mode prevents an application from stopping the operating system because of an error, and it can address 4GB of memory.
- Virtual 8086 mode allows the operating system to divide the 80386 into several virtual 8086 microprocessors, all running with their own 1MB of space, and all running a separate program.

80486 Also called the 486 or i486. A 32-bit microprocessor introduced by Intel in April 1989. The 80486 adds several notable features to the 80386, including an on-board cache, a built-in floating-point processor, and a memory management unit (MMU), as well as advanced provisions for multiprocessing and a pipelined execution scheme.

A

A+ Certification A certification program from the CompTIA (Computer Technology Industry Association) designed to measure competence in basic computer repair and aimed at the computer technician. Certification requires passing two tests: a core exam to test general knowledge of PCs, including configuration, installation and upgrading, diagnosis, repair, maintenance, customer interaction, and safety, and at least one specialty exam that tests operating system knowledge.

A+B signaling A type of in-band signaling used in T1 transmission; 1 bit from each of the 24 subchannels in every sixth frame is used to carry dialing and other control information. A+B signaling reduces the T1 bandwidth from 1.544Mbps to 1.536 Mbps.

See also T1.

a-b box A switching box that allows two or more computers to share a peripheral device such as a printer. It can be switched manually or through software.

A/UX A version of the Unix operating system that runs on the Macintosh. A/UX is based on the System V release 2 of Unix and includes a number of Apple features, such as support for the Macintosh Toolbox. This support allows applications running under A/UX to use the familiar Macintosh user interface. You need a Macintosh II with a

Motorola 68020 or higher microprocessor and at least 4MB of memory to use A/UX.

See also Unix.

AAL *See* ATM Adaptation Layer.

AASE *See* Associate Accredited Systems Engineer.

ABCP *See* Associate Business Continuity Professional.

abend Contraction of *abnormal end*. A message issued by an operating system when it detects a serious problem, such as a hardware failure or major software damage.

ABI *See* Application Binary Interface.

ABR *See* Available Bit Rate.

ABUI *See* Association of Banyan Users International.

accelerator board An add-in, printed circuit board that replaces the main processor with a higher-performance processor. Using an accelerator board can reduce upgrading costs substantially, because you don't need to replace the monitor, case, keyboard, and so on. However, the main processor is not the only component that affects the overall performance of your system. Other factors, such as disk-access time and video speed, contribute to a system's performance.

See also graphics accelerator board.

access To use, write to, or read from a file, or to log in to a computer system or network.

AccessBuilder Remote access software from 3Com Corporation that lets you access network resources over a dial-up connection from a remote location.

access control entry Abbreviated ACE. The basic unit of security in Microsoft Windows 2000 that controls access to the file system, to Active Directory objects, to printers and other network resources, and to the Registry.

An ACE consists of a security identifier (SID) and an access mask that defines the access rights of that SID. A collection of ACEs that control access to an object is known as an access control list.

See also access control list; security identifier.

access control list Abbreviated ACL. A list or table containing information about the users, processes, and objects that can access a specific file or object. ACLs are usually attached to file-system directories, and they specify access permissions such as read, write, execute, delete, and so on.

ACLs are implemented in Novell NetWare, Microsoft Windows 2000, and Unix:

- In Novell Directory Services, ACLs are associated with every object in the NDS tree, storing the list of rights for each trustee that can access the object.
- In the Unix Network File System, ACLs include the name of the user or group,

along with the rights granted to that user or group.

- In Windows 2000, everything is an object, and every object has an associated ACL.

See also Active Directory; authentication; NDS tree; Novell Directory Services; rights; security.

access method The set of rules that determines which node in a network has access to the transmission media at any moment.

Attempts at simultaneous access are either managed by a collision detection mechanism such as CSMA/CD or prevented by use of a token-passing method.

access protocol The set of rules that workstations use to avoid collisions when sending information over shared network media. Also known as the media access control protocol.

access rights *See* rights.

access server A computer that provides access for remote users who dial in to the system and access network resources as though their computers were directly attached to the network.

See also communications/modem server; mobile computing.

access time The period of time that elapses between a request for information from disk or memory and the arrival of that information at the requesting device.

Memory-access time refers to the time it takes to transfer a character between memory and the processor. Disk-access time

refers to the time it takes to place the read/write heads over the requested data. RAM may have an access time of 80 nanoseconds or less, while hard-disk access time could be 10 milliseconds or less.

access token In Microsoft Windows 2000, an object that contains the security identifier (SID) of a running process.

When a process is started by another process, the second process inherits the starting process's access token. This access token is then checked against each object's access control list to confirm that the appropriate permissions are in place to permit any services requested by the process.

See also access control list; permissions; process; rights.

account On LANs or multiuser operating systems, an account is set up for each user. Accounts are usually kept for administrative or security reasons. For communications and online services, accounts identify a subscriber for billing purposes.

See also user account.

accounting The process of tracking the resources on a network. The network administrator can charge for files accessed, connect time, disk space used for file storage, and service requests by assigning account balances to users. The users can then draw from their account balances as they use network services.

account lockout In Microsoft Windows 2000 and other operating systems, a count of the number of invalid logon attempts allowed before a user is locked out.

See also authentication; user account.

account policy On networks and multiuser operating systems, the set of rules that defines whether a new user is permitted access to the system and whether an existing user is granted additional rights or expanded access to other system resources. Account policy also specifies the minimum length of passwords, the frequency with which passwords must be changed, and whether users can recycle old passwords and use them again.

Accredited Systems Engineer Abbreviated ASE. A certification from Compaq designed to evaluate and recognize expertise in installing and administering Compaq workstations and servers running both Microsoft Windows 2000 and Novell NetWare network operating systems.

See also Associate Accredited Systems Engineer.

ACE *See* access control entry.

Acer Group One of the top five PC makers in the world, with factories in Malaysia, the Netherlands, the Philippines, Taiwan, and the United States. The Acer Group bought Texas Instrument's notebook computer business in 1997 and has formed business alliances with companies, including 3Com and Hitachi, for the development of advanced digital consumer products such as PC-ready televisions and DVD systems.

For more information on the Acer Group, see www.acer.com.tw.

ACF *See* Advanced Communications Function.

ACK *See* acknowledgment.

acknowledgment Abbreviated ACK. In communications, ACK is a control code, ASCII 06, sent by the receiving computer to indicate that the data has been received without error and that the next part of the transmission may be sent.

See also NAK.

ACL *See* access control list.

ACONSOLE A Novell NetWare 3.x workstation utility that controls a modem attached to the workstation. ACONSOLE is used to establish an asynchronous remote console connection to a server. The RS232 NetWare Loadable Module (NLM) must be loaded on the server to which you want to connect. In NetWare 4.x, use RCONSOLE to perform this function.

acoustic coupler A modem that includes a pair of rubber cups that fit over the mouthpiece and earpiece of a standard telephone handset (to prevent external noise from being picked up). An acoustic coupler allows you to connect your computer to a telephone system that does not have the standard RJ-11 connections used with conventional modems.

ACPI *See* Advanced Configuration and Power Interface.

ACR *See* Available Cell Rate.

across-the-wire migration A method of migrating file-server data, trustee rights, and other information to a Novell NetWare server using the NetWare Migration utility. You can also use across-the-wire

migration to upgrade from LAN Manager, LAN Server, and earlier versions of NetWare; a similar process known as BMI-GRATE allows users to migrate from Banyan VINES.

ACS *See* Advanced Communications Service.

ACTE *See* Ascend Certified Technical Expert.

Active Desktop In Microsoft Windows, a user interface feature that makes the Desktop look and behave just like a Web page, with underlined icons and a hand mouse pointer. Active Desktop is offered as an alternative to the classic Windows user interface; you can use Active Desktop, you can use the classic Windows interface, or you can swap back and forth between the two.

Active Directory In Microsoft Windows 2000, a system for large-scale network management that views the network as a hierarchy of objects. Active Directory does the following:

- Provides a hierarchy for the management of all network objects, including users, servers, services, file shares, Web pages, printers, and so on.
- Divides administration and security into subdomains, domains, and trees of domains.
- Scales to 10 million users per domain.
- Implements MIT's Kerberos authentication system based on private key encryption and also supports public key encryption for authentication of clients and business partners.

- Emulates Windows NT 4.x directory services for backward compatibility.
- Uses DNS rather than WINS, and requires all user and host names to be in DNS form.
- Uses LDAP rather than a proprietary protocol so that non-Microsoft applications can query the name database.
- Interoperates with Novell NetWare Directory Services.

See also forest; Kerberos; Lightweight Directory Access Protocol; NetWare Directory Services; Microsoft Windows 2000; tree.

Active Directory Service Interface

Abbreviated ADSI. In Microsoft Active Directory, an application programming interface (API) designed to simplify access to Active Directory objects.

See also Active Directory; application programming interface.

Active Directory Users and Computers

Abbreviated ADUC. In Microsoft Windows 2000 Server, the main administrative tool used to manage user accounts, security groups, organizational units, and policies.

ADUC is a Microsoft Management Console (MMC) snap-in and replaces Windows NT User Manager.

See also Active Directory; Microsoft Management Console; snap-in.

active hub A device that amplifies transmission signals in a network, allowing signals to be sent over a much greater distance than is possible with a passive hub.

An active hub may have ports for coaxial, twisted-pair, or fiber-optic cable connections, as well as LEDs to show that each port is operating correctly.

See also repeater.

Active Monitor The first station to be started on a Token Ring network. The Active Monitor is responsible for passing and maintaining the token and detects error conditions. The Active Monitor's performance is constantly monitored by the Standby Monitor

See also Standby Monitor.

Active Server Pages Abbreviated ASP. In Microsoft Internet Information Server, a script interpreter and execution environment that supports VBScript and JavaScript and is compatible with other scripting languages such as Perl, REXX, Tcl, and Python through add-ins from third-party developers.

ASP allows you to combine HTML, scripts, and ActiveX components on the same Web server; all the code runs on the server and presents the results of this dynamic process to the client browser as a standard HTML page.

See also JavaScript; Perl; Tcl; VBScript.

active termination A technique used to terminate a SCSI. Active termination reduces electrical interference in a long string of SCSI devices.

See also forced perfect termination; passive termination.

active window The window currently accepting mouse clicks and keyboard input. The title bar of the active window is always a different color from that of the other open windows.

ActiveX The latest development of Microsoft's COM, the foundation that supported OLE. By adding network capabilities (and so creating DCOM, or Distributed COM) and by reducing the scope of OLE to create ActiveX, Microsoft has created a comprehensive set of component-based Internet- and intranet-oriented applications.

In an attempt to promote ActiveX as a standard, in 1996 Microsoft turned over control of ActiveX to the Open Group to manage future developments.

See also ActiveX control; Distributed Component Object Model; Java.

ActiveX control The basic building block of Microsoft's ActiveX specification. An ActiveX control is a software module that cannot run by itself, but requires an ActiveX container such as a Web browser, a word processor, or a spreadsheet. Each control delivers a specific function such as database access, user-interface elements, or file access and can communicate with another ActiveX control, an ActiveX container, or the underlying Windows operating system.

Unlike Java applets, which for security reasons run in a sandbox designed to protect the file system from unauthorized access, ActiveX can directly access files. To provide a measure of security, ActiveX controls are packaged with digital certificates that prove the origin of the control.

See also ActiveX; certificate; Java; Java applet; Java Virtual Machine; sandbox.

AD *See* Active Directory.

Adaptec, Inc. A leading manufacturer of high-performance networking and connectivity products, including SCSI adapters, RAID products, Fast Ethernet adapters, ATM network interface cards, and server management software. In 1998, Adaptec acquired Ridge Technologies, a manufacturer of RAID and other storage solutions for Microsoft Windows 2000.

For more information on Adaptec, see www.adaptec.com.

adapter A printed circuit board that plugs into a computer's expansion bus to provide added capabilities.

Common adapters include video adapters, joy-stick controllers, and I/O adapters, as well as other devices, such as internal modems, CD-ROMs, and network interface cards. One adapter can often support several different devices. Some of today's PC designs incorporate many of the functions previously performed by these individual adapters on the motherboard.

adaptive equalization *See* adaptive speed leveling.

adaptive routing A mechanism that allows a network to reroute messages dynamically, using the best available path, if a portion of the network fails.

See also alternative route.

adaptive speed leveling A modem technology that allows a modem to respond

to changing line conditions by changing its data rate. As line quality improves, the modem attempts to increase the data rate; as line quality declines, the modem compensates by lowering the data rate. Also known as adaptive equalization.

ADCCP See Advanced Data Communications Control Procedures.

address 1. The precise location in memory or on disk where a piece of information is stored. Each byte in memory and each sector on a disk has its own unique address.

2. The unique identifier for a specific node on a network. An address may be a physical address specified by switches or jumpers on the network interface card hardware, or it can be a logical address established by the network operating system.

3. To reference or manage a storage location.

4. In Unix, an IP address as specified in the `/etc/hosts` file.

5. Information used by a network or the Internet to specify a specific location in the form `username@hostname`; `username` is your user name, logon name, or account name or number, and `hostname` is the name of the Internet Service Provider (ISP) or computer system you use. The `hostname` may consist of several parts, each separated from the next by a period.

See also address bus; Domain Name Service; e-mail address; Internet Service Provider; IP address; memory address.

address bus The electronic channel, usually from 20 to 64 lines wide, used to

transmit the signals that specify locations in memory.

The number of lines in the address bus determines the number of memory locations that the processor can access, because each line carries one bit of the address. A 20-line address bus (used in early Intel 8086/8088 processors) can access 1MB of memory, a 24-line address bus can access 16MB, and a 32-line address bus can access more than 4GB. A 64-line address bus (used in the DEC Alpha APX) can access 16EB.

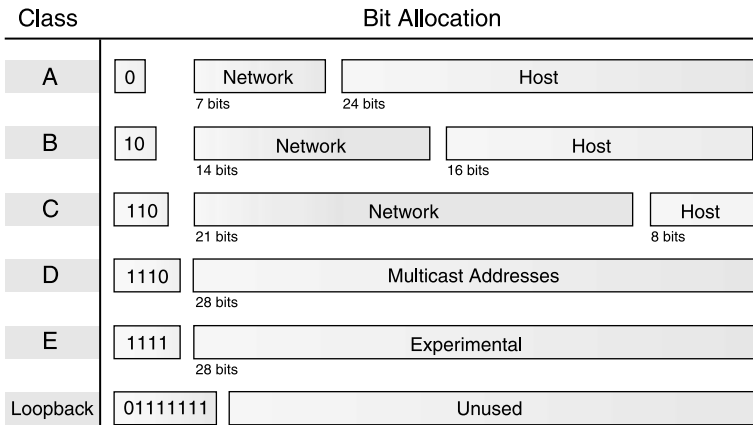
address classes In a 32-bit IP address, which is shown in the accompanying illustration, the number of bits used to identify the network and the host vary according to the network class of the address, as follows:

- Class A is used only for very large networks. The high-order bit in a Class A network is always zero, leaving 7 bits available to define 127 networks. The remaining 24 bits of the address allow each Class A network to hold as many as 16,777,216 hosts. Examples of Class A networks include General Electric, IBM, Hewlett-Packard, Apple Computer, Xerox, Digital Equipment Corporation, and MIT. All the Class A networks are in use, and no more are available.
- Class B is used for medium-sized networks. The 2 high-order bits are always 10, and the remaining bits are used to define 16,384 networks, each with as many as 65,535 hosts attached. Examples of Class B networks include Microsoft and Exxon. All Class B networks are in use, and no more are available.

addressing space

- Class C is for smaller networks. The 3 high-order bits are always 110, and the remaining bits are used to define 2,097,152 networks, but each network can have a maximum of only 254 hosts. Class C networks are still available.
 - Class D is a special multicast address and cannot be used for networks. The 4 high-order bits are always 1110, and the remaining 28 bits allow for more than 268 million possible addresses.
 - Class E is reserved for experimental purposes. The first four bits in the address are always 1111.
- See also Classless Inter-Domain Routing; IP address; subnet mask.

IP ADDRESS STRUCTURE



addressing space The amount of RAM available to the operating system running on a server.

address mask See subnet mask.

Address Resolution Protocol Abbreviated ARP. A protocol within TCP/IP (Transmission Control Protocol/Internet Protocol) and AppleTalk networks that allows a host to find the physical address of a

node on the same network when it knows only the target’s logical or IP address.

Under ARP, a network interface card contains a table (known as the address resolution cache) that maps logical addresses to the hardware addresses of nodes on the network. When a node needs to send a packet, it first checks the address resolution cache to see if the physical address information is already present. If so, that address is used, and network traffic is reduced; otherwise, a

normal ARP request is made to determine the address.

See also IP address.

address translation gateway Abbreviated ATG. A function in Cisco DECnet routing software capable of establishing a user-specified address translation mechanism for certain selected nodes on the network.

adjacency A term describing the relationship formed between certain neighboring routers for the purpose of swapping routing information. Adjacency is based on the use of a common network segment.

administrative alerts In Windows 2000, informational messages sent to specific accounts, groups, or computers to announce security events, impending shutdown due to loss of server power, performance problems, and printer errors.

When a server generates an administrative alert, the appropriate message is sent to a predefined list of users and computers.

See also Alerter service.

administrative distance A term used by Cisco Systems, Inc., to express the integrity of a routing-information source. Administrative distance is expressed as a value in the range 0 through 255; the higher the value, the lower the quality of the routing information.

Administrator account In Microsoft Windows 2000, a special account with the maximum authority and permissions that can assign any permission to any user or group.

The Administrator account cannot be deleted, but it can be renamed, which is probably a good security policy.

See also permissions.

ADMIN object A NetWare Directory Services User object, created during the installation of NetWare, that has special privileges, including the supervisory rights to create and manage other objects.

ADMIN has Supervisor rights and can, therefore, manage the NetWare Directory Services tree and add or delete Directory objects.

ADSI *See* Active Directory Service Interface.

ADSL *See* Asymmetric Digital Subscriber Line.

ADUC *See* Active Directory Users and Computers.

Advanced Communications

Function Abbreviated ACF. A set of program packages from IBM that allows computer resources to be shared over communications links using the concepts of SAA (Systems Application Architecture).

For example, ACF/TCAM (Advanced Communications Functions/Telecommunications Access Method) and ACF/VTAM (Advanced Communications Functions/Virtual Telecommunications Access Method) allow the interconnection of two or more domains into one multiple-domain network.

Advanced Communications Service

Abbreviated ACS. A large data-communications network established by AT&T.

Advanced Configuration and Power Interface Abbreviated ACPI. An interface specification developed by Intel, Microsoft, and Toshiba for controlling power use on the PC and all other devices attached to the system. A BIOS-level hardware specification, ACPI depends on specific hardware that allows the operating system to direct power management and system configuration.

See also Advanced Power Management.

Advanced Data Communications Control Procedures Abbreviated ADCCP. A bit-oriented, link-layer, ANSI-standard communications protocol.

See also High-level Data Link Control.

Advanced Interactive Executive Abbreviated AIX. A version of Unix from IBM that runs on its RS/6000 workstations and on minicomputers and mainframes.

Although AIX is based on Unix System V Release 3, it contains many of the features available in System V Release 4, is POSIX-compliant, and meets the Trusted Computer Base (TCB) Level C2 security.

One of the major enhancements of AIX is Visual Systems Management (VSM), a graphical interface into the older Systems Management Interface Tool (SMIT). VSM contains four main elements: Print Manager, Device Manager, Storage Manager, and Users and Groups Manager.

Advanced Micro Devices, Inc. Abbreviated AMD. The fifth largest manufacturer of integrated circuits, flash memory, and microprocessors, specializing in clones of Intel's popular PC chips, including the

AMD386, AMD486, AMDK5, and the AMDK6.

For more information about AMD, see www.amd.com.

See also Cyrix; Intel Corporation; Pentium; Pentium II; Pentium III.

Advanced Mobile Phone Service Abbreviated AMPS. Currently the cellular telephone standard in the United States; an analog, cellular communications system developed by AT&T. AMPS uses frequency-division multiplexing (FDM) and operates in the 825 to 890MHz range.

See also Cellular Digital Packet Data.

Advanced Peer-to-Peer Internet-working An SNA routing scheme proposed by Cisco Systems and eventually abandoned.

Advanced Peer-to-Peer Networking Abbreviated APPN. IBM's SNA (Systems Network Architecture) protocol, based on APPC (Advanced Program-to-Program Communications). APPN allows nodes on the network to interact without a main-frame host computer and implements dynamic network directories and dynamic routing in an SNA network.

APPN can run over a variety of network media, including Ethernet, token ring, FDDI, ISDN, X.25, SDLC, and higher-speed links such as B-ISDN or ATM.

See also Asynchronous Transfer Mode; Customer Information Control System; Systems Network Architecture.

Advanced Power Management Abbreviated APM. An API specification from

Microsoft and Intel intended to monitor and extend battery life on a laptop computer by shutting down certain system components after a period of inactivity.

See also Advanced Configuration and Power Interface.

Advanced Program-to-Program

Communications Abbreviated APPC. A set of protocols developed by IBM as a part of its SNA (Systems Network Architecture), designed to allow applications running on PCs and mid-range hosts to exchange data easily and directly with mainframes. APPC can be used over an SNA, Ethernet, X.25, or Token Ring network and is an open, published communications protocol.

APPC/PC is a PC-based version of APPC used over a Token Ring network.

advanced run-length limited

encoding Abbreviated ARLL. A technique used to store information on a hard disk that increases the capacity of run-length limited (RLL) storage by more than 25 percent and increases the data-transfer rate to 9Mbps.

See also RLL encoding.

Advanced Technology Attachment

Abbreviated ATA. The ANSI X3T10 standard for the disk-drive interface usually known as Integrated Drive Electronics (IDE).

See also Integrated Drive Electronics.

advertising The process by which services on a network inform other devices on the network of their availability. Novell Net-

Ware uses the Service Advertising Protocol (SAP) for this purpose.

AFP *See* AppleTalk Filing Protocol.

AFS *See* Andrews File System.

aftermarket The market for related hardware, software, and peripheral devices created by the sale of a large number of computers of a specific type.

agent 1. A program that performs a task in the background and informs the user when the task reaches a certain milestone or is complete.

2. A program that searches through archives looking for information specified by the user. A good example is a spider that searches Usenet articles. Sometimes called an intelligent agent.

3. In SNMP (Simple Network Management Protocol), a program that monitors network traffic.

4. In client-server applications, a program that mediates between the client and the server.

Aggregate Route-Based IP

Switching Abbreviated ARIS. A scheme from IBM used to establish switched paths through networks that act as virtual circuits, transmitting data packets through the network without the need to make routing decisions at every step. ARIS uses tagging techniques to add information to the data packets that can be used to guide the packets through the virtual circuits based on information already established by protocols such as Open Shortest Path First (OSPF) and Border Gateway Protocol (BGP).

AIX *See* Advanced Interactive Executive.

alert log In Microsoft Windows 2000, a System Monitor log used to monitor counters that exceed user-specified limits. When such a limit is exceeded, the event is logged into the alert log.

See also counter log; System Monitor; trace log.

Alerter service A Microsoft Windows 2000 service that warns a predefined list of users and computers of an administrative alert. The Alerter service is used by the Server service and requires the Messenger service.

See also administrative alerts; Messenger service; service.

Alias object In Novell NetWare, a leaf object that references the original location of an object in the directory. Using Alias objects, one object can appear in several containers at the same time, allowing users to locate and use the object quickly and easily.

See also leaf object.

Allowed Cell Rate *See* Available Cell Rate.

alphanumeric Consisting of letters, numbers, and sometimes special control characters, spaces, and other punctuation characters.

See also American Standard Code for Information Interchange; Extended Binary Coded Decimal Interchange Code; Unicode.

alpha testing The first stage in testing a new hardware or software product, usually performed by the in-house developers or programmers.

See also beta testing.

alternative route A secondary communications path to a specific destination. An alternative route is used when the primary path is not available.

See also adaptive routing.

alt newsgroups A set of Usenet newsgroups containing articles on controversial subjects often considered outside the mainstream. Alt is an abbreviation for alternative.

These newsgroups were originally created to avoid the rigorous process required to create an ordinary newsgroup. Some alt newsgroups contain valuable discussions on subjects ranging from agriculture to wolves, others contains sexually explicit material, and others are just for fun. Not all ISPs and online services give access to the complete set of alt newsgroups.

See also mailing list; moderated newsgroup; newsgroup; unmoderated newsgroup.

AMD *See* Advanced Micro Devices, Inc.

American National Standards

Institute Abbreviated ANSI. A nonprofit organization of more than 1000 business and industry groups, founded in 1918, devoted to the development of voluntary standards.

ANSI represents the United States in the International Organization for Standardization (ISO) and is affiliated with CCITT.

ANSI committees have developed many important standards, including the following:

- ANSI X3J11: Standard for the C programming language, including language semantics, syntax, execution environment, and the definition of the library and header files.
- ANSI X3J16: Standard for the C++ programming language.
- ANSI X3J3: Definition of the Fortran programming language compiler.
- ANSI X3.131-1986: Definition of the SCSI standard. The X3T9.2 standard contains the extensions for SCSI-2.
- ANSI X3T9.5: The working group for the FDDI definition.

American Standard Code for Information Interchange Abbreviated ASCII, pronounced “as-kee.” A standard coding scheme that assigns numeric values to letters, numbers, punctuation characters, and control characters to achieve compatibility among different computers and peripheral devices. In ASCII, each character is represented by a unique integer value in the range 0 through 255. See Appendix C.

See also ASCII extended character set; ASCII file; ASCII standard character set; double-byte character set; Extended Binary Coded Decimal Interchange Code; Unicode.

American Wire Gauge Abbreviated AWG. A measurement system that specifies copper wire by thickness; as thickness increases, the AWG number decreases. Some common conductor gauges are:

- RS-232-C: 22 or 24 AWG

- Thick Ethernet: 12 AWG
- Thin Ethernet: 20 AWG

See also cabling standards; EIA/TIA 586; Type 1–9 cable.

America Online, Inc. Abbreviated AOL. The world’s largest online service, headquartered in Vienna, Virginia, with more than 15 million subscribers. AOL provides a gateway to the Internet, as well as its own news, sports, e-mail, chat rooms, and other fee-based services. In 1997, AOL bought CompuServe, and in 1998, Netscape Communications.

For more information about America Online, see www.aol.com.

ampersand (&) 1. In Unix, a command suffix used to indicate that the preceding command should be run in the background.

2. In Unix, a root user command used to start a daemon that is to keep running after logout.

3. In HTML, a special character entry in a document.

See also daemon; HyperText Markup Language.

AMPS *See* Advanced Mobile Phone Service.

analog Describes any device that represents changing values by a continuously variable physical property, such as a voltage in a circuit. Analog often refers to transmission methods developed to transmit voice signals rather than high-speed digital signals.

Andrews File System Abbreviated AFS. A protocol developed at Carnegie Mellon University; used to share remote files across systems using TCP/IP.

AFS has certain advantages over NFS in that it only allows users to access files linked to AFS rather than giving access to all files, it has a built-in cache that helps to reduce the demands made on the system, and system administrators can allocate disk space on the fly as required.

See also Distributed File System; Network File System; Transmission Control Protocol/Internet Protocol.

angle brackets The less-than (<) and greater-than (>) symbols used to identify a tag in an HTML document.

Also used to identify the return address in an e-mail message header.

See also HyperText Markup Language; tag.

ANI *See* automatic number identification.

anonymous FTP A method used to access an Internet computer that does not require you to have an account on the target computer system. Simply log on to the Internet computer with the user name *anonymous*, and use your e-mail address as your password. This access method was originally provided as a courtesy so that system administrators could see who had logged on to their systems, but now it is often required to gain access to an Internet computer that has FTP service.

You cannot use anonymous FTP with every computer on the Internet, only with those systems set up to offer the service. The system administrator decides which files and directories will be open to public access, and the rest of the system is considered off limits and cannot be accessed by anonymous FTP users. Some sites only allow you to download files; as a security precaution, you are not allowed to upload files.

See also File Transfer Protocol; Telnet.

anonymous posting In a Usenet newsgroup, a public message posted via an anonymous server in order to conceal the identity of the original sender.

anonymous remailer *See* anonymous server.

anonymous server A special Usenet service that removes from a Usenet post all header information that could identify the original sender and then forwards the message to its final destination. If you use an anonymous server, be sure to remove your signature from the end of the message; not all anonymous servers look for and then strip a signature. Also known as an anonymous remailer.

ANSI *See* American National Standards Institute.

answer mode A function that allows a modem to answer an incoming call, detect the protocol being used by the calling modem, and synchronize with that protocol.

See also auto-answer; auto-dial.

antivirus program A program that detects or eliminates a computer virus. Some antivirus programs can detect suspicious activity on your computer as it happens; others must be run periodically as part of your normal housekeeping activities.

An antivirus program locates and identifies a virus by looking for characteristic patterns or suspicious activity in the system, such as unexpected disk access or .EXE files changing in some unusual way. It recognizes the virus by comparing information from the system against a database of known viruses, which is kept on disk.

Be sure you test an antivirus program carefully on your network before you employ it everywhere; some programs impose an enormous overhead on normal network operations.

See also file-infecting virus; macro virus; multipart virus; polymorphic virus; stealth virus; Trojan Horse; vaccine; virus.

AnyNet A family of gateway products from IBM used to integrate SNA, TCP/IP, and NetBIOS networks with products running on IBM's AIX/6000, OS/2, and OS/400 and with Microsoft Windows.

AOL *See* America Online, Inc.

Apache HTTP Server A freeware Web server, supported by the Unix community, in use on almost half of the Web sites on the Internet. So called because the original university-lab software was patched with new features and fixes until it became known as "a patchy server."

Apache dominates the Web because of its low cost, excellent performance, good scalability, and great flexibility. Don't expect easy graphical configuration programs and hypertext help; you'll get the command line and the man pages instead, so it certainly helps to have staff with Unix experience.

Apache Server is available as part of the Red Hat Software Linux distribution, which also provides developers with full support for CGI, Perl, Tcl, a C or C++ compiler, an Apache server API, and a SQL database.

For more information on Apache Server, see The Apache Group Web site at www.apache.org.

See also Linux; Red Hat Software.

API *See* application programming interface.

APM *See* Advanced Power Management.

app *See* application.

APPC *See* Advanced Program-to-Program Communications.

APPI *See* Advanced Peer-to-Peer Inter-networking.

Apple Computer, Inc. Manufacturer of the successful Macintosh and Quadra series of computers based on Motorola chips. The company was founded by Steve Wozniak and Steve Jobs in a garage on April 1, 1976.

In 1993, Apple entered the consumer electronics marketplace with a personal digital assistant known as Newton, combining fax, electronic mail, and other functions into a unit small enough to fit into a pocket.

In 1994, Apple launched a new series of computers called the Power Macintosh (or Power Mac), based on the PowerPC, capable of running either the Macintosh operating system or Windows programs under software emulation.

Apple always kept the architecture of the Mac proprietary, a move that has cost the company considerable market share; nevertheless, Apple has always had a strong following among musicians and graphical designers.

In 1997, Steve Jobs rejoined Apple and, after realigning Apple's product line, led the development and launch of the popular and capable Internet-ready iMac computer.

For more information on Apple Computer, Inc., see www.apple.com.

Apple Desktop Bus A serial communications link that connects low-speed input devices, such as a mouse or a keyboard, to the computer on the Macintosh SE, II, IIfx, IIfx, and SE/30.

Light pens, trackballs, and drawing tablets may also be connected via the Apple Desktop Bus. Most Apple Desktop Bus devices allow one device to be daisy-chained to the next, up to a maximum of 16 devices.

Apple Macintosh See Macintosh.

AppleShare Network software from Apple Computer that requires a dedicated Macintosh computer acting as a centralized server and includes both server and workstation software. AppleShare uses the AppleTalk Filing Protocol (AFP).

AppleTalk An Apple Macintosh network protocol, based on the OSI Reference Model, which gives every Macintosh networking capabilities. AppleTalk can run under several network operating systems, including Apple Computer's AppleShare, Novell NetWare for the Macintosh, and TOPS from Sun Microsystems.

AppleTalk includes specifications for the data-link layer as LocalTalk, EtherTalk, FDDI-Talk, or TokenTalk, and the network layer as Datagram Delivery Protocol. The transport layer contains four protocols:

- Routing Table Maintenance Protocol (RTMP)
- AppleTalk Echo Protocol (AEP)
- AppleTalk Transaction Protocol (ATP)
- Name Binding Protocol (NBP)

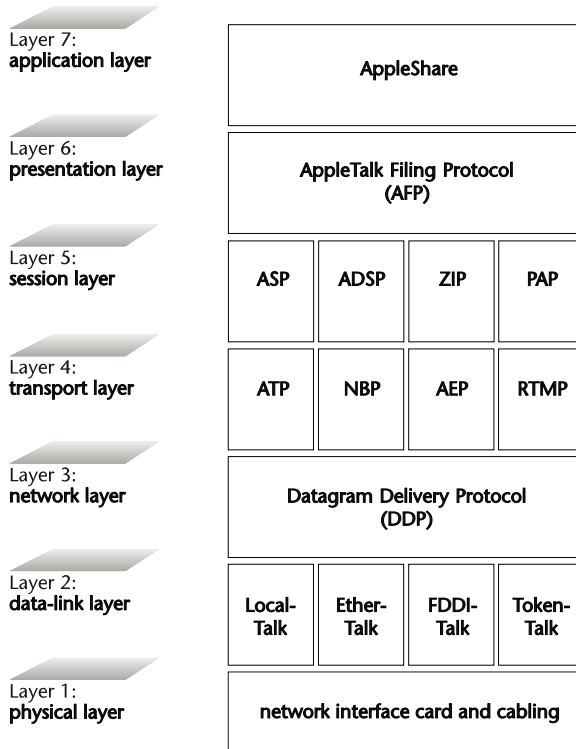
The session layer includes

- AppleTalk Data Stream Protocol (ADSP)
- AppleTalk Session Protocol (ASP)
- Printer Access Protocol (PAP)
- Zone Information Protocol (ZIP)

The presentation layer adds the AppleTalk Filing Protocol (AFP) for access to remote files on shared disks.

AppleTalk Filing Protocol Abbreviated AFP. AFP is located in the presentation and application layers of the AppleTalk protocol stack. AFP lets users access remote files as though they were local, as well as providing security features that can restrict user access to certain files.

APPLETALK PROTOCOL STACK



application Abbreviated app, or if the application is a small one, it is referred to as an applet. A computer program designed to perform a specific task, such as accounting, scientific analysis, word processing, or desktop publishing.

In general, applications can be distinguished from system software, system utilities, and computer language compilers, and they can be categorized as either stand-alone or

network applications. Stand-alone applications run from the hard disk in an independent computer, so only one user at a time can access the application. Network applications run on networked computers and can be shared by many users. Advanced applications such as groupware and e-mail allow communications between network users.

See also application metering; client/server architecture; LAN-aware.

Application Binary Interface

Application Binary Interface Abbreviated ABI. A specification that aims to ensure binary compatibility between applications running on the same family of processors or CPUs using Unix System V Release 4.

Applications developed using ABI can run on hardware from different manufacturers without being recompiled; any system calls needed for specific hardware are maintained in libraries.

The specification was originally developed by AT&T and Sun Microsystems and includes a test and verification suite used to determine if a system complies with the standard.

See also application programming interface.

application layer The seventh, or highest, layer in the OSI Reference Model for computer-to-computer communications. This layer uses services provided by the lower layers but is completely insulated from the details of the network hardware. It describes how applications interact with the network operating system, including database management, electronic mail, and terminal emulation programs.

See also OSI Reference Model.

application-level filter A category of firewall that provides a high degree of security but at the cost of lower speed and greater complexity. Typical application-level filters can provide proxy services for applications and protocols such as Telnet, FTP, HTTP, and SMTP.

See also firewall; packet-level filter; stateless filter.

application log In Microsoft Windows 2000, a system log that contains events generated by applications or by alert logs. You can use the Event Viewer to display the contents of the application log.

See also Event Viewer; security log; system log.

application metering The process of counting the number of executions of the copies of an application in use on the network at any given time and ensuring that the number does not exceed preset limits.

Application metering is usually performed by a network management application running on the file server. Most application metering software will allow only a certain number of copies (usually that number specified in the application software license) of an application to run at any one time and will send a message to any users who try to exceed this limit.

See also concurrent license.

Application object In Novell Directory Services (NDS), a leaf object that represents a network application in a NetWare Directory tree.

See also Computer object; container object; leaf object; Novell Directory Services.

application programming interface Abbreviated API. The complete set of all operating system functions that an application can use to perform such tasks as managing files and displaying information.

An API provides a standard way to write an application, and it also describes how the application should use the functions it provides. Using an API is quicker and easier

than developing functions from scratch and helps to ensure some level of consistency among all the applications developed for a specific operating system.

In operating systems that support a graphical user interface, the API also defines functions to support windows, icons, drop-down menus, and other components of the interface. In network operating systems, an API defines a standard method that applications can use to take advantage of all the network features.

application server A special-purpose file server that is optimized for a specific task, such as communications or a database application, and that uses higher-end hardware than a typical file server.

See also superserver.

application-specific integrated circuit Abbreviated ASIC. A computer chip developed for a specific purpose, designed by incorporating standard cells from a library rather than created from scratch. Also known as gate arrays, ASICs are found in all sorts of appliances, including modems, security systems, digital cameras, and even microwave ovens and automobiles.

APPN *See* Advanced Peer-to-Peer Networking.

ARB *See* Area Border router.

arbitration The set of rules used to manage competing demands for a computer resource, such as memory or peripheral devices, made by multiple processes or users.

See also contention.

Archie A system used on the Internet to locate files available by anonymous FTP. Once a week, special programs connect to all the known anonymous FTP sites on the Internet and collect a complete listing of all the publicly available files. This listing of files is kept in an Internet Archive Database, and when you ask Archie to look for a file, it searches this database rather than the whole Internet; you then use anonymous FTP to retrieve the file.

See also anonymous FTP.

architecture 1. The overall design and construction of all or part of a computer, particularly the processor hardware and the size and ordering sequence of its bytes.

2. The overall design of software, including interfaces to other software, the operating system, and to the network.

See also client/server architecture; closed architecture; complex instruction set computing; open architecture; reduced instruction set computing.

archive 1. To transfer files to some form of long-term storage, such as magnetic tape or large-capacity disk, when the files are no longer needed regularly but must be maintained for periodic reference.

2. On the Internet, a site containing a collection of files available via anonymous FTP.

3. A compressed file.

archive file A single file that contains one or more files or directories that may have been compressed to save space. Archives are often used as a way to transport large numbers of related files across the Internet.

An archive file created under Unix may have the filename extension .TAR (for tape archive), .GZ (for gzip), or .Z (for compress or pack). Those created in Windows may have the filename extension .ZIP from the PKZIP or WinZip programs. Archive files created on a Macintosh will have the filename extension .SAE or .SIT from the StuffIt program.

An Internet host that provides access to large numbers of archive files is known as an archive site.

ARCNet Acronym for Attached Resources Computing Network. A network available from the Datapoint Corporation and other vendors that can connect a wide variety of PCs and workstations (up to a maximum of 255) on coaxial, twisted-pair, or fiber-optic cable. ARCnet uses a proprietary token-passing access method at speeds of 2.5Mbps. ARCNet Plus is Datapoint's proprietary product that runs at 20Mbps.

ARCNet was popular for smaller networks, because it is relatively easy to set up and to operate and also because the components are inexpensive and widely available. These days, however, it is showing its age and is no longer sold by the major vendors.

See also Token Ring network.

Area Border router Abbreviated ARB. In an internetwork that uses link state routing protocols such as Open Shortest Path First (OSPF) protocols, a router that has at least one connection to another router in a different part of the internetwork. Also known as a Border router.

See also Autonomous System Border router; Open Shortest Path First.

ARIS *See* Aggregate Route-Based IP Switching.

ARLL *See* advanced run-length limited encoding.

ARP *See* Address Resolution Protocol.

ARPAnet Acronym for Advanced Research Projects Agency Network. A research network funded by the Defense Advanced Research Projects Agency (DARPA) to link universities and government research agencies, originally built by BBN, Inc., in 1969. It was the backbone for the now huge Internet. TCP/IP protocols were pioneered on ARPAnet. In 1983, the military communications portion was split off into the MILnet.

article An e-mail message posted to one of the Usenet newsgroups, accessible by anyone with a newsreader and a connection to the Internet. Also called a post.

See also newsreader.

AS/400 A series of mid-range mini-computers from IBM, first introduced in 1988, that replaces the System/36 and System/38 series of computers. The AS/400 can serve in a wide variety of network configurations: as a host or an intermediate node to other AS/400 and System/3x computers, as a remote system to System/370-controlled networks, or as a network server to a group of PCs.

ASBR *See* Autonomous System Border router.

Ascend Certified Technical Expert Abbreviated ACTE. A certification scheme from Ascend Communications, Inc., aimed

at technical professionals with experience in installing, configuring, and troubleshooting Ascend remote-access products. Two written exams, the Networking and Telecommunications exam, and the Remote Access exam, are followed by a hands-on lab test administered at Ascend's headquarters in Alameda, CA.

Ascend Communications, Inc. A leading provider of solutions for telecommunications carriers, ISPs, and corporate customers, Ascend manufactures products for remote access, for wide area networking, and for linking telephone switches, network connections, and videoconferencing facilities to phone company networks.

Ascend recently acquired Cascade Communications, establishing the company as one of the largest suppliers of frame-relay and ATM equipment. In 1999, Ascend was bought by Lucent Technologies.

For more information on Ascend, see www.ascend.com.

ASCII *See* American Standard Code for Information Interchange.

ASCII extended character set The second group of characters, from 128 through 255, in the ASCII character set. The extended ASCII character set is assigned variable sets of characters by computer hardware manufacturers and software developers, and it is not necessarily compatible between different computers. The IBM extended character set used in the PC (see Appendix C) includes mathematics symbols and characters from the PC line-drawing set.

See also American Standard Code for Information Interchange; ASCII file; ASCII

standard character set; double-byte character set; Extended Binary Coded Decimal Interchange Code; Unicode.

ASCII file A file that contains only text characters from the ASCII character set. An ASCII file can include letters, numbers, and punctuation symbols, but does not contain any hidden text-formatting codes. Also known as a text file or an ASCII text file.

See also American Standard Code for Information Interchange; ASCII extended character set; ASCII standard character set; binary file.

ASCII standard character set A character set that consists of the first 128 (from 0 through 127) ASCII characters. The values 0 through 31 are used for nonprinting control codes (see Appendix C), and the range 32 through 127 is used to represent the letters of the alphabet and common punctuation symbols. The entire set from 0 through 127 is referred to as the standard ASCII character set. All computers that use ASCII can understand the standard ASCII character set.

See also American Standard Code for Information Interchange; ASCII file; ASCII extended character set; double-byte character set; Extended Binary Coded Decimal Interchange Code; Unicode.

ASCII text file *See* ASCII file.

ASE *See* Accredited Systems Engineer.

ASIC *See* application-specific integrated circuit.

ASP *See* Active Server Pages.

assembly language A low-level programming language in which each program statement must correspond to a single machine language instruction that the processor can execute.

Assembly languages are specific to a given microprocessor and, as such, are not portable; programs written for one type of processor must be rewritten before they can be used on another type of processor.

You use assembly language for two reasons:

- To wring as much performance out of the processor as possible
- To gain access to specific characteristics of the hardware that might not be possible from a higher-level language

See also compiler; interpreter; machine language; microcode

Associate Business Continuity

Professional Abbreviated ABCP. A certification from Disaster Recovery Institute International (DRII) that covers basic information on business continuity planning and disaster recovery.

See also Certified Business Continuity Professional; Master Business Continuity Professional.

Associated Accredited Systems

Engineer Abbreviated AASE. A certification from Compaq designed to evaluate and recognize basic knowledge of PC architecture and operations. An AASE may choose to specialize in Microsoft Windows 2000 or Novell NetWare operation.

See also Accredited Systems Engineer.

Association of Banyan Users

International Abbreviated ABUI. The Banyan user group, with 1700 members worldwide, concerned with all hardware and software related to the Banyan system, including Banyan VINES.

See also Banyan VINES.

asterisk In several operating systems, you can use the asterisk (*) as a wildcard character to represent one or more unknown characters in a filename or filename extension.

See also question mark; star-dot-star.

AST Research One of the world's top ten computer manufacturers, AST Research makes desktop, laptop, notebook, and hand-held computers, as well as monitors, graphics cards, and memory products. A subsidiary of Samsung Electronics Company Limited.

For more information on AST Research, see www.ast.com.

asymmetrical multiprocessing A multiprocessing design in which the programmer matches a specific task to a certain processor when writing the program.

This design makes for a much less flexible system than SMP (symmetrical multiprocessing) and may result in one processor being overworked while another stands idle. SMP allocates tasks to processors as the program starts up, on the basis of current system load and available resources. Needless to say, asymmetrical multiprocessing systems are easier to design, code, and test than symmetrical multiprocessing systems.

Asymmetric Digital Subscriber Line

Abbreviated ADSL. A high-speed data transmission technology originally developed by Bellcore and now standardized by ANSI as T1.413. ADSL delivers high bandwidth over existing twisted-pair copper telephone lines. Also called Asymmetric Digital Subscriber Loop.

ADSL supports speeds in the range of 1.5 to 9Mbps in the downstream direction (from the network to the subscriber) and supports upstream speeds in the range of 16 Kbps to 640 Kbps; hence, the term *asymmetric*.

See also Digital Subscriber Line; High-Bit-Rate Digital Subscriber Line; Rate-Adaptive Digital Subscriber Line; Single-Line Digital Subscriber Line; Very-High-Bit-Rate Digital Subscriber Line.

Assymetric Digital Subscriber Loop

See Assymetric Digital Subscriber Line.

asynchronous communications *See* asynchronous transmission.

asynchronous communications

server A LAN server that allows a network user to dial out of the network into the public switched telephone system or to access leased lines for asynchronous communications. Asynchronous communications servers may also be called dial-in/dial-out servers or modem servers.

asynchronous time-division

multiplexing Abbreviated ATDM. An asynchronous and adaptive version of time-

division multiplexing in which time slots are allocated dynamically based on demand.

See also time-division multiplexing.

Asynchronous Transfer Mode Abbreviated ATM. A method used for transmitting voice, video, and data over high-speed LANs and WANs. ATM uses continuous bursts of fixed-length packets called cells to transmit data. The basic packet consists of 53 bytes, 5 of which are used for control functions and 48 for data.

ATM is a connection-oriented protocol, and two kinds of connections are possible:

- Permanent virtual circuits (PVCs), in which connections are created manually
- Switched virtual circuits (SVCs), in which connections are made automatically

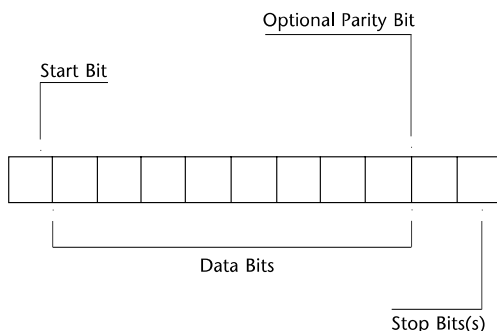
Speeds of up to 2.488Gbps have been achieved in testing. ATM will find wide acceptance in the LAN and WAN arenas as a solution to integrating disparate networks over large geographical distances. Also known as cell relay.

See also permanent virtual circuit; switched virtual circuit.

asynchronous transmission A method of data transmission that uses start bits and stop bits to coordinate the flow of data so that the time intervals between individual characters do not need to be equal. Parity also may be used to check the accuracy of the data received.

See also communications parameters; data bits; synchronous transmission.

ASYNCHRONOUS TRANSMISSION



ATA See Advanced Technology Attachment.

AT command set A set of standard instructions used to activate features on a modem. Originally developed by Hayes Microcomputer Products, the AT command set is now used by almost all modem manufacturers.

See also modem.

ATDM See asynchronous time-division multiplexing.

ATG See address translation gateway.

ATM See Asynchronous Transfer Mode.

ATM25 A low-cost version of Asynchronous Transfer Mode (ATM) that runs on Category 3 unshielded twisted-pair cable and provides 25Mbps transmissions in both directions.

See also Asynchronous Transfer Mode; IP over ATM; LAN Emulation; Multiprotocol over ATM.

ATM Adaptation Layer Abbreviated AAL. A service-dependent layer in Asynchronous Transfer Mode (ATM) that provides the protocol translation between ATM and the other communications services involved in a transmission.

AAL has several service types and classes of operation to handle different kinds of traffic, depending on how data is transmitted, the bandwidth required, and the types of connection involved.

See also convergence sublayer; segmentation and reassembly sublayer.

ATM LAN Emulation An Asynchronous Transfer Mode (ATM) variation that uses software to make the network operating system see an ATM adapter as an Ethernet or Token Ring adapter.

See also IP over ATM; LAN Emulation; Multiprotocol over ATM.

at symbol The separating character (@) between account name and domain name in an Internet e-mail address.

See also bang path.

AT&T The parent company of Bell Laboratories, the original developers of Unix. For many years Bell Labs was one of the two major development centers for Unix (the other being the Computer Systems Research Group at the University of California at Berkeley), but in 1990, AT&T formed Unix Systems Laboratories, or USL, to continue the development of Unix. In 1993, USL was sold to Novell, and in 1995, Novell sold the rights to SCO (Santa Cruz Operation).

See also Baby Bells; Regional Bell Operating Companies.

attach To establish a connection between a workstation and a network file server; particularly, to access additional servers after logging on to one server.

attachment *See* enclosure.

Attachment Unit Interface Abbreviated AUI. A 15-pin socket used by some Ethernet devices. AUI connections adapt between two different cabling types and work with a wide range of wiring schemes. Also known as a DIX (for Digital, Intel, Xerox) connector.

attack An attempt to circumvent the security measures in place on a network either to gain unauthorized access to the system or to force a denial of service.

See also brute-force attack; dictionary attack; social engineering.

attenuation The decrease in power of a signal with increasing distance. Attenuation is measured in decibels, and it increases as the power of the signal decreases. The best cables (those exhibiting the least attenuation) are fiber-optic lines, and the worst cables are unshielded, untwisted-pair lines, such as the silver, flat-satin cables used in short-run telephone and modem lines.

In a LAN, attenuation can become a problem when cable lengths exceed the stated network specification; however, the useful length of a cable may be extended by the use of a repeater.

attribute 1. A file attribute is a technique for describing access to and properties of files and directories within a file system. You may see the term *attribute* used interchangeably with the term *property*.

2. A screen attribute controls a character's background and foreground colors, as well as other characteristics, such as underlining, reverse video, and blinking or animation.

3. In operating systems, a characteristic that indicates whether a file is a read-only file, a hidden file, or a system file or has changed in some way since it was last backed up.

4. In markup languages such as HTML and SGML, a name-value pair within a tagged element that modifies certain characteristics of that element.

5. In a database record, the name or structure of a field.

See also tag.

AUDITCON A Novell NetWare and IntranetWare workstation utility that creates

a log file to allow an independent auditor to verify that network transactions are accurate and that confidential information is secure. When auditing is enabled, an auditor can track when files or directories are created, deleted, modified, salvaged, moved, or renamed. Changes to security rights can also be tracked.

audit policy In Microsoft Windows 2000, the policy that defines the security events to track and report to the network administrator.

See also security log.

audit trail An automatic feature of certain programs or operating systems that creates a running record of all transactions. An audit trail allows you to track a piece of data from the moment it enters the system to the moment it leaves and to determine the origin of any changes to that data.

auditing The process of scrutinizing network security-related events and transactions to ensure that they are accurate, particularly reviewing attempts to create, access, and delete files and directories and reviewing security violations. Records of these events are usually stored in a security log file, which can only be examined by users with special permissions.

AUI *See* Attachment Unit Interface.

authentication In a network operating system or multiuser system, the process that validates a user's login information.

Authentication may involve comparing the user name and password to a list of authorized users. If a match is found, the user can

log on and access the system in accordance with the rights or permissions assigned to his or her user account.

See also authorization; Kerberos; password; user; user account.

authoring The process of preparing a multimedia presentation or a Web page. This involves not only writing the text of the presentation or Web page, but also the production of the slides, sound, video, and graphical components.

authorization The provision of rights or permissions based on identity. Authorization and authentication go hand in hand in networking; your access to services is based on your identity, and the authentication processes confirm that you are who you say you are.

See also authentication.

auto-answer A feature of a modem that allows it to answer incoming calls automatically.

See also answer mode; dialback modem.

auto-dial A feature of a modem that allows it to open a telephone line and start a call. To auto-dial, the modem sends a series of pulses or tones that represent a stored telephone number.

See also callback modem.

AUTOEXEC.BAT A contraction of Automatically Executed Batch. A special MS-DOS batch file, located in the root directory of the startup disk, that runs automatically every time you start or restart your computer. The commands contained

in AUTOEXEC.BAT are executed one by one, just as if you typed them at the system prompt. An AUTOEXEC.BAT file can be used to load hardware device drivers, set the system prompt, change the default drive to the first network drive, and log the user in to the file server.

In OS/2, you can select any batch file to be used as AUTOEXEC.BAT for a specific MS-DOS session, so you can tailor specific environments for separate MS-DOS sessions, each using a different AUTOEXEC.BAT file.

See also AUTOEXEC.NCF; boot; bootstrap; CONFIG.SYS.

AUTOEXEC.NCF A Novell NetWare batch file usually located on the NetWare partition of the server's hard disk, used to set the NetWare operating system configuration. AUTOEXEC.NCF loads the LAN drivers, the NLMs, and the settings for the network interface boards and then binds the protocols to the installed drivers.

Automatic Client Upgrade A mechanism used to upgrade Novell client software during the logon process by executing four separate programs called by the logon script. Automatic Client Upgrade can be very useful when all client workstations use standard configurations.

automatic forwarding A feature of many e-mail programs that automatically retransmits incoming messages to another e-mail address.

automatic number identification Abbreviated ANI. A method of passing a caller's telephone number over the network to the recipient so that the caller can be iden-

tified. ANI is often associated with ISDN and is sometimes known as caller ID.

automatic rollback In a Novell NetWare network, a feature of the Transaction Tracking System (TTS) that abandons the current transaction and returns a database to its original condition if the network fails in the middle of a transaction. Automatic rollback prevents the database from being corrupted by information from incomplete transactions.

See also backing out.

Autonomous System Border router

Abbreviated ASBR. In an internetwork that uses link state routing protocols such as Open Shortest Path First (OSPF) protocols, a router that has at least one connection to a router in an external network.

See also Area Border router; Open Shortest Path First.

AutoPlay A feature of Microsoft Windows that automatically executes an application from a CD-ROM or automatically plays an audio CD when the disk is inserted into the CD-ROM drive.

Available Bit Rate A Type 3 or Type 4 Asynchronous Transfer Mode Adaption Layer (AAL) service designed for non-time-critical applications such as LAN emulation and LAN internetworking.

See also Constant Bit Rate; Unspecified Bit Rate; Variable Bit Rate.

Available Cell Rate Abbreviated ACR. A measure of the bandwidth in Asynchronous Transfer Mode (ATM) networks. The ACR value represents the number of ATM

cells available or allowed per second for a specific quality of service (QoS) class. Available Cell Rate is also known as Allowed Cell Rate.

See also Asynchronous Transfer Mode; Minimum Cell Rate; Peak Cell Rate; Sustainable Cell Rate.

avatar **1.** In Unix, another name for the superuser account; an alternative to the name root.

2. A visual representation of a user in a shared virtual-reality environment.

AWG *See* American Wire Gauge.

B

Baan Advanced Certification Abbreviated BAC. A certification from Baan available in four specialties covering the Baan IV suite of products: Enterprise Logistics, Enterprise Finance, Enterprise Tools, and Enterprise Modeler.

See also Baan Basic Certification.

Baan Basic Certification Abbreviated BBC. A certification from Baan designed to evaluate basic proficiency with the Baan IV suite of products; a prerequisite qualification to taking the Baan Advanced Certification exams.

See also Baan Advanced Certification.

Baan Company A leading provider of enterprise and inter-enterprise business software used for managing finance, manufacturing, inventory, distribution, transportation, and administrative functions for large companies.

For more information on Baan, see www2.baan.com.

Baby Bells A slang term for the 22 Regional Bell Operating Companies (RBOC), formed when AT&T was broken up in 1984.

See also Bell Labs.

BAC *See* Baan Advanced Certification.

backbone That portion of the network that manages the bulk of the traffic. The backbone may connect several locations or buildings, and other, smaller networks may be attached to it. The backbone often uses a higher-speed protocol than the individual LAN segments.

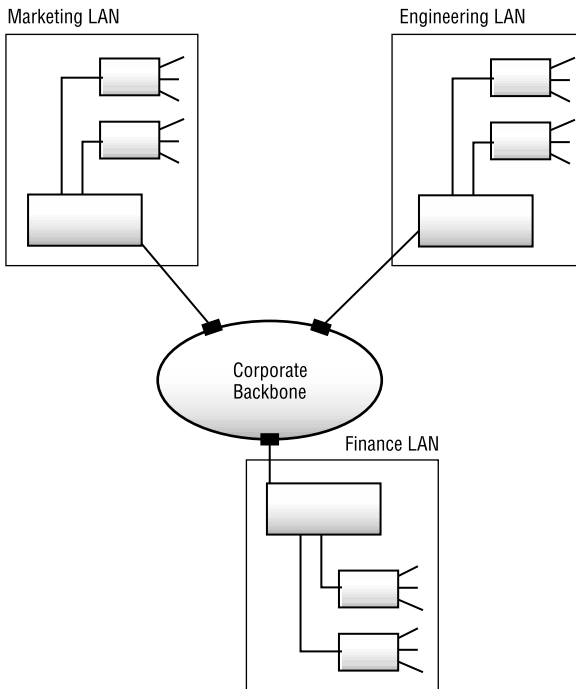
back-end processor A secondary processor that performs one specialized task very effectively, freeing the main processor for other, more important work.

back-end system The server part of a client/server system that runs on one or more file servers and provides services to the front-end applications running on networked workstations. The back-end system accepts query requests sent from a front-end application, processes those requests, and returns the results to the workstation.

Back-end systems may be PC-based servers, superservers, midrange systems, or mainframes.

See also client/server architecture.

BACKBONE



background 1. On a computer screen, the color on which characters are displayed; for example, white characters may appear on a blue background.

2. In an operating system, a process that runs in the background generally runs at a lower level of priority than a foreground task and does not accept input from the user. Only multitasking operating systems support true background and foreground processing, but some applications can mimic it. For example, many word processors

can print a document while still accepting input from the keyboard. In older systems, a process spends its entire existence in either the background or the foreground; in newer systems, you can change the processing environment and move foreground programs into the background, and vice versa.

See also ampersand; foreground; multitasking.

background authentication The process used to give a workstation access to a particular server. For example, in Novell

NetWare, password authentication uses a public key encryption scheme to protect password information before it is transmitted over the network. All this activity takes place in the background, and all the user has to do is enter his or her password.

background noise Any unwanted signal that enters a line, channel, or circuit; for example, electrical interference on a telephone circuit.

See also noise.

backing out The process of abandoning the current transaction and returning a database to its original condition if the network fails during the transaction. This process prevents the database from being corrupted by information from the incomplete transaction.

See also automatic rollback; Transaction Tracking System.

backlink In Novell Directory Services, an attribute used by a directory to indicate the external references to a property.

backslash **1.** In MS-DOS, OS/2, and other operating systems, you must use the backslash character (\) to separate directory or subdirectory names in a path statement or when changing to another directory.

2. A shorthand name for the root directory. Sometimes called the reverse slash or backslant.

See also slash.

backup An up-to-date copy of all your files. You make a backup for several reasons:

- Insurance against possible hard-disk or file-server failure. Hard disks often fail

completely, taking all your work with them. If this failure occurs, you can reload your files and directories from the backup copy. A backup is your insurance against disk failure affecting the thousands or possibly tens of thousands of files you might have on your file server.

- Protection against accidental deletion of files or directories. Again, if you mistakenly delete a file or directory, you can retrieve a copy from your last backup.
- Protection against the new version of software you are about to install not working to your expectations; make a backup before installing new software.
- As an archive at the end of a project, when a person leaves your company, or at the end of a financial period such as year-end close.

Your decision when or how often to make a backup depends on how frequently important data on your system changes. If you rely on certain files always being available on your system, it is crucial that you make regular, consistent backups. Here are some backup tips:

- Keep multiple copies; redundancy should be a part of your backup plan.
- Test your backups to make sure they are what you think they are before you bring the server into service, and make sure you can reload the information you need.
- Store your backups in a secure off-site location; do not leave them right next to the computer (if the computer is damaged by an accident, the backup may be damaged as well).
- Replace your backup media on a regular basis.

backup browser

- Consider making incremental backups of critical data at more frequent intervals.

It all comes down to one very simple rule: back up all the files that you cannot afford to lose. Do it now.

See also archive; differential backup; disk duplexing; disk mirroring; full backup; incremental backup.

backup browser In Microsoft Windows 2000 Server, a computer that maintains a list of computers and services available on the network. This list is provided by the master browser and distributed to a workgroup or a domain by the backup browser.

See also browser; master browser.

backup domain controller In Microsoft Windows NT, a server containing accurate replications of the security and user databases.

The backup domain controller receives a copy of the domain's directory database, containing all the account and security information for the domain, from the primary domain controller. This copy is periodically synchronized with the original master database. A domain can contain several backup domain controllers.

See also domain controller; primary domain controller.

backup program An application that you use to make archive or backup copies of important data files.

Most operating systems offer commands for making backups, but many are limited in capability. The most cost-effective backup programs are from third-party vendors.

See also disk duplexing; disk mirroring; full backup; incremental backup; tape cartridge; tape drive.

backward compatibility Full compatibility with earlier versions of the same application or computer system.

Backward Error Correction Abbreviated BEC. An error-correction method in which an error can be detected by the recipient, which immediately requests a retransmission of the affected data.

See also error detection and correction.

BACP *See* Bandwidth Allocation Control Protocol.

bad sector An area on a hard disk or floppy disk that cannot be used to store data because of a manufacturing defect or accidental damage.

The operating system will find, mark, and isolate bad sectors. Almost all hard disks have some bad sectors, often listed in the bad track table. Usually, bad sectors are a result of the manufacturing process and not a concern; the operating system will mark them as bad, and you will never even know they are there.

bad track table A list of the defective areas on a hard disk, usually determined during final testing of the disk at the factory. Some disk-preparation programs ask you to enter information from this list to reduce the time that a low-level format takes to prepare the disk for use by the operating system.

balun A contraction of *balanced* *unbalanced*. A small device used to connect

a balanced line (such as a twisted-pair cable) to an unbalanced line (such as a coaxial cable). The balun matches impedances between the two media.

bandwidth 1. In communications, the difference between the highest and lowest frequencies available for transmission in any given range.

2. In networking, the transmission capacity of a computer or a communications channel, stated in megabits per second (Mbps). For example, FDDI (Fiber Distributed Data Interface) has a bandwidth of 100Mbps. To relate this to a real-world example, a complete page of text, in English, is approximately 16,000 bits.

Bandwidth Allocation Control

Protocol Abbreviated BACP. A proposed Internet protocol designed to manage a combination of dial-up links, usually over ISDN connections.

BACP provides bandwidth on demand and can combine two or more circuits into a single circuit with a higher data rate.

See also bandwidth on demand.

bandwidth on demand A technique that allows the user to add additional bandwidth as the application requires it.

Most network traffic does not flow in steady and easily predictable streams, but in short bursts, separated by longer periods of inactivity. This pattern makes it difficult to predict peak loads. Bandwidth on demand is useful for applications such as backups and videoconferencing and allows the user to pay for only the amount of bandwidth used.

See also virtual data network.

bandwidth reservation A technique used to reserve channel capacity for special time-sensitive transmissions such as videoconferencing and real-time audio transmissions.

bandwidth throttling A feature in some Web server software that allows the system administrator to control or alter the proportions of bandwidth available to the services that the server provides.

bang A name given in the Unix and Internet worlds to the exclamation point (!) character; also known as *pling* in the United Kingdom.

See also bang path.

bang path An old-style uucp e-mail address that uses exclamation points to separate the sequence of computer names needed to get to the addressee.

Bang paths list addresses—general to specific—from left to right, which is the reverse of the sequence used by other addressing schemes.

See also at symbol; domain name; e-mail address; uucp.

Banyan Systems, Inc. A leading provider of enterprise network products, including the Banyan VINES network operating system, products based on the Banyan StreetTalk Directory, and Beyond-Mail messaging technologies.

For more information on Banyan, see www.banyan.com.

Banyan VINES A network operating system from Banyan Systems. VINES (a contraction of Virtual Networking Software) is

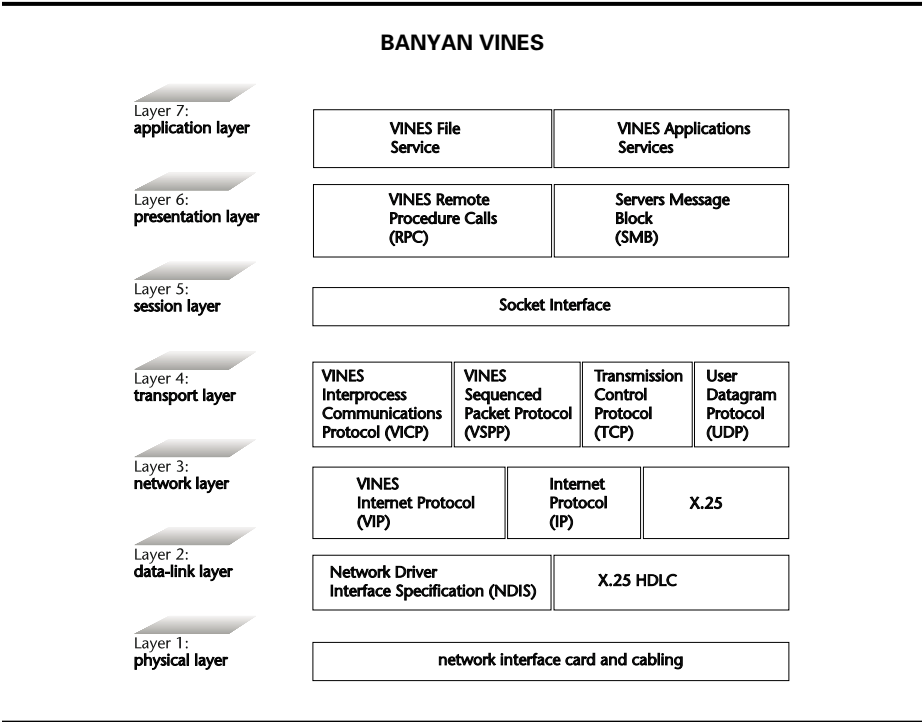
based on a special version of the Unix System V operating system.

The following illustration shows the internal organization of VINES. The Unix layer is hidden from view by VINES and is not available for other applications. VINES provides all server functions, including those of a communications/modem server, and offers many options for connecting to minicomputers, mainframes, and other network file servers. VINES automatically manages protocol binding and translations required between the network interface cards for routing to

different LAN segments. A complete set of network management tools is also available.

Workstations can run MS-DOS, Microsoft Windows, Unix, or OS/2, and they can store native-form files on the server. Macintosh computers can also attach to the network. VINES offers special support for very large LANs and WANs with multiple file servers. VINES also allows PCs that support multiple processors to use multiprocessing to divide the file-server processing load.

See also Enterprise Network Services; StreetTalk.



base 10 radix *See* decimal.

base64 An encoding scheme used by MIME-compliant mail systems to convert binary files into a text format that can be processed and sent as e-mail.

See also binary file; Multipurpose Internet Mail Extension.

baseband modem *See* line driver.

baseband network A technique for transmitting signals as direct-current pulses rather than as modulated signals. The entire bandwidth of the transmission medium is used by a single digital signal, so computers in a baseband network can transmit only when the channel is not busy. However, the network can use techniques such as multiplexing to allow channel sharing.

A baseband network can operate over relatively short distances (up to 2 miles if network traffic is light) at speeds from 50Kbps to 100Mbps. Ethernet, AppleTalk, and most PC local-area networks (LANs) use baseband techniques.

See also bandwidth; broadband network; frequency-division multiplexing; statistical multiplexing; time-division multiplexing.

baseline The process of determining and documenting network throughput and other performance information when the network is operating under what is considered a normal load. Measured performance characteristics might include error-rate and data-transfer information, along with

information about the most active users and their applications.

Bash In Unix, a popular command interpreter. Bash, the Bourne-Again Shell, was first released in 1989 by Brian Fox and Chet Ramey, as part of the Free Software Foundation GNU Project.

Bash provides features found in the other Unix shells, particularly the Bourne shell, the C shell, and the Korn shell, and includes Bourne shell syntax, redirection and quoting, C-shell command-line editing and tilde expansion, job control, and command history. Bash also includes built-in commands and variables, as well as aliases from the Korn shell.

See also Bourne shell; C shell; Korn shell; Linux; Unix shell.

basic disk In Microsoft Windows 2000 Server, a normal hard disk, available from any operating system, that uses primary and extended partitions to divide the available space into logical units.

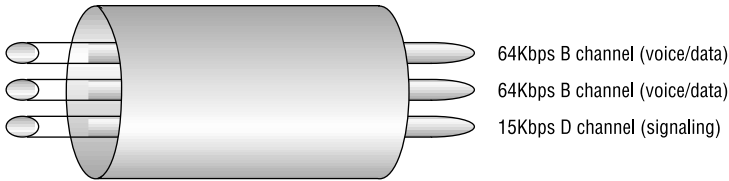
See also dynamic disk.

Basic Rate ISDN Abbreviated BRI. An ISDN service that offers two 64Kbps B channels used for data transfer and one 16Kbps D channel used for signaling and control information.

Each B channel can carry a single digital voice call or can be used as a data channel; the B channels can also be combined into a single 128Kbps data channel.

See also 23B+D; 2B+D; Primary Rate ISDN.

BASIC RATE ISDN



bastion host A computer system that acts as the main connection to the Internet for users of a LAN. A bastion host is usually configured in such a way as to minimize the risk of intruders gaining access to the main LAN. It gets its name from the fortified projections on the outer walls of medieval European castles.

See also firewall; proxy server.

batch file An ASCII file that contains operating system commands and other commands supported by the batch processor. Batch files are used to automate repetitive tasks. The commands in the file are executed one line at a time, just as if you had typed them at the system prompt.

baud A measurement of data-transmission speed. Originally used in measuring the speed of telegraph equipment, it now usually refers to the data-transmission speed of a modem or other serial device.

See also baud rate.

baud rate In communications equipment, a measurement of the number of state changes (from 0 to 1 or vice versa) per sec-

ond on an asynchronous communications channel.

Baud rate is often assumed to correspond to the number of bits transmitted per second, but baud rate and bits per second (bps) are not always the same. In high-speed digital communications systems, one state change can be made to represent more than one data bit.

See also baud; bits per second.

Bay Networks Certified Expert A certification from Bay Networks, Inc., designed to recognize technical expertise in constructing enterprise-wide networking solutions. Currently, this certification is available in Router, Hub, and Network Management technologies.

Bay Networks Certified Specialist A certification from Bay Networks, Inc., designed to evaluate a basic level of technical expertise in one or more Bay Networks product areas. Currently, this certification is available in Router, Hub, Network Management, Remote Access, and Switching technologies.

Bay Networks, Inc. A leading manufacturer of networking products such as hubs, routers, and switches, now venturing into the areas of network management and virtual private networks. In 1998, the company announced a merger with Northern Telecom.

For more information on Bay Networks, see www.baynetworks.com.

BBC See Baan Basic Certification.

BBS See bulletin board system.

bcc See blind carbon copy.

BCD See binary coded decimal.

beaconing In a token-ring network, the process of informing other nodes that token passing has been suspended because of a severe error condition, such as a broken cable. Communication cannot resume until the condition is resolved.

BEC See Backward Error Correction.

Bell communications standards A set of data-transmission standards developed by AT&T in the 1980s that rapidly became the de facto standard for modem manufacturers. Although several of these standards are still widely used in the United States, the CCITT V series definitions are now generally accepted as the defining standards for modem use, data compression, and associated hardware.

Bell Labs The research arm of AT&T and the birthplace of the Unix operating system and the C programming language in the 1970s.

See also Berkeley Software Distribution Unix; Computer Systems Research Group.

benchmark A specific standard against which some aspect of computer performance can be compared.

A benchmark is a test that attempts to quantify hardware, software, or system performance—usually in terms of speed, reliability, or accuracy. One of the major problems in determining performance is deciding which of the many benchmarks available actually reflects the way you intend to use the system.

See also benchmark program.

benchmark program A program that attempts to provide a consistent measurement of system performance. Here are some examples:

- Dhrystone, which measures microprocessor and memory performance
- Whetstone, which measures speed of arithmetic operations
- Khornerstone, which measures overall system performance, including disk-drive access speed, memory access speed, and processor performance

The Systems Performance Evaluation Cooperative (SPEC) developed a set of ten tests to measure performance in actual application environments. The results of these tests are known as SPECmarks.

BER See bit error rate; error rate.

Berkeley Software Distribution Unix Abbreviated BSD Unix, and also known as Berkeley Unix. BSD Unix was developed at the University of California at Berkeley by

researchers working in the Computer Systems Research Group (CSRG) from the 1970s to 1993 when the group finally closed its doors. BSD added many significant advanced features to Unix, including the C shell, the vi editor, TCP/IP networking additions, and virtual memory.

Because the CSRG was an academic group producing state-of-the-art software with no support obligations, BSD Unix was not supported in the normal way; bug fixes were sometimes made available, but it was a hit-and-miss process at best. For this reason, BSD Unix appealed to the research community and scientific users rather than to commercial users who tended to use Unix from AT&T.

BSD Unix 4.1 through to the last release, version 4.4, and the related commercial products, including those from Sun Microsystems, DEC, and Mt Xinu, are still popular and in use in universities and commercial institutions all over the world.

See also FreeBSD; Linux; NetBSD; Unix.

Berkeley Unix *See* Berkeley Software Distribution Unix.

beta site A location where beta testing is performed before a hardware or software product is formally released for commercial distribution.

See also beta software; beta testing.

beta software Software that has been released to a cross-section of typical users for testing before the commercial release of the package.

See also beta testing.

beta testing The process of field testing new hardware or software products before the product's commercial or formal release. Beta testing is usually done by a cross-section of users, not just programmers. The purpose of beta testing is to expose the new product to as many real-life operating conditions as possible.

If the beta tests indicate a higher-than-expected number of bugs, the developer usually fixes the problems and sends the product out again for another round of beta testing. Preliminary versions of the product documentation are also circulated for review during the beta testing.

See also alpha testing.

BGP *See* Border Gateway Protocol.

Big Blue A nickname for International Business Machine Corporation (IBM), which uses blue as its corporate color.

big endian A computer architecture in which the most significant byte has the lowest address and so is stored big end first.

Many processors, including those from Motorola and Sun, certain RISC processors, the PDP-11, and the IBM 3270 series are all big endian. The term comes from Jonathan Swift's *Gulliver's Travels*, in which wars were fought over whether boiled eggs should be opened at the big end or the little end.

See also holy wars; little endian.

binaries A slang term for a group of binary files.

binary Any scheme that uses two different states, components, conditions, or conclusions.

In mathematics, the binary or base-2 numbering system uses combinations of the digits 0 and 1 to represent all values. The more familiar decimal system has a base of 10 (0–9).

Unlike computers, people find binary numbers that consist of long strings of zeros and ones difficult to read, so most programmers use hexadecimal (base-16) or octal (base-8) numbers instead.

Binary also refers to an executable file containing a program.

binary coded decimal Abbreviated BCD. A simple system for converting decimal numbers into binary form, in which each decimal digit is converted into binary and then stored as a single character.

binary file A file consisting of binary information. Usually, a binary file is a program or data file in machine-readable form rather than in human-readable ASCII text. You can convert a binary file into a text-based form so that you can transmit it over the Internet.

See also base64; Multipurpose Internet Mail Extension.

binary license A license granted to a user by a software developer, entitling the user to run a specific software package under well-defined circumstances, using the binary files provided by the developer.

A binary license does not entitle the user to a copy of the source code for the package,

although a certain level of configuration can be done, and the user does not have the right to modify the software in any way beyond this simple configuration.

See also open source software; source license.

BIND 1. A Novell NetWare server utility used to bind a protocol to a network interface card or device driver.

2. Abbreviation for Berkeley Internet Name Domain, a BSD client/server program that manages host and IP addresses by matching the host name with the IP dotted decimal address.

See also Domain Name Service; dotted decimal.

bindery A database maintained by older Novell NetWare network operating systems. The bindery contains information about users, servers, and other important network configuration details in a flat database. The bindery is crucial to NetWare's operation and is constantly consulted by the operating system.

In NetWare 4.x and later systems, the bindery is replaced by NetWare Directory Services (NDS).

See also NetWare Directory Services.

bindery emulation A NetWare 4.x and later feature that allows bindery-based utilities to work with NetWare Directory Services (NDS) on the same network. Bindery emulation applies only to leaf objects within the Organizational container object.

See also container object; leaf object; NetWare Directory Services.

Bindery object

Bindery object A leaf object that represents an object placed in the Directory tree by an upgrade utility, but that cannot be further identified by NetWare Directory Services (NDS). The Bindery object is available for the purposes of backward compatibility with older, bindery-based utilities and applications.

See also leaf object; NetWare Directory Services.

bindery services *See* bindery emulation.

binding The process of establishing communications between the protocol device driver and the network interface card driver.

BIOS Acronym for basic input/output system, pronounced “bye-os.” In the PC, the BIOS is a set of instructions that tests the hardware when the computer is first turned on, starts to load the operating system, and lets the computer’s hardware and operating system communicate with applications and peripheral devices, such as hard disks, printers, and video adapters. These instructions are stored in ROM as a permanent part of the computer.

As new hardware is developed, new BIOS routines must be created to service those devices. For example, BIOS support has been added for power management and for ever-larger hard disks. If you are experiencing problems accessing such devices after adding them to an existing system, your computer’s BIOS may be out of date. Contact

your computer supplier for information about BIOS updates.

bis A term describing a secondary CCITT (Consultative Committee for International Telephony and Telegraphy) recommendation that is an alternative or extension to the primary recommendation. For example, the CCITT V.42 standard refers to error correction, and the V.42 bis standard refers to data compression.

See also ter.

B-ISDN *See* Broadband Integrated Services Digital Network.

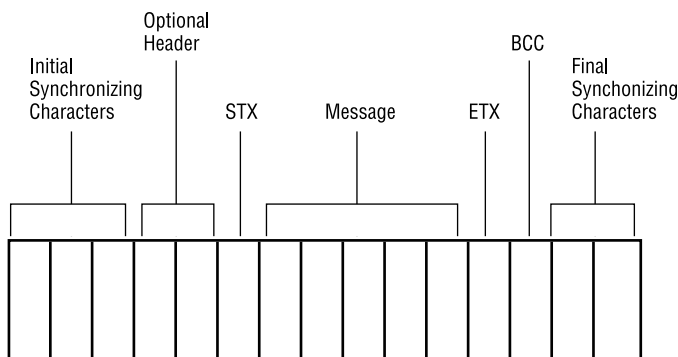
bisynchronous communications A protocol used extensively in mainframe computer networks. With bisynchronous communications, both the sending and receiving devices must be synchronized before data transmission begins.

Data is collected into a package known as a *frame*. Each frame contains leading and trailing characters that allow the computers to synchronize their clocks. The structure of a bisynchronous communications frame is shown in the accompanying illustration. The STX and ETX control characters mark the beginning and end of the message. BCC is a set of characters used to verify the accuracy of the transmission.

A more modern form of this kind of protocol is SDLC (Synchronous Data Link Control), which is used in IBM’s proprietary networking scheme, SNA.

See also asynchronous transmission; Synchronous Data Link Control.

BISYNCHRONOUS COMMUNICATIONS



bit Contraction of binary digit. A bit is the basic unit of information in the binary numbering system, representing either 0 (off) or 1 (on). Bits can be grouped to form larger storage units; the most common grouping is the 7- or 8-bit byte.

See also octet.

bit error rate Abbreviated BRI. The number of erroneous bits in a data transmission or in a data transfer, such as from CD-ROM to memory.

bit-oriented protocol A communications protocol in which data is transmitted as a stream of bits rather than as a stream of bytes.

A bit-oriented protocol uses specific sequences of bits as control codes, unlike a byte-oriented protocol, which uses reserved characters. HDLC (High-level Data Link Control) and IBM's SDLC (Synchronous

Data Link Control) are both bit-oriented protocols.

See also byte-oriented protocol.

bit rate The rate at which bits are transmitted over a communications channel, described in terms of bits per second (bps).

See also baud rate.

bits per inch Abbreviated bpi. The number of bits that a tape or tape cartridge can store per inch of length.

bits per second Abbreviated bps. The number of binary digits, or bits, transmitted every second during a data-transfer procedure. Bits per second is a measurement of the speed of operation of equipment, such as a computer's data bus or a modem that connects a computer to a communications circuit.

bit stuffing A technique used to ensure that a specific bit pattern never occurs in

blackout

the data transmitted over a communications channel. Additional bits are added at the transmitting end and removed at the receiving end of the channel as the data is processed.

blackout A total loss of commercial electric power. To avoid loss of computer data due to a blackout, use a battery-backed UPS (uninterruptible power supply).

See also brownout; power conditioning.

blind carbon copy Abbreviated bcc. A list of recipients of an e-mail message whose names do not appear in the normal message header, so the original recipient of the message does not know that copies have been forwarded to other locations. Sometimes called blind courtesy copy.

block In communications, a unit of transmitted information that includes header codes (such as addresses), data, and error-checking codes.

See also checksum; error; error detection and correction.

blocks Sections of a tape or disk that are read or written at the same time; units of storage that are transferred as single units.

block size The largest contiguous amount of disk space allocated by the file system. Files larger than the file system's block size may be broken into smaller fragments when they are stored, with sections of the file being stored in different loca-

tions across the disk. This is known as *fragmentation* and, in some cases when taken to an extreme, can severely limit system performance.

The file system's block size is often different from the hard disk's physical block size.

See also fragmentation.

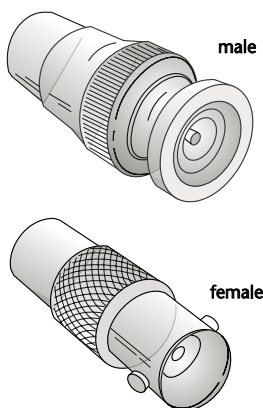
block suballocation A mechanism used in Novell NetWare that allows files to share the same block space by dividing each 8K hard-disk block into smaller 512-byte segments. Files needing extra space can use these smaller segments rather than wasting a whole new disk block, making for more efficient disk-space use.

blue screen of death Abbreviated BSOD. An affectionate name for the screen displayed when Microsoft Windows encounters an error so serious that the operating system cannot continue to run. Windows may also display information about the failure and may perform a memory dump and an automatic system restart.

BNC connector A small connector with a half-turn locking shell for coaxial cable, used with thin Ethernet and RG-62 cabling. The accompanying illustration shows both male and female connectors.

BNCE *See* Bay Networks Certified Expert.

BNCS *See* Bay Networks Certified Specialist.

BNC CONNECTOR

bookmark 1. In the Microsoft Windows NT Performance Monitor, a feature that allows you to mark a point of interest in a log file and then return to that same point in the file at a later time.

2. An option available in a Web browser that lets you mark a Web page so that you can identify and return to a favorite site quickly and easily without having to retype the URL or even remember how you got there.

Boolean Any variable that can have a logical value of true or false. Named after George Boole, the developer of a branch of algebra based on the values of true and false, Boolean works with logical rather than numeric relationships.

Boolean operators include AND (logical conjunction), OR (logical inclusion), XOR (exclusive or), and NOT (logical negation)

and are sometimes described as logical operators.

Many popular Internet search engines allow Boolean searches.

See also search engine.

boot To load an operating system into memory, usually from a hard disk, although occasionally from a floppy disk. Booting is generally an automatic procedure that begins when you turn on or reset your computer. A set of instructions contained in ROM begins executing. The instructions run a series of power-on self tests (POSTs) to check that devices such as hard disks are in working order, locate and load the operating system, and finally pass control over to that operating system.

Boot may be derived from the expression “pulling yourself up by your own bootstraps” and is sometimes called bootstrap.

See also BIOS; cold boot; warm boot.

bootable disk

bootable disk Any disk capable of loading and starting the operating system. Bootable floppy disks are becoming less common because operating systems are growing larger. In some cases, all the files needed to start the operating system will not fit on even the largest-capacity floppy disk, which makes it impossible to boot from a floppy disk.

BOOTCONF.SYS A Novell NetWare configuration file that specifies how a diskless workstation boots the operating system from the file server. If the file server contains several remote boot image files, BOOTCONF.SYS determines which one will be loaded onto the workstation.

See also diskless workstation.

BOOTP *See* Bootstrap Protocol.

boot ROM A type of ROM that allows a workstation to communicate with the network file server and to read an image file containing an operating system program. In this way, a workstation without a local disk can boot the operating system from the file server.

See also diskless workstation.

boot sector virus A virus that infects the master boot record of a computer by overwriting the original boot code with infected boot code. This kind of virus is usually spread to a hard disk from an infected floppy disk being used as a boot disk.

See also antivirus program; file-infecting virus; macro virus; multipart virus; polymorphic virus; stealth virus; Trojan Horse; vaccine; virus.

bootstrap *See* boot.

Bootstrap Protocol Abbreviated BOOTP. An Internet protocol that provides network configuration information to a diskless workstation.

When the workstation first boots, it sends out a BOOTP message on the network. This message is received by the server, which obtains the appropriate configuration information and returns that information to the workstation. This information includes the workstation's IP address, the IP address of the server, the host name of the server, and the IP address of a default router.

See also Dynamic Host Configuration Protocol.

border A security perimeter formed by logical boundaries that can only be crossed at specifically defined locations known as border gateways.

See also border gateway.

border gateway A router that connects a private LAN to the Internet. A border gateway is a security checkpoint used to force all network traffic, either inbound or outbound, through a single point of control.

See also border; firewall; Novell Border-Manager; router.

Border Gateway Protocol Abbreviated BGP. A routing protocol designed to replace EGP (External Gateway Protocol) and interconnect organizational networks. BGP, unlike EGP, evaluates each of the possible routes for the best one.

Border router *See* Area Border router.

Borland International, Inc. *See* Inprise Corporation.

bot *See* robot.

bounce The return of an e-mail message to its original sender due to an error in delivery. You may have made a simple spelling mistake in the e-mail address, the recipient's computer system may be down, or the recipient may no longer subscribe to or have an account on the system. A returned e-mail message will usually contain a description of why the message bounced.

Bourne shell In Unix, a popular command interpreter with a built-in programming language.

The Bourne shell, developed by Dr. Steven Bourne of Bell Labs, is the oldest Unix shell still in popular use and features a built-in command set for writing shell scripts, background execution of commands, input and output redirection, job control, and a set of shell variables to allow environment customization.

See also Bash; C shell; Korn shell; Linux; Unix shell.

bpi *See* bits per inch.

bps *See* bits per second.

brain damaged An expression used to describe any poorly designed program or piece of hardware that does not include those features most users would consider

essential. The implication is that the designer should have known better than to leave those features out of the product.

breach A loss of network security as a result of a successful attack.

See also brute-force attack; dictionary attack; social engineering.

breakout box A small device that can be connected into a multicore cable for testing the signals in a transmission. Small LEDs in the breakout box indicate when a signal is transmitted over one of the lines. Switches or short jumper cables can be used to re-route these signals to other pins as required for troubleshooting.

See also sniffer.

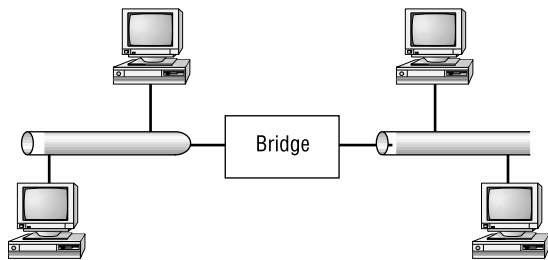
BRI *See* Basic Rate ISDN.

bridge A hardware device used to connect LANs so that they can exchange data. Bridges can work with networks that use different wiring or network protocols, joining two or more LAN segments to form what appears to be a single network.

A bridge operates at the data-link layer of the OSI Reference Model for computer-to-computer communications. It manages the flow of traffic between the two LANs by reading the address of every packet of data that it receives.

See also brouter; gateway; OSI Reference Model; router.

BRIDGE



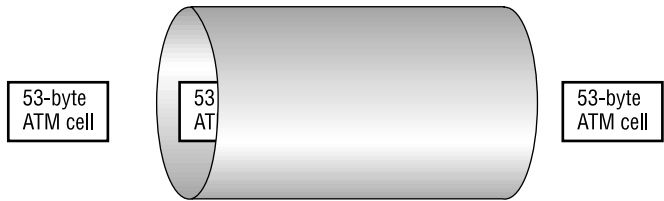
BrightWorks A package of network management utilities from McAfee Associates that includes hardware and software inventory, application metering, remote control of clients, virus detection, and a help-desk utility.

Broadband Integrated Services Digital Network Abbreviated B-ISDN. A high-speed communications standard for WANs

that handles high-bandwidth applications, such as video, voice, data, and graphics. SMDS (Switched Multimegabit Data Services) and ATM (Asynchronous Transfer Mode) are two BISDN services that can provide a huge bandwidth.

See also Asynchronous Transfer Mode; SONET; Switched Multimegabit Data Services.

BROADBAND INTEGRATED SERVICES DIGITAL NETWORK



broadband network A technique for transmitting a large amount of information, including voice, data, and video, over long distances using the same communications

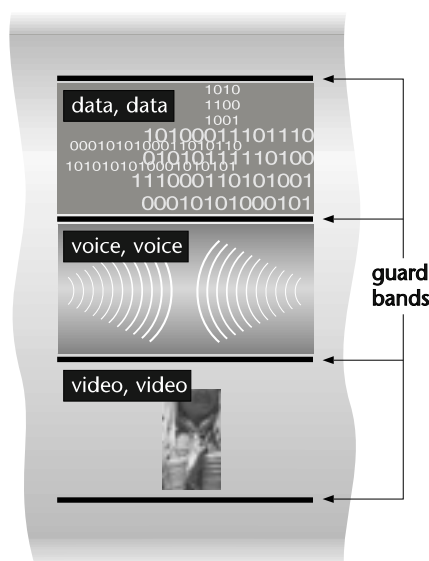
channel. Sometimes called wideband transmission, it is based on the same technology used by cable television.

The transmission capacity is divided into several distinct channels that can be used concurrently by different networks, normally by frequency-division multiplexing (FDM). The individual channels are pro-

ected from each other by guard channels of unused frequencies. A broadband network can operate at speeds of up to 20Mbps.

See also baseband network; multiplexer.

BROADBAND NETWORK



broadcast To send a message to all users currently logged in to the network.

See also multicast.

broadcast storm Congestion on a network that occurs when a large number of frames are transmitted by many workstations in response to a transmission from one workstation.

broker A Novell Distributed Print Services (NDPS) service providing management services for network printers, including event notification and storage of printer resources such as device drivers.

brouter A networking device that combines the attributes of a bridge and a router. A brouter can route one or more specific protocols, such as TCP/IP, and bridge all

others, for example, and can operate at either the data-link layer or the network layer of the OSI Reference Model.

See also gateway.

brownout A short period of low voltage, often the result of an unusually heavy demand for power, that may cause your computer to crash. If your area experiences frequent brownouts, consider using a UPS (uninterruptible power supply) as a battery backup system.

browse list A list of computers and services available on a network.

browser 1. An application program used to explore Internet resources. A browser lets you wander from Web site to Web site without concern for the technical details of the links between them or the specific methods used to access them and presents the information—text, graphics, sound, or video—as a document on the screen.

2. A small application used to scan a database or a list of files.

3. In Windows NT networking, a mechanism used as a name service.

See also Web browser; World Wide Web.

brute-force attack A technique employed by intruders that checks every password in a password file against every possible password generated sequentially. A brute-force attack is very clumsy, is usually considered the last resort in an attack, takes a long time to run, and is easily detected by even the most modest security precautions. Also called a keyspace attack.

See also denial of service attack; dictionary attack; social engineering; Trojan Horse.

BSD Unix *See* Berkeley Software Distribution Unix.

BSOD *See* blue screen of death.

buffer An area of memory set aside for temporary storage of data. Often, the data remains in the buffer until some external event finishes. A buffer can compensate for the differences in transmission or processing speed between two devices or between a computer and a peripheral device, such as a printer.

Buffers are implemented in a variety of ways, including first-in-first-out (FIFO) used for pipes and last-in-last-out used for stacks and circular buffers such as event logs.

See also pipe.

buffered repeater Any device that amplifies and retransmits a signal so that it can travel greater distances. A buffered repeater can also control the flow of information to prevent collisions.

See also repeater.

bug A logical or programming error in hardware or software that causes a malfunction of some sort. If the problem is in software, it can be fixed by changes to the program. If the fault is in hardware, new circuits must be designed and constructed. Some bugs are fatal and may cause a program to stop responding or cause data loss, others are just annoying, and many are not even noticeable. The term apparently originates from the days of the first electromechanical computers, when a problem was

traced to a moth caught between two contacts inside the machinery.

See also bug fix.

bug fix A release of hardware or software that corrects known bugs but does not contain new features. Such releases are usually designated by an increase in the decimal portion of the revision number; for example, the revision level may advance from 2.0 to 2.01 or 2.1, rather than from 2.0 to 3.0.

built-in groups The default groups provided with some operating systems that define a collection of rights and permissions for members. Using built-in groups is an easy way of providing access to commonly used network resources.

bulletin board system Abbreviated BBS. A computer system equipped with one or more modems, serving as a message-passing system or centralized information source, usually for a particular special interest group. Bulletin board systems were often established by software vendors and by PC user groups in the past, but have been replaced in recent times by Web sites.

bundled software Programs combined into a single package sold for a single price. Sales aimed at a specific target, such as the medical profession, usually bundle hardware with application-specific software suited to that profession.

See also unbundled software.

burned-in address The hardware address on a network interface card (NIC). This address is assigned by the manufacturer

of the interface card, which ensures that every card has a unique address.

See also hardware address.

burst mode A method of data transmission in which information is collected and then sent in one single high-speed transmission, rather than one packet or character at a time.

Systems that use multiplexers to serve several channels often use burst mode to service each channel in turn. Much LAN traffic can be considered burst mode transmission; long periods of inactivity punctuated by short bursts of intense activity. In Novell NetWare, burst mode is known as packet burst.

bus An electronic pathway along which signals are sent from one part of a computer to another. A PC contains several buses, each used for a different purpose:

- The address bus allocates memory addresses.
- The data bus carries data between the processor and memory.
- The control bus carries signals from the control unit.

See also architecture; Extended Industry Standard Architecture; Industry Standard Architecture; local bus; Microchannel Architecture; Peripheral Component Interconnect local bus.

bus mastering A technique that allows certain advanced bus architectures to delegate control of data transfers between the CPU and associated peripheral devices to an add-in board. This technique gives network

bus network

interface cards (NICs) greater system bus access and higher data-transfer speeds.

In the PC, bus mastering is supported by all the common architectures except for the older Industry Standard Architecture.

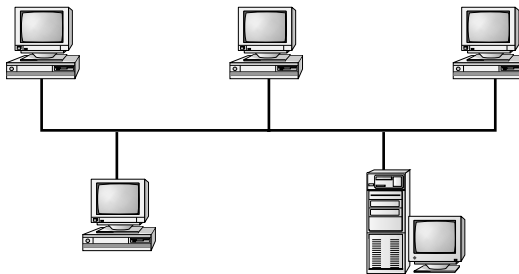
See also Extended Industry Standard Architecture; Industry Standard Architecture; local bus; MicroChannel Architecture;

Peripheral Component Interconnect local bus.

bus network In networking, a topology that allows all network nodes to receive the same message through the network cable at the same time.

See also ring network; star network; token-ring network.

BUS NETWORK



byte Contraction of binary digit eight. A group of bits. In computer storage terms, a byte usually holds a single character, such as a number, letter, or symbol. A byte usually contains 8 bits, but on some older systems, a byte may only have 7 bits or may have as many as 11.

Because bytes represent a very small amount of storage, they are usually grouped into kilobytes (1,024 bytes), megabytes (1,048,576 bytes), and gigabytes (1,073,741,824 bytes) for convenience when describing hard-disk capacity or computer memory size.

See also octet.

bytecode An intermediate form of computer code produced by Java and other programming languages.

Most language compilers create code that is ready to run on a specific kind of processor. Java creates the bytecode in an abstract, processor-independent form, which requires further processing before it can actually execute on a computer. When a byte-code file is downloaded into your computer from a Web page, it provides 70 to 80 percent of the data needed to run the Java applet; the other 20 to 30 percent is provided by the Java run-time environment, which tells the applet how to perform on the target computer system.

See also Java; Java applet; just-in-time compiler; sandbox.

byte-oriented protocol A communications protocol in which data is transmitted as a series of bytes, or characters. In order to distinguish the data from the control in-

formation, the protocol uses control characters that have a special meaning for the transmitting and receiving stations. Most of the common asynchronous communications protocols used in conjunction with modems are byte-oriented protocols.

See also bit-oriented protocol.

C

C2 One of a series of seven levels of computer security defined by the National Security Agency.

C2-level security requires that users are logged in and tracked during their session, that all resources have owners, that objects such as files be protected from processes that might damage them, that events can be audited, and that the system has adequate protection against intrusion.

C2 security applies to a stand-alone system, so in theory a networked computer cannot be C2-compliant; following the guidelines can certainly help you to set up a more secure system.

CA See certificate authority.

cable modem A modem that sends and receives signals through a coaxial cable connected to a cable-television system, rather than through conventional telephone lines.

Cable modems, with speeds of up to 500Kbps, are faster than current conventional modems, but are subject to performance changes as system load increases. Theoretical data rates are much higher than those achieved with conventional modems; downstream rates of up to 36Mbps are possible, with 3Mbps to 10Mbps likely, and upstream rates up to 10Mbps.

Cabletron Systems A leading producer of Internet and intranet hardware and

management solutions, Cabletron is also moving into Gigabit Ethernet and data, voice, and video systems.

For more information about Cabletron, see www.cabletron.com.

cabling standards National cabling standards, concerned with the performance of cables and connectors under conditions of actual use, are specified by the National Electric Code, American National Standards Institute, and Underwriters Laboratories. Other standards have been specified by the Electronics Industry Association/Telecommunications Industries Association (EIA/TIA). Standards include:

- ANSI/EIA/TIA-568-1991 Commercial Building Telecommunications Wiring.
- EIA/TIA TSB-36 Additional Cable Specifications for UTP Cables. 1991.
- EIA/TIA TSB-40 Telecommunications Systems Bulletin—Additional Transmission Specifications for UTP Connecting Hardware. 1992.
- ANSI/EIA/TIA-568A 1995 revises the original 568 document and adds material from TSB-36 and TSB-40.
- ANSI/EIA/TIA-569-1990 Commercial Building Standard for Telecommunications Pathways and Spaces.
- ANSI/EIA/TIA-570-1991 Residential and Light Commercial Telecommunications Wiring Standard.

- ANSI/EIA/TIA-606-1993 Administration Standard for the Telecommunications Infrastructure of Commercial Buildings.
- ANSI/EIA/TIA-607-1994 Commercial Building Grounding and Bonding Requirements for Telecommunications.

Local codes and standards may impose additional requirements.

Underwriters Laboratories (UL) tests cable and other devices to determine the conditions under which the device will function safely. Two important tests for cable performance are:

- UL-910, which tests smoke emission and flame spread for plenum cable
- UL-1666, which tests smoke emission and flame spread for riser cable

CAC See Connection Admission Control.

cache Pronounced “cash.” A special area of memory, managed by a cache controller, that improves performance by storing the contents of frequently accessed memory locations and their addresses.

A memory cache and a disk cache are not the same. A memory cache is implemented in hardware and speeds up access to memory. A disk cache is software that improves hard-disk performance.

When the processor references a memory address, the cache checks to see if it holds that address. If it does, the information is passed directly to the processor, so RAM access is not necessary. A cache can speed up operations in a computer whose RAM access is slow compared with its processor speed, because cache memory is always faster than normal RAM.

There are several types of caches:

- **Direct-mapped cache** A location in the cache corresponds to several specific locations in memory, so when the processor calls for certain data, the cache can locate it quickly. However, since several blocks in RAM correspond to that same location in the cache, the cache may spend its time refreshing itself and calling main memory.
- **Fully associative cache** Information from RAM may be placed in any free blocks in the cache so that the most recently accessed data is usually present; however, the search to find that information may be slow because the cache has to index the data in order to find it.
- **Set-associative cache** Information from RAM is kept in sets, and these sets may have multiple locations, each holding a block of data; each block may be in any of the sets, but it will only be in one location within that set. Search time is shortened, and frequently used data are less likely to be overwritten. A set-associative cache may use two, four, or eight sets.

See also disk cache; wait state; write-back cache; write-through cache.

cache buffer A Novell NetWare implementation of a disk cache used to speed server disk accesses, thereby allowing workstations to access data more quickly. Reading data from cache memory is much faster than reading data from the hard disk. NetWare uses cache buffers for a variety of purposes:

- For use by NetWare Loadable Modules (NLMs), such as LAN drivers, database servers, communications servers, and print servers

cache buffer pool

- To cache each volume's FAT
- To cache files currently in use
- To build a hash table of directory information

See also disk cache.

cache buffer pool In Novell NetWare, the amount of memory available for use after the SERVER.EXE file has been loaded into memory. Memory in the pool can be used for a variety of purposes, including caching the file allocation tables for each volume and creating a hash table of directory information.

See also hash table.

cache controller Pronounced “cash controller.” A special-purpose processor whose sole task is to manage cache memory. On newer processors, such as the Intel Pentium II, cache management is integrated directly into the processor.

See also cache.

cache memory Pronounced “cash memory.” A relatively small section of very fast memory (often static RAM) reserved for the temporary storage of the data or instructions likely to be needed next by the processor.

Cache memory integrated directly onto the microprocessor is called primary cache or L1 cache, and cache memory located in an external circuit is known as secondary cache or L2 cache.

See also cache.

caddy The flat plastic container used to load a compact disc into certain CD-ROM

disk drives. Most current CD-ROM drives do not require a caddy.

CAE *See* Common Application Environment.

CAI *See* Computer Associates International.

Caldera, Inc. A software company in Provo, Utah, that repackages one of the most popular versions of Linux under the name of OpenLinux.

For more information on Caldera, see www.caldera.com.

callback modem Also known as a dial-back modem. A special modem that does not answer an incoming call, but instead requires the caller to enter a code and hang up so that the modem can return the call. As long as the entered code matches a previously authorized number, the modem dials the number. Callback modems are useful in installations for which communications lines must be available for remote users but data must be protected from intruders.

caller ID *See* automatic number identification.

call packet A block of data that carries addressing information, as well as any other information needed to establish an X.25 switched virtual circuit.

campus network A network that connects LANs from multiple departments inside a single building or set of buildings. Campus networks are LANs because they do not include WAN services, even though they may extend for several miles.

canonical The usual standard Unix way of doing something. This term has a more

precise meaning in mathematics, in which rules dictate the way that formulas are written, but in Unix, it tends to mean “according to ancient or religious law.”

See also holy wars.

Canon, Inc. One of the world’s leading suppliers of imaging products, electronic equipment, computer printers, fax machines, and scanners.

For more information on Canon, see www.canon.com.

CAPI *See* Cryptography API.

card A printed circuit board or adapter that you plug into a computer to add support for a specific piece of hardware.

See also expansion board; expansion bus.

card services Part of the software support needed for PCMCIA hardware devices in a portable computer. Card services control the use of system interrupts, memory, and power management.

When an application wants to access a PCMCIA card, it always goes through the card services software and never communicates directly with the underlying hardware. For example, if you use a PCMCIA modem, it is the card services, not the applications program, that establishes which communications port and which interrupts and I/O addresses are in use.

See also device driver; PC Memory Card International Association; socket services.

carriage return A control character (ASCII 13) that signals the print head or

display cursor to return to the first position of the current line.

See also line feed.

carrier An analog signal of fixed amplitude and frequency that is combined with a data-carrying signal to produce an output signal suitable for transmitting data.

See also carrier signal.

carrier detect Abbreviated CD. An electrical signal sent from a modem to the attached computer to indicate that the modem is online.

See also Data Carrier Detect; RS-232-C.

Carrier Sense Multiple Access/Collision Detection Abbreviated CSMA/CD. A baseband protocol with a built-in collision-detection technique. Each node on the network listens first and transmits only when the line is free. If two nodes transmit at exactly the same time and a collision occurs, both nodes stop transmitting. Then, to avoid a subsequent collision, each node waits for a different random length of time before attempting to transmit again. Ethernet and 802.3 LANs use CSMA/CD access methods.

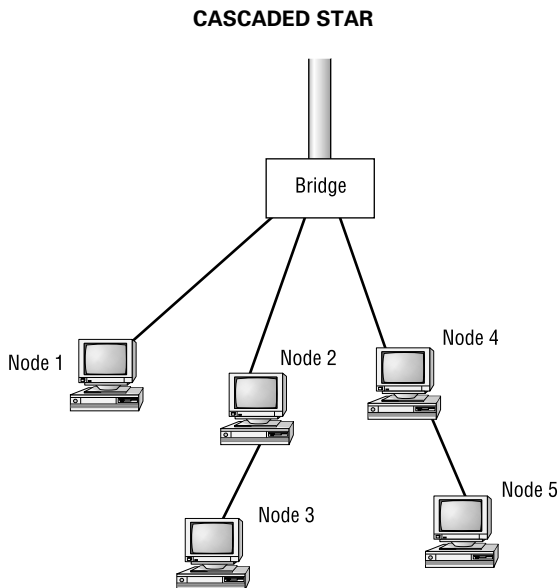
See also collision; demand priority; Fast Ethernet; token passing.

carrier signal A signal of chosen frequency generated to carry data; often used for long-distance transmissions. A carrier signal does not convey any information until the data is added to the signal by modulation and then decoded on the receiving end by demodulation.

cascaded star

cascaded star A network topology in which multiple hubs or data centers are con-

nected in a succession of levels, which permits many more connections than a single level.



cascading menus In Microsoft Windows 2000, a function that presents the contents of network drives or special folders such as Control Panel as menu items that can be selected by name.

See also personalized menus.

Castanet A collection of software-update tools from Marimba, Inc. that operate over the Internet. Based on Java technology, Castanet can automatically deliver software updates not only for Java programs, but also for those written in C, C++, and Visual Basic, and because only the updated content is downloaded, performance is optimized.

For more information on Castanet, see www.marimba.com.

See also server push.

Category 1–5 The Electronics Industry Association/Telecommunications Industry Association (EIA/TIA) 586 cabling standards, sometimes abbreviated CAT 1-5, as follows:

- **Category 1** For unshielded twisted-pair (UTP) telephone cable. This cable may be used for voice, but is not suitable for data transmissions.

- **Category 2** For UTP cable use at speeds up to 4Mbps. Category 2 cable is similar to IBM Cabling System Type 3 cable.
- **Category 3** For UTP cable use at speeds up to 10Mbps. Category 3 cable is the minimum requirement for 10BaseT and is required for Token Ring. This cable has four pairs of conductors and three twists per foot.
- **Category 4** For the lowest acceptable grade of UTP cable for use with 16Mbps Token Ring.
- **Category 5** For 100-ohm, four-wire twisted-pair copper cable for use at speeds up to 100Mbps with Ethernet or ATM (Asynchronous Transfer Mode). This cable is low-capacitance and shows low crosstalk when installed according to specifications.

See also cabling standards; Type 1–9 cable.

CAU *See* Controlled Access Unit.

CAV *See* constant angular velocity.

CBCP *See* Certified Business Continuity Professional.

CBE *See* Certified Banyan Engineer.

CBR *See* Constant Bit Rate.

CBS *See* Certified Banyan Specialist.

CCDA *See* Cisco Certified Design Associate.

CCDP *See* Cisco Certified Design Professional.

CCIE *See* Cisco Certified Internetworking Expert.

CCITT *See* Consultative Committee for International Telephony and Telegraphy.

CCITT Groups 1–4 A set of four standards for facsimile transmissions. Groups 1 and 2, which are no longer used, define analog facsimile transmissions. Groups 3 and 4 describe digital systems, as follows:

- **CCITT Group 3** Specifies a 9600bps modem to transmit standard images of 203 dpi horizontally by 98 dpi vertically in standard mode, and 203 dpi by 198 dpi in fine mode.
- **CCITT Group 4** Supports images up to 400 dpi for high-speed transmission over a digital data network (for example, ISDN), rather than over a dial-up telephone line.
- CCITT is now known as the International Telecommunication Union.

See also International Telecommunication Union.

CCITT V Series A set of recommended standards for data communications, including transmission speeds and operational modes, issued by CCITT, now known as the International Telecommunication Union.

Each standard is assigned a number, although not in chronological order. Higher numbers do not always indicate a newer standard. A second or revised version is indicated by *bis*, and *ter* indicates a third version.

See also International Telecommunication Union.

CCITT X Series A set of recommended standards issued by CCITT to standardize protocols and equipment used in public and private computer networks. The standards include transmission speeds, interfaces to and between networks, and operation of user hardware. CCITT is now known as the International Telecommunication Union.

See also International Telecommunication Union.

CCNA *See* Cisco Certified Network Associate.

CCNP *See* Cisco Certified Network Professional.

CCP *See* Certified Computing Professional.

CCP *See* Compression Control Protocol.

CD *See* carrier detect.

CDA *See* Certified Database Administrator.

CDDI *See* Copper Distributed Data Interface.

CDE *See* Common Desktop Environment.

CD-I *See* Compact Disc-Interactive.

CDIA *See* Certified Document Imaging Architect.

CDMA *See* Code Division Multiple Access.

CDP *See* Cisco Discovery Protocol.

CDPD *See* Cellular Digital Packet Data.

CD-R *See* CD-Recordable.

CD-Recordable Abbreviated CD-R. Using CD-R, you can write to the disc only once; after that, the disc can only be read from and not written to.

From a functional point of view, a CD-R and a CD-ROM are identical; you can read CD-R discs using almost any CD-ROM drive, although the processes that create the discs are slightly different. Low-cost CD-R drives are available from several manufacturers, including Kao, Kodak, Mitsui, Phillips, Ricoh, Sony, TDK, 3M, and Verbatim.

See also CD ReWritable; digital video disc; Magneto-optical storage; WORM.

CD ReWritable Abbreviated CD-RW. A CD format that can be written to and erased as many as 1000 times.

From a functional point of view, a CD-RW and a CD-ROM are identical, but not all CD-ROM drives can read CD-RW discs. Low-cost CD-RW drives are available from several manufacturers, including Kodak, Mitsui, Phillips, and Sony.

See also CD-Recordable; digital video disc; Magneto-optical storage; WORM.

CD-ROM *See* Compact Disc—Read-Only Memory.

CD-ROM disk drive A disk device that uses compact disc technology for information storage. Many CD-ROM disk drives also have headphone jacks, external speaker jacks, and a volume control.

CD-ROM disk drives designed for computer use are more expensive than audio CD players, because CD-ROM disk drives

are manufactured to much higher tolerances. If a CD player misreads a small amount of data, the human ear probably will not detect the difference; if a CD-ROM disk drive misreads a few bytes of a program, the program will not run.

The two most popular CD-ROM drive interface cards are SCSI and ATAPI (AT Attachment Packet Interface). ATAPI is part of the Enhanced IDE specification introduced by Western Digital in 1994 and lets you plug an IDE CD-ROM directly into an IDE controller on the system's motherboard. Other CD-ROM drives may use the computer's parallel port or a PCMCIA connection.

See also Compact Disc—Read-Only Memory.

CD-ROM Extended Architecture Abbreviated CD-ROM/XA. An extension to the CD-ROM format, developed by Microsoft, Phillips, and Sony, that allows for the storage of audio and visual information on compact disc so that you can play the audio at the same time you view the visual data.

CD-ROM/XA is compatible with the High Sierra specification, also known as ISO standard 9660.

CD-ROM/XA *See* CD-ROM Extended Architecture.

CD-RW *See* CD ReWritable.

cell Any fixed-length data packet. For example, Asynchronous Transfer Mode (ATM) uses 53-byte cells, consisting of 48 bytes of data and 5 bytes of header information.

Cell Loss Priority Abbreviated CLP. In an Asynchronous Transfer Mode (ATM) cell, a field contained in the 5-byte cell header that defines how to drop a cell if network congestion occurs. The value holds priority values, with 0 indicating a cell with the highest priority.

See also Header Error Control; Payload Type Identifier; Virtual Channel Identifier; Virtual Path Identifier.

cell relay A form of packet transmission used in Broadcast Integrated Services Digital Network (B-ISDN) networks that uses a fixed-length, 53-byte cell over a packet-switched network. Also known as Asynchronous Transfer Mode (ATM).

cell switching A term that describes how a cellular telephone system switches from one cell to the next as the signal strength fades. The switch takes about 300 milliseconds to complete and is not noticeable by the user.

Cellular Digital Packet Data Abbreviated CDPD. A method used in cellular communications and wireless modems for sending data more efficiently by using any idle cellular channel. Capacity becomes available for data transmissions when a voice call is terminated or when a call is switched from cell to cell.

CDPD uses voice channels, but can switch to a new frequency if a voice transmission begins in the cell currently in use.

central processing unit Abbreviated CPU. The computing and control part of the computer. The CPU in a mainframe computer may be contained on many printed circuit

boards. In a minicomputer, the CPU may be contained on several boards. The CPU in a PC is usually contained in a single, extremely powerful microprocessor.

Centrex Acronym formed from Central Exchange. Services provided to a company by the local telephone company. All the switching takes place at the telephone company's central office rather than at the customer site, so Centrex services are easy to expand.

Centronics parallel interface A standard 36-pin interface used to connect a PC to a peripheral device, such as a printer; originally developed by the printer manufacturer Centronics, Inc. The standard defines eight parallel data lines, plus additional lines for status and control information.

See also parallel port.

CERN *See* Conseil Européen pour la Recherche Nucléaire.

CERT *See* Computer Emergency Response Team.

certificate An encrypted digital signature used for authentication to prove that you are who you claim to be, either as an individual, the provider of a service, the vendor of a product, or a corporation, or to guarantee that an e-mail message is actually from the person you think it is from and that it has not been altered in any way during transmission.

See also authentication; certificate authority; Secure Sockets Layer.

certificate authority Abbreviated CA. A trusted organization that validates and

issues certificates; often called a “trusted authority.”

Certified Application Developer for Developer/2000 A certification from Oracle consisting of a set of exams covering Structured Query Language, the creation of procedures using Oracle Procedure Builder, using Developer/2000, and managing the user interface.

Certified Banyan Engineer Abbreviated CBE. A premium certification from Banyan designed to evaluate technical knowledge of the Banyan VINES network operating system. A candidate must complete two required courses and pass two exams.

Certified Banyan Specialist Abbreviated CBS. A basic certification from Banyan designed to evaluate technical knowledge of the Banyan VINES network operating system. A candidate must complete one required course and pass three exams. An additional certification concentrates on the integration of Banyan's StreetTalk onto Microsoft Windows 2000 servers; to complete this certification, you must first become a Microsoft Certified Professional.

Certified Business Continuity Professional Abbreviated CBCP. A certification from Disaster Recovery Institute International (DRII) that covers intermediate information on business continuity planning and disaster recovery and associated work experience and board certification.

See also Associate Business Continuity Professional; Master Business Continuity Professional.

Certified Computing Professional

Abbreviated CCP. A certification from the Institute for Certification of Computing Professionals designed for experienced professionals with more than four years experience in a wide variety of computing and related tasks.

Certified Database Administrator

1. Abbreviated CDA. A certification from Oracle that covers knowledge of Structured Query Language, administration of Oracle products, along with backup and recovery, and system performance tuning.

2. Abbreviated CDA. A certification from Sybase that covers designing, building, and supporting Sybase SQL Server databases.

See also Certified Performance and Tuning Specialist.

Certified Document Imaging Architect

Abbreviated CDIA. A certification from the Computer Technology Industry Association (CompTIA) that assesses skills in document management, including scanning and storing documents in digital form and using optical character recognition software.

Certified Information System Auditor

Abbreviated CISA. A certification from the Information Systems Audit and Control Association (ISACA) that covers ethics, security, system organization and management, and system development, acquisition, and maintenance.

Certified Information Systems Security Professional

Abbreviated CISSP. A certification from the International Information System Security Certification

Consortium (ISC)² designed for system security experts with at least three years of practical experience. The exam covers access control systems, operations security, cryptography, applications and systems planning, business continuity and disaster planning, telecommunications and network security, ethics, and the law.

Certified Java Developer Abbreviated CJD. An advanced certification from Sun Microsystems designed to evaluate Java programming language skills.

See also Certified Java Programmer.

Certified Java Programmer Abbreviated CJP. A certification from Sun Microsystems designed to evaluate Java programming language skills.

See also Certified Java Developer.

Certified Lotus Professional Abbreviated CLP. A certification from Lotus designed to evaluate a wide range of skills and knowledge of Lotus products. Several certifications are available, including CLP: Application Developer, CLP: Principal Application Developer, CLP: System Administrator, CLP: Principal System Administrator, CLP: cc:Mail System Administrator.

See also Certified Lotus Specialist.

Certified Lotus Specialist Abbreviated CLS. A certification from Lotus designed to evaluate knowledge of a single Lotus product. Options include certification in Lotus Domino, Lotus 1-2-3, Lotus Notes, and cc:Mail.

See also Certified Lotus Professional.

Certified Network Professional Abbreviated CNP. A certification from the Network Professional Association (NPA) that involves meeting requirements in the areas of client operating systems, networking hardware, network operating systems, communications protocols, and network topologies. Two years of relevant work experience and two vendor-based certifications are required to complete the program.

Certified Novell Administrator Abbreviated CNA. A Novell certification program for network administrators responsible for the day-to-day operation of a network. Within the CNA program, a candidate can specialize in one or more Novell products, including IntranetWare, NetWare, or GroupWise.

See also Certified Novell Engineer; Master Certified Novell Engineer.

Certified Novell Engineer Abbreviated CNE. A Novell certification program for technical professionals concerned with network system design, implementation, and maintenance. Within the CNE program, a candidate can specialize in one or more Novell products, including IntranetWare, NetWare, or GroupWise.

See also Certified Novell Administrator; Master Certified Novell Engineer.

Certified Novell Engineer Professional Association Abbreviated CNEPA. An association of Certified Novell Engineers (CNEs) that provides benefits, such as workshops demonstrating how to configure and troubleshoot Novell products, as

well as admission to network-related events and complementary subscriptions.

See also Certified Novell Engineer.

Certified Novell Instructor Abbreviated CNI. A Novell certification program for trainers who want to teach Novell courses. Once certified, by passing the appropriate exam at the instructor level, a CNI can teach any instructor-led Novell course, including those for Certified Novell Administrator (CNA), Certified Novell Engineer (CNE), and Master CNE (MCNE).

See also Master Certified Novell Instructor.

Certified Performance and Tuning Specialist Abbreviated CPTS. A certification from Sybase that tests for proficiency across a wide range of database administration, tuning, and performance concepts.

See also Certified Database Administrator.

Certified PowerBuilder Developer Associate Abbreviated CPDA. A certification from Sybase designed for experienced PowerBuilder developers that tests knowledge in the areas of client/server architecture, object-oriented programming, relational databases, and Structured Query Language.

See also Certified PowerBuilder Developer Professional.

Certified PowerBuilder Developer Professional Abbreviated CPDP. A certification from Sybase designed for experienced PowerBuilder developers that requires hands-on experience building PowerBuilder applications as well as a demonstration of good development practices.

See also Certified PowerBuilder Developer Associate.

Certified Solutions Expert Abbreviated CSE. A pair of Internet certifications from IBM. CSE: Net.Commerce is aimed at the developers of Internet store fronts and other financial transactions over the Internet. CSE: Firewall is a security-related certification.

Certified Unicenter Engineer Abbreviated CUE. A certification from Computer Associates designed to evaluate expertise in the Unicenter TNG product line.

CGI *See* Common Gateway Interface.

Challenge-Handshake Authentication Protocol Abbreviated CHAP. A method of authentication that you can use when connecting to an ISP that allows you to log on automatically.

See also Password Authentication Protocol.

challenge-response authentication A method of authentication used by Microsoft Windows 2000 and other operating systems.

When a user contacts a server, the server responds with a challenge, upon which the user then performs a cryptographic operation and returns the result to the server. The server then performs the same operation, and if the two results are the same, the user is considered authentic.

See also clear text authentication.

channel **1.** In communications, any connecting path that carries information from a sending device to a receiving device. A

channel may refer to a physical medium (for example, a coaxial cable) or to a specific frequency within a larger channel.

2. In Internet Relay Chat (IRC), a named forum where you can chat in real time with other users; also known as a chat room.

channelization The process of dividing the bandwidth of a communications circuit into smaller increments.

See also T1.

channelized T1 *See* T1.

Channel Service Unit Abbreviated CSU. A device that functions as a certified safe electrical circuit, acting as a buffer between the customer's equipment and a public carrier's WAN.

A CSU prevents faulty CPE (customer-premises equipment), such as DSUs (data service units), from affecting a public carrier's transmission systems and ensures that all signals placed on the line are appropriately timed and formed. All CSU designs must be approved and certified by the FCC.

See also Data Service Unit.

CHAP *See* Challenge-Handshake Authentication Protocol.

character A symbol that corresponds to a key on the keyboard. A character can be a letter, a number, punctuation, or a special symbol and is usually stored as a single byte. A collection of related characters is known as a *character set*, and the most common character set on PC systems is the American Standard Code for Information Interchange (ASCII). Some larger IBM systems still use Extended Binary Coded Decimal Interchange

character-based interface

Code (EBCDIC). In an attempt to rationalize the many international character sets in use these days, some systems use more than one byte to store a character.

See also American Standard Code for Information Interchange; Extended Binary Coded Decimal Interchange Code; Unicode.

character-based interface An operating system or application that uses text characters rather than graphical techniques for the user interface.

See also command line; graphical user interface.

character code A code that represents one specific alphanumeric or control character in a set of characters.

See also American Standard Code for Information Interchange; Extended Binary Coded Decimal Interchange Code; Unicode.

character mode A mode in which the computer displays characters on the screen using the built-in character set, but does not show any graphics characters or a mouse pointer. Also known as text mode.

character set A standard group of letters, numbers, punctuation marks, special symbols, and control characters used by a computer.

See also American Standard Code for Information Interchange; Extended Binary Coded Decimal Interchange Code; Unicode.

characters per second Abbreviated cps. The number of characters, or bytes, transmitted every second during a data transfer. A measurement of the speed of operation of equipment, such as serial printers and terminals.

character string Any group of alphanumeric characters treated as a single unit. Also known as a string.

cheapernet wire *See* thin Ethernet.

checkpointing The process of moving transactions from the transaction log to their permanent disk location.

checksum A method of providing information for error detection, usually calculated by summing a set of values.

The checksum is usually appended to the end of the data that it is calculated from so that they can be compared. For example, Xmodem, a popular file-transfer protocol, uses a 1-byte checksum calculated by adding all the ASCII values for all 128 data bytes and ignoring any numeric overflow. The checksum is added to the end of the Xmodem data packet. This type of checksum does not always detect all errors. In later versions of the Xmodem protocol, cyclical redundancy check (CRC) is used instead for more rigorous error control.

See also cyclical redundancy check; error detection and correction.

child domain In Microsoft Windows 2000 Server, a secondary domain beneath the top-level domain in a Windows 2000 domain tree.

See also first-layer domain; parent domain; root domain.

chip A slang expression for integrated circuit.

See also integrated circuit.

choke packet A packet used for flow control. A node that notices congestion on the network generates a choke packet and sends it toward the source of the congestion, which is then required to reduce its sending rate.

CHRP *See* Common Hardware Reference Platform.

CICS *See* Customer Information Control System.

CIDR *See* Classless Inter-Domain Routing.

CIFS *See* Common Internet File System.

CIOS *See* Internetworking Operating System.

circuit 1. A communications channel or path between two devices capable of carrying electrical current.

2. A set of components connected to perform a specific task.

circuit switching A temporary communications connection established as required between the sending and receiving nodes. Circuit switching is often used in modem communications over dial-up telephone lines. It is also used in some privately maintained communications networks.

See also message switching; packet switching; virtual circuit.

Cirrus Logic, Inc. A leading manufacturer of integrated circuits, particularly semiconductor wafers, Cirrus Logic has also moved into the areas of 56K modem chip sets and sound-card chip sets.

For more information on Cirrus Logic, see www.cirrus.com.

CIS *See* CompuServe.

CISA *See* Certified Information System Auditor.

CISC *See* complex instruction set computing.

ciscoBus The bus used in routers from Cisco Systems, Inc.

See also Cisco Extended Bus; switch processor.

ciscoBus controller *See* switch processor.

Cisco Certified Design Associate Abbreviated CCDA. A certification from Cisco designed to evaluate knowledge of relatively simple networks.

See also Cisco Certified Design Professional.

Cisco Certified Design Professional

Abbreviated CCDP. A certification from Cisco designed to evaluate knowledge of complex networks based on Cisco LAN and WAN routers and LAN switches.

See also Cisco Certified Design Associate.

Cisco Certified Internetworking

Expert Abbreviated CCIE. An advanced certification from Cisco offered in three

areas: CCIE-Routing and Switching, CCIE-Internet Service Provider (ISP) Dial, and CCIE-WAN Switching.

Cisco Certified Network Associate

Abbreviated CCNA. A certification from Cisco designed to evaluate network support knowledge.

See also Cisco Certified Network Professional.

Cisco Certified Network Professional

Abbreviated CCNP. An advanced certification from Cisco designed to evaluate network support knowledge.

See also Cisco Certified Network Associate.

Cisco Discovery Protocol Abbreviated CDP. A proprietary Cisco protocol that runs on all hardware made by Cisco, including routers, switches, access servers, and bridges.

CDP is both protocol- and media-independent. A router running CDP can advertise its existence to other routers and can discover a directly connected neighbor's port and hostname information, as well as hardware model number and system capabilities.

Cisco Extended Bus Abbreviated Cx-Bus. The extended bus used in routers from Cisco Systems, Inc.

See also ciscoBus; switch processor

Cisco Systems, Inc. The world's leading manufacturer of routers and internet-working hardware and software products. More than 80 percent of the backbone routers currently in use on the Internet were

made by Cisco, and its Internetwork Operating System (IOS) is quickly becoming an industry standard that other vendors are incorporating into their own products.

For more information on Cisco, see www.cisco.com.

CiscoView Device management software for Cisco networking devices. CiscoView provides troubleshooting and device monitoring functions and can be used in conjunction with Simple Network Management Protocol (SNMP) products.

See also Simple Network Management Protocol.

Ciscoworks A set of management applications from Cisco Systems, Inc., designed for use with Cisco routers.

CISSP *See* Certified Information Systems Security Professional.

CIX *See* Commercial Internet Exchange.

CJD *See* Certified Java Developer.

CJP *See* Certified Java Programmer.

cladding The transparent material, usually glass, that surrounds the core of an optical fiber. Cladding has a lower refractive index than the core and so prevents the light signal from spreading out due to modal dispersion, by reflecting the signal back into the central core. This helps to maintain the signal strength over long distances.

See also dispersion.

Class A certification An FCC certification for computer equipment, including mainframe computers and minicomputers

destined for industrial, commercial, or office use, rather than for personal use at home. The Class A commercial certification is less restrictive than the Class B certification for residential use, because it assumes that most residential areas are more than 30 feet from any commercial computer equipment.

See also Class B certification.

Class A network In the IP addressing scheme, a very large network. The high-order bit in a Class A network is always zero, leaving 7 bits available to define 127 networks. The remaining 24 bits of the address allow each Class A network to hold as many as 16,777,216 hosts. Examples of Class A networks include General Electric, IBM, Hewlett-Packard, Apple Computer, Xerox, Digital Equipment Corporation, and MIT. All the Class A networks are in use, and no more are available.

See also address classes; IP address.

Class B certification An FCC certification for computer equipment, including PCs, laptops, and portables destined for use in the home rather than in a commercial setting. Class B levels of radio frequency interference (RFI) must be low enough so that they do not interfere with radio or television reception when there is more than one wall and 30 feet separating the computer from the receiver. Class B certification is more restrictive than the commercial Class A certification.

See also Class A certification.

Class B network In the IP addressing scheme, a medium-sized network. The 2 high-order bits are always 10, and the remaining

bits are used to define 16,384 networks, each with as many as 65,535 hosts attached. Examples of Class B networks include Microsoft and Exxon. All Class B networks are in use, and no more are available.

See also address classes; IP address.

Class C network In the IP addressing scheme, a smaller network. The 3 high-order bits are always 110, and the remaining bits are used to define 2,097,152 networks, but each network can have a maximum of only 254 hosts. Class C networks are still available.

See also address classes; IP address.

Class D network In the IP addressing scheme, a special multicast address that cannot be used for networks. The 4 high-order bits are always 1110, and the remaining 28 bits allow access to more than 268 million possible addresses.

See also address classes; IP address.

Class E network In the IP addressing scheme, a special address reserved for experimental purposes. The first 4 bits in the address are always 1111.

See also address classes; IP address.

classes In Novell Directory Services, an object can be defined as an instance of an object class. Classes include User, Group, Printer, Print Server, Computer, and so on.

Classless Inter-Domain Routing Abbreviated CIDR, pronounced “cider.” An interim solution to the problem that the Internet is running out of addresses.

Blocks of Class C addresses are assigned to a site based on the number of addresses that site requires, to prevent wasted addresses. Also, the Class C address space is divided into four major areas—Europe, North America, Central and South America, and Asia and the Pacific—and each zone is assigned 32 million addresses.

CIDR networks are often described as “slash x ” networks; the x represents the number of bits in the IP address range controlled by the granting authority. For example, a Class C network in CIDR terms becomes a slash 24 network.

See also address classes; IP address.

cleartext Text that has not been encrypted in any way and that can be intercepted and read easily while in transit; usually applied to an unencrypted password.

See also plaintext.

clear text authentication An authentication method that encodes user name and password information according to a freely available 64-bit encoding utility.

See also challenge-response authentication; encryption.

Clear to Send Abbreviated CTS. A hardware signal defined by the RS-232-C standard that indicates that the transmission can proceed.

See also RS-232-C; Request to Send.

CLEC *See* Competitive Local Exchange Carrier.

client A device or application that uses the services provided by a server.

A client may be a PC or a workstation on a network using services provided from the network file server, or it may be that part of an application program that runs on the workstation supported by additional software running on the server.

One of the most familiar clients is the Web browser.

See also client/server architecture; DOS client; Macintosh client; OS/2 client; Unix client; Windows client.

client application In OLE, the application that starts a server application to manipulate linked or embedded information.

client pull A mechanism used on the Internet whereby a client application, usually a Web browser, initiates a request for services from a Web site.

See also server push.

Client Services for NetWare A software package included with Microsoft Windows 2000 that connects a Windows 2000 client to a Novell NetWare file server.

client/server architecture A computing architecture that distributes processing between clients and servers on the network.

In the past, traditional computing has relied on a hierarchical architecture based on non-programmable dumb terminals connected to a mainframe computer. In this scheme, the database was on the same computer that was running the application. A client/server approach replaces this structure by dividing the application into two separate parts: a front-end client and a back-end

server, usually referred to as a *client* and a *server*.

The client component provides the user with the power to run the data-entry part of the application, and this part of the client is usually optimized for user interaction with the system.

The server component, which can be either local or remote, provides the data management, administration, and system security features and manages information sharing with the rest of the network.

In other words, clients request information from the servers, and the servers store data and programs and provide network services to clients.

Client/server architecture can sustain several levels of organizational complexity, including the following:

- Stand-alone (non-networked) client applications, such as local word processors
- Applications that run on the client but request data from the server, such as spreadsheets
- Programs that use server capabilities to share information among network users, such as electronic mail systems
- Programs in which the physical search of records takes place on the server, while a much smaller program running on the client handles all user-interface functions, such as database applications

Client/server computing lightens the processing load for the client PCs, but increases the load on the server. For this reason, server computers tend to have larger and faster hard-disk drives and much more memory installed than conventional file servers. The

server may also be a minicomputer or a mainframe computer.

Typically, a client/server approach reduces network traffic, because relatively small amounts of data are moved over the network. This is in sharp contrast to the typical network, in which entire files are constantly being transmitted between the workstation and the file server.

Database applications were some of the first to embrace the client/server concept, particularly those using Structured Query Language (SQL). SQL has grown into an industry standard database language; it is relatively easy to implement, it is robust and powerful, and it is easy for users to learn.

See also network computer; thin client.

client-side caching *See* offline files.

Clipper chip A low-cost encryption device backed by the U.S. federal government.

The chip would allow businesses to transmit encoded messages, but at the same time, allow certain government agencies to intercept and decode the messages if criminal activities were suspected. Needless to say, this proposal has generated a lot of intense discussion, particularly from civil rights groups concerned with an individual's right to privacy and other ethical issues; other potential users want access to the best available encryption systems, not just those put forward by the government.

CLNP *See* Connectionless Network Protocol.

clock An electronic circuit that generates regularly spaced timing pulses at speeds up

clock-multiplying

to millions of cycles per second. These pulses are used to synchronize the flow of information through the computer's internal communications channels.

See also clock speed.

clock-multiplying A mechanism used by some Intel processors that allows the chip to process data and instructions internally at a speed different from that used by the rest of the system.

clock speed The internal speed of a computer or processor, normally expressed in megahertz (MHz). Also known as clock rate. The faster the clock speed, the faster the computer will perform a specific operation (assuming the other components in the system, such as disk drives, can keep up with the increased speed).

The Intel 8088 processor used in the original IBM PC had a clock speed of 4.77MHz—painfully slow when compared with speeds used by current processors, which can run at clock speeds of several hundred MHz.

clone Hardware that is identical in function to an original.

For example, an IBM clone is a PC that uses an Intel (or similar) microprocessor and functions in the same way as the IBM PC standard. A Macintosh clone functions in the same way as a computer manufactured by Apple Computer, Inc.

Although most clones do perform as intended, small internal differences can cause problems in some cases. It can be difficult to ensure consistency of components and level of operation when using a number of clones purchased over a long period of time.

See also Advanced Micro Devices, Inc.; Cyrix.

closed architecture A design that does not allow for easy, user-supplied additions. This term is often used to describe some of the early Macintosh computers, which did not allow easy expansion of the system with add-in cards. Closed architecture can also refer to a computer design whose specifications are not published or generally available, making it impossible for third-party companies to provide products that work with the computer.

See also open architecture.

CLP *See* Cell Loss Priority; Certified Lotus Professional.

CLS *See* Certified Lotus Specialist.

CLTP *See* Connectionless Transport Protocol.

cluster controller An IBM or IBM-compatible device located between a group of 3270 terminals and the mainframe computer. The cluster controller communicates between the computer and the terminals using SDLC (Synchronous Data Link Control) or a bisynchronous communications protocol.

clustering A fault-tolerant technology designed to keep server availability at a very high level.

Clustering groups servers and other network resources into a single system; if one of the servers in the cluster fails, the other servers can take over the workload. Clustering software also adds a load-balancing feature to make sure that processing is

distributed in such a way as to optimize system throughput.

CLV *See* constant linear velocity.

CMIP *See* Common Management Information Protocol.

CMIS *See* Common Management Information Services.

CMOS *See* Complementary Metal-Oxide Semiconductor.

CNA *See* Certified Novell Administrator.

CNE *See* Certified Novell Engineer.

CNEPA *See* Certified Novell Engineer Professional Association.

CNI *See* Certified Novell Instructor.

CNP *See* Certified Network Professional.

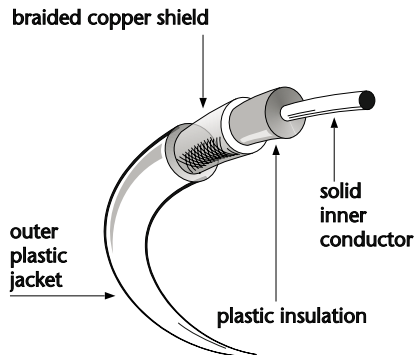
coax *See* coaxial cable.

coaxial cable Abbreviated coax, pronounced “co-ax.” A high-capacity cable used in networking that contains a solid inner copper conductor surrounded by plastic insulation, and an outer braided copper or foil shield.

Coaxial cable is used for broadband and baseband communications networks (and for cable television), because the cable is usually free from external interference and permits high transmission rates over long distances.

See also fiber-optic cable; RG-58; RG-59; RG-62; thick Ethernet; thin Ethernet.

COAXIAL CABLE



codec 1. Acronym for coder/decoder, pronounced “coe-deck.” A device that converts analog signals (such as voice or video) into a digital bit stream suitable for

transmission and then converts those digital signals back into analog signals at the receiving end.

Code Division Multiple Access

2. Acronym for compression/decompression, pronounced “coe-deck.” An overall term to describe the hardware and software used in processing animation, digital video, and stereo-quality audio.

See also lossless compression; lossy compression.

Code Division Multiple Access Abbreviated CDMA, also known as spread spectrum. A digital cellular standard approved by the Telecommunications Industry Association (TIA) in 1993 and known as IS-95.

CDMA combines both data and voice into a single wireless network and can provide users with digital voice services, voice mail, text messaging, and caller ID. CDMA also increases system capacity up to 10 times that of analog systems.

See also Advanced Mobile Phone Service; Cellular Digital Packet Data; wireless communications.

cold boot The computer startup process that begins when you turn on power to the computer. You are doing a cold boot when you first turn on your computer. A cold boot might also be necessary if a program or the operating system crashes and freezes entirely. If your keyboard is operational, a warm boot may suffice.

See also boot; warm boot.

collaboration software A set of network-based applications that let users share information quickly and easily.

See also whiteboard.

collision In networking or communications, an attempt by two nodes to send a

message on the same channel at exactly the same moment.

See also Carrier Sense Multiple Access/Collision Detection; token-ring network.

colon The symbol used after the protocol name in a URL.

See also Uniform Resource Locator.

COM *See* Common Object Model; Component Object Model.

COM 1–4 *See* COM port.

command interpreter *See* command processor.

command line Any interface between the user and the command processor that allows you to enter commands from the keyboard for execution by the operating system.

See also graphical user interface; shell; text mode.

command-line argument A parameter that alters the default mode of a command. In many operating systems, a command-line argument is one or more letters or numbers preceded by the / (slash) character. In Unix, a command-line argument may be called an option or a flag and is usually a single character preceded by a hyphen (as in -r). With some commands, you can group several switches. Sometimes called a command-line switch.

See also command line.

command processor The part of the operating system that displays the command prompt on the screen, interprets and

executes all the commands and filenames that you enter, and displays error messages when appropriate. Also called the command interpreter. The command processor also contains the system environment.

command prompt A symbol (character or group of characters) on the screen that lets you know that the operating system is available and ready to receive input.

Commercial Internet Exchange Abbreviated CIX, pronounced “kicks.” A connection point between ISPs.

A location where top-tier ISPs maintain the routers used to route packets between their respective network segments.

Common Application Environment

Abbreviated CAE. A set of standards, developed by X/Open for application development, including standards for the operating system, compilers, software development tools, data management, networking, and the graphical user interface.

common carrier A communications company, such as AT&T or MCI, that provides data and voice telecommunication services to the general public.

See also Postal Telephone and Telegraph.

Common Desktop Environment Abbreviated CDE. A set of specifications developed by the Common Open Software Environment (COSE) that defines an API for a common Unix graphical user interface. The specifications cover the interoperability of applications across different hardware platforms, multimedia and networking operations, as well as object-

oriented technology and system administration issues.

Common Gateway Interface Abbreviated CGI. A standard way that programs can interface with Web servers and allow them to run applications such as search engines and to access databases and other back-end applications.

CGI defines the field specifics and control tags to be placed in an HTML document, the environment variables where the Web server places information for use by scripts, and the flow of information between client Web browser, the server, and the Web server scripts. CGI is platform-independent.

See also HyperText Markup Language; script; Web server.

Common Internet File System Abbreviated CIFS. A file system supported by Microsoft, DEC, Data General, SCO, and others, which allows users and organizations to run file systems over the Internet.

CIFS is an extension to Microsoft's Server Message Blocks (SMB) file-sharing protocol and allows users to share files over the Internet in the same way that they share files using networking services on Windows clients.

See also WebNFS.

Common Hardware Reference

Platform Abbreviated CHRP. An open hardware architecture, based on the PowerPC and originally defined by IBM, that ensures compatibility between systems made by different manufacturers.

Common Management Information Protocol

Common Management Information Protocol

Abbreviated CMIP. The Open Systems Interconnection (OSI) management information protocol for network monitoring and control information, designated ISO 9596.

CMIP includes fault management, configuration management, performance management, and security and accounting management. It is not widely available.

See also Common Management Information Services; Simple Network Management Protocol.

Common Management Information Services

Abbreviated CMIS. The Open Systems Interconnect (OSI) standard functions for network monitoring and control.

See also Common Management Information Protocol.

Common name In Novell Directory Services, every object has a name that is unique within its context in the directory structure. This name is known as the Common name when it refers to users, nodes, or servers.

See also Distinguished Name.

Common Object Model Abbreviated COM. A specification from Microsoft and DEC to provide cross-platform interoperability across non-Windows platforms.

COM was developed to allow networks using Microsoft's OLE technology to communicate with networks using DEC's Object-Broker technology.

Common Object Request Broker

Architecture Abbreviated CORBA. A standard from the Object Management Group (OMG), whose members include Sun Microsystems, Hewlett-Packard, DEC, and IBM, that enables communications between distributed object-oriented applications, regardless of the language they are written in and the hardware platform on which they run.

CORBA uses Object Request Brokers to set up communications between objects and to invoke methods on behalf of these objects.

CORBA competes with Microsoft's Distributed Component Object Model (DCOM) and ActiveX technology.

See also Distributed Component Object Model.

Common Open Software Environ-

ment Abbreviated COSE, pronounced "cosy." An industry group consisting of almost 100 members, organized to develop a standard graphical user interface for Unix, known as the Common Desktop Environment, or CDE. Original members included Sun Microsystems, Hewlett-Packard, IBM, SCO, and the UNIX Systems Group.

Common Programming Interface for

Communications Abbreviated CPI-C. A cross-platform API from IBM that interfaces to the Advanced Program-to-Program Communications (APPC) environment.

CPI-C is designed to support a common environment for application execution across several IBM operating systems, including MVS, VS, OS/400, and OS/2-based systems.

See also Advanced Peer-to-Peer Networking; Advanced Program-to-Program Communications; Advanced Interactive Executive; Multiple Virtual Storage.

common share In Microsoft Windows 2000, system elements automatically shared by default when the operating system is installed or upgraded. Many common shares are also hidden shares.

See also default shares; hidden share; share; sharing.

communications/modem server In a network, a server equipped with a bank of modems, which can be shared by users for outgoing calls.

See also access server.

communications parameters Any of several settings required to allow computers to communicate successfully. In asynchronous transmissions, commonly used in modem communications, the settings for baud rate, number of data bits, number of stop bits, and parity parameters must all be correct.

communications protocol 1. A standard way of communicating between computers or between computers and terminals. Communications protocols vary in complexity, ranging from Xmodem, a simple file-transfer protocol used to transfer files from one PC to another, to the seven-layer OSI Reference Model used as the theoretical basis for many large, complex computer networks.

2. A hardware interface standard, such as RS-232-C.

communications satellite A satellite in geostationary orbit, acting as a microwave relay station. The satellite receives signals from a groundstation, amplifies them, and retransmits them on a different frequency to another groundstation.

See also downlink; propagation delay; uplink.

compact disc Abbreviated CD. A non-magnetic, polished, optical disc used to store large amounts of digital information. A CD can store approximately 650MB of information, equivalent to more than 1700 low-density floppy disks. This storage capacity translates into approximately 300,000 pages of text or 72 minutes of music, all on a single 4.72-inch disc.

Digital information is stored on the compact disc as a series of microscopic pits and smooth areas that have different reflective properties. A beam of laser light shines on the disc so that the reflections can be detected and converted into digital data.

See also CD-Recordable; CD ReWritable; CD-ROM Extended Architecture; Compact Disc-Interactive; Compact Disc—Read-Only Memory; digital video disc.

Compact Disc-Interactive Abbreviated CD-I. A hardware and software standard disc format for data, text, audio, still video images, and animated graphics. The standard also defines methods of encoding and decoding compressed data, as well as displaying data.

See also compact disc; Compact Disc—Read-Only Memory.

Compact Disc—Read-Only Memory

Abbreviated CD-ROM. A high-capacity, optical storage device that uses the same technology used to make ordinary music discs to store large amounts of information. A single 4.72-inch disc can hold up to 650MB.

CD-ROMs are important components of multimedia applications. They are also used to store encyclopedias, dictionaries, and other large reference works, as well as libraries of fonts and clip art for desktop publishing. CD-ROMs have replaced floppy disks as the distribution mechanism for software packages, including network operating systems and large applications; you can load the whole package from a single compact disc, and you can load an operating system from a set of discs.

A CD-ROM uses the constant linear velocity data encoding scheme to store information in a single, spiral track, divided into many equal-length segments. To read data, the CD-ROM disk drive must increase the rotational speed as the read head gets closer to the center of the disk and must decrease as the head moves back out.

See also CD-Recordable; Compact Disc-Interactive; constant angular velocity; High Sierra specification.

Compaq Computer Corporation One of the world's largest computer manufacturers, Compaq recently acquired the Digital Equipment Corporation (DEC), which boosted the company into the top three along with IBM and Hewlett-Packard.

By acquiring DEC, Compaq gained a world-wide service and support structure,

as well as access to DEC's highly regarded hardware and software products.

Compaq also shook up the computer industry when it introduced the first PC that cost less than \$1,000.

For more information about Compaq, see www.compaq.com.

compatibility The extent to which a given piece of hardware or software conforms to an accepted standard, regardless of the original manufacturer.

In hardware, compatibility is often expressed in terms of widely accepted models—this designation implies that the device will perform in the same way as the standard device.

In software, compatibility is usually described as the ability to read data file formats created by another vendor's software or the ability to work together and share data.

See also plug-compatible.

Competitive Local Exchange Carrier

Abbreviated CLEC. A term coined from the Telecommunications Act of 1996 to describe the deregulated, competitive phone companies that will be able to offer local exchange service as well as long distance and international services, Internet access, and cable and video on-demand services.

See also Incumbent Local Exchange Carrier; Local Exchange Carrier.

compiler A program that converts a set of program language source code statements into a machine-readable form suitable for execution by a computer.

Most compilers do much more than this, however; they translate the entire program into machine language, while at the same time, they check your source code syntax for errors and then post error messages or warnings as appropriate.

See also just-in-time compiler.

Complementary Metal-Oxide

Semiconductor Abbreviated CMOS, pronounced “See-moss.” A type of integrated circuit used in processors and for memory. Because CMOS devices operate at very high speeds and use little power, they generate little heat. In the PC, battery-backed CMOS memory is used to store operating parameters, such as the hard disk type, when the computer is switched off.

complex instruction set computing

Abbreviated CISC, pronounced “sisk.” A processor that can recognize and execute more than 100 different assembly-language, or low-level, instructions. CISC processors can be powerful, but the instructions take a high number of clock cycles to execute.

This complexity is in contrast to the simplicity of reduced instruction set computing (RISC) processors, in which the number of available instructions has been cut to a minimum. RISC processors are common in workstations and can be designed to run up to 70 percent faster than CISC processors.

See also assembly language; reduced instruction set computing.

Component Object Model Abbreviated COM. A specification from Microsoft that defines how objects interact in the Windows environment.

COM components can be written in any programming language and can be added to or removed from a program without requiring recompilation. COM is the foundation of the Object Linking and Embedding (OLE) and ActiveX specifications.

See also ActiveX; Common Object Request Broker Architecture; Distributed Component Object Model.

COM port In MS-DOS and Microsoft Windows, the device name used to denote a serial communications port. In versions of MS-DOS after 3.3, four COM ports are supported: COM1, COM2, COM3, and COM4. Earlier versions support only COM1 and COM2.

compound document A document file that consists of information created by two or more applications, for example, a database document embedded within a word-processing document.

Object Linking and Embedding (OLE) can link and embed documents and can be used to start the appropriate application program.

See also Object Linking and Embedding.

compressed file A file that has been processed by a special utility so that it occupies as little hard-disk space as possible. When the file is needed, the same program decompresses the file back into its original form so that it can be read by the computer.

Popular compression techniques include schemes that replace commonly occurring sequences of characters by tokens that take up less space. Some utilities use Huffman

Compression Control Protocol

coding to shrink a file, and others use adaptive Lempel-Ziv coding.

See also data compression; file compression; PKZip; WinZip; ZIP file.

Compression Control Protocol Abbreviated CCP. A protocol used with Point-to-Point Protocol (PPP) to configure, enable, and disable data compression algorithms at both ends of the point-to-point connection. CCP can support different compression algorithms in each direction of the connection.

See also Point-to-Point Protocol.

CompuServe Abbreviated CIS. A major provider of online services including e-mail, file downloading, forums on a variety of topics, chat rooms, and Internet access, as well as commercial services offered to a large number of business users. In 1998, CompuServe was bought by America Online.

For more information on CompuServe, see www.compuserve.com.

computation bound A condition in which the speed of operation of the processor actually limits the speed of program execution. The processor is limited by the number of arithmetic operations it can perform.

See also input/output bound.

computer account In Microsoft Windows 2000, an object in the Security Accounts Manager that describes a specific computer within a network domain. A computer account is added for each node added to the domain.

Computer Associates International

Abbreviated CAI. A leading supplier of enterprise applications and services to the corporate world, including defense contractors and Fortune 500 companies.

CAI's rapid expansion in recent years has been due to an aggressive acquisitions policy that has led to the incorporation of Legent and Cheyenne Software and to a joint venture with Fujitsu that created the object-oriented database Jasmine.

CAI currently enjoys success with Unicenter TNG, a collection of applications and tools used to manage enterprise computing.

For more information on CAI, see www.cai.com.

Computer Emergency Response Team

Abbreviated CERT. Founded in 1988 at Carnegie-Mellon University, CERT works with the Internet community to increase awareness of security issues; it conducts research into improving existing systems and provides 24-hour technical assistance service for responding to security incidents.

Computer Management In Microsoft Windows 2000, the administrative tool used to manage a local or remote computer.

Computer Management is a Microsoft Management Console (MMC) snap-in and replaces several Windows NT administrative tools, including Server Manager, User Manager for Domains, and Disk Administrator.

See also Active Directory; Microsoft Management Console; snap-in.

computer name 1. In Microsoft Windows NT, a name of up to 15 uppercase letters that identifies a specific computer to the other computers on the network. The computer name is created during installation and must be unique within the workgroup or domain; you can change the computer name using the Network applet in Control Panel.

2. In NetBIOS, a name of up to 15 characters that uniquely identifies a workstation to computers and users. Once a computer is named, NetBIOS can establish sessions between computers and use those links to exchange data between systems. These exchanges can be either NetBIOS requests or Server Message Block (SMB) data. Network applications use NetBIOS names to locate resources, although the Windows Sockets API is now more commonly used.

See also NetBIOS; WinSock.

Computer object In NetWare Directory Services, a leaf object representing a computer on the network. The Computer object's properties can contain information such as the computer's physical serial number and the name of the person to whom the computer is currently assigned.

See also leaf object; Novell Directory Services.

Computer Systems Research Group

Abbreviated CSRG. The University of California, Berkeley, group responsible for the development of the Berkeley Software Distribution (BSD).

CSRG was home to some remarkable programmers over the years and introduced into Unix many of the fundamental features we take for granted. The last BSD release, 4.4BSD, was made during 1993, and the group was disbanded shortly afterward.

concentrator A repeater or hub that joins communications channels from several network nodes. Concentrators are not just simple wire-concentration centers, but also provide bridging, routing, and other management functions.

concurrent When two or more programs (processes) have access to the processor at the same time and must share the system resources, they are said to be "running concurrently." Because a computer can perform operations so quickly, the processes seem to be occurring at the same time, although actually they are not.

See also multiprocessing; task; task switching; thread.

concurrent license A software license that allows more than one person at a company to share an application over a network, providing that, at any given time, only one person is using it.

Different versions of concurrent licensing allow a fixed number of people in an office to share one copy of an application and allow the application to be used on both desktop and portable PCs, rather than run only from the file server.

See also application metering.

CONFIG.SYS In MS-DOS and OS/2, a special text file containing settings that control the way the operating system works.

In MS-DOS, the CONFIG.SYS file may contain 10 to 20 lines of entries. In OS/2, it is likely to contain between 70 and 100 lines of configuration information. CONFIG.SYS must be in the root directory of the default boot disk, normally drive C, and is read once as the system starts running.

configuration The process of establishing your own preferred setup for an application, expansion board, computer system, or network. Most current software can establish a configuration for you automatically, although you may need to adjust that configuration to get the best results.

See also Desktop Management Interface; Plug and Play.

configuration file A file, created by an application or an operating system, containing configuration information specific to your own computing environment. Application configuration files may have a filename extension of CFG or SET; Windows configuration files use the INI filename extension.

If you accidentally erase an application's configuration file, the program will return to using its default settings. Although the program will continue to function, its configuration settings may not be suitable for your system.

See also AUTOEXEC.BAT; CONFIG.SYS; NET.CFG.

configuration management A term covering a wide range of network

administration tasks, often performed by the network administrator, including:

- Maintaining a hardware database containing details of routers, bridges, and connections so that changes in the network can be made quickly in the event of a failure
- Adding and removing workstations and users to the network as needed
- Adding and configuring new servers and cabling systems as the network expands

See also Desktop Management Interface.

configuration register A 16-bit, user-configurable value that determines how a Cisco router functions during router initialization. You can configure information in hardware by using jumpers and in software by specifying a hexadecimal number in configuration commands.

congestion An excessive amount of traffic on the network, causing messages to be blocked for long periods of time and adversely affecting network performance.

You may see a very slow response from a server, or you may see an error message telling you that no ports are available at the present time for the service or host you are requesting.

See also Ethernet meltdown.

connectionless A protocol in which the source and destination addresses are included inside each packet so that a direct connection between sender and receiver or an established session between nodes is not required for communications. In a connectionless service, data packets may not reach their destination in the same order in which

they were sent. UDP is a connectionless transport service.

See also connection-oriented; User Datagram Protocol.

Connectionless Network Protocol

Abbreviated CLNP. An Open Systems Interconnect (OSI) protocol that provides the OSI Connectionless Network Service for the delivery of data. It uses datagrams that include addressing information to route network messages. CLNP is used for LANs rather than WANs. CLNP is the OSI equivalent of IP (Internet Protocol).

See also Connection-Oriented Network Service.

Connectionless Transport Protocol

Abbreviated CLTP. An Open Systems Interconnect (OSI) protocol that provides end-to-end transport data addressing and error correction, but does not guarantee delivery or provide any flow control. CLTP is the OSI equivalent of UDP.

See also Connection-Oriented Network Service; User Datagram Protocol.

Connection Admission Control Abbreviated CAC. An Asynchronous Transfer Mode (ATM) function that determines whether a virtual circuit connection request will be accepted.

See also Asynchronous Transfer Mode.

connection number A number assigned to a workstation that attaches to a server; it may be a different number each time the workstation attaches. Connection numbers are also assigned to print servers, as well as

other applications and processes that use the server connections.

connection-oriented A term used to describe a communications model that goes through three well-defined stages: establishing the connection, transferring the data, and releasing the connection. Analogous to a voice telephone call. In a connection-oriented service, data packets always reach their destination in the same order in which they were sent. TCP is a connection-oriented transport service.

See also connectionless; Transmission Control Protocol.

Connection-Oriented Network

Service Abbreviated CONS. A data transmission service in which data is only transmitted once a connection has been established. Asynchronous Transfer Mode (ATM) is a connection-oriented service.

See also Connectionless Network Protocol; Connectionless Transport Protocol.

connection speed The speed of a data communications circuit. Some circuits are symmetrical and can maintain the same speed in both directions; others are asymmetrical and use a faster speed in one direction, usually the downstream side. Table C.1 compares the connection speeds available for several technologies.

See also Asymmetric Digital Subscriber Line; cable modem; High-Bit-Rate Digital Subscriber Line; Integrated Services Digital Network; modem; Single-Line Digital Subscriber Line; Very-High-Bit-Rate Digital Subscriber Line.

TABLE C.1 CONNECTION SPEEDS

Technology	Speed	Distance Limitation
Conventional modem	56Kbps downstream, up to 33.6Kbps upstream	None
ISDN	Up to 128Kbps symmetric	18,000 ft
Cable modem	Up to 30Mbps downstream, to 10Mbps upstream	30 mi
ADSL	1.5 to 8Mbps downstream, to 1.544Mbps upstream	18,000 ft
SDSL	1.544 to 2.048Mbps symmetric	10,000 ft
HDSL	1.544 to 2.048Mbps symmetric (over 3 phone lines)	15,000 ft
VDSL	13 to 52Mbps downstream, 1.5 to 2.3Mbps upstream	4,500 ft

connectivity The degree to which any given computer or application can cooperate with other network components purchased from other vendors, in a network environment in which resources are shared.

connect time The period of time during which a user is logged on to the network.

ConnectView An application with a graphical user interface used to manage NetWare Connect and Novell Internet Access Server (NIAS) communication servers.

CONS See Connection-Oriented Network Service.

cons See console.

Conseil Européen pour la Recherche Nucléaire Abbreviated CERN. The European Laboratory for Particle Physics located in Geneva, Switzerland, where Tim Berners-Lee and associates created the communications protocols that led to the World Wide Web.

For more information on CERN, see www.cern.ch.

console Sometimes abbreviated cons. The monitor and keyboard from which the server or host computer activity can be monitored.

Certain operating system commands and utilities must be executed from the console

device; they will not operate from a workstation. In some systems, the console is a virtual device that can be invoked from any workstation by a network administrator with the appropriate rights and privileges.

constant angular velocity Abbreviated CAV. An unchanging speed of rotation. Hard disks use a CAV encoding scheme. The constant rate of rotation means that sectors on the disk are at the maximum density along the inside track of the disk. As the read/write heads move outward, the sectors must spread out to cover the increased track circumference, and therefore the data-transfer rate falls off.

See also constant linear velocity.

Constant Bit Rate Abbreviated CBR. A type of Asynchronous Transfer Mode (ATM) service reserved for voice or video or other data that must be transmitted at a constant rate and are intolerant of data loss.

See also Asynchronous Transfer Mode; Available Bit Rate; Unspecified Bit Rate; Variable Bit Rate.

constant linear velocity Abbreviated CLV. A changing speed of rotation. CD-ROM disk drives use a CLV encoding scheme to make sure that the data density remains constant. Information on a compact disc is stored in a single, spiral track, divided into many equal-length segments. To read the data, the CD-ROM disk drive must increase the rotational speed as the read head gets closer to the center of the disc and decrease as the head moves back out.

See also constant angular velocity.

Consultative Committee for International Telephony and Telegraphy Abbreviated CCITT. An organization based in Geneva that develops worldwide data communications standards. Three main sets of standards have been established:

- CCITT Groups 1–4 standards apply to facsimile transmissions.
- CCITT V series of standards apply to modems and error detection and correction methods.
- CCITT X series standards apply to LANs.

Recommendations are published every four years. In 1993, after a reorganization, the name was changed to International Telecommunication Union (ITU), and even though ITU now creates recommendations and standards, you will still hear the CCITT standards mentioned.

See also International Telecommunication Union.

container One of the Java programming language classes that can contain graphical user interface components. Components in a container usually appear within the boundaries of that container. For example, the classes `Dialog`, `Frame`, and `Window` are all containers.

See also Java.

container administrator In Novell Directory Services, an administrator who is granted rights to a container object and all the objects that the container holds. A container administrator can be exclusive, indicating that no other administrator is allowed access to that container.

See also container object; leaf object.

container object In Novell Directory Services, an object that can contain other objects and that is used to represent a logical or physical organizational element within a company, such as a department or a division. The Tree, Country, Organization, and Organizational Unit objects are all container objects.

See also container administrator; Country object; leaf object; Organization object; Organizational Unit object; Tree object.

container security equivalence *See* implied security equivalence.

contention The competition between transmitting nodes for access to communications lines or network resources. The first device to gain access to a channel takes control of the channel. In the event of a collision, when two nodes attempt to transmit at the same time, some arbitration scheme must be invoked.

See also Carrier Sense Multiple Access/ Collision Detection; token passing.

context In Novell Directory Services, an object's location within the Directory tree. The context is the full path to the container object in which the object is placed. If an object is moved from one container to another, it has changed contexts.

See also container object; leaf object.

context switching Switching from one program to another without ending the first program. Context switching allows you to operate several programs at the same time; but it differs from true multitasking in that when you are using one program, all the

other programs loaded onto your system must halt.

control character A nonprinting character with a special meaning.

Control characters, such as Carriage Return, Line Feed, Bell, or Escape, perform a specific operation on a terminal, printer, or communications line. They are grouped together as the first 32 characters in the ASCII character set; see Appendix C for details.

You can type a control character from the keyboard by pressing and holding the Ctrl key while you simultaneously press another key. For example, if you press and hold the Ctrl key and then press C, you generate Ctrl+C, also known as Break. Control character sequences are often used inside application programs as menu command shortcuts.

See also American Standard Code for Information Interchange; Extended Binary Coded Decimal Interchange Code; Unicode.

control code A sequence of one or more characters used for hardware control; also known as setup strings or escape sequences. Control codes are used with printers, modems, and displays. Printer control codes often begin with an escape character, followed by one or more characters that the printer interprets as commands it must perform rather than as text it must print.

Control Panel In Microsoft Windows, a special system folder that contains applets used to look at or change configuration information. Each applet manages a single task such as adding or removing a program

from your system, setting up a connection to the Internet, or changing display settings.

See also Network and Dial-up Connections.

Controlled Access Unit Abbreviated CAU. An intelligent Multistation Access Unit (MAU) or multiport wiring hub for a token-ring network that allows ports to be switched on and off.

controllerless modem A modem that shifts all the protocol management, error detection and correction, and data compression onto software running on the system's CPU. This allows the modem manufacturer to make a much cheaper modem that does not require the memory or processing power of a traditional modem. Also known as a soft modem.

control set That portion of the Microsoft Windows Registry that contains information required to boot the operating system and restore the last known good configuration.

conventional memory The amount of memory accessible by MS-DOS in PCs using an Intel processor operating in real mode; normally the first 640KB.

The designers of the original IBM PC made 640KB available to the operating system and applications and reserved the remaining space for internal system use, the BIOS, and video buffers. Although 640KB may not seem like much memory space now, it was ten times the amount of memory available in other leading personal computers available at the time. Since then, applications have increased in size to the point that 640KB is inadequate.

See also expanded memory; extended memory; high memory area; memory management; protected mode.

convergence The synchronization process that a network must go through immediately after a routing change takes place on the network. Convergence time is the time required to update all the routers on the network with routing information changes.

See also routing table.

convergence sublayer Abbreviated CS. One of two sublayers that make up the ATM Adaptation Layer; the other being the segmentation and reassembly sublayer (SAR).

The convergence sublayer is the upper layer that provides the interface for the various ATM services, pads the cells, and adds headers and trailers before passing the cell to the SAR.

See also ATM Adaptation Layer; segmentation and reassembly sublayer.

cookie 1. A block of data sent from a server to a client in response to a request by the client.

2. On the World Wide Web, a block of data stored by the server on the system running the browser or client software, which can be retrieved by the server during a future session. A cookie contains information that can identify the user for administrative reasons or to prepare a custom Web page.

cooperative multitasking A form of multitasking in which all running applications must work together to share system resources.

Copper Distributed Data Interface

The operating system supports cooperative multitasking by maintaining a list of the active applications and the order in which they execute. When control is transferred to an application, the other applications cannot run until that application returns control back to the operating system once again.

A cooperative multitasking system differs from a preemptive multitasking system, in which the operating system executes each application in turn for a specific period of time (depending on priority) before switching to the next application, regardless of whether the applications themselves return control to the operating system.

See also context switching; time-slice multitasking.

Copper Distributed Data Interface

Abbreviated CDDI. A version of the FDDI standard designed to run on shielded and unshielded twisted-pair cable rather than on fiber-optic cable. CDDI is capable of 100Mbps data transfer rates.

See also Fiber Distributed Data Interface.

coprocessor A secondary processor used to speed up operations by taking over a specific part of the main processor's work. The most common type of coprocessor is the math, or floating-point, coprocessor, which is designed to manage arithmetic calculations many times faster than does the main processor.

copy left The copyright or General Public License of the Free Software Foundation (FSF), which states that any of the software developed using free software from the FSF

must be distributed to others without charge.

See also open source software.

CORBA *See* Common Object Request Broker Architecture.

COSE *See* Common Open Software Environment.

counter log In Microsoft Windows 2000, a System Monitor log used to monitor data from local or remote computers about hardware use and system service activity.

See also alert log; System Monitor; trace log.

Country object In Novell Directory Services, a container object placed directly under the Root object in the Directory tree that defines the country for a specific part of your network. Country object names are defined by the International Organization for Standardization (ISO) and follow a standard naming convention. You must specify the Country object if you want to connect to external networks using X.500 directory services.

CPDA *See* Certified PowerBuilder Developer Associate.

CPDP *See* Certified PowerBuilder Developer Professional.

CPE *See* customer-premises equipment.

CPI-C *See* Common Programming Interface for Communications.

cps *See* characters per second.

CPTS *See* Certified Performance and Tuning Specialist.

CPU *See* central processing unit.

cracker An unauthorized person who breaks into a computer system planning to do harm or damage or with criminal intent. The popular press often portrays crackers as people with exceptional talent for eluding detection, and some of them are, but most of them use a set of well-worn tricks to exploit common security weaknesses in the systems they target.

See also attack; hacker; intruder.

crash An unexpected program halt, sometimes due to a hardware failure but most often due to a software error, from which there is no recovery. You usually need to reboot the computer to recover after a crash.

See also blue screen of death.

CRC *See* cyclical redundancy check.

critical error An error in a program that forces the program to stop until the user corrects the error condition. Examples of this kind of error are attempts to write to a floppy disk when there is no disk in the drive or to print to a printer that has run out of paper.

crosstalk In communications, any interference from a physically adjacent channel that corrupts the signal and causes transmission errors.

See also far-end crosstalk; near-end crosstalk.

Cryptography API Abbreviated CAPI. An API first introduced in Microsoft

Windows NT 4 that provides encryption and decryption functions for application developers.

CS *See* convergence sublayer.

CSE *See* Certified Solutions Expert.

C shell In Unix, a popular command interpreter; pronounced “sea shell.”

Developed at the University of California at Berkeley as part of the BSD development as an alternative to the Bourne shell. In addition to the features found in the Bourne shell, the C shell adds integer arithmetic, a history mechanism that can recall past commands in whole or in part, aliasing of frequently used commands, job control, and a built-in set of operators based on the C programming language used for writing shell scripts.

See also Bash; Bourne shell; Korn shell; Linux; Unix shell.

CSMA/CD *See* Carrier Sense Multiple Access/Collision Detection.

CSRG *See* Computer Systems Research Group.

CSU *See* Channel Service Unit.

Ctrl+Alt+Del A three-key combination used to reset the machine and reload the operating system. By pressing Ctrl+Alt+Del, you initiate a warm boot, which restarts the computer without going through the power-on self tests (POSTs) normally run when the computer goes through a cold boot.

In Windows 98 and 2000, the sequence opens a dialog box from which you can either end a task or shut down the

computer. Sometimes called the three-finger salute.

See also warm boot.

Ctrl+Break *See* Ctrl+C.

Ctrl+C 1. A key combination recognized by Unix, MS-DOS, and other operating systems as a user-initiated interruption. Pressing Ctrl+C stops a batch file, macro, or command (for example, a directory listing, a search, or a sort).

2. A keyboard shortcut recognized by many programs as the instruction to copy the selected item.

Ctrl key A key on the keyboard that, when pressed at the same time as another key, generates a nonprinting control character.

On some keyboards, this key is labeled Control rather than Ctrl, but it produces the same function.

CTS *See* Clear to Send.

CUE *See* Certified Unicenter Engineer.

current directory In many operating systems, the directory that will be searched first for any file you request, and the directory in which any new files will be stored (unless you specifically designate another directory). The current directory is not the same as the default directory, which is the directory that an application uses unless you specify another.

See also dot; dot dot; period and double-period directories.

current drive In many operating systems, the disk drive that is being used for reading and writing files. The current drive is not the same as the default drive, which is the drive that an application uses unless you specify another.

See also drive mapping.

cursor A special character displayed on a monitor to indicate where the next character will appear when it is typed. In text or character mode, the cursor is usually a blinking rectangle or underline. In a graphical user interface, the mouse cursor can take many shapes, depending on the current operation and its screen location.

cursor-movement keys The keys on the keyboard that move the cursor; also called cursor-control keys. These keys include the four labeled with arrows and the Home, Pg Up, End, and Pg Dn keys.

On full-size keyboards, cursor-movement keys are often found on the numeric keypad; laptops and notebooks often have separate cursor-movement keys.

CU-SeeMe A popular videoconferencing and videophone product that works over the Internet.

CU-SeeMe was originally developed at Cornell University and is available free for the PC and the Macintosh. An enhanced commercial version that adds an electronic chalkboard is available from White Pine Software. The software is designed for personal use and for use in instruction and in business communications.

See also White Pine Software.

Customer Information Control

System Abbreviated CICS. An IBM-mainframe client/server program that manages transaction processing in IBM's VM and MVS operating systems and that is scalable to thousands of users. It also provides password security, transaction logging for backup and recovery, and an activity log that can be used to analyze session performance, as well as facilities for creating, using, and maintaining databases.

customer-premises equipment Abbreviated CPE. Communications equipment, either leased or owned, used at a customer site.

cut through A technique used by some Ethernet hardware to speed up packet forwarding. Only the first few bytes of the packet are examined before it is forwarded or filtered. This process is much faster than looking at the whole packet, but it does allow some bad packets to be forwarded.

See also store-and-forward.

cut-through switching A type of switching used on a Token Ring network in which data is forwarded as soon as the first 20 or 30 bytes in a data frame have been read.

After the header information has been read, the connection is established between input and output ports, and the transmission begins immediately. Sometimes known as on-the-fly switching.

See also store-and-forward.

CxBus *See* Cisco Extended Bus.

cyclical redundancy check Abbreviated CRC. A complex calculation method used to check the accuracy of a digital transmission over a communications link or to ensure the integrity of a file stored on a hard disk.

The sending computer uses one of several formulas to calculate a value from the information contained in the data, and this value is appended to the message block before it is sent. The receiving computer performs the same calculation on the same data and compares this number with the received CRC. If the two CRCs do not match, indicating a transmission error, the receiving computer asks the sending computer to retransmit the data.

This procedure is known as a redundancy check because each transmission includes extra or redundant error-checking values as well as the data itself.

As a security check, a CRC may be used to compare the current size of an executable file against the original size to determine if the file has been tampered with or changed in some way.

See also checksum; Kermit; Xmodem; Ymodem; Zmodem.

Cyrix A designer of microprocessors, including clones of popular Intel chips such as the 6x86MX and the MII (pronounced M-two). Cyrix was bought by National Semiconductor in 1998.

For more information about Cyrix, see www.cyrix.com.

See also Advanced Micro Devices, Inc.; Pentium; Pentium II; Pentium III.

D

DACL *See* Discretionary Access Control List.

daemon Pronounced “dee-mon.” A background program that runs unattended and is usually invisible to the user, providing important system services.

Daemons manage all sorts of tasks, including e-mail, networking, and Internet services. Some daemons are triggered automatically by events to perform their work; others operate at timed intervals. Because daemons spend so much of their time idle, waiting for something to happen, they do not consume large amounts of system resources.

DAP *See* Directory Access Protocol.

DAS *See* dual-attached station.

DAT *See* digital audio tape.

data Information in a form suitable for processing by a computer, such as the digital representation of text, numbers, graphic images, or sounds. Strictly speaking, data is the plural of the Latin word *datum*, meaning an item of information; but the term is commonly used in both plural and singular constructions.

database A collection of related objects, including tables, forms, reports, queries, and scripts, created and organized by a database management system (DBMS). A database can contain information of almost any type, such as a list of magazine subscribers,

personal data on the space shuttle astronauts, or a collection of graphical images and video clips.

See also database management system; database model; table.

database management system Abbreviated DBMS. Software that controls the data in a database, including overall organization, storage, retrieval, security, and data integrity. A DBMS can also format reports for printed output and can import and export data from other applications using standard file formats. A data-manipulation language is usually provided to support database queries.

See also database; database model; query language.

database model The method used by a database management system (DBMS) to organize the structure of the database. The most common database model is the relational database.

See also relational database.

database server Any database application that follows the client/server architecture model, which divides the application into two parts: a front-end running on the user's workstation and a back-end running on a server or host computer. The front-end interacts with the user and collects and displays the data. The back-end performs all

the computer-intensive tasks, including data analysis, storage, and manipulation.

Database Specialist: Informix Dynamic Server

A certification from Informix designed for computer professionals who implement and manage Informix Dynamic Server databases. The associated exams cover relational database design, Structured Query Language (SQL), and management and optimization of Informix Dynamic Server databases.

See also Informix-4GL Certified Professional; system administration.

data bits In asynchronous transmissions, the bits that actually make up the data. Usually, seven or eight data bits are grouped together. Each group of data bits in a transmission is preceded by a start bit and followed by an optional parity bit as well as one or more stop bits.

See also communications parameters; parity; start bit; stop bit(s).

Data Carrier Detect Abbreviated DCD. A hardware signal defined by the RS-232-C standard that indicates that the device, usually a modem, is online and ready for transmission.

data communication The transfer of information from one computer to another over a communications link. The transfer can be occasional, continuous, or a combination of both.

data communications equipment

Abbreviated DCE. In communications, any device that connects a computer or terminal

to a communications channel or public network; usually a modem.

See also data terminal equipment.

data compression Any method of encoding data so that it occupies less space than it did in its original form, thus allowing that data to be stored, backed up, retrieved, or transmitted more efficiently.

Data compression is used in fax and many other forms of data transmission, CD-ROM publishing, still-image and video-image manipulation, and database management systems.

See also Huffman coding; Joint Photographic Experts Group; lossless compression; lossy compression; Moving Pictures Experts Group.

data connector (Type 1) A connector for use with Type 1 cable, designed by IBM for use in Token Ring network wiring centers.

data-encoding scheme The method used by a hard-disk controller to store information onto a hard disk or a floppy disk. Common encoding schemes include the run-length limited (RLL) and advanced run-length limited (ARLL) methods.

See also advanced run-length limited encoding; RLL encoding.

Data Encryption Standard Abbreviated DES. A standard method of encrypting and decrypting data, developed by the U.S. National Bureau of Standards. DES works by a combination of transposition and substitution. It is used by the federal government and most banks and money-transfer

systems to protect all sensitive computer information.

See also encryption; Pretty Good Privacy.

data file A file that contains information—text, graphics, or numbers—rather than executable program code.

datagram A message unit that contains source and destination address information, as well as the data itself, which is routed through a packet-switching network.

The data held in the datagram is often referred to as the payload, and the addressing information is usually contained in the header. Because the destination address is contained in all datagrams, they do not have to arrive in consecutive order. Datagrams are commonly used in connectionless transmission systems. IP (Internet Protocol) and IPX (Internetwork Packet Exchange) are both datagram services.

See also frame; data packet.

Datagram Delivery Protocol Abbreviated DDP. A routing protocol developed by Apple Computer as a part of its AppleTalk network.

data-link layer The second of seven layers of the OSI Reference Model for computer-to-computer communications. The data-link layer validates the integrity of the flow of data from one node to another by synchronizing blocks of data and controlling the flow of data.

The Institute of Electrical and Electronic Engineers (IEEE) has divided the data-link layer into two other layers—the logical link

control (LLC) layer sits above the media access control (MAC) layer.

See also OSI Reference Model.

Data Link Switching Abbreviated DLSw. A standard for encapsulating or tunneling IBM Systems Network Architecture (SNA) and NetBIOS applications across IP networks.

See also Advanced Peer-to-Peer Networking; Systems Network Architecture.

data mining The process of displaying historical commercial data in a multidimensional form so that previously hidden relationships are exposed through the use of advanced statistical tools, making them easier to group and summarize.

See also data warehousing; online analytical processing; online transaction processing.

data packet One unit of information transmitted as a discrete entity from one node on the network to another. More specifically, a packet is a transmission unit of a fixed maximum length that contains a header with the destination address, a set of data, and error control information.

See also frame.

data processing Abbreviated DP. Also called electronic data processing (EDP). A term used to describe work done by minicomputers and mainframe computers in a data center or business environment.

data protection Techniques used by network operating systems to ensure the integrity of data on the network, including

protecting data against surface defects developing on the disk and storing redundant copies of important system data, such as file indices and file allocation tables (FATs). Disk duplexing, disk mirroring, a well thought out backup scheme, and RAID techniques all provide different levels of data protection.

See also diskless workstation; disk striping; disk striping with parity; fault tolerance; Hot Fix; intruder; redundant array of inexpensive disks; virus.

Data Service Unit Abbreviated DSU. A device that connects DTE (data terminal equipment) to digital communications lines. A DSU formats the data for transmission on the public carrier WANs and ensures that the carrier's requirements for data formats are met.

See also Channel Service Unit.

Data Set Ready Abbreviated DSR. A hardware signal defined by the RS-232-C standard to indicate that the device is ready to operate.

See also Clear to Send.

data terminal equipment Abbreviated DTE. In communications, any device, such as a terminal or a computer, connected to a communications device, channel, or public network.

See also data communications equipment.

Data Terminal Ready Abbreviated DTR. A hardware signal defined by the RS-232-C standard sent from a computer to a modem to indicate that the computer is ready to receive a transmission.

data-transfer rate 1. The speed at which a disk drive can transfer information from the drive to the processor, usually measured in megabits or megabytes per second.

2. The rate of information exchange between two systems. For example, an Ethernet LAN may achieve 10Mbps, and a Fiber Distributed Data Interface (FDDI) system may reach 100Mbps.

See also connection speed.

data warehousing A method of storing very large amounts of data, usually historical transaction processing data, for later analysis and reporting.

The data warehouse is accessed by software capable of extracting trends from the raw data and creating comparative reports.

See also data mining.

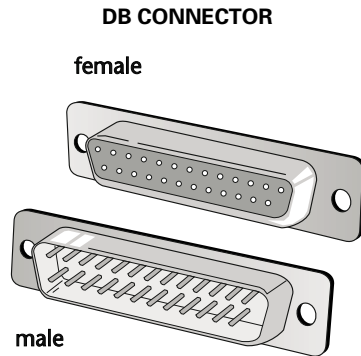
dB *See* decibel.

DB connector Any of several types of cable connectors used for parallel or serial cables. The number following the letters DB (for data bus) indicates the number of pins that the connector usually has; a DB-25 connector can have a maximum of 25 pins, and a DB-9 connector can have as many as 9. In practice, not all the pins (and not all the lines in the cable) may be present in the larger connectors. If your situation demands that all the lines be present, make sure you buy the right cable. Common DB connectors include the following:

- **DB-9** Defined by the RS-449 standard as well as the ISO (International Organization for Standardization).

- **DB-25** A standard connector used with RS-232-C wiring, with 25 pins (13 on the top row and 12 on the bottom).
- **DB-37** Defined as the RS-449 primary channel connector.

DB-15, DB-19, and DB-50 connectors are also available. The accompanying illustration shows a male and female DB-25 connector.



DBCS See double-byte character set.

DBMS See database management system.

DC-2000 A quarter-inch tape minicartridge used in some tape backup systems. DC-2000 has a capacity of up to 250MB when some form of data compression is used.
See also quarter-inch cartridge.

DCD See Data Carrier Detect.

DCE See data communications equipment.

DCE See Distributed Computing Environment.

D channel The channel in ISDN that is used for control signals and customer data.

In the Base Rate ISDN (BRI), the D channel operates at 16Kbps; in the Primary Rate ISDN (PRI), it operates at 64Kbps.

DCOM See Distributed Component Object Model.

DCPROMO In Microsoft Windows 2000 Server, the Active Directory Installation Wizard that promotes a member server or stand-alone server to a domain controller. When run on a domain controller, this program can demote the computer to a member server or a stand-alone server.

DDCMP See Digital Data Communications Message Protocol.

DDD See direct distance dialing.

DDE See Dynamic Data Exchange.

DDNS *See* Dynamic DNS.

DDP *See* Datagram Delivery Protocol.

deadlock An error condition or stalemate that occurs when two programs or devices are each waiting for a signal from the other before they can continue.

DEC *See* Digital Equipment Corporation.

DEC Alpha Also called the DEC Alpha AXP or the DECchip 21264. A 64-bit, RISC (reduced instruction set computing) microprocessor from Digital Equipment Corporation (DEC), first introduced in 1992.

The Alpha is a superscalar, superpipelined design, which allows the processor to execute more than one instruction per clock cycle; it can execute as many as six instructions per clock cycle and can sustain four instructions per clock cycle.

It has data and instruction caches, a floating-point processor, 64-bit registers, 64-bit data and address buses, and a 128-bit data path between the processor and memory. The internal architecture is symmetrical multiprocessing (SMP) compliant, meaning that it can be used in multiprocessing configurations. The chip is available in several models with operating frequencies in the 300 to 700MHz range; a 1GHz model is expected by the year 2000.

decapsulation A process used in networking in which the receiving system looks at the header of an arriving message to determine if the message contains data. If the message does contain data, the header is removed and the data decoded.

See also encapsulation.

decibel Abbreviated dB. One-tenth of a bel, a unit of measurement common in electronics that quantifies the loudness or strength of a signal. A decibel is a relative measurement derived by comparing a measured level against a known reference.

decimal The base-10 numbering system that uses the familiar numbers 0–9; also known as the base 10 radix or the decimal radix.

See also binary; hexadecimal.

decimal radix *See* decimal.

DECnet A series of communications and networking products from Digital Equipment Corporation (DEC). DECnet is compatible with Ethernet, as well as with WANs using baseband and broadband private and public communications channels. DECnet is built into the VAX VMS operating system.

See also Digital Network Architecture.

decode **1.** To decompress a video file after receipt so that you can view it. Most decoding is done by the client browser.

2. To convert coded data back into its original form, usually as readable text.

See also codec; decryption; encode; uuencode.

decryption The process of converting encrypted data back into its original form.

See also encryption.

dedicated circuit *See* dedicated line; direct connection.

dedicated line

dedicated line A communications circuit used for one specific purpose and not used by or shared between other users. You need only dial a dedicated line to restore service after an unscheduled interruption. Also known as a dedicated circuit.

See also leased line.

dedicated router A router that performs no functions other than routing tasks.

dedicated server A computer on the network that functions only as a server performing specific networking tasks, such as storing files, printing, or managing external communications.

Dedicated Token Ring An IEEE 802.5r Token Ring specification that allows for full-duplex connections at a speed of up to 32Mbps. By enabling full-duplex communications, the token-passing mechanism is not used, and communications can take place between a device and a switch at any time.

See also 802.x; token-ring network.

default A standard setting, used until an alternative is chosen. The default server is the first server that you log on to. The default drive is the drive that a workstation is currently using.

A default is usually a relatively safe course of action to try first; many programs provide defaults you can use until you know enough about the program to specify your own settings.

default directory A standard directory or set of directories used by the operating system or by an application.

Some operating systems create several default directories when the file system is created. For example, Novell NetWare creates the following directories when the SYS volume is created:

- SYS:SYSTEM, for the NetWare operating system files
- SYS:PUBLIC, for utility and user programs
- SYS:LOGIN, for programs allowing users to log in to the server
- SYS:MAIL, a directory used by NetWare-compatible electronic-mail applications

default server In Novell NetWare, the server that responds to the Get Nearest Server request made as a user starts the logon process. Novell Directory Services has replaced the need for the default server destination with the default context.

See also Context.

default shares In Microsoft Windows 2000, resources shared by default when the operating system is first installed.

See also common share; hidden share; share; sharing.

defense in depth A term borrowed from the military used to describe defensive measures that reinforce each other, hiding the defenders activities from view and allowing the defender to respond to an attack quickly and effectively.

In the network world, defense in depth describes an approach to network security that uses several forms of defense against an intruder and that does not rely on one single defensive mechanism.

defragmentation The process of reorganizing and rewriting files so that they occupy one large area on a hard disk rather than several smaller areas.

When a file on a hard disk is updated, it may be written into different areas all over the disk. This outcome is particularly likely when the hard disk is continuously updated over a long period of time. This file fragmentation can lead to significant delays in loading files, but its effect can be reversed by defragmentation.

See also disk optimizer.

defragmenter Any utility that rewrites all the parts of a fragmented file into contiguous areas on a hard disk. A defragmenter (such as the Microsoft Windows 98 utility Disk Defragmenter) can restore performance lost because of file fragmentation.

See also defragmentation; disk optimizer.

delay In communications, a pause in activity, representing the time during which transmission-related resources are unavailable for relaying a message.

See also propagation delay.

delay distortion The distortion of a signal caused by the relative difference in speed of the various components of that signal; in particular, the distortion of the high-frequency component. Also called envelope delay.

delete To remove a file from a disk or to remove an item from a file. Files can be deleted using operating system commands or directly from within an application.

When a file is deleted from a disk, the file is not physically removed; although it is

hidden, it is still there on the disk until it is overwritten. In certain circumstances it is possible to undelete or recover the original information with utilities designed for that purpose. If you find you have deleted an important file by accident, do not write any other files to that disk so that you do not overwrite the deleted file. Some network operating systems use a delete inhibit attribute to prevent accidental deletions; other operating systems rely on a read-only attribute.

delimiter 1. Any special character that separates individual items in a data set or file. For example, in a comma-delimited file, the comma is placed between each data value as the delimiter.

2. In a token-ring network, a delimiter is a bit pattern that defines the limits of a frame or token on the network.

Dell Computer Corporation One of the world's top five PC manufacturers. Dell pioneered direct sales of the PC using a configure-it-yourself Web site.

Best known for PC and laptop sales, Dell is also pursuing the server market and has recently expanded into the enterprise storage market with the PowerVault line of SCSI-based storage subsystems, tape backup systems, and scalable disk subsystems.

For more information on Dell, see www.dell.com.

demand paging A common form of virtual memory management in which pages of information are read into memory from disk only when required by the program.

See also swapping.

demand priority A technique used in 100VG-AnyLAN to arbitrate access to the network and avoid collisions. Demand priority replaces CSMA/CD, which is used in slower Ethernet networks.

Demand priority can also prioritize specific network traffic such as video and other time-critical data, giving it a higher precedence; if multiple requests are received, the highest priority is always serviced first.

See also Fast Ethernet.

demodulation In communications, the process of retrieving the data from a modulated carrier signal; the reverse of modulation.

See also modem.

denial of service attack An attack by an intruder that prevents a computer system from providing a service.

A denial-of-service attack will typically involve opening and dropping a large number of TCP/IP connections very quickly so that the target system spends all its time dealing with the connection overhead to the point that it cannot respond to valid user requests. Other attacks may exploit known software security holes to crash servers.

A denial-of-service attack is much easier to execute than an attempt at unauthorized access, because the denial-of-service attack never actually requires access to the system.

See also brute-force attack; dictionary attack; mail bombing; social engineering; Trojan Horse.

departmental LAN A local-area network used by a relatively small group of

people working on common tasks; it provides shared local resources, such as printers, data, and applications.

DES *See* Data Encryption Standard.

descendant key In Microsoft Windows, all the subkeys that appear when you expand a key in the Windows Registry.

See also key; Registry.

DeskSet A collection of graphical desktop applications bundled with Sun Microsystem's Solaris.

DeskSet includes a file manager with options for copying, moving, renaming, and deleting files, a terminal emulator, text editor, calculator, clock, and calendar, as well as special programs and utilities.

See also Solaris.

desktop management The process of managing desktop workstation hardware and software components automatically, often from a central location.

See also total cost of ownership; Zero Administration for Windows.

Desktop Management Interface Abbreviated DMI. A standard API for identifying desktop workstation hardware components automatically, without intervention from the user.

At a minimum, DMI identifies the manufacturer, component name, version, serial number (if appropriate), and installation time and date of any component installed in a networked workstation. This information is designed to help network administrators resolve configuration problems quickly and

easily and to indicate when and where system upgrades should be applied. PCs, Macintosh computers, and Unix systems are all covered by DMI.

DMI is backed by Digital Equipment Corporation (DEC), IBM, Intel, Microsoft, Novell, Sun, and more than 300 other vendors.

See also Plug and Play; total cost of ownership; Web-based Enterprise Management; Wired for Management; Zero Administration for Windows.

desktop video The combination of video capture hardware and application software that controls the display of video or television pictures on a desktop PC.

Desktop video is becoming increasingly important with the sharp increase in video-conferencing applications now available.

destination address The address portion of a packet or datagram that identifies the intended recipient station.

See also source address.

destination host A computer system on the network that is the final destination for a file transfer or for an e-mail message.

device A general term used to describe any computer peripheral or hardware element that can send or receive data.

Some examples are modems, printers, serial ports, disk drives, routers, bridges, and concentrators. Some devices require special software, or device drivers, to control or manage them; others have built-in intelligence.

In Unix and certain other operating systems, all peripherals are treated as though they were files. When Unix writes information to

your terminal, it is actually writing to a special file that represents your terminal.

See also device driver.

device dependence The requirement that a specific hardware component be present for a program to work. Device-dependent software is often difficult to move or port to another computer because of its reliance on specific hardware.

See also device independence.

device driver A small program that allows a computer to communicate with and control a device. Each operating system contains a standard set of device drivers for the keyboard, the monitor, and so on. When you add specialized peripheral devices, such as a CD-ROM disk drive or a network interface card, you must install the appropriate device driver so that the operating system knows how to manage the device.

See also Plug and Play.

device independence The ability to produce similar results in a variety of environments, without requiring the presence of specific hardware.

The Java programming language and the PostScript page-description language are examples of device independence. Java runs on a wide range of computers, from the PC to a Cray; PostScript is used by many printer manufacturers.

See also device dependence.

device name The name used by the operating system to identify a computer-system component. For example, COM1 is the Windows device name for the first serial port on the PC.

device number A unique number assigned to a device so that it can operate on the network. Devices are identified by three numbers:

- A physical address set by jumpers on the adapter board
- A device code determined by the physical address
- A logical address determined by the order in which the drivers are loaded and by the physical address of the adapter

See also Ethernet address; hardware address.

Dfs *See* Distributed file system.

DHCP *See* Dynamic Host Configuration Protocol.

diagnostic program A program that tests computer hardware and peripheral devices for correct operation. Some faults, known as hard faults, are relatively easy to find, and the diagnostic program will diagnose them correctly every time. Other faults, called soft faults, can be difficult to find, because they occur under specific circumstances rather than every time the memory location is tested.

Most computers run a simple set of system checks when the computer is first turned on. The PC tests are stored in ROM and are known as power-on self tests (POSTs). If a POST detects an error condition, the computer stops and displays an error message on the screen.

dialback modem *See* callback modem.

dial-in/dial-out server *See* asynchronous communications server.

Dialogic Corp A major manufacturer of high-performance telephony products, including those used in voice, data, fax, speech synthesis, ISDN networking, and call center management applications.

For more information on Dialogic, see www.dialogic.com.

dialup line A nondedicated communication line in which a connection is established by dialing the destination code and then broken when the call is complete.

See also dedicated line; leased line.

Dial-Up Networking Software provided with Microsoft's Windows products that allows the clients to dial out and establish Point-to-Point Protocol (PPP) connections.

dictionary attack An attack by an intruder that checks passwords in a password file against a list of words likely to be used as passwords. Some versions of this attack check the entire language lexicon.

See also brute-force attack; denial of service attack; mail bombing; social engineering; Trojan Horse.

differential backup A backup of a hard disk that includes only the information that has changed since the last complete backup.

A differential backup assumes that a full backup already exists and that in the event of an accident, this complete backup will be restored before the differential backup is reloaded.

differential SCSI A Small Computer System Interface (SCSI) bus wiring scheme that uses two wires for each signal on the bus. One wire carries the signal, while the other carries its inverse. Differential SCSI minimizes the effects of external interference and so allows longer SCSI cable lengths to be used.

See also single-ended SCSI; Small Computer System Interface.

digest A collection of Internet mailing list posts collected together and sent out as a single large message rather than as a number of smaller messages. Using a digest is a good way to cut down on the number of noncritical e-mail messages you receive.

See also LISTSERV; listserver; mailing list.

digital Describes any device that represents values in the form of binary digits or bits.

See also analog.

digital audio tape Abbreviated DAT. A method of recording information in digital form on a small audio tape cassette, originally developed by Sony and Hewlett-Packard. The most common format is a 4-millimeter, helical-scan drive, which can hold more than 3GB of information. DATs can be used as backup media; however, like all tape devices, they are relatively slow.

Digital Data Communications Message Protocol Abbreviated DDCMP. A byte-oriented, link-layer synchronous protocol from Digital Equipment Corporation (DEC), used as the primary data-link component of DECnet.

Digital Equipment Corporation Abbreviated DEC. A major manufacturer of minicomputers and mainframe computers, founded in 1957, long recognized for its high-quality computer systems. DEC's most popular product line, the VAX series, ranges from small desktop systems to large mainframes suitable for scientific and commercial processing. DEC was bought by Compaq Computer Corporation in 1998.

For more information on DEC, see www.digital.com.

See also DEC Alpha.

Digital Network Architecture Abbreviated DNA. The framework within which Digital Equipment Corporation (DEC) designs and develops all its communications products.

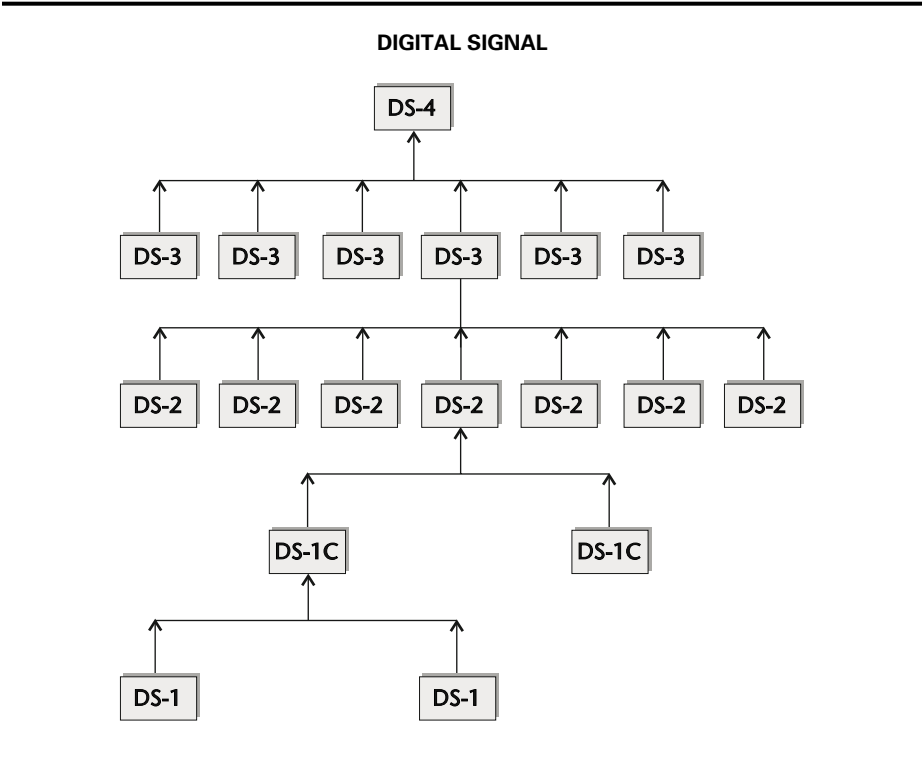
digital service *See* digital signal.

digital signal Abbreviated DS; also known as digital service. There are several levels of common carrier digital transmission service:

- **DS-0** 64Kbps.
- **DS-1** 1.544Mbps (T1).
- **DS-1C** Two DS-1 channels are multiplexed into a single DS-1C 3.152Mbps channel.
- **DS-2** Two DS-1C channels are multiplexed into one DS-2 6.312Mbps (T2) channel.
- **DS-3** Seven DS-2 channels are multiplexed into a single 44.736Mbps (T3) channel.
- **DS-4** Six DS-3 channels are multiplexed into one 274.176Mbps (T4) DS-4 channel.

The higher-capacity channels are constructed by multiplexing the lower-bandwidth channels together, with some additional framing and administrative overhead.

DS-0 is also referred to as fractional T1, because it bridges the gap between 56-Kbps direct dial service (DDS) and a full T1 implementation.



Digital Signal Processing Abbreviated DSP. An integrated circuit used in high-speed data manipulation. You will find DSP chips integrated into sound cards, modems, and video-conferencing hardware for use in communications, image manipulation, and other data-acquisition applications.

See also codec; desktop video.

digital signature An electronic signature that cannot be forged. A digital signature confirms that the document or e-mail in question originated from the individual or company whose name appears on the document and that the document has not been altered or tampered with in any way since it was signed.

See also certificate.

Digital Subscriber Line Abbreviated DSL. A high-speed data transmission technology originally developed by Bellcore that delivers high bandwidth over existing twisted-pair copper telephone lines.

There are several DSL services, providing data rates from 16Kbps to 52Mbps. The services can be symmetrical, with the same data rate in both upstream and downstream directions, or asymmetrical, with the downstream capacity greater than the upstream capacity. Asymmetric services are particularly suitable for Internet users, because more information is downloaded than is uploaded.

As DSL data rates increase, the distance over which the service is provided decreases; certain users who are located too far from the telephone company's central office may not be able to obtain the higher speeds or, in some cases, may not be able to receive the service at all.

See also Asymmetric Digital Subscriber Line; High-Bit-Rate Digital Subscriber Line; Rate-Adaptive Digital Subscriber Line; Single-Line Digital Subscriber Line; Very-High-Bit-Rate Digital Subscriber Line.

digital versatile disc *See* digital video disc.

digital video disc Abbreviated DVD; sometimes called digital versatile disc. A compact disc format. A standard single-layer single-sided disc can currently store 4.7GB of information; a two-layer standard increases this to 8.5GB, and eventually double-sided discs are expected to store 17GB

per disc. DVD drives can also read conventional compact discs.

digital video disc-erasable Abbreviated DVD-E. An extension to the digital video disc format to allow multiple re-recordings.

See also digital video disc-recordable.

digital video disc-recordable Abbreviated DVD-R. An extension to the digital video disc format to allow one-time recording.

See also digital video disc-erasable.

digital video disc-ROM Abbreviated DVD-ROM. A computer-readable form of digital video disc with either 4.7 or 8.5GB of storage per side.

dimmed command In a graphical user interface, a command that is not currently available. Also known as a grayed command, because it is often displayed in light gray rather than the usual black. For example, a command to perform a network action will be dimmed until you log on to the network.

DIN connector A connector that meets the specification of the German national standards body, Deutsche Industrie Norm (DIN).

Several models of Macintosh computers use 8-pin DIN connectors as the serial port connector. On many IBM-compatible computers, a 5-pin DIN connector connects the keyboard to the system unit.

DIP *See* dual in-line package.

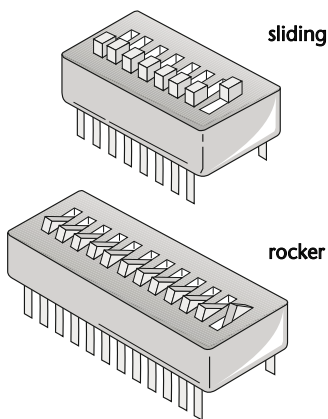
DIP switch A small switch used to select the operating mode of a device, mounted

direct access

as a dual in-line package. DIP switches can be either sliding or rocker switches, and they are often grouped for convenience.

They are used in printed circuit boards, dot-matrix printers, modems, and many other peripheral devices.

DIP SWITCH



direct access *See* random access.

direct connection A communications circuit used for one specific purpose and not used by or shared with other users. Also known as a dedicated circuit or a direct line. An ISDN qualifies as a direct connection.

See also Integrated Services Digital Network.

direct distance dialing Abbreviated DDD. Use of the normal long-distance telephone system without the need for the intervention of an operator.

direct distance dial network *See* public network.

direct line *See* direct connection.

direct manipulation In a graphical user interface, the process of working with objects using a mouse or other pointing device, rather than using menu selections to manipulate the objects. Using drag-and-drop to print a file or using the mouse to adjust the size of a window are both examples of direct manipulation.

direct memory access Abbreviated DMA. A method of transferring information directly from a mass-storage device, such as a hard disk, into memory without the information passing through the processor. Because the processor is not involved in the transfer, DMA is usually fast.

Directory 1. In Novell Directory Services (NDS), the Directory database contains and organizes all the NDS objects.

2. In the Microsoft Active Directory structure, a Directory contains information about users, groups of users, computers, and so on.

See also Active Directory; Directory object; Directory Schema; directory tree; Novell Directory Services.

directory In a hierarchical file system, a convenient way of organizing and grouping files and other directories on a disk. Sometimes called a folder. The beginning directory is known as the root directory, from which all other directories must branch.

Directories inside another directory are called subdirectories.

Depending on the operating system, you can list the files in a directory in a variety of ways: by name, by creation date and time, by file size, or by icon if you use a graphical user interface.

See also current directory; default directory; directory services; file allocation table; parent directory; period and double-period directories.

Directory Access Protocol Abbreviated DAP. A mail standard used to access white page directories containing names, addresses, e-mail addresses, and telephone numbers. Because of its complexity, DAP has been largely replaced by Lightweight Directory Access Protocol (LDAP).

See also Lightweight Directory Access Protocol.

directory caching A feature of Novell NetWare that copies the file allocation table (FAT) and the directory entry table into the network server's memory. When file requests are made, information is retrieved from the cache rather than from the hard disk, thus speeding the retrieval process significantly. As the directory cache fills up, the least-used directory entries are eliminated from the cache.

directory hashing A feature of Novell NetWare that indexes file locations on disk, thus speeding file retrieval.

Directory Map object In Novell Directory Services (NDS), a leaf object that refers to a directory on a volume. The Directory Map object allows a drive to be mapped to an application or to a login script without requiring the actual path and volume where the application is physically located. This is done so that login scripts don't have to be rewritten when a drive path changes; you can simply change the Directory Map object.

See also Novell Directory Services.

Directory object In Novell Directory Services (NDS), a set of properties stored in the Directory database. A Directory object can represent a physical or a logical network resource, but it does not actually contain the resource.

See also Directory; Novell Directory Services.

directory path *See* path.

Directory replica In Novell NetWare, a copy of the NetWare Directory partition

that allows the NetWare Directory Database to be stored on several servers on the network without having to duplicate the entire database on each server. Directory replicas remove a single point of failure from the network and thereby increase fault tolerance.

See also NetWare Directory Database.

directory replication A process that copies a master set of directories from a server, called an export server, to other specified servers or workstations, known as import computers. This process simplifies the task of maintaining and synchronizing identical sets of directories and files, because only one master copy of the data is maintained. A file is replicated when it is added to an exported directory, as well as when changes are made to the file.

Directory Schema In Novell Directory Services (NDS), a set of rules that defines how information can be stored in the Directory database. The Schema contains four major definitions:

- **Attribute information** Describes the kinds of information that can be associated with an object.
- **Inheritance** Defines which objects can inherit the rights and properties of other objects.
- **Naming** Determines the structure of the Directory tree.
- **Subordination** Specifies the location of objects in the Directory tree.

See also Novell Directory Services.

directory service client Abbreviated DSCLIENT. In Microsoft Windows 95/98, an add-on that makes a Windows 95/98 client computer aware of Active Directory.

See also Active Directory.

Directory Service log In Microsoft Windows 2000 Server, a special log that records any events associated with running the Directory Service, any problems connecting to the global catalog, and any events associated with Active Directory (AD) on the network.

See also DNS Server log; Event Viewer; File Replication Service log.

Directory services In the Microsoft Active Directory structure, a directory as well as the services it provides, such as security and replication.

See also Active Directory.

directory services A listing of all users and resources on a network, designed to help clients locate network users and services. Some examples are the OSI's X.500, Novell Directory Services (NDS), Microsoft's Active Directory, and Banyan's StreetTalk.

See also domain directory services; global directory services.

directory structure duplication A technique that maintains duplicate copies of the file allocation table (FAT) and directory entry table in separate areas of a hard disk. If the first copy is damaged or destroyed, the second copy is immediately available for use.

See also directory verification.

directory tree A method of representing the hierarchical structure of the directories, subdirectories, and files on a disk. The term is often used in graphical user interfaces. In object-oriented systems, this structure may also represent a group of objects, as in Novell Directory Services (NDS).

directory verification A process that performs a consistency check on duplicate sets of file allocation tables (FATs) and directory entry tables to verify that they are identical. The verification occurs every time the server is started.

See also directory structure duplication.

disable To turn a function off or prevent something from happening. In a graphical user interface, disabled menu commands are often shown in gray to indicate that they are not available.

See also dimmed command; enable.

disaster recovery The process used to restore services after a major interruption in computing or in communications. Large events such as earthquakes or fires can interrupt networked computing activities, but so can more mundane events such as hard-disk failure or a construction worker accidentally cutting through a power or telephone line. The whole point to disaster recovery is to plan for it before anything happens and then follow the plan when restoring services.

See also backup; disk duplexing; disk mirroring; redundant array of inexpensive disks.

Discretionary Access Control List Abbreviated DACL. In Microsoft Windows NT, a list of user and group accounts that have permission to access an object's services. The DACL has as many Access Control Entries as there are user or group accounts with access to the object.

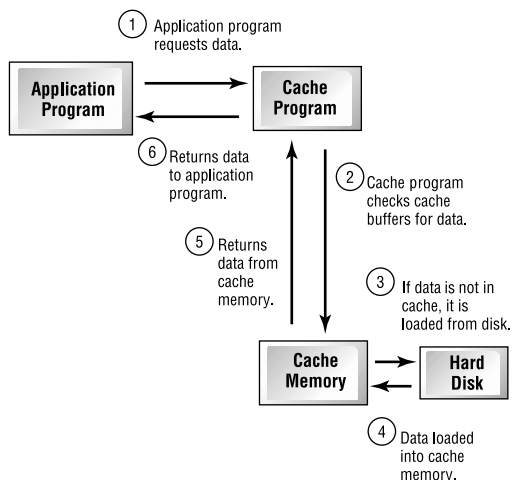
disk cache Pronounced "disk cash." An area of computer memory where data is temporarily stored on its way to or from a disk.

When an application asks for information from the hard disk, the cache program first checks to see if that data is already in the cache memory. If it is, the disk cache program loads the information from the cache memory rather than from the hard disk. If the information is not in memory, the cache program reads the data from the disk, copies it into the cache memory for future reference, and then passes the data to the requesting application. This process is shown in the accompanying illustration.

A disk cache program can significantly speed most disk operations. Some network operating systems also cache other often accessed and important information, such as directories and the file allocation table (FAT).

See also directory caching.

DISK CACHE



disk controller The electronic circuitry that controls and manages the operation of floppy disks and hard disks.

A single disk controller may manage more than one hard disk. Many disk controllers also manage floppy disks and compatible tape drives. In the Macintosh, the disk controller is built into the system. In the PC, the disk controller may be a printed circuit board inserted into the expansion bus, or it may be part of the hard disk drive itself.

disk coprocessor board See host bus adapter.

disk drive A peripheral storage device that reads and writes magnetic or optical disks. When more than one disk drive is installed on a computer, the operating system assigns each drive (or logical drive) a unique name.

Several types of disk drives are in common use: floppy disk drives, hard disk drives, compact disc drives, digital video disc drives, Zip drives, and magneto-optical disc drives.

disk duplexing A fault-tolerant technique that writes the same information simultaneously onto two hard disks.

Each hard disk uses a different disk controller to provide greater redundancy. If one disk or disk controller fails, information from the other system can be used to continue operations. Disk duplexing is offered by most major network operating systems. It is designed to protect the system against a single disk failure, not multiple disk failures. Disk duplexing is no substitute for a well-planned series of disk backups.

See also data protection; disk mirroring; redundant array of inexpensive disks.

Disk Manager In Microsoft Windows 2000, the administrative tool used to manage hard disks and other storage media. Disk Manager replaces NT's Disk Administrator.

See also basic disk; dynamic disk.

disk quotas In Microsoft Windows 2000, a feature that allows an administrator to limit the amount of hard disk space available to a user. You can set quotas only on NTFS hard disks.

diskless workstation A networked computer that does not have any local disk storage capability. The computer boots and loads all its programs from the network file server. Diskless workstations are particularly valuable when sensitive information is processed; information cannot be copied from the file server onto a local disk, because the workstation does not have one.

See also dumb terminal; network computer; thin client.

disk mirroring A fault-tolerant technique that writes the same information simultaneously onto two hard disks or two hard-disk partitions, using the same disk controller. If one disk or partition fails, information from the other can be used to continue operations. Disk mirroring is offered by most major network operating systems. It is designed to protect the system against a single disk failure, not multiple disk failures. Disk mirroring is no substitute for a well-planned series of disk backups.

See also data protection; disk duplexing; redundant array of inexpensive disks.

disk optimizer A utility that rearranges files and directories on a disk for optimum performance. By reducing or eliminating file fragmentation (storage in pieces in different locations on the hard disk), a disk optimizer can restore the original level of performance of your disk system. Also, it is usually easier to undelete, or recover, an unfragmented file than a fragmented one.

Many disk optimizers will not only rewrite files as contiguous files, but will also place specific unchanging files in particular locations on the disk, optimize directories, and even place specific applications on the disk so that they load more quickly.

See also defragmentation.

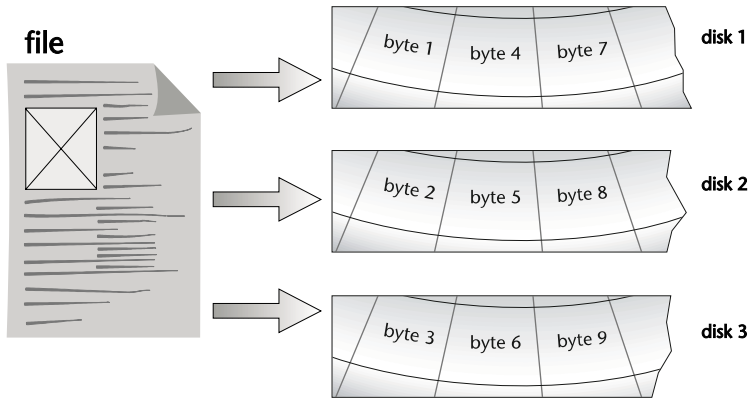
disk striping The technique of combining a set of disk partitions located on different hard disks into a single volume, creating a virtual "stripe" across the partitions that the operating system recognizes as a single drive. Disk striping can occur at the bit level or at the sector level and allows multiple concurrent disk accesses that can improve performance considerably.

See also disk striping with parity; redundant array of inexpensive disks; stripe set.

disk striping with parity The addition of parity information across a disk stripe so that if a single disk partition fails, the data on that disk can be re-created from the information stored across the remaining partitions in the disk stripe.

See also disk striping; redundant array of inexpensive disks; stripe set.

DISK STRIPING



dispersion The degree of scattering that takes place as a beam of light travels along a fiber-optic cable.

See also cladding.

distance vector algorithm A family of routing algorithms that calculate the best-path route to use for data transmission from information present in adjacent nodes on the network. Routing information is broadcast periodically rather than only when a change occurs, which makes the method bandwidth intensive. For this reason, distance vector algorithm is best used in relatively small networks with few inter-router connections.

Distinguished Name In Novell Directory Services (NDS), the full name of an object, including the object's Common name and its context. Sometimes called the Full Distinguished Name.

See also Common name; Context; Relative Distinguished Name.

Distributed Component Object Model

Abbreviated DCOM. A specification from Microsoft that enables communications between distributed objects.

DCOM extends the Component Object Model (COM) so that clients can communicate directly with other processes on different computers on a LAN or on the Internet.

See also ActiveX; Component Object Model.

distributed computing *See* distributed processing.

Distributed Computing Environment

Abbreviated DCE. The Open Software Foundation's (OSF) architecture for developing application software for use on different networks.

DCE hides the differences between products, technologies, and standards, providing independence from the operating system and underlying network. No particular communications protocol is specified, so the network can run on IP (Internet Protocol), IPX (Inter-network Packet Exchange), or SNA (Systems Network Architecture).

distributed database A database managed as a single system even though it includes many clients and many servers at both local and remote sites. A distributed database requires that data redundancy is managed and controlled.

Distributed Director Web site load-balancing software from Cisco Systems.

A popular Web site can be overwhelmed by requests from users, and to avoid long wait times, you can configure additional Web servers to hold duplicate information. Web requests are then sent to the Web server best able to handle the request. Distributed Director runs on Cisco routers and uses a routing protocol to send user requests to the server that is closest to them.

Distributed file system Abbreviated Dfs. In Microsoft Windows 2000 Server, a hierarchical file system that combines resources from all over the network into a single logical view.

distributed file system Any file system in which files and programs are located on more than one computer or server. Users can access files and applications as though they were stored on a single local system.

Distributed System Object Model

Abbreviated DSOM. IBM's extension to System Object Model (SOM) that allows objects to communicate in a distributed processing environment.

distributed processing A computer system in which processing is performed by several separate computers linked by a communications network.

The term often refers to any computer system supported by a network, but more properly refers to a system in which each computer is chosen to handle a specific workload and the network supports the system as a whole. Each computer contributes to the completion of a common task by completing one or more subtasks independently of its peers and then reporting the results from these subtasks. All this is totally transparent to the users; all they see are the results of the process.

See also client/server architecture.

distribution groups In Microsoft Windows 2000 Server, a collection of recipient e-mail addresses. Distribution groups have nothing to do with system security. They do not have security identifiers, and they do not appear on access control lists.

See also security groups.

distribution medium The type of data-storage device used to distribute original software or software updates.

Tapes used to be the favorite distribution medium, but compact discs are rapidly gaining popularity.

DIX connector *See* Attachment Unit Interface.

DLL *See* Dynamic Link Library.

DLSw *See* Data Link Switching.

DMA *See* direct memory access.

DMI *See* Desktop Management Interface.

DNA *See* Digital Network Architecture.

DNS *See* Domain Name Service.

DNS alias A host name that the Domain Name Service (DNS) server knows points to another host. Computers always have one real name, but they can also have several aliases. A DNS alias is sometimes called a CNAME or a canonical name.

See also Domain Name Service.

DNS name server A server containing information that is part of the Domain Name Service (DNS) distributed database, which makes computer names available to client programs querying for name resolution on the Internet.

See also Domain Name Service.

DNS Server log In Microsoft Windows 2000 Server, a special log that records any events associated with running the Domain Name Service (DNS) Server.

See also Directory Service log; Event Viewer; File Replication Service log.

docking station A hardware system into which a portable computer fits so that it can be used as a full-fledged desktop computer.

Docking stations vary from simple port replicators that allow you access to parallel and serial ports and a mouse to complete

systems that give you access to network connections, CD-ROMs, and even a tape backup system or PCMCIA ports. The portable computer and docking station are designed as two parts of the same system; you cannot swap computers and docking stations from different manufacturers or even from different models.

See also port replicator.

document database A carefully organized collection of related documents; for example, a set of technical support bulletins.

document instance In Standard Generalized Markup Language (SGML), the text component of a document as distinct from the structure of the document.

See also document type definition; Standard Generalized Markup Language.

document management The cataloging, storage, and retrieval of documents in a networked environment. In this context, a document may be text, scanned graphics, a spreadsheet, a form, a Web page, or any other unique file.

Each file is tagged with information that includes the name of the original author, document description, creation date, and the name of the application used to create the document.

See also groupware; Lotus Notes; workflow software.

document root On a Web server, a directory that contains the files, images, and data you want to present to all users who access the server with a browser.

document type definition Abbreviated DTD. In Standard Generalized Markup Language (SGML), the structural component of a document as distinct from the actual data or content of the document.

See also document instance; Standard Generalized Markup Language.

documentation The instructions, tutorials, specifications, troubleshooting advice, and reference guides that accompany a computer program or a piece of hardware. Documentation can be in printed or online format. Early system documentation was often written by programmers and engineers and was usually filled with technical jargon. Today's documentation is generally better written and easier to understand.

domain A description of a single computer, a whole department, or a complete site, used for naming and administrative purposes.

Top-level domains must be registered to receive mail from outside the organization; local domains have meaning only inside their own enterprise. Depending on the context, the term can have several slightly different meanings:

- On the Internet, a domain is part of the Domain Name Service (DNS).
- In Novell NetWare, a domain is a special area of memory where a new NetWare Loadable Module (NLM) can be tested without the risk of corrupting the operating system memory.
- In IBM's Systems Network Architecture (SNA), a domain represents all the terminals and other network resources controlled by a single processor or processor group.

- In Microsoft Windows NT, a group of computers, users, and network peripherals managed with a single set of account descriptions and security policies. A user can log on to the local computer and be authenticated to access just that one system, or a user can log on to a domain and be authenticated to access other servers within that domain.

- In Lotus Notes, a domain is one or more Notes servers that share the same Public Name and Address Book database. This database contains information about the users within the domain, including their e-mail addresses and other information.

See also domain name.

domain component In the Microsoft Active Directory structure, a portion of a domain name. For example, in computer .sybex.com, each element of the name—computer, sybex, and com—is a domain component.

See also Active Directory.

domain controller In Microsoft Windows NT, a server that stores and shares domain information, including the central database of users, passwords, and permissions.

A domain controller can be a primary domain controller, which stores the master copy of the domain information, or a backup domain controller, which uses a replicated version of this data to verify user logons and rights.

domain directory services Directory services that consist of one or more linked servers. Each domain within a network must

domain name

be managed and administered separately. Windows NT Server and IBM's LAN Server both use domain directory services.

See also global directory services.

domain name In the Domain Name Service (DNS), an easy-to-remember name that identifies a specific Internet host, as opposed to the hard-to-remember numeric IP address.

See also bang path.

Domain Name Service Abbreviated DNS, sometimes referred to as Domain Naming System. A distributed addressing system that resolves the domain name into the numeric IP address. DNS lets you use the Internet without having to remember long lists of cryptic numbers.

The most common high-level domains on the Internet include:

- .com A commercial organization
- .edu An educational establishment such as a university
- .gov A branch of the U.S. government
- .int An international treaty organization
- .mil A branch of the U.S. military
- .net A network provider
- .org A nonprofit organization

Most countries also have unique domains named after their international abbreviation—for example, .uk for the United Kingdom and .ca for Canada.

See also IP address; name resolution.

DOS Acronym for Disk Operating System. An operating system originally developed by Microsoft for the IBM PC.

DOS existed in two similar versions: MS-DOS, developed and marketed by Microsoft for use with IBM-compatible computers, and PC-DOS, supported and sold by IBM for use on computers manufactured by IBM.

See also MS-DOS.

DOS client A workstation that boots MS-DOS and gains access to the network using either a NetWare shell or the NetWare DOS Requester software.

DOS prompt *See* MS-DOS prompt.

dot A synonym for the name of the current directory, usually invisible as its name begins with a period.

In Unix, a file whose name begins with a dot usually contains configuration information; you can customize your environment by creating the appropriate dot file in the current directory or in your home directory.

See also current directory; dot dot; period and double-period directories.

dot dot A synonym for the name of the parent directory to the current directory, usually invisible as its name begins with a period.

See also current directory; dot; period and double-period directories.

dotted decimal A method of representing an IP address as four decimal numbers

separated by dots, or periods; for example, 194.65.87.3.

See also IP address.

double-byte character set Abbreviated DBCS. A method that uses two bytes to hold the definition of the character rather than the single byte used by the American Standard Code for Information Interchange (ASCII) or the Extended Binary Coded Decimal Interchange Code (EBCDIC).

By utilizing two bytes instead one, the many international character sets in use these days can be managed much more easily.

See also ASCII standard character set; ASCII extended character set; Unicode.

double-slash Notation used with a colon to separate the communications protocol from the host computer name in a Uniform Resource Locator (URL) as in <http://www.sybex.com>.

See also Uniform Resource Locator.

downlevel trust In Microsoft Windows 2000 Server, a trust explicitly established between a Windows 2000 domain and a Windows NT 4 domain.

See also explicit trust; shortcut trust; trust relationship; two-way transitive trust.

downlink The transmission of information from a satellite to an earth station.

See also uplink.

download 1. In communications, to transfer a file or files from one computer to another over a network or using a modem.

2. To send information, such as font information or a PostScript file, from a computer to a printer.

downsizing The redesign of mainframe-based business applications to applications capable of running on smaller, less-expensive systems, often PC LANs. Client/server architecture is the model most often implemented during downsizing.

When applications are moved from large computer systems to PCs, security, integrity, and overall control may be compromised, and development and training costs can be high. However, a collection of appropriately configured PCs, networked together, can provide more than ten times the power for the same cost as a mainframe computer supporting remote terminals.

A more accurate term might be *rightsizing*, to match the application requirements of the corporation to the capabilities of the hardware and software systems available.

See also outsourcing.

downtime The amount of time during which a computer system is not available to users, because of a hardware or software failure.

downward compatibility *See* backward compatibility.

DP *See* data processing.

drag-and-drop In a graphical user interface, to move a selected object onto another object with the mouse to initiate a process. For example, if you drag a document icon and drop it onto a word processor's icon, the program will run, and the document

will be opened. To print a file, you can drag the file to the printer icon using the mouse and then release the mouse button. You can also use drag-and-drop to copy a file from one disk to another or to move a marked block of text to a new location inside a word-processed document.

DRAM See dynamic RAM.

drive array The group of hard disk drives used in one of the RAID (redundant array of inexpensive disks) configurations. See also redundant array of inexpensive disks.

drive letter A designation used to specify a PC disk drive. For example, the first floppy disk drive is usually referred to as drive A, and the first hard disk drive is usually referred to as drive C.

drive mapping The technique of assigning a drive letter to represent a complete directory path statement. Novell NetWare supports four types of drive mapping:

- **Local drive mapping** Maps drives to local hard disk and floppy disk drives
- **Network drive mapping** Maps drives to volumes and directories on the network
- **Network search drive mapping** Maps drives to directories that should be searched if an application or file is not found in the current directory
- **Directory map object** Maps drives to directories that contain frequently used files, such as applications, without having to specify the actual physical location of the file

See also Directory Map object.

driver Jargon for device driver.

See also device driver.

drop 1. The location in a multidrop line where a tap is inserted to allow the connection of a new device.

2. To lose part of a signal, usually unintentionally, as in dropping bits.

drop cable The cable used in thick Ethernet to connect a network device to a Multistation Access Unit (MAU). The maximum cable length is 50 meters (165 feet). Sometimes called a transceiver cable.

DS See digital signal.

DSCLIENT See directory service client.

DSL See Digital Subscriber Line.

DSOM See Distributed System Object Model.

DSP See Digital Signal Processing.

DSR See Data Set Ready.

DSU See Data Service Unit.

DTD See document type definition.

DTE See data terminal equipment.

DTR See Data Terminal Ready; Dedicated Token Ring.

dual-attached station Abbreviated DAS. In the Fiber Distributed Data Interface (FDDI), a device attached to both of the dual, counter-rotating rings. Concentrators, bridges, and routers often use DAS connections to provide fault tolerance. In contrast, a single attached station (SAS) connects to only one ring.

dual-homed host A network server configured with two network interface cards, each connected to a different network. Dual-homed hosts are often used with firewalls to increase network security.

See also firewall; proxy server.

dual homing In Fiber Distributed Data Interface (FDDI), a method of cabling concentrators and stations in a tree configuration, providing an alternative route to the FDDI network should the primary connection fail.

dual in-line package Abbreviated DIP. A standard housing constructed of hard plastic commonly used to hold an integrated circuit. The circuit's leads are connected to two parallel rows of pins designed to fit snugly into a socket; these pins may also be soldered directly to a printed circuit board.

See also DIP switch.

dumb terminal A combination of keyboard and screen that has no local computing power, used to input information to a large, remote computer, often a minicomputer or a mainframe. The remote computer provides all the processing power for the system.

See also diskless workstation; intelligent terminal; network computer; thin client.

duplex In asynchronous transmissions, the ability to transmit and receive on the same channel at the same time; also referred to as full duplex. Half-duplex channels can transmit only or receive only.

See also communications parameters.

DVD *See* digital video disc.

DVD-E *See* digital video disc-erasable.

DVD-R *See* digital video disc-recordable.

DVD-ROM *See* digital video disc-ROM.

dynamic adaptive routing *See* dynamic routing.

dynamic bandwidth allocation A method of bandwidth allocation that subdivides the available bandwidth between multiple applications almost instantaneously, to provide each application with just the amount of bandwidth that it currently needs.

Dynamic Data Exchange Abbreviated DDE. A technique used for application-to-application communications, available in several operating systems, including Microsoft Windows, Macintosh, and OS/2.

When two or more programs that support DDE are running at the same time, they can exchange data and commands, by means of conversations. A DDE conversation is a two-way connection between two applications, used to transmit data by each program alternately.

DDE is used for low-level communications that do not need user intervention. For example, a communications program might feed stock market information into a spreadsheet program, where that data can be displayed in a meaningful way and recalculated automatically as it changes.

DDE has largely been superseded by Object Linking and Embedding (OLE).

See also Object Linking and Embedding.

dynamic disk In Microsoft Windows 2000, a hard disk that is available only to Windows 2000 and not available from any other operating system. A dynamic disk does not use primary and extended partitions to divide the available space into logical units.

To create a dynamic disk, use the Disk Management utility to convert a basic disk into a dynamic disk. Dynamic disks allow you to create, delete, and extend fault-tolerant and multidisk volumes without restarting the operating system. Windows 2000 RAID (redundant array of inexpensive disks) services are available only on dynamic disks.

See also basic disk.

Dynamic DNS Abbreviated DDNS. In Microsoft Windows 2000 Server, a mechanism that allows Dynamic Host Configuration Protocol (DHCP) and Windows 2000 clients to update Domain Name Service (DNS) records dynamically, rather than by the traditional method of manually adding the new records to static DNS zone files.

See also Domain Name Service.

Dynamic Host Configuration Protocol

Abbreviated DHCP. A system based on network interface card addresses that is used to allocate IP addresses and other configuration information automatically for networked systems. DHCP is an update of the Bootstrap Protocol.

See also Bootstrap Protocol; hardware address; IP address; Transmission Control Protocol/Internet Protocol.

Dynamic Link Library Abbreviated DLL. A program module that contains executable code and data that can be used by applications, or even by other DLLs, in performing a specific task.

DLLs are used extensively throughout the family of Microsoft Windows products.

DLLs may have filename extensions of .DLL, .DRV, or .FON.

The DLL is linked into the application only when the program runs, and it is unloaded again when no longer needed. If two DLL applications are running at the same time and both perform a particular function, only one copy of the code for that function is loaded, for more efficient use of limited memory. Another benefit of using dynamic linking is that the .EXE files are not as large as they would be, because frequently used routines can be put into a DLL rather than repeated in each .EXE file that uses them. A smaller .EXE file means saved disk space and faster program loading.

dynamic RAM Abbreviated DRAM, pronounced “dee-ram.” A common type of computer memory that uses capacitors and transistors storing electrical charges to represent memory states. These capacitors lose their electrical charge, so they need to be refreshed every millisecond, during which time they cannot be read by the processor.

DRAM chips are small, simple, cheap, easy to make, and hold approximately four times as much information as a static RAM (SRAM) chip of similar complexity. However, they are slower than SRAM.

dynamic routing A routing technique that allows the route that a message takes to

change, as the message is in transit through the network, in response to changing network conditions. Conditions forcing a route change might include unusually heavy traffic on a particular section of the network or a cable failure. Also known as dynamic adaptive routing.

dynamic volume In Microsoft Windows 2000, a logical hard disk volume created by Disk Management.

Dynamic volumes can include simple, striped, spanned, and RAID-5 components and can exist only on dynamic disks.

See also basic disk; dynamic disk.

DynaText A form of electronic document and viewer used in Novell NetWare for online manuals. You can use DynaText directly from the CD, or you can install it on the server or on a workstation. Replaces Electrotex.

E

E See *exa-*.

E1 A point-to-point, dedicated, 2.048Mbps communications circuit capable of supporting thirty-two 64Kbps channels used as 30 voice channels, 1 control channel, and 1 synchronization and framing channel. The European equivalent of North America's T1.

E1 circuits carry more channels than the 24 channels used in T1 systems, and repeaters are required every 6000 feet when copper wire is used.

E2 A European point-to-point, dedicated, 8.848Mbps communications circuit equivalent to four E1 circuits. E2 is the European equivalent of North America's T2 and is rarely used.

E3 A European point-to-point, dedicated, 34.368Mbps communications circuit equivalent to 16 E1 circuits. The European equivalent of North America's T3.

E4 A European point-to-point, dedicated, 139.26Mbps communications circuit equivalent to 4 E3 or 64 E1 circuits.

E5 A European point-to-point, dedicated, 565.148Mbps communications circuit equivalent to 4 E4 or 256 E1 circuits.

EAP See *Extensible Authentication Protocol*.

eavesdropping The process of gathering information about a target network by listening in on transmitted data.

EB See *exabyte*.

EBCDIC See *Extended Binary Coded Decimal Interchange Code*.

echo 1. A transmitted signal that is reflected back to the sender strongly enough so that it can be distinguished from the original signal; often encountered on long-distance telephone lines and satellite links.

2. A form of repetition, used as a mechanism in testing network nodes, in which each receiving station on the network echoes a message back to the main server or host computer.

echo cancellation A mechanism used to control echoes on communications links such as satellite links.

The modem checks for a delayed duplication of the original signal and adds a reversed version of this transmission to the channel on which it receives information. This process effectively removes the echo without affecting the incoming signal.

ECNE See *Master Certified Novell Engineer*.

EDI See *electronic data interchange*.

EDO RAM Abbreviation for *extended data out RAM*. A type of RAM that keeps data available to the processor while the next memory access is being initialized, thus speeding overall access times. EDO RAM is significantly faster than conventional dynamic RAM.

EDP *See* data processing.

EEMS *See* Enhanced Expanded Memory Specification.

effective rights In Novell Directory Services (NDS), any rights an object can use to look at or change a specific directory, file, or other object.

Effective rights are recalculated every time an object attempts such an operation and are controlled by the Inherited Rights Filter, the trustee assignment, and the specified security restrictions. Effective rights are of two types:

- Object rights determine what a user can do with an object.
- Property rights control a user's access to that object.

See also Inherited Rights Filter; Novell Directory Services; rights; trustee assignments.

EFS *See* Encrypted File System.

EGP *See* External Gateway Protocol.

EIA *See* Electronic Industries Association.

EIA/TIA 586 A standard, jointly defined by the Electronic Industries Association and the Telecommunications Industry Association (EIA/TIA), for telecommunications wiring used in commercial buildings.

The standard is designed to do the following:

- Specify a generic wiring system for all commercial buildings

- Define media types, as well as connections and terminations
- Provide a basis for interoperation between competing products and services in wiring, design, installation, and management
- Allow for the wiring of a building before the definition of the products that will use that wiring, and allow for elegant future expansion

EIA/TIA 586 applies to all unshielded twisted-pair wiring that works with Ethernet, Token Ring, ISDN, and other networking systems.

See also cabling standards; Category 1–5.

EIDE *See* Enhanced IDE.

EIGRP *See* Enhanced Interior Gateway Routing Protocol.

EISA *See* Extended Industry Standard Architecture.

electromagnetic interference Abbreviated EMI. Any electromagnetic radiation released by an electronic device that disrupts the operation or performance of another device.

EMI is produced by many sources commonly found in an office environment, including fluorescent lights, photocopiers, and motors such as those used in elevators. EMI is also produced by natural atmospheric or solar activity.

See also Class A certification; Class B certification; Federal Communications Commission; radio frequency interference.

electronic commerce The buying and selling of goods and services over the Internet. Electronic commerce may also involve business-to-business transactions in the exchange of purchase orders, invoices, and other electronic documents.

electronic data interchange Abbreviated EDI. A method of electronically exchanging business documents, including bills of materials, purchase orders, and invoices.

Customers and suppliers can establish an EDI network by means of Open Systems Interconnect (OSI) standards or by using proprietary products. Widely accepted standards include ANSI X.12, ISO 9735, and CCITT X.435.

electronic data processing *See* data processing.

Electronic Industries Association Abbreviated EIA. A trade association representing American manufacturers in standards organizations. The EIA has published and formalized several important standards, including RS-232-C, RS-422, RS-423, RS-449, RS-485, and RS-530. Standards having to do with communications are produced jointly with the Telecommunications Industry Association.

For more information on the EIA, see www.eia.org.

See also EIA/TIA 586.

electronic mail The use of a network to transmit text messages, memos, and reports; usually referred to as e-mail. Users can send a message to one or more individ-

uals, to a predefined group, or to all users on the system. When you receive a message, you can read, print, forward, answer, or delete it.

An e-mail system may be implemented on a peer-to-peer network, a client/server architecture, a mainframe computer, or on a dial-up service, such as America Online. E-mail is by far the most popular Internet application, with well over 80 percent of Internet users taking advantage of the service.

E-mail has several advantages over conventional mail systems, including:

- E-mail is fast—very fast when compared with conventional mail.
- If something exists on your computer as a file—text, graphical images, even program files and video segments—you can usually send it as e-mail.
- E-mail is very extensive. You can now send e-mail to well over half the countries in the world.

The problems associated with e-mail are similar to those associated with online communications in general, such as security, privacy (always assume that your e-mail is not private), and the legal status of documents exchanged via e-mail.

See also mailbox; Multipurpose Internet Mail Extension; voice mail.

element A unit of structure in HyperText Markup Language (HTML), such as a title or a list. Some elements have start and stop tags; others have only a single tag. Certain elements can be nested within other elements.

See also HyperText Markup Language; tag.

elevator seeking A technique that allows the server hard disk head to access files in the direction that the head is already traveling across the disk, rather than in the order in which they were requested. This feature allows the drive heads to operate continuously and thus improves disk performance and minimizes disk-head seek times.

ELF *See* extremely low-frequency emission.

EMA *See* Enterprise Memory Architecture.

emacs A popular Unix editor, written by Richard Stallman, founder of the Free Software Foundation.

The name *emacs* is a contraction of “editing macros,” but it is much more than a simple text editor and includes extensions for all sorts of common tasks, ranging from compiling and debugging programs to reading and sending e-mail. You can even extend emacs yourself as the editing commands are written in the Lisp programming language.

See also Free Software Foundation; vi.

e-mail *See* electronic mail.

e-mail address The addressing information required for an e-mail message to reach the correct recipient.

See also bang path; Internet address; mailbox.

Emergency Repair Disk Abbreviated ERD. In Microsoft Windows 2000, a floppy disk created by the Backup program. An ERD contains information about the cur-

rent system configuration that can be used to repair a system that will not restart.

See also Recovery Console; safe mode.

EMI *See* electromagnetic interference.

EMM *See* expanded memory manager.

emoticon A collection of text characters often used in e-mail and posts to newsgroups to signify emotions.

An emoticon can be as simple as including <g> or <grin> in your text, an indication that the writer is joking, and as complex as some of the smiley faces, which are all designed to be read sideways, such as the wink ;-) or the frown :-(.

See also Internet abbreviations.

EMS *See* Expanded Memory Specification.

emulator A device built to work exactly like another device—hardware, software, or a combination of both.

For example, a terminal emulation program lets a PC pretend to be a terminal attached to a mainframe computer or to an online service by providing the control codes that the remote system expects to receive. In printers, some brands emulate popular models such as Hewlett-Packard’s LaserJet line.

enable To turn a function on or allow something to happen. When a function is enabled, it is available for use. In a graphical user interface, enabled menu commands are often shown in black type.

See also disable.

Encapsulated PostScript

Encapsulated PostScript Abbreviated EPS. The file format of the PostScript page-description language.

The EPS standard is device independent, so images can easily be transferred between applications, and they can be sized and output to different printers without any loss of image quality or distortion.

The EPS file contains the PostScript commands needed to recreate the image, but the image itself cannot be displayed on a monitor unless the file also contains an optional preview image stored in TIFF or PICT format.

You can print an EPS file only on a PostScript-compatible laser printer, and the printer itself determines the final printing resolution. A laser printer might be capable of 600 dpi, whereas a Linotronic printer is capable of 2450 dpi.

encapsulation The process of inserting the frame header and data from a higher-level protocol into the data frame of a lower-level protocol.

See also tunneling.

enclosure A term for a file—text, fax, binary, or image—sent as a part of an e-mail message. Sometimes called an attachment.

See also Multipurpose Internet Mail Extension.

encode 1. To compress a video file using a codec so that the file can be transmitted in the shortest possible time.

2. To convert a binary file into a form suitable for data transmission.

See also codec; decode; decryption; uuencode.

Encrypted File System In Microsoft Windows 2000, a feature that lets mobile users, who are concerned with the security of their files in the case of unauthorized access to their computers, encrypt designated files and directories using a public key encryption scheme.

Encrypting File System Abbreviated EFS. In Microsoft Windows 2000, a file system used to encrypt files and folders on an NTFS 5 hard disk.

encryption The process of encoding information in an attempt to make it secure from unauthorized access, particularly during transmission. The reverse of this process is known as decryption.

Two main encryption schemes are in common use:

- **Private (Symmetrical) Key** An encryption algorithm based on a private encryption key known to both the sender and the recipient of the information. The encrypted message is unreadable and can be transmitted over nonsecure systems.
- **Public (Asymmetrical) Key** An encryption scheme based on using the two halves of a long bit sequence as encryption keys. Either half of the bit sequence can be used to encrypt the data, but the other half is required to decrypt the data.

See also Data Encryption Standard; Pretty Good Privacy; ROT-13.

encryption key A unique and secret number used to encrypt data to protect it from unauthorized access.

end-of-file Abbreviated EOF. A special code placed after the last byte in a file that indicates to the operating system that no more data follows.

An end-of-file code is needed because disk space is assigned to a file in blocks, and the file may not always terminate at the end of a block. In the ASCII system, an EOF is represented by the decimal value 26 or by the Ctrl+Z control character.

end node A networked node such as a PC that can only send and receive information for its own use; it cannot route or forward information to another node.

end-of-text Abbreviated ETX. A character used in computer communications to indicate the end of a text file.

In the ASCII system, an ETX is represented by the decimal value 3 or by the Ctrl+C control character. A different symbol, end-of-transmission (EOT, ASCII 4, or Ctrl+D) is used to indicate the end of a complete transmission.

end-of-transmission Abbreviated EOT. A character used in computer communications to indicate the end of a transmission. In the ASCII system, an EOT is represented by the decimal value 4 or by the Ctrl+D control character.

end user Often refers to people who use an application to produce their own results on their own computer or workstation.

During the mainframe computer era, end users were people who received output from the computer and used that output in their work. They rarely, if ever, actually saw the computer, much less learned to use it themselves. Today, end users often write macros to automate complex or repetitive tasks and sometimes write procedures using command languages.

Enhanced Expanded Memory

Specification Abbreviated EEMS. A revised version of the original Lotus-Intel-Microsoft Expanded Memory Specification (LIM EMS) that lets MS-DOS applications use more than 640KB memory space.

See also Expanded Memory Specification.

Enhanced IDE An extension to the Integrated Drive Electronics (IDE) interface standard, which supports hard disks as large as 8.4GB (IDE supports hard disks of up to 528MB) and transfer rates of up to 13.3MBps (IDE allows rates of up to 3.3MBps).

See also Integrated Drive Electronics; Small Computer System Interface.

Enhanced Interior Gateway Routing

Protocol Abbreviated EIGRP. A proprietary link-state distance-vector routing protocol from Cisco, first released in 1994.

EIGRP improves performance by using information derived from distance-vector protocols, but also includes an algorithm for removing transient loops in possible routes, which reduces the complexity of the calculations required.

See also distance vector algorithm; Interior Gateway Routing Protocol.

Enhanced Small Device Interface Abbreviated ESDI. A popular hard disk, floppy disk, and tape drive interface standard, capable of a data-transfer rate of 10 to 20Mbps. ESDI is most often used with large hard disks.

See also Integrated Drive Electronics; Small Computer System Interface.

ENS *See* Enterprise Network Services.

Enter key Also known as the Return key, short for carriage return. The key that indicates the end of a command or the end of user input from the keyboard.

enterprise A term used to encompass an entire business group, organization, or corporation, including all local, remote, and satellite offices.

Enterprise CNE *See* Master Certified Novell Engineer.

Enterprise Memory Architecture Abbreviated EMA. In Microsoft Windows 2000, a mechanism that lets applications address as much as 64GB of virtual memory.

enterprise network A network that connects every computer in every location of a business group, organization, or corporation and runs the company's mission-critical applications.

In many cases, an enterprise network includes several types of computers running several different operating systems.

Enterprise Network Services Abbreviated ENS. A software product based on Banyan Systems' StreetTalk Directory Service for VINES that brings global directory

service features to other networks. ENS includes StreetTalk Directory Assistance, the Banyan Security Service, and Banyan Network Management.

Specific versions of ENS are available for Novell NetWare, SCO Unix, and HP-UX so that servers running those operating systems can interoperate and share management in a network with VINES servers.

See also Banyan VINES.

Enterprise Systems Connection Abbreviated ESCON. A set of products and services from IBM that provide direct channel-to-channel connections between ES/9000 mainframes and peripheral devices over 10 to 17Mbps fiber-optic links.

envelope delay *See* delay distortion.

environment 1. The complete set of hardware and software resources made available to any user of a system.

2. The operating system that a program needs in order to execute. For example, a program may be said to be running in the Unix environment.

EOF *See* end-of-file.

EOT *See* end-of-transmission.

EPS *See* Encapsulated PostScript.

equalization The process of balancing a circuit by reducing frequency and phase distortion so that it passes all expected frequencies with equal efficiency.

ERD *See* Emergency Repair Disk.

error The difference between the expected and the actual.

In computing, the way that the operating system reports unexpected, unusual, impossible, or illegal events is by displaying an error number or error message. Errors range from trivial, such as an attempt to write a file to a disk drive that does not contain a disk, to fatal, such as when a serious operating system bug renders the system useless.

In communications, errors are often caused by line noise and signal distortion. Parity or cyclical redundancy check (CRC) information is often added as overhead to the data stream, and techniques such as error detection and correction are employed to detect and correct as many errors as possible.

See also attenuation; crosstalk; error handling; error message; error rate; parity error.

error detection and correction A mechanism used to determine whether transmission errors have occurred and, if so, to correct those errors.

Some programs or transmission protocols simply request a retransmission of the affected block of data if an error is detected. More complex protocols attempt to both detect and determine at the receiving end what the correct transmission should have been.

See also checksum; cyclical redundancy check; forward error correction; Hamming code; parity.

error handling The way that a program copes with errors or exceptions that occur as the program is running.

Good error handling manages unexpected events or wrongly entered data gracefully, usually by opening a dialog box to prompt the user to take the appropriate action or to enter the correct information. Badly written programs may simply stop running when the wrong data is entered or when an unanticipated disk error occurs.

error message A message from the program or the operating system that contains information about a condition that requires some human intervention to solve.

Error messages can indicate relatively trivial problems, such as a disk drive that does not contain a disk, as well as fatal problems, such as when a serious operating system bug renders the system useless and requires a system reboot.

error rate In communications, the ratio between the number of bits received incorrectly and the total number of bits in the transmission, also known as bit error rate (BER).

Some methods for determining error rate use larger or logical units, such as blocks, packets, or frames. In these cases, the measurement of error rate is expressed in terms of the number of units found to be in error out of the total number of units transmitted.

escape code *See* escape sequence.

Escape key The key on the keyboard labeled Esc. The Escape key generates an escape code, ASCII 27. In most applications, pressing the Escape key cancels the current command or operation.

escape sequence A sequence of characters, beginning with Escape (ASCII 27) and

followed by one or more other characters, that performs a specific function. Sometimes called an escape code.

Escape sequences are often used to control printers or monitors, which treat them as commands and act upon them rather than processing them as characters to print or display.

ESCON See Enterprise Systems Connection.

ESDI See Enhanced Small Device Interface.

Ethernet A popular network protocol and cabling scheme with a transfer rate of 10Mbps, originally developed at Xerox in 1970 by Dr. Robert Metcalf. Ethernet uses a bus topology, and network nodes are connected by either thick or thin coaxial cable, fiber-optic cable, or twisted-pair cable.

Ethernet uses CSMA/CD (Carrier Sense Multiple Access/Collision Detection) to prevent network failures or collisions when two devices try to access the network at exactly the same time.

The original DIX (Digital Equipment, Intel, Xerox), or Blue Book, standard has evolved into the slightly more complex IEEE 802.3 standard and the ISO's 8802.3 specification.

The advantages of Ethernet include:

- It's easy to install at a moderate cost.
- Technology is available from many sources and is very well known.
- It offers a variety of cabling options.
- It works very well in networks with only occasional heavy traffic.

And the disadvantages include:

- Heavy traffic can slow down the network.
- A break in the main cable can bring down large parts of the network.
- Troubleshooting a bus topology can prove difficult.

See also 10/100; 100VG-AnyLAN; 10Base2; 10Base5; 10BaseF; 10BaseT; demand priority; Fast Ethernet; Gigabit Ethernet.

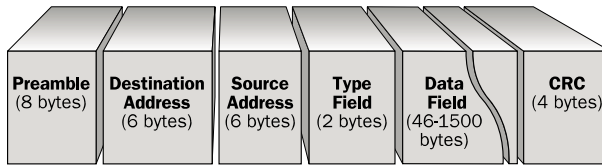
Ethernet address The address assigned to a network interface card by the original manufacturer or by the network administrator if the card is configurable.

This address identifies the local device address to the rest of the network and allows messages to reach the correct destination. Also known as the media access control (MAC) or hardware address.

Ethernet meltdown An event that causes saturation on an Ethernet-based system, often the result of illegal or misdirected packets. An Ethernet meltdown usually lasts for only a short period of time.

Ethernet packet A variable-length unit in which information is transmitted on an Ethernet network.

An Ethernet packet consists of a synchronization preamble, a destination address, a source address, a field that contains a type code indicator, a data field that can vary from 46 to 1500 bytes, and a cyclical redundancy check (CRC) that provides a statistically derived value used to confirm the accuracy of the data.

ETHERNET PACKET


EtherTalk An implementation of the Ethernet LAN developed for Apple computers, designed to work with the AppleShare network operating system.

EtherTalk operates over coaxial cable at the Ethernet transfer rate of 10Mbps, much faster than the 230.4Kbps rate available with AppleTalk. Each networked Macintosh computer must be supplied with a special EtherTalk network interface card.

ETX *See* end-of-text.

Eudora A popular and widely used e-mail application originally developed by Steve Dorner at the University of Illinois and now available from Qualcomm, Inc.

European Laboratory for Particle Physics *See* Conseil Européen pour la Recherche Nucléaire.

even parity *See* parity.

Event Log service A Microsoft Windows 2000 service that logs important application, security, and system events into the event log.

Event Viewer In Microsoft Windows 2000, a utility that displays the contents of the system logs.

See also application log; Directory Service log; DNS Server log; File Replication Service log; security log; system log.

exa- Abbreviated E. A prefix meaning one quintillion, or 10^{18} . In computing, the prefix means 1,152,921,504,606,846,976, or the power of 2 closest to one quintillion (2^{60}).

exabyte Abbreviated EB. 1 quintillion bytes, or 1,152,921,504,606,846,976 bytes.

exclusive container administrator In Novell NetWare, a special type of container administrator who is given the rights to a container and all the objects within that container. The Inherited Rights Filter prevents other administrators from having rights in the container.

See also container administrator; Inherited Rights Filter.

executable file Any file that can be executed by the operating system. Some executable files contain compiler binary instructions, and others are text files containing commands or shell scripts.

execute **1.** To run a program, command, or script.

2. One of the permissions assigned to a file or directory; you can only execute a file when the execute permission is enabled.

execute permission An access permission on a file or directory that gives you permission to execute the file or to access the contents of the directory.

expandability The ability of a system to accommodate expansion. In hardware, this may include the addition of more memory, more or larger disk drives, and new adapters. In software, expandability may include the ability of a network to add users, nodes, or connections to other networks.

expanded memory An MS-DOS mechanism by which applications can access more than the 640KB of memory normally available to them. The architecture of the early Intel processors restricted the original IBM PC to accessing 1MB of memory, 640KB of which was available for applications; the remaining 384KB was reserved for system use, the BIOS, and the video system. At that time, 640KB was more than ten times the amount of memory available in other personal computers. However, as both applications and MS-DOS grew, they began to run out of room.

The Expanded Memory Specification LIM 4.0 let programs running on the Intel family of processors access as much as 32MB of expanded memory. The expanded memory manager (EMM) creates a block of addresses into which data (held in memory above the 1MB limit) is swapped in and out as needed by the program. In other words, a 64KB segment of addressable memory creates a small

window through which segments of expanded memory can be seen, but only one segment at a time.

expanded memory manager Abbreviated EMM. A device driver that supports the software portion of the Expanded Memory Specification (EMS) in an IBM-compatible computer.

Expanded Memory Specification Abbreviated EMS. The original version of the Lotus-Intel-Microsoft Expanded Memory Specification (LIM EMS), which let MS-DOS applications use more than 640KB of memory space.

These days Microsoft Windows running in protected mode on 80386 and later processors is free of this limitation.

expansion board A printed circuit board that plugs into a computer's expansion bus to provide added capabilities. Also called an adapter.

Common expansion boards include video adapters, joy-stick controllers, and input/output (I/O) adapters, as well as other devices, such as internal modems, CD-ROMs, and network interface cards (NICs). One expansion board can often support several different devices. Some current PC designs incorporate many of the functions previously performed by these individual adapters on the motherboard.

expansion bus An extension of the main computer bus that includes expansion slots for use by compatible adapters, such as memory boards, video adapters, hard-disk

controllers, and SCSI (Small Computer System Interface) interface cards.

expansion slot One of the connectors on the expansion bus that gives an adapter access to the system bus. You can install as many additional adapters as there are expansion slots inside your computer.

For portable computers, an expansion slot is often supplied by a PCMCIA connection designed to accept a PC Card.

expansion unit An external housing available with certain portable computers designed to contain additional expansion slots and maintain a connection to the main expansion bus in the computer's system unit.

See also port replicator.

explicit rights In Novell Directory Services (NDS), any rights granted directly to a user for a directory or other NDS object. Explicit rights always override inherited rights.

See also inherited rights.

explicit security equivalence In Novell Directory Services (NDS), a technique used to give one trustee the same rights as another trustee. Explicit security equivalence can be assigned with group membership, an Organizational Role, or the trustee's Security Equal To property.

See also implied security equivalence; security equivalence.

explicit trust In Microsoft Windows 2000 Server, a trust explicitly established between two Windows 2000 domains. This is in addition to the built-in transitive trusts

automatically established if they are in the same forest

See also downlevel trust; forest; shortcut trust; trust relationship; two-way transitive trust.

extended ASCII character set *See* ASCII extended character set.

Extended Binary Coded Decimal Interchange Code Abbreviated EBCDIC; pronounced "eb-se-dic." EBCDIC is the character set commonly used on large IBM mainframe computers, most IBM minicomputers, and computers from many other manufacturers.

It is an 8-bit code, allowing 256 different characters (see Appendix D). Unlike ASCII, the placement of the letters of the alphabet in EBCDIC is discontinuous. Also, there is no direct character-to-character match when converting from EBCDIC to ASCII; some characters exist in one set but not in the other.

See also American Standard Code for Information Interchange; double-byte character set; ISO 10646; Unicode.

Extended Industry Standard

Architecture Abbreviated EISA, pronounced "ee-sah." A PC bus standard that extends the traditional AT-bus to 32 bits and allows more than one processor to share the bus.

EISA was developed by the so-called Gang of Nine (AST Research, Compaq Computer Corporation, Epson, Hewlett-Packard, NEC, Olivetti, Tandy, Wyse Technology, and Zenith Data Systems) in reply to IBM's introduction of its proprietary MCA (Microchannel

Architecture). EISA maintains compatibility with the earlier ISA (Industry Standard Architecture), and it also provides for additional features introduced by IBM in the MCA standard. EISA accepts ISA expansion cards and so, unlike MCA, is compatible with earlier systems. EISA has been eclipsed by Peripheral Control Interconnect-based systems.

See also local bus; PCI local bus; PC Memory Card International Association.

extended LAN A term used to describe a network that consists of a series of LANs connected by bridges.

extended memory Memory beyond 1MB on computers using the Intel 80286 and later processors, not configured for expanded memory.

PCs based on the early Intel processors could access only 1MB of memory, of which 640KB was available for applications, and the remaining 384KB was reserved for MS-DOS, the BIOS, and video settings.

Later processors can access more memory, but it was the 80386 with its ability to address 4GB of memory that really made extended memory usable. Also, Microsoft Windows and other operating systems running on Intel processors using the protected mode of the 80386 and later processors can access all the available system memory in the same way.

extended memory manager A device driver that supports the software portion of the Extended Memory Specification in an IBM-compatible computer.

Extended Memory Specification Abbreviated XMS. A standard developed by Microsoft, Intel, Lotus, and AST Research that became the preferred way of accessing extended memory in the PC. MS-DOS includes the extended memory device driver HIMEM.SYS, and this command or an equivalent must be present in CONFIG.SYS to allow you to access extended memory.

Extensible Authentication Protocol

Abbreviated EAP. In Microsoft Windows 2000 Server, a mechanism that allows future authentication schemes to be added to the operating system quickly and easily.

See also Challenge-Handshake Authentication Protocol.

Extensible Markup Language Abbreviated XML. A technology, based on Standard Generalized Markup Language (SGML), that allows the data on a Hypertext Markup Language (HTML) page to be described in terms of the information it represents.

See also document type definition; Hypertext Markup Language.

external command A command that is a separate program, and not built in to the operating system. FORMAT, BACKUP, and FDISK are examples of external commands.

See also internal command.

External Gateway Protocol Abbreviated EGP. A routing protocol used to exchange network availability information among organizational networks. EGP indicates whether a given network is reachable,

but it does not evaluate that information or make routing or priority decisions.

See also Border Gateway Protocol.

external modem A stand-alone modem, separate from the computer and connected to it by a serial cable. LEDs on the front of the chassis indicate the current modem status. An external modem can be used with different computers at different times and also with different types of computers.

external reference In Novell NetWare, a pointer to a NetWare Directory Services (NDS) object that is not located on the current server.

extranet Originally coined as a term to describe any network in which part of the network was protected behind a firewall and part of the network was accessible from the Internet.

In current usage, an extranet describes a technology that allows different corporate intranets to communicate for the purposes

of electronic commerce and collaboration. Those parts of an extranet outside the firewall contain their own set of security safeguards, allowing only limited access for specific purposes.

See also intranet.

extremely low-frequency emission

Abbreviated ELF. Radiation emitted by a computer monitor and other common electrical appliances.

ELF emissions fall into the range of 5 to 2000 hertz and decline with the square of the distance from the source. Emissions are not constant around a monitor; they are higher from the sides and rear and weakest from the front of the screen. Low-emission models are available, and laptop computers with an LCD display do not emit any ELF fields.

See also electromagnetic interference; radio frequency interference; very low-frequency emission.

F

facsimile *See* fax.

fading In both electrical and wireless systems, a decrease in a signal's strength.

Fading may be due to physical obstructions of the transmitter or receiver, to distance from the source of the transmission, or to some form of external interference from other signals or from atmospheric conditions.

fail-safe system Any computer system that is designed to keep operating, without losing data, when part of the system seriously malfunctions or fails completely.

fail-soft system Any computer system that is designed to fail gracefully, with the minimum amount of data or program destruction, when part of the system malfunctions. Fail-soft systems close down non-essential functions and operate at a reduced capacity until the problem is resolved.

fake root A subdirectory that functions as the root directory.

Fake roots can be useful with network applications that must be installed in the root directory. You can install the application in a subdirectory and then map a fake root on the file server to the subdirectory containing the application. Many Web servers use a fake root.

fall back A technique used by modems to adjust their data rate in response to changing line conditions.

FAQ *See* frequently asked questions.

far-end crosstalk Abbreviated FEXT. Interference that occurs when signals on one twisted-pair are coupled with another pair as they arrive at the far end of a multi-pair cable system.

FEXT becomes a problem on short loops supporting high-bandwidth services such as Very-High-Bit-Rate Digital Subscriber Line (VDSL) because of the high carrier frequencies used.

See also crosstalk; near-end crosstalk.

Fast Ethernet A term applied to the IEEE 802.3 Higher Speed Ethernet Study Group proposals, which were originally developed by Grand Junction Networks, 3Com, SynOptics, Intel, and others. Also known as 100BaseT.

Fast Ethernet modifies the existing Ethernet standard to allow speeds of 10Mbps or 100Mbps or both and uses the CSMA/CD access method.

The official standard defines three physical-layer specifications for different cabling types:

- 100BaseTX for two-pair Category 5 unshielded twisted-pair
- 100BaseT4 for four-pair Category 3, 4, or 5 unshielded twisted-pair
- 100BaseFX for fiber-optic cable

See also 100VG-AnyLAN; demand priority.

Fast IP A technology from 3Com Corporation that gives certain types of network

traffic, such as real-time video, a higher priority than other, less urgent network traffic such as e-mail messages.

3Com achieves this using what it calls policy management. A workstation requests a certain level of priority, tags the data accordingly, and begins the transmission. When the transmission is complete, the workstation indicates that to the network, and previously reserved resources are freed up for use.

See also IP over ATM; IP switching; quality of service.

Fast IR A 4Mbps extension to the Serial Infrared Data Link Standard that provides wireless data transmission between IrDA-compliant devices.

See also Infrared Data Association.

Fast SCSI A version of the SCSI-2 interface that can transfer data 8 bits at a time at data rates of up to 10MBps. The Fast SCSI connector has 50 pins.

See also Fast/Wide SCSI; SCSI-2; Small Computer System Interface; Ultra SCSI; Ultra Wide SCSI; Wide SCSI.

Fast/Wide SCSI A version of the SCSI-2 interface that can transfer data 16 bits at a time at data rates of up to 20MBps. The Fast/Wide SCSI connector has 50 pins.

See also Fast SCSI; SCSI-2; Small Computer System Interface; Ultra SCSI; Ultra Wide SCSI; Wide SCSI.

FAT *See* file allocation table.

FAT16 In Microsoft Windows, a file allocation table that uses a 16-bit cluster

addressing scheme which restricts the maximum hard-disk size to 2.6GB. Also, FAT16 is inefficient in disk-space utilization as the default cluster size can be as large as 32KB.

See also FAT32; file allocation table.

FAT32 In Microsoft Windows 95 (release 2) and later versions of Windows, a file allocation table that uses a 32-bit cluster addressing scheme to support hard disks larger than 2.6GB, as well as a default cluster size of as small as 4KB. FAT32 can support hard disks of up to 2 terabytes in size.

See also FAT16; file allocation table.

fatal error An operating system or application program error from which there is no recovery without reloading the program or rebooting the operating system.

fault management One of the five basic types of network management defined by the International Organization for Standardization (ISO) and CCITT. Fault management is used in detecting, isolating, and correcting faults on the network.

fault tolerance A design method that ensures continued system operation in the event of individual failures by providing redundant elements.

At the component level, the design includes redundant chips and circuits and the capability to bypass faults automatically. At the computer-system level, any elements that are likely to fail, such as processors and large disk drives, are replicated.

Fault-tolerant operations often require backup or UPS (uninterruptible power

supply) systems in the event of a main power failure. In some cases, the entire computer system is duplicated in a remote location to protect against vandalism, acts of war, or natural disaster.

See also clustering; data protection; disk duplexing; disk mirroring; redundant array of inexpensive disks; System Fault Tolerance.

fax Short for facsimile. The electronic transmission of copies of documents for reproduction at a remote location. The term *fax* can be used as a verb for the process and as a noun for the machine that does the work and also for the item that is actually transmitted.

The sending fax machine scans a paper image and converts the image into a form suitable for transmission over a telephone line. The receiving fax machine decodes and prints a copy of the original image. Each fax machine includes a scanner, modem, and printer.

Originally, facsimile machines were rotating drums (CCITT Groups 1 and 2); then came modems (CCITT Group 3), and eventually they will be completely digital (CCITT Group 4).

See also CCITT Groups 1–4; fax modem.

fax board *See* fax modem.

fax modem An adapter that fits into a PC expansion slot providing many of the capabilities of a full-sized fax machine, but at a fraction of the cost. Some external modems also have fax capabilities.

The advantages of a fax modem include ease of use and convenience; the main disadvantage is that the material you want to fax must be present in digital form in the computer. Unless you have access to a scanner, you cannot fax handwritten notes, line art, or certain kinds of graphics. Most faxes sent directly from a PC using a fax modem are text files.

fax server A dedicated server that provides fax sending and receiving services to users on the network.

FCC *See* Federal Communications Commission.

FCC certification Approval by the FCC (Federal Communications Commission) that a specific computer model meets its standards for radio frequency interference (RFI) emissions. There are two levels of certification:

- Class A certification, which is for computers used in commercial settings, such as mainframes and minicomputers
- The more stringent Class B certification, which is for computers used in the home and in home offices, such as PCs, laptops, and portables

See also extremely low-frequency emission; radio frequency interference; very low-frequency emission.

FDDI *See* Fiber Distributed Data Interface.

FDDI-II A variation of the Fiber Distributed Data Interface standard. The FDDI-II version is designed for networks transmitting real-time full-motion video (or other

information that cannot tolerate any delays) and requires that all nodes on the network use FDDI-II; otherwise, the network automatically reverts to FDDI.

FDDI-II divides the bandwidth into 16 dedicated circuits operating at from 6.144Mbps to a maximum of 99.072Mbps. Each of these channels can be further subdivided for a total of 96 separate 64Kbps circuits.

FDDI-II is not widely used because it is incompatible with FDDI and because Fast Ethernet and Asynchronous Transfer Mode (ATM) both provide better solutions.

FDM See frequency-division multiplexing.

FDX See full-duplex.

Federal Communications

Commission Abbreviated FCC. A U.S. government regulatory body for radio,

television, all interstate telecommunications services, and all international services that originate inside the United States. All computer equipment must be certified by the FCC before it can be offered for sale in the United States. The certification ensures that the equipment meets the legal limits for conductive and radio frequency interference, which could otherwise interfere with commercial broadcasts.

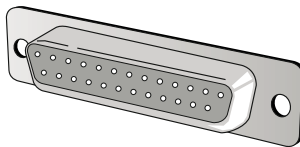
For more information on the FCC, see www.fcc.gov.

See also FCC certification.

female connector Any cable connector with receptacles designed to receive the pins on the male connector.

See also male connector.

FEMALE CONNECTOR



FEXT See far-end crosstalk.

FF See form feed.

Fiber Distributed Data Interface Abbreviated FDDI. The ANSI X3T9.5 specification for fiber-optic networks transmitting at a speed of up to 100Mbps over a dual, counter rotating, token-ring topology.

FDDI's 100Mbps speed is close to the internal speed of most computers, which makes it a good choice to serve as a super backbone for linking two or more LANs or as a fiber-optic bus connecting high-performance engineering workstations. FDDI is suited to systems that require the transfer of large amounts of information, such as medical

fiber-optic cable

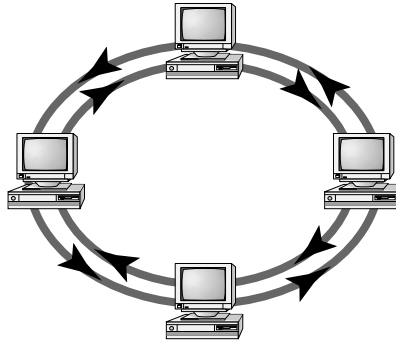
imaging, three-dimensional seismic processing, and oil reservoir simulation. The FDDI-II version of the standard is designed for networks transmitting real-time full-motion video (or other information that cannot tolerate any delays) and requires that all nodes on the network use FDDI-II; otherwise, the network automatically reverts to FDDI.

A FDDI network using multimode fiber-optic cable can include as many as 500 sta-

tions up to 2 kilometers (1.25 miles) apart; with single-mode fiber, run length increases up to 60 kilometers (37.2 miles) between stations. This type of network can also run over shielded and unshielded twisted-pair cabling (when it is known as CDDI, or Copper Distributed Data Interface) for shorter distances.

See also FDDI-II.

FIBER DISTRIBUTED DATA INTERFACE



fiber-optic cable A transmission technology that sends pulses of light along specially manufactured optical fibers.

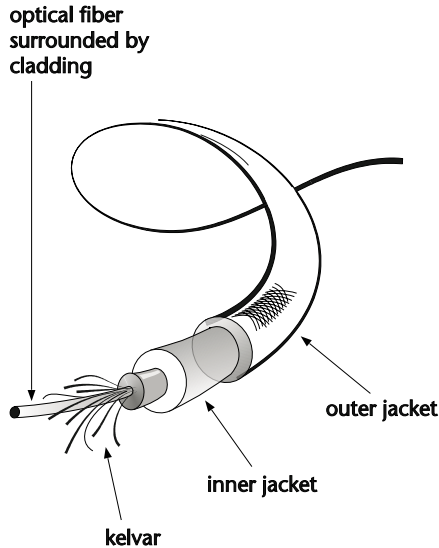
Each fiber consists of a core, thinner than a human hair, surrounded by a sheath with a much lower refractive index. Light signals introduced at one end of the cable are conducted along the cable as the signals are reflected from the sheath.

Fiber-optic cable is lighter and smaller than traditional copper cable, is immune

to electrical interference, offers better security, and has better signal-transmitting qualities. However, it is more expensive than traditional cables and is more difficult to repair. Fiber-optic cable is often used for high-speed backbones, but as prices drop, we may even see fiber-optic cable running to the desktop.

See also multimode fiber; single-mode fiber.

FIBER-OPTIC CABLE



Fibre Channel An interconnection standard designed to connect peripherals, mass storage systems, archiving and imaging systems, and engineering workstations.

Fibre Channel provides bandwidth from 100Mbps to 800Mbps over a variety of cable types, including multimode fiber, coaxial cable, and shielded twisted-pair.

The Fibre Channel Interconnect Standard was developed by the American National Standards Institute (ANSI) X3T9 committee.

file A named collection of information that appears to the user as a single entity and is stored on disk.

A file can contain a program or part of a program, just data, or a user-created document. Files may actually be fragmented or stored in many different places across the disk. The operating system manages the task of locating all the pieces when a request is made to read the file.

file allocation table Abbreviated FAT, pronounced “fat.” A table, maintained by the operating systems, that lists all the blocks of disk space available on a disk.

The FAT includes the location of each block, as well as whether it is in use, available for use, or damaged in some way and therefore unavailable. Because files are not necessarily stored in consecutive blocks on

a disk, the FAT also keeps track of which pieces belong to which file.

See also FAT16; FAT32; file fragmentation.

file compression A technique that shrinks program or data files so that they occupy less disk space. The file must then be extracted or decompressed before use. Some types of files, such as word processor documents, can be compressed by 50 percent or more. Recompressing an already compressed file usually makes the file slightly larger because of the compression overhead.

File compression can be automatic and performed by the operating system, or it can be manual and performed by a file-compression program.

See also file-compression program.

file-compression program An application that compresses files so that they take up less space on the disk. Some file-compression programs are individual, stand-alone applications; others are built into the operating system.

Individual file-compression programs can compress one or more files at a time. The utilities PKZIP and WinZip (for Microsoft Windows) and StuffIt (for the Macintosh) operate that way. When compression is built into the operating system, it usually compresses all the files on a specific disk, disk partition, or volume.

Many of the stand-alone file-compression programs, such as PKZIP, LHArc, and StuffIt, are available as shareware, and many popular utility packages, such as the

Norton Utilities from Symantec, also contain file-compression programs.

file-conversion program An application that converts a file from one format to another without altering the contents of the file.

The conversion may be between the formats of two applications that use the same operating system (such as between two different Microsoft Windows word processors) or between the formats of applications from different operating systems. Applications are also available that convert a graphical image from one file format to another.

Many of the applications that change a Macintosh file into a PC-compatible file, or vice versa, consist of two programs running simultaneously on two physically connected computers. MacLink Plus from Data Viz and LapLink Mac from Traveling Software are two examples of this type of file-conversion program.

file format A file structure that defines the way information is stored in the file and how the file appears on the screen or on the printer.

The simplest file format is a plain ASCII file. Some of the more complex formats are DCA (Document Content Architecture) and RTF (Rich Text Format), which include control information for use by a printer; TIFF (Tagged Image File Format) and EPS (Encapsulated PostScript), which hold graphics information; and DBF (Xbase database file) and DB (Paradox file), which are database formats. Word processing programs, such as Microsoft Word, also create files in special formats.

file fragmentation Storage of files in pieces scattered on a disk. As files grow on a hard disk, they can be divided into several small pieces. By fragmenting files, the operating system makes reasonable use of the disk space available. The problem with file fragmentation is that the disk heads must move to different locations on the disk to read or write to a fragmented file. This process takes more time than reading the file as a single piece. To speed up file operations, you can use a disk optimizer or defragmenter.

See also FAT16; FAT32; defragmenter.

file indexing A technique used in Novell NetWare to speed up accesses to large files by indexing file allocation table (FAT) entries.

file-infecting virus Any virus that infects files on disk, usually executable files with filename extensions of .COM, .EXE, and .OVL. An unexpected change in the file size may indicate an infection. In certain cases, the original program is replaced with a new file containing an infected program.

See also antivirus program; boot sector virus; infection; macro virus; multipart virus; polymorphic virus; stealth virus; Trojan Horse; vaccine; virus.

file locking *See* file and record locking.

filename The name of a file on a disk used so that both you and the operating system can find the file again. Every file in a directory must have a unique name, but files in different directories can share the same name.

In MS-DOS, file and directory names have two parts. They can have up to eight characters in the name and up to three characters in the optional filename extension, separated from the name by a period. Many applications take over the extension part of the filename, using specific groups of characters to designate a particular file type.

In the Macintosh operating system, filenames can be up to 32 characters and can contain any character except a colon (:), which is used to separate elements of a path name.

Microsoft Windows allows 255-character filenames including spaces, but the name cannot include any of the following characters: \ / : * ? " < > |.

filename extension In the MS-DOS file allocation table (FAT) file system, an optional three-character suffix added to the end of a filename and separated from the name by a period.

file permissions A set of permissions associated with a file (or a directory) that specifies who can access the file and in what way. There are three basic permissions:

- Read permission lets you read files.
- Write permission lets you write (or overwrite) files.
- Execute permission lets you execute files.

Additional file permissions vary according to the operating system in use and the security system in place. For example, Novell NetWare has the following file permissions: Access Control, Create, Erase, File Scan, Modify, Read, Supervisor, and Write. Microsoft Windows 2000 has Add and Read,

Change Permissions, Delete, Full Control, No Access, and Take Ownership.

See also rights.

File and Print Services for NetWare

An add-on product for Microsoft Windows 2000 that allows Novell NetWare clients to access Windows servers as though they were NetWare servers.

FILER A Novell NetWare workstation utility used to manage the NetWare file system. Almost any task related to the file system can be performed using FILER. Administrators can view information about files, directories, and volumes; modify and view attributes, rights, and trustee assignments for files and directories; search for files and directories; copy files; and recover and purge deleted files. In NetWare 3.x, these last two functions are found in the SALVAGE and PURGE commands.

file and record locking A method of controlling file access in a multiuser environment, where there is always a possibility that two users will attempt to update the same file at the same time but with different information. The first user to access the file locks out all the other users, preventing them from opening the file. After the file is updated and closed again, the next user can gain access.

File locking is a simple way to prevent simultaneous updates, but it can seriously degrade system performance if many users attempt to access the same files time after time. To prevent this slowdown, many database management systems use record locking instead. Record locking limits access to individual records within the database file.

file recovery The process of recovering deleted or damaged files from a disk. A file can be deleted accidentally or can become inaccessible when part of the file's control information is lost. In many operating systems, a deleted file still exists on disk until the space it occupies is overwritten with something else or until the file is purged.

See also undelete.

File Replication Service In Microsoft Windows 2000 Server, the replacement for Windows NT Directory Replication.

File Replication Service log In Microsoft Windows 2000 Server, a special log that records any events associated with domain controller updates.

See also Directory Service log; DNS Server log; Event Viewer; File Replication Service.

file server A networked computer used to store files for access by other client computers on the network.

On larger networks, the file server runs a special network operating system. On smaller installations, the file server may run a PC operating system supplemented by peer-to-peer networking software.

See also client; server.

file sharing The sharing of files over a network or between several applications running on the same workstation.

Shared files can be read, reviewed, and updated by more than one individual. Access to the file or files is often regulated by password protection, account or security clearance, or file locking to prevent simultaneous changes by multiple users.

See also file and record locking.

filespec A contraction of *file specification*, commonly used to denote the complete drive letter, path name, directory name, and filename needed to access a specific file.

file system In an operating system, the structure by which files are organized, stored, and named.

Some file systems are built-in components of the operating system; others are installable. For example, OS/2, Unix, and Microsoft Windows are all capable of supporting several different file systems at the same time. Other, installable file systems provide support for specific devices such as CD-ROM or DVD.

File Transfer Access and Management Abbreviated FTAM. The Open Systems Interconnect (OSI) protocol for transferring and remotely accessing files on different makes and models of computers that are also using FTAM.

file-transfer program An application used to move a file from a computer of one type to a computer of another type. The file format itself may also be changed during this transfer.

Many of the applications that change a Macintosh file into a PC-compatible file, or vice versa, consist of two programs running simultaneously on two physically connected computers. MacLink Plus from Data Viz and LapLink Mac from Traveling Software are two file-transfer programs that also offer a wide variety of popular file-format conversions.

See also file-conversion program; File Transfer Protocol; Kermit; Xmodem; Ymodem; Zmodem.

File Transfer Protocol Abbreviated FTP. The TCP/IP Internet protocol used when transferring single or multiple files from one computer system to another.

FTP uses a client/server model, in which a small client program runs on your computer and accesses a larger FTP server running on an Internet host. FTP provides all the tools needed to look at directories and files, change to other directories, and transfer text and binary files from one system to another.

See also anonymous FTP; ftp; Telnet.

FILTCFG A Novell NetWare 4.x NetWare Loadable Module (NLM) that allows you to set up and configure filters for Inter-network Packet eXchange (IPX), TCP/IP, and AppleTalk protocols. Filters help to control the type and amount of information sent to and received by a router, to limit traffic to certain segments of the network, and to provide security.

filtering 1. The mechanism that prevents certain source and destination addresses from crossing a bridge or router onto another part of the network.

2. The process of automatically selecting specific frequencies and discarding others.

finger A utility found on many Internet systems that displays information about a specific user, including full name, logon time, and location. Originated in the Unix world.

FIRE PHASERS

The finger utility may also display the contents of the user's .plan or .profile file, and there are those who exploit this in novel ways to display such varied information as instructions for using a university's Coke-vending machine, posting sports scores, and listing earthquake activity.

FIRE PHASERS A Novell NetWare login script command that makes a noise using the workstation's speaker.

The sound is supposed to resemble the phasers on the USS *Enterprise*. With fire phasers, you can have the computer emit up to nine sounds or blasts.

firewall A barrier established in hardware or in software, or sometimes in both, that monitors and controls the flow of traffic between two networks, usually a private LAN and the Internet.

A firewall provides a single point of entry where security can be concentrated. It allows access to the Internet from within the organization and provides tightly controlled access from the Internet to resources on the organization's internal network.

See also application-level filter; dual-homed host; Intrusion Detection System; packet-level filter; proxy server; stateless filter.

FireWire *See* 1394.

firmware Any software stored in a form of read-only memory (ROM), erasable programmable read-only memory (EPROM), or electrically erasable programmable read-only memory (EEPROM) that maintains its contents when power is removed.

first-layer domain In Microsoft Windows 2000 Server, a Windows 2000 domain whose parent domain is the root domain.

See also domain; parent domain.

flame A deliberately insulting e-mail message or post to a Usenet newsgroup, often containing a personal attack on the writer of an earlier post.

See also flame bait; flame war.

flame bait An insulting or outrageous e-mail post to a Usenet newsgroup specifically designed to provoke other users into flaming the originator.

See also flame; flame war.

flame war In a Usenet newsgroup, a prolonged series of flames, which may have begun as a creative exchange of views but which quickly descended into personal attacks and crude name-calling.

See also flame bait.

flash memory A special form of ROM that can be erased at signal levels commonly found inside the PC.

This ability allows the contents to be reprogrammed without removing the chips from the computer. Also, once flash memory has been programmed, you can remove the expansion board it is mounted on and plug it into another computer without loss of the new information.

See also PC Card.

flavor A slang expression meaning type or kind, as in "Unix comes in a variety of flavors."

floating-point processor A special-purpose, secondary processor designed to perform floating-point calculations much faster than the main processor.

Many processors, such as the Intel 80386, had matched companion floating-point processors. However, the current trend in processor design is to integrate the floating-point unit into the main processor, as in the Intel Pentium.

flooding A denial of service attack in which a huge number of nuisance connection attempts are made in order to consume all the available processing time.

See also brute-force attack; denial of service attack; dictionary attack; social engineering; Trojan Horse.

flow control 1. In communications, control of the rate at which information is exchanged between two computers over a transmission channel. Flow control is needed when one of the devices cannot receive the information at the same rate as it can be sent, usually because some processing is required on the receiving end before the next transmission unit can be accepted. Flow control can be implemented either in hardware or in software.

2. In networking, control of the flow of data throughout the network, ensuring that network segments are not congested. A router controls data flow by routing around any trouble spots.

See also handshaking.

followup A reply to a post in a Usenet newsgroup or to an e-mail message. A followup post may quote the original post so

that other readers are reminded of the discussion so far, and this quoted part of the post is usually indicated by greater than symbols (>). If you do quote from a previous post, only quote the minimum amount to get your point across; never quote the whole message as this is considered a waste of time and resources.

See also thread.

footprint The amount of desktop space or floor space occupied by a computer, printer, or monitor.

forced perfect termination A technique used to terminate a Small Computer System Interface (SCSI). Forced perfect termination actively monitors the bus to ensure that no signal reflection occurs.

See also active termination; passive termination.

foreground In an operating system, a process that runs in the foreground is running at a higher level of priority than is a background task.

Only multitasking operating systems support true foreground and background processing; however, some application programs can mimic it. For example, many word processors will print a document while still accepting input from the keyboard.

See also background.

forest In Microsoft Windows 2000, a collection of Active Directory trees that do not share a contiguous namespace, but do share a common schema and global catalog.

For example, if `acme.com` and `widget.com` were linked via a trust relationship but shared a common schema and global catalog, they would be considered a forest; in other words, a forest is a group of trees that trust each other.

See also Active Directory; Global Catalog; schema; tree.

formatting The process of initializing a new, blank floppy disk or hard disk so that it can be used to store information.

form feed Abbreviated FF. A printer command that advances the paper in the printer to the top of the next page. The Form Feed button on the printer also performs this same function. An application can also issue the command. In the ASCII character set, a form feed has the decimal value of 12.

forward error correction A technique used to control errors that insert extra or redundant bits into the data stream. The receiving device uses the redundant bits to detect and, if possible, correct the errors in the data.

See also error detection and correction.

forwarding The process of passing data on to an intermediate or final destination. Forwarding takes place in network bridges, routers, and gateways.

four-wire circuit A transmission system in which two half-duplex circuits, consisting of two wires each, are combined to create one full-duplex circuit.

fps *See* frames per second.

FQDN *See* fully qualified domain name.

fractional T1 One portion of a T1 circuit. A T1 circuit has a capacity of 1.544Mbps, the equivalent of twenty-four 64Kbps channels. Customers can lease as many of these 64Kbps channels as they need; they are not required to lease the entire 1.544Mbps circuit.

fragmentation *See* file fragmentation.

frame 1. A block of data suitable for transmission as a single unit; also referred to as a packet or a block. Some media can support multiple frame formats.

2. In digital video, one screen of information, including both text and graphics.

See also frames per second.

frame relay A CCITT standard for a packet-switching protocol, running at speeds of up to 2Mbps, that also provides for bandwidth on demand. Frame relay is less robust than X.25 but provides better efficiency and higher throughput.

Frame relay is available from many companies, including AT&T, CompuServe, Sprint, WilTel, and the Bell companies.

See also Asynchronous Transfer Mode.

frames per second Abbreviated fps. The number of video frames displayed each second. Although 24fps is considered the slowest frame rate that provides convincing motion to the human eye, most Internet video runs at between 5 and 15fps. To maintain a 15fps rate, you need a fast Pentium-based system with a 56Kbps modem.

framing The process of dividing data for transmission into groups of bits and adding a header as well as a checksum to form a frame. In asynchronous communications, framing is the process of inserting start bits and stop bits before and after the data to be transmitted.

FreeBSD A free implementation of Unix for the Intel family of processors, derived from the 4.4BSD Lite releases.

The distribution is free, but there may be a small charge to cover the distribution media and packaging. FreeBSD also includes XFree86, a port of the X Window system to the Intel processors.

Most of FreeBSD is covered by a license that allows redistribution as long as the code acknowledges the copyright of the Regents of the University of California and the FreeBSD Project. Those parts of FreeBSD that include GNU software are covered separately by the Free Software Foundation license.

See also copy left; Free Software Foundation; freeware; GNU; Linux; open source software.

free memory Any area of memory not currently in use. Often refers to the memory space remaining for applications to use after the operating system and the system device drivers have been loaded.

Free Software Foundation Abbreviated FSF. An organization founded by Richard Stallman that develops the freely available GNU software.

The FSF philosophy is that all software should be free for everyone to use and that

source code should always accompany the software. The theory being, that if you make a modification or fix an error, the change can be sent out to all the other users, saving time and preventing duplication of effort.

Also, any software developed under the FSF General Public License (GPL) must also be covered by the same terms of the GPL; in other words, you cannot use the free software to develop a commercial product for sale.

For more information on the Free Software Foundation, see www.gnu.org/fsf.

See also copy left; FreeBSD; freeware; GNU; open source software.

freeware A form of software distribution in which the author retains copyright to the software, but makes the program available to others at no cost. The program cannot be resold by a third party for profit.

See also copy left; Free Software Foundation; GNU; open source software.

frequency-division multiplexing Abbreviated FDM. A method of sharing a transmission channel by dividing the bandwidth into several parallel paths, defined and separated by guard bands of different frequencies designed to minimize interference. All signals are carried simultaneously.

FDM is used in analog transmissions, such as in communications over a telephone line.

See also inverse multiplexing; statistical multiplexing; time-division multiplexing; wavelength division multiplexing.

frequently asked questions

frequently asked questions Abbreviated FAQ, pronounced “fack.” A document that contains answers to questions that new users often ask when they first subscribe to a newsgroup or first access a Web site.

The FAQ contains answers to common questions that the seasoned users have grown tired of answering. New users should look for and read the FAQ before posting their question, just in case the FAQ contains the answer.

fried A slang expression for burned-out hardware, especially hardware that has suffered from a power surge. Also applied to people, as in “My brain is fried; I haven’t slept since last weekend.”

front-end application An application running on a networked workstation that works in conjunction with a back-end system running on the server. Examples are e-mail and database programs.

See also client/server architecture.

front-end processor A specialized processor that manipulates data before passing it on to the main processor.

In large computer-to-computer communications systems, a front-end processor is often used to manage all aspects of communications, leaving the main computer free to handle the data processing.

See also back-end processor.

FRS *See* File Replication Service.

Frye Utilities A package of network management utilities from Frye Computer Systems that includes hardware and software inventory, NetWare server monitoring, traffic monitoring, application metering, and software distribution.

FSF *See* Free Software Foundation.

FTAM *See* File Transfer Access and Management.

FTP *See* File Transfer Protocol.

ftp A command used to transfer files to and from remote computers using the File Transfer Protocol. You can use ftp to log on to an Internet computer and transfer text and binary files.

When you use ftp, you start a client program on your computer that connects to a server program on the Internet computer. The commands that you give to ftp are translated into instructions that the server program executes for you.

The original ftp program started life as a Unix utility, but versions are now available for all popular operating systems; ftp is also built into all the major Web browsers.

See also anonymous ftp; File Transfer Protocol.

full backup A backup that includes all files on a hard disk or set of hard disks. A network administrator must decide how often to perform a full backup, balancing the need for security against the time taken for the backup.

See also differential backup; incremental backup.

full-duplex Abbreviated FDX. The capability for simultaneous transmission in two directions so that devices can be sending and receiving data at the same time.

See also four-wire circuit; half-duplex.

full-page display Any monitor capable of displaying a whole page of text. Full-page displays are useful for graphical art and desktop publishing applications, as well as medical applications.

fully qualified domain name A host name with the appropriate domain name appended. For example, on a host with the host name `wallaby` and the Domain Name Service (DNS) domain name `my-company.com`,

the fully qualified domain name becomes `wallaby.my-company.com`.

See also Domain Name Service.

function keys The set of programmable keys on the keyboard that can perform special tasks assigned by the current application.

Most keyboards have 10 or 12 function keys (F1 to F10 or F1 to F12), some of which are used by an application as short-cut keys. For example, many programs use F1 to gain access to the Help system. In some programs, the use of function keys is so complex that special plastic key overlays are provided as guides for users.

G

G Abbreviation for giga-, meaning 1 billion, or 10^9 .

See also gigabyte.

Gartner Group, Inc. An independent research organization for the computer hardware, software, communications, and related industries.

Gartner Group first developed the concept of total cost of ownership (TCO) and now provides TCO software tools since the acquisition of Interpose. Gartner Group also owns Datapro Information Services and the market analysis company Dataquest.

For more information on Gartner Group, see www.gartner.com.

See also total cost of ownership.

gate array *See* application-specific integrated circuit.

gateway A shared connection between a LAN and a larger system, such as a main-frame computer or a large packet-switching network, whose communications protocols are different. Usually slower than a bridge or router, a gateway is a combination of hardware and software with its own processor and memory used to perform protocol conversions.

See also bridge; brouter; router.

Gateway, Inc. A leading direct-marketer of PCs, servers, and related peripherals, originally known as Gateway 2000. Also an ISP offering Internet access through gateway.net.

For more information on Gateway, Inc., see www.gateway.com.

gateway server A communications server that provides access between networks that use different access protocols.

Gateway Services for NetWare A Microsoft Windows NT Server service that allows an NT server to act as a gateway to a NetWare network. NT clients can access a NetWare server using the same methods used to access an NT server.

gauge A measurement of the physical size of a cable. Under the American Wire Gauge (AWG) standards, higher numbers indicate thinner cable.

See also cabling standards.

GB *See* gigabyte.

Gb *See* gigabit.

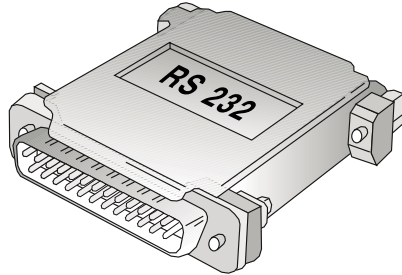
Gbit *See* gigabit.

GDS *See* Global Directory Service.

gender changer A special intermediary connector for use with two cables that both have only male connectors or only female connectors.

GENDER CHANGER

male to male



geostationary Also known as geosynchronous. The type of orbit required to keep a communications satellite in a fixed position relative to the earth.

The satellite's angular rate and direction of rotation are matched to those of the earth, and the satellite orbits the earth every 24 hours at about 36,000 kilometers (22,350 miles). Three satellites in geostationary orbit can cover 95 percent of the earth's surface (the remaining 5 percent is above the Arctic Circle).

geosynchronous See geostationary.

GIF See Graphics Interchange Format.

giga- A prefix meaning 1 billion, or 10^9 .
See also gigabyte.

gigabit Abbreviated Gbit and Gb. Usually 1,073,824 binary digits or bits of data. Sometimes used as equivalent to one billion bits.

See also kilobit; megabit.

Gigabit Ethernet A 1Gbps (1,000Mbps) extension of the IEEE 802.3 Ethernet standard, known as 1000Base-X.

This standard has been developed by the IEEE 802.3z Task Group and a number of interested companies collectively known as the Gigabit Ethernet Alliance. Gigabit Ethernet runs over multimode fiber-optic cable and is intended for use as a backbone and a way to connect high-speed routers, switches, and hubs.

See also Ethernet; Fast Ethernet.

gigabyte Abbreviated GB. Strictly speaking, one billion bytes; however, in computing, in which bytes are most often counted in powers of 2, a gigabyte becomes 2^{30} , or 1,073,741,824 bytes.

global account In Microsoft Windows NT Server, a user or group account defined on a primary domain controller that can be used from all the computers in the domain.

Global Catalog In Microsoft Active Directory, the storage of selected properties for all the objects in the Active Directory.

The Global Catalog allows users in an organization with multiple sites to locate resources quickly, without having to cross WAN links. In this way, users can examine a local source to find the location of a network resource.

See also Active Directory; forest; tree.

Global Directory Service Abbreviated GDS. An implementation of the X.500 directory service for managing remote users and addresses.

See also X.500.

global directory services Directory services that view the entire network as a single entity. A global directory system allows the network administrator to define all network resources—users, printers, and servers—at one time.

Banyan's StreetTalk and Novell's NetWare Directory Services (NDS) are examples of global directory services.

See also domain directory services.

global group In Microsoft Windows NT Server, user accounts granted server and workstation rights in their own and other domains whose security systems allow access. Global groups are a means of providing rights and permissions to resources inside and outside the domain to a group of users within a single domain.

See also local group.

global login A mechanism that permits users to log on to the network, rather than repeatedly logging on to individual servers. A global logon can provide access to all network resources.

globally unique identifier Abbreviated GUID. In Microsoft Active Directory, a unique 128-bit number that identifies an Active Directory object.

See also Active Directory.

global network An international network that spans all departments, offices, and subsidiaries of the corporation.

Global networks bring their own set of problems, including those of different time zones, languages, established standards, and PTT (Postal Telephone and Telegraph) companies.

GNU Pronounced “ga-noo.” A Free Software Foundation (FSF) project devoted to developing a complete, freely available Unix system that contains no AT&T code. The name GNU is a recursive acronym for “GNU’s not Unix!”

Many of the tools and utilities developed for this project have been released and are very popular with users of 4.4BSD, FreeBSD, and Linux.

For more information on GNU, see www.gnu.org.

See also 4.4BSD Lite; FreeBSD; Hurd; Linux; open source software.

Gopher A client/server application that presents Internet text resources as a series of menus, shielding the user from the

underlying mechanical details of IP addresses and different access methods.

Gopher menus may contain documents you can view or download, searches you can perform, or additional menu selections. When you choose one of these items, Gopher does whatever is necessary to obtain the resource you requested, either by downloading a document or by jumping to the selected Gopher server and presenting its top-level menu.

Gopher clients are available for most popular operating systems, including the Macintosh, MS-DOS, Windows, and Unix.

See also Gopherspace; World Wide Web.

Gopherspace A collective term used to describe all the Internet resources accessible using Gopher. Gopher is so good at hiding the mechanical details of the Internet that this term was coined to represent all the resources reachable using Gopher.

GOSIP Acronym formed from Government Open System Interconnection Profile. A suite of standards intended for use in government projects and based on the Open Systems Interconnect (OSI) reference model. Some measure of GOSIP compliance is required for government networking purchases. GOSIPs exist in many countries, including the United States, Canada, France, Germany, Australia, and the United Kingdom.

GPO *See* group policy object.

grace login Allows a user to finish logging on using an expired password without

changing it. You can set the number of grace logons a user is allowed.

graphical user interface Abbreviated GUI, pronounced “gooey.” A graphics-based user interface that allows users to select files, programs, and commands by pointing to pictorial representations on the screen rather than by typing long, complex commands from a command prompt.

Applications execute in windows, using a consistent set of drop-down menus, dialog boxes, and other graphical elements, such as scroll bars and icons. This consistency among interface elements is a major benefit for the user, because as soon as you learn how to use the interface in one program, you can use it in all other programs running in the same environment.

The use of graphical elements in a user interface was pioneered at Xerox Corporation's Palo Alto Research Center (PARC) in the early 1970s. Unfortunately, at that time the hardware needed to support such a user interface was well beyond the reach of most users. In 1979, Steve Jobs of Apple Computer visited PARC and recognized the importance of the user-interface work being done; this visit led to the development of the interface for the ill-fated Apple Lisa computer, and eventually to the Apple Macintosh series of computers. Since then, GUIs have been developed for most computing environments.

graphics accelerator board A specialized expansion board containing a graphics coprocessor as well as all the other circuitry found on a video adapter. Sometimes called a video accelerator board.

Transferring most of the graphics processing tasks from the main processor to the graphics accelerator board improves system performance considerably, particularly for Microsoft Windows users.

graphics coprocessor A fixed-function graphics chip, designed to speed up the processing and display of high-resolution images. Popular coprocessors include S3 Inc's 86C911 and Weitek's W5086 and W5186.

Graphics Interchange Format Abbreviated GIF; pronounced "gif." A graphics file format, originating on CompuServe, that results in relatively small graphics files.

An image may contain as many as 256 colors, including a transparent color, and a lossless compression method reduces the size of the file. A graphic in this format can be used as an inline image on a Web page.

A revision of this format, known as GIF89a, adds animation features, transparent backgrounds, and image interleaving.

See also lossless compression; Joint Photographic Experts Group.

graphics mode A video adapter mode in which everything displayed on the screen is created pixel by pixel. Text mode, by contrast, uses ready-made characters from a built-in character set.

grayed command *See* dimmed command.

group A collection of network users who all have the same level of security and can all be managed collectively.

group account An account containing other member accounts. All the rights and permissions accorded to the group are also granted to the group members, making group accounts a convenient way to provide a common set of capabilities to collections of user accounts.

group identifiers Security identifiers containing the set of permissions granted to the group. If a user account is part of a group, the group identifier is appended to that user's security identifier, granting the user all the permissions granted to that group.

See also security identifier.

Group object In Novell NetWare Directory Services (NDS), an object that contains a list of user object names.

See also leaf object; NetWare Directory Services.

group policy In Microsoft Windows 2000 Server, a central point of administration, allowing administrators to install software and apply standard settings to multiple users and computers throughout an organization.

group policy object Abbreviated GPO. In Microsoft Windows 2000 Server, a collection of group policy settings defined at the local machine, site, domain, or organizational unit level.

groupware Network software designed for use by a group of people all working on the same project or who need access to the same data.

See also Lotus Notes; workflow software; workgroup.

guard band A small-frequency band used to separate multiple bands in a broad-band transmission and prevent interference between the communications channels.

guest account In Microsoft Windows 2000, a built-in account available to users who do not have an account on the system.

Allowing people to use the guest account may breach company security policies.

GUI *See* graphical user interface.

GUID *See* globally unique identifier.

guru An operating system expert with a reputation for being helpful to other, less knowledgeable users.

H

H.323 A videoconferencing standard developed by the International Telecommunication Union (ITU) that defines videoconferencing from the desktop over LANs, intranets, and the Internet.

H.323 specifies techniques for compressing and transmitting real-time voice, video, and data between a pair of videoconferencing workstations. It also describes signaling protocols for managing audio and video streams, as well as procedures for breaking data into packets and synchronizing transmissions across communications channels.

See also Real-time Transport Protocol; T.120.

hack Originally, an expedient, although short-term, solution to a programming problem. This solution often bypassed some of the more traditional software-development processes. Now the word *hack* is often used to describe a well-crafted piece of work that produces just what is needed; it does not imply malicious intent to break into other people's systems for gain.

See also hacker; kluge.

hacker In the programming community, where the term originated, this term describes a person who pursues knowledge of computer systems for its own sake—someone willing to “hack through” the steps of putting together a working program.

More recently, in popular culture at large, the term has come to mean a person who breaks into other people's computers with malicious intent (what programmers call a “cracker”). Many countries now treat convicted crackers in the same way that they treat conventional breaking-and-entering criminals.

See also intruder.

HAL *See* Hardware Abstraction Layer.

half-duplex Abbreviated HDX. In asynchronous transmissions, the ability to transmit on the same channel in two directions, but only in one direction at a time.

See also communications parameters; duplex; full-duplex.

Hamming code A forward-error correction technique used to detect and correct single-bit errors during transmission.

The Hamming code adds three verification bits to the end of each four bits of data. The receiving device performs a similar process to ensure that the four data bits were received correctly and to detect any missing bits.

See also error detection and correction.

hand-held computer A portable computer that is small enough to be held in one hand, such as the enormously popular PalmPilot from 3Com Corporation.

handshaking The exchange of control codes or particular characters to maintain and coordinate data flow between two devices so that data is only transmitted when the receiving device is ready to accept the data.

Handshaking can be implemented in either hardware or software, and it occurs between a variety of devices. For example, the data flow might be from one computer to another computer or from a computer to a peripheral device, such as a modem or a printer.

See also flow control; XON/XOFF.

hang 1. When a program waits for an event that never occurs, as in “the program hangs waiting for a character from the keyboard.”

2. A slang expression used when attaching a new piece of hardware to a system, usually an external device attached by one or more cables, as in “I’m going to hang a new tape drive on the server this afternoon.”

See also deadlock.

hard-coded A description of software written in a way that does not allow for flexibility or future expansion. For example, when program variables are placed directly in the code rather than supplied as input from the user, the entire program must be recompiled to change the value, an obvious waste of resources.

See also hard-wired.

hard disk The part of a hard disk drive that stores data, rather than the mechanism for reading and writing to it. Sometimes called a platter.

hard-disk controller An expansion board that contains the necessary circuitry to control and coordinate a hard disk drive.

Many hard-disk controllers can manage more than one hard disk, as well as floppy disks and tape drives. On some PCs, the hard-disk controller is built into the motherboard, and in the case of an Integrated Drive Electronics hard disk, the controlling circuitry is mounted on the drive itself, eliminating the need for a separate controller.

hard disk drive A storage device that uses a set of rotating, magnetically coated disks called platters to store data or programs. In everyday use, the terms *hard disk*, *hard disk drive*, and *hard drive* are used interchangeably, because the disk and the drive mechanism are a single unit.

A typical hard-disk platter rotates at several thousand revolutions per minute, and the read/write heads float on a cushion of air from 10 to 25 millionths of an inch thick so that the heads never come into contact with the recording surface. The whole unit is hermetically sealed to prevent airborne contaminants from entering and interfering with these close tolerances.

Hard disks range in storage capacity from a few tens of megabytes to several terabytes. The more storage space on the disk, the more important your backup strategy becomes. Hard disks are reliable, but they do

hard-disk interface

fail, and usually at the most inconvenient moment.

See also high-capacity storage system; mini-hard disk; redundant array of inexpensive disks; single large expensive disk.

hard-disk interface A standard way of accessing the data stored on a hard disk. Several hard-disk interface standards have evolved over time, including the ST506 Interface, the Enhanced Small Device Interface (ESDI), the Integrated Drive Electronics Interface (IDE), and the SCSI (Small Computer System Interface).

See also PC Memory Card International Association.

hard-disk type A number stored in a personal computer's CMOS random access memory area that defines certain hard-disk characteristics, such as the number of read/write heads and the number of cylinders on the disk. This number is not directly accessible from the operating system. Some PCs require a special configuration program to access the hard-disk type; others permit access via the computer's built-in ROM BIOS setup program.

hard reset A system reset made by pressing the computer's Reset button or by turning the power off and then on again. A hard reset is used only when the system has crashed so badly that pressing Ctrl+Alt+Del to reboot does not work.

See also hang; deadlock.

hardware All the physical electronic components of a computer system, including peripheral devices, printed-circuit

boards, displays, and printers. If you can stub your toe on it, it must be hardware.

See also firmware; liveware; software.

Hardware Abstraction Layer Abbreviated HAL. The lowest level of the Microsoft Windows NT operating system, which is specifically tailored to the type of hardware used in the server. If the hardware changes, changes also have to be made to the HAL.

hardware address The address assigned to a network interface card (NIC) by the original manufacturer or by the network administrator if the interface card is configurable.

This address identifies the local device address to the rest of the network and allows messages to find the correct destination. Also known as the physical address, media access control (MAC) address, or Ethernet address.

Hardware Compatibility List Abbreviated HCL. A list of all the hardware devices supported by Microsoft Windows NT and Windows 2000. Items on this list have actually been tested and verified to work properly with Windows.

hardware dependent The requirement that a specific hardware component be present for a program to work. Hardware-dependent software is often difficult to move or port to another computer.

See also hardware independent.

hardware independent The ability to produce similar results in a wide variety of environments, without requiring the

presence of specific hardware. The Java programming language and the PostScript page-description language are both examples of hardware independence. Java runs on a wide range of computers, from the PC to a mainframe; PostScript is used by many printer manufacturers.

See also hardware dependent.

hardware interrupt An interrupt or request for service generated by a hardware device, such as a keystroke or a tick from the clock. Because the processor may receive several such signals simultaneously, hardware interrupts are usually assigned a priority level and processed according to that priority.

See also interrupt request; software interrupt.

hard-wired Describes a system designed in a way that does not allow for flexibility or future expansion. May also refer to a device that is connected directly to the network, such as a printer.

See also hard-coded.

hash function A function that maps a data item to a numeric value by use of a transformation. A hash function can convert a number that has meaning to a user, such as a key or other identifier, into a value for the location of that data in a structure such as a table.

See also hashing; hash table.

hashing The process of creating or recreating a hash table by recalculating the search index code assigned to each piece of data in the table.

See also hash function; hash table.

hash table A method of representing data so that it can be found again very quickly.

A hash table assigns a special index code to each piece of data, and specially designed software uses this code to locate the data, rather than repeating what might be a very lengthy search each time the data is requested.

See also hash function; hashing.

Hayes-compatible modem Any modem that recognizes the commands in the industry-standard AT command set, originally defined by Hayes Microcomputer Products, Inc.

HBA *See* host bus adapter.

H-channel A set of ISDN Primary Rate Interface (PRI) services with predefined speeds, designed to carry videoconferencing data streams.

H-channel services are created from multiple 64Kbps B channels. Three service levels are commonly available:

- H0 operating at 384Kbps
- H11 operating at 1,536Kbps
- H12 operating at 1,920Kbps

See also Integrated Services Digital Network; Primary Rate ISDN.

HCL *See* Hardware Compatibility List.

HCSS *See* high-capacity storage system.

HDLC *See* High-level Data Link Control.

HDSL See High-Bit-Rate Digital Subscriber Line.

HDX See half-duplex.

header Information placed at the beginning of a file, a data transmission, or an archive.

In an e-mail message, the header contains information about the message, including sender and recipient information, and information about the route the message took as it was being delivered.

In a data transmission, the header may contain source and destination address information, as well as other control data.

In an archive, the header is a block that contains information describing the contents of the archive.

Header Error Control In an Asynchronous Transfer Mode (ATM) cell header, an 8-bit field used for detecting errors and correcting single-bit errors.

See also Asynchronous Transfer Mode.

headless server A server computer with no monitor attached.

heartbeat An Ethernet signal quality test function. This signal proves that a component is working and can detect collisions. Also known as signal quality error (SQE).

HEC See Header Error Control.

Hello protocol In Asynchronous Transfer Mode (ATM), a link state protocol used by nodes to determine who and where their neighbors are.

helper A program launched or used by a Web browser to process a file type that the browser cannot handle. Sometimes called a plug-in.

A helper may view an image, play a sound file, or expand a compressed file. A helper that deals with video, graphics, or animation is called a viewer; a helper that deals with sound files is called a player.

See also player; viewer; Web browser.

hertz Abbreviated Hz. A unit of frequency measurement; 1 hertz equals one cycle per second.

See also megahertz.

heterogeneous network A network that consists of workstations, servers, network interface cards, operating systems, and applications from many vendors, all working together as a single unit. The network may also use different media and different protocols over different network links.

See also enterprise network; homogeneous network.

Hewlett-Packard Company Abbreviated HP. A major manufacturer of hand-held calculators, personal computers, servers, minicomputers, mainframes, scientific and medical equipment, test and measurement equipment, laser and ink jet printers, plotters, and software.

Founded by William Hewlett and David Packard in 1939 in a garage, the company is now headquartered in Palo Alto, California. HP has a widely diversified product line of more than 10,000 items, and it has a

well-earned reputation for building rugged and reliable equipment.

For more information on Hewlett-Packard, see www.hp.com.

hex See hexadecimal.

hexadecimal Abbreviated hex. The base-16 numbering system that uses the digits 0 through 9, followed by the letters A through F, which are equivalent to the decimal numbers 10 through 15.

Hex is a convenient way to represent the binary numbers that computers use internally because it fits neatly into the 8-bit byte. All the 16 hex digits 0 through F can be represented in 4 bits, and 2 hex digits (1 digit for each set of 4 bits) can be stored in a single byte. This means that 1 byte can contain any one of 256 different hex numbers, from 0 through FF.

See also binary; decimal.

HFS See Hierarchical File System.

hidden file In many operating systems, any file that has the hidden attribute set, which indicates to the operating system that information about the file should not appear in normal directory listings. There may also be further restrictions on a hidden file, and users may not be able to delete, copy, or display the contents of such a file.

hidden share In Microsoft Windows 2000, a share whose name contains a dollar sign as the last character. The names of hidden shares are not displayed under normal circumstances, although they are always visible in Computer Management.

See also common share; default shares; share; sharing.

Hierarchical File System Abbreviated HFS. A tree-structured file system used on the Macintosh; designed for use with hard disks.

Hierarchical Storage Management

Abbreviated HSM. A combination of several types of file-storage systems, managed by intelligent software.

In HSM, data is moved from one type of storage to another depending on how frequently the data is accessed. Active data is held on hard disks, less frequently used data is held in near-line storage such as an optical disk system, and data used only infrequently is stored in a tape backup.

See also archive; high-capacity storage system; jukebox.

High-Bit-Rate Digital Subscriber Line

Abbreviated HDSL. A high-speed data transmission technology originally developed by Bellcore that delivers high bandwidth over existing twisted-pair copper telephone lines.

HDSL is the most common Digital Subscriber Line (DSL) service and provides T1 data rates of 1.544Mbps over lines of up to 3.6 kilometers, or 12,000 feet, in length. HDSL is symmetric, providing the same data rate in each direction.

The service is not intended for residential purposes, but is used in the telephone company's own private data networks, Internet servers, and interexchange connections.

See also Asymmetric Digital Subscriber Line; Digital Subscriber Line; Rate-Adaptive Digital Subscriber Line; Single-Line Digital Subscriber Line; Very-High-Bit-Rate Digital Subscriber Line.

high-capacity storage system Abbreviated HCSS. A data-storage system that extends the storage capacity of a Novell NetWare server by integrating an optical-disk library, or jukebox, into the NetWare file system.

Network users and applications can access files and directories on the jukebox with the same NetWare commands and function calls used to access files from a hard disk. The most frequently used HCSS files may be cached temporarily on the server hard disk to speed up access times. HCSS can also access a magnetic tape system.

See also archive.

high-end An expensive, full-featured product from the top of a company's product list.

See also low-end.

High-level Data Link Control Abbreviated HDLC. An international protocol defined by the ISO (International Organization for Standardization), included in CCITT X.25 packet-switching networks. HDLC is a bit-oriented, synchronous protocol that provides error correction at the data-link layer. In HDLC, messages are transmitted in variable-length units known as frames.

See also Synchronous Data Link Control.

high-level language Any machine-independent programming language that uses English-like syntax in which each statement corresponds to many assembly language instructions. High-level languages free programmers from dealing with the underlying machine architecture and allow

them to concentrate on the logic of the problem at hand.

See also low-level language.

high memory area Abbreviated HMA. In a computer running MS-DOS, the first 64KB of extended memory above the 1MB limit of 8086 and 8088 addresses.

Programs that conform to the Extended Memory Specification (EMS) can use this memory as an extension of conventional memory. However, only one program can use or control HMA at a time. If MS-DOS is loaded into the HMA, approximately 50KB more conventional memory becomes available for use by applications.

See also expanded memory; extended memory; memory management.

High-Performance File System Abbreviated HPFS. A file system available in OS/2 and Microsoft Windows NT that supports the following:

- Long, mixed-case filenames of up to 255 characters
- As much as 64KB of extended attributes per file
- Faster disk access with an advanced disk cache for caching files and directory information
- Highly contiguous file allocation that eliminates file fragmentation
- Hard disks of up to 64GB in size

MS-DOS does not recognize the HPFS file structure, it cannot be used on a floppy disk, and it is not supported in Windows NT 4 or later.

HIGH-LEVEL DATA LINK CONTROL FRAME
**High-Performance Parallel Interface**

Abbreviated HPPI. A parallel interface standard from the American National Standards Institute (ANSI) known as X3T9.3, originally developed as an interface between supercomputers and fast peripherals such as disk arrays and frame buffers.

Recently, HPPI has been extended for use in networked computers and high-end workstations. It provides a data rate of 800Mbps over 32-pair twisted-pair copper wire, known as single HPPI, and 1,600Mbps over 64 pairs, known as double HPPI. Connection length is limited to 25 meters (82 feet).

High-Performance Routing Abbreviated HPR. An internetworking protocol from IBM, intended as an upgrade to its Advanced Peer-to-Peer Networking (APPN) package. HPR provides internetworking capabilities similar to those of TCP/IP.

See also Advanced Peer-to-Peer Networking; Data Link Switching.

High Sierra specification A specification for CD-ROM data that served as the basis for the ISO (International Organization for Standardization) 9660 standard. It is called High Sierra because it was defined at a meeting held near Lake Tahoe in November 1985.

High Speed Serial Interface Abbreviated HSSI. A serial data communications interface optimized for speeds of up to 52Mbps. Often used for connecting an Asynchronous Transfer Mode (ATM) switch to a T3 Data Service Unit.

See also Asynchronous Transfer Mode; Data Service Unit; T3.

hijacking An attack on a computer system in which an established TCP/IP session is redirected in mid-session to an unauthorized host system.

See also spoofing.

hit On the World Wide Web, a request from a browser for a file on the server. A hit on a Web page occurs whenever any file is accessed, whether it is a text document, a graphic, a script, or an audio or video clip. If you access three files on a Web page, you generate three hits, so a hit is a poor measure of the number of people visiting a Web site, as it simply reflects the number of file requests made.

hive A major logical division within the Microsoft Windows Registry. The Registry is divided into several hives:

- HKEY_CLASSES_ROOT contains information about file associations.

- **HKEY_CURRENT_USER** contains information about the currently logged on user.
- **HKEY_LOCAL_MACHINE** contains information about the computer running Windows, including details of applications, device drivers, and hardware.
- **HKEY_USERS** contains information about each active user with a user profile.
- **HKEY_CURRENT_CONFIG** contains details about the current system configuration.
- **HKEY_DYN_DATA** contains information about plug-and-play devices on the system.

You can view and edit the Registry using the `regedit.exe` program.

See also key; sub-key.

HMA *See* high memory area.

holy wars A fundamentally unresolvable computer-related argument, in which the participants spend most of their time trying to establish often wildly personal choices as carefully thought out and deeply considered technical evaluations. Topics might include any version of Unix versus any other version of Unix, the C programming language versus C++ (or any other programming language), big endian systems versus little endian systems, and so on.

home directory In Unix, a directory that contains the files for a specific user. The name of your home directory is kept in the password file, and when you log in, your current directory is always set to be your home directory.

home page On the World Wide Web, an initial starting page. A home page may be prepared by an individual or by a corporation and is a convenient jumping-off point to other Web pages or Internet resources.

See also portal.

homogeneous network A network that consists of one type of workstation, server, network interface card, and operating system, with a limited number of applications, all purchased from a single vendor. All nodes use the same protocol and the same control procedures.

See also enterprise network; heterogeneous network.

hooked vector An intercepted interrupt vector that now points to a replacement interrupt service routine (ISR) rather than to the original service routine.

hop A single link between two computer systems that an e-mail message must cross on its way to its destination. A message may have to pass over many hops to reach its ultimate destination; if it must pass between five computers, it is said to have taken four hops to reach its destination.

See also hop count.

hop count In routing, the number of links that must be crossed to get from any given source node to any given destination node. Hop count is often used as a metric for evaluating a route for a least-cost routing algorithm.

horizontal application Any application software that is broad in scope and not designed for use in one specific industry or

setting. Word-processing software falls into this category, but software specifically designed to manage a medical practice does not.

See also vertical application.

host The central or controlling computer in a networked or distributed processing environment which provides services that other computers or terminals can access via the network.

host bus adapter Abbreviated HBA. A board acting as an interface between the processor and the hard-disk controller in a network server, used by Novell NetWare to relieve the main processor of data-storage and retrieval tasks. Also known as a disk coprocessor board (DCB).

The HBA and its disk subsystems make up a disk channel. NetWare can access five disk channels, with four controllers on each channel, and eight hard-disk drives attached to each controller.

Hot Fix A Novell NetWare feature that marks defective disk blocks dynamically so that the operating system will not use them.

Data is redirected from any faulty blocks to a small portion of disk space set aside as the Hot Fix redirection area. Hot Fix then marks the defective area as bad, and the server will not attempt to store data there again. By default, 2 percent of a disk partition is set aside as the Hot Fix redirection area.

See also fault tolerance.

HotJava A highly interactive Web browser from Sun Microsystems that is written in the Java programming language.

HP *See* Hewlett-Packard Company.

HPFS *See* High-Performance File System.

HP OpenView Certified Consultant A certification from Hewlett-Packard that recognizes technical competency to both sell and support HP OpenView, a network management system used to monitor and manage networks consisting of equipment and software from multiple vendors.

Three specializations are available:

- Unix
- Windows NT
- Unix and Windows NT combined

HPR *See* High-Performance Routing.

HP-UX A version of Unix that runs on Hewlett-Packard computers. HP-UX includes BSD extensions, including the network commands, the Korn shell, and a version of emacs. Visual User Environment (VUE) is HP's graphical user interface, with workspaces for different tasks, drag-and-drop functions, a text editor, a color icon editor, and other productivity tools.

HP-UX also includes System Administration Manager (SAM) for common administrative tasks, such as adding new users, installing and configuring peripherals, managing processes, and scheduling jobs.

HSM *See* Hierarchical Storage Management.

HSSI *See* High Speed Serial Interface.

HTML See HyperText Markup Language.

HTTP See Hypertext Transfer Protocol.

hub A device used to extend a network so that additional workstations can be attached. There are two main types of hubs:

- **Active hubs** amplify transmission signals to extend cable length and ports.
- **Passive hubs** split the transmission signal, allowing additional workstations to be added, usually at a loss of distance.

In some star networks, a hub is the central controlling device.

Huffman coding In data compression, a method of encoding data on the basis of the relative frequency of the individual elements. Huffman coding is often used with text files; the coding is based on how frequently each letter occurs, because it is a lossless compression method. Huffman coding is used in fax transmissions.

See also data compression; lossless compression; lossy compression.

Hurd A project from the Free Software Foundation (FSF) to develop and distribute a free version of the Unix operating system for many different hardware platforms.

Hurd (or sometimes HURD) is considered a collection of all the GNU software, compilers, editors, and utilities, as well as the operating system.

See also Free Software Foundation; GNU.

hybrid network A network that uses a collection of different technologies, such as frame relay, leased lines, and X.25.

See also homogeneous network; heterogeneous network.

hybrid organization In Novell Directory Services (NDS), an organizational strategy that combines two or more of the methods—locational, divisional, and workgroup. A hybrid organization makes most sense in a large organization.

hypermedia A term used to describe nonsequential applications that have interactive, hypertext links between different multimedia elements—graphics, sound, text, animation, and video.

If an application relies heavily on text-based information, it is known as hypertext; however, if full-motion video, animation, graphics, and sound are used, the application is considered hypermedia.

hypertext A method of presenting information so that it can be viewed by the user in a nonsequential way, regardless of how the topics were originally organized.

Hypertext was designed to make a computer respond to the nonlinear way that humans think and access information—by association, rather than according to the linear organization of film, books, and speech.

In a hypertext application, you can browse through the information with considerable flexibility, choosing to follow a new path each time you access the information. When you click a highlighted word, you activate a link to another hypertext document, which may be on the same Internet host or on a completely different system thousands of miles away. These links depend on the care

that the document originator used when assembling the document; unfortunately, many links turn into dead-ends.

See also link rot.

HyperText Markup Language Abbreviated HTML. A standard document formatting language used to create Web pages and other hypertext documents. HTML is a subset of Standardized General Markup Language (SGML).

HTML defines the appearance and placement on the page of elements such as fonts, graphics, text, links to other Web sites, and so on; it has nothing to do with the actual material presented. Hypertext documents often have the filename extension .htm or .html.

The published HTML standards have been revised several times. HTML version 2 was the first version widely used on the World Wide Web and supported by the popular Web browsers of the day. Subsequent revisions to the standard have added new HTML elements such as tables, text flow around images, frames, applets, and style sheets.

Future HTML revisions will be developed by the World Wide Web Consortium

(W3C), at least in theory; the manufacturers of the most popular browsers have driven this process in the past by creating their own non-standard HTML elements.

HTML has been vital in the development of the World Wide Web; however, the functions that it performs via the Web browser are becoming restrictive. In part, this has led to the development of other technologies, such as Java, Virtual Reality Modeling Language (VRML), and Extensible Markup Language (XML).

See also Extensible Markup Language; Java; Secure HTTP; Virtual Reality Modeling Language.

Hypertext Transfer Protocol Abbreviated HTTP. The command and control protocol used to manage communications between a Web browser and a Web server.

When you access a Web page, you see a mixture of text, graphics, and links to other documents or other Internet resources. HTTP is the mechanism that opens the related document when you select a link, no matter where that document is located.

Hz *See* hertz.

I2O *See* Intelligent Input/Output.

IAB *See* Internet Architecture Board.

IAC *See* Inter-Application Communication.

IANA *See* Internet Assigned Numbers Authority.

IBM *See* International Business Machines Corporation.

IBM 3270 A general name for a family of IBM system components—printers, terminals, and terminal cluster controllers—that can be used with a mainframe computer by an SNA (Systems Network Architecture) link.

Software that emulates a 3270 terminal is available for all major operating systems.

IBM cabling systems *See* Type 1-9 cable.

IBM Certified Advanced Technical Expert One of the many certifications from IBM, available in many specializations, including RS/6000 AIX.

IBM Certified AIX User One of the many certifications from IBM. This one is available to the AIX user.

IBM Certified Expert One of the many certifications from IBM, available in a range of specializations, including OS/2 Warp Server and OS/2 LAN Server.

IBM Certified Specialist One of the many certifications from IBM, available in a range of specializations, including:

- AIX System Administration
- AIX Support
- AS/400 Associate System Operator
- AS/400 Professional System Operator
- AS/400 Associate System Administrator
- AS/400 Professional System Administrator
- OS/2 Warp Server Administration
- LAN Server 4 Administration

IBM Certified Systems Expert One of the many certifications from IBM, available in a range of specializations, including OS/2 Warp and OS/2 Warp 4.

IBM RS/6000 A set of seven or nine separate 32-bit chips used in IBM's line of reduced instruction set computing (RISC) workstations.

With as many as 7.4 million transistors, the RS/6000 uses a superscalar design with four separate 16KB data-cache units and an 8KB instruction cache. The joint venture announced between IBM, Apple, and Motorola in late 1991 specified the development of a single-chip version of the RS/6000 architecture called the PowerPC.

IBM Suite for Windows NT A software suite from IBM for Microsoft Windows NT that includes five major modules:

- Intel's LANDesk Manager

- Lotus Domino
- IBM's DB2 Universal Database (UDB)
- IBM's eNetwork Communications Server
- IBM's ADSTAR Distributed Storage Manager (ADSM)

The suite is available in both department and enterprise versions.

IBM ThinkPad A series of innovative and popular notebook computers from IBM.

The ThinkPad first introduced the touch-sensitive dual-button pointing stick (called a TrackPoint), the pencil-eraser-like device, found between the G, H, and B keys. It is now included on many portable computers and replaces the mouse.

The top-of-the-line ThinkPad 770 runs a 300MHz Pentium II processor, an 8.1GB hard disk, a 56K modem, a 14.1-inch SVGA screen with resolutions up to 1280 × 1024, and as much as 320MB of memory.

IC *See* integrated circuit.

ICA *See* Independent Computing Architecture.

ICMP *See* Internet Control Message Protocol.

ICP *See* Internet Content Provider.

Icon A general-purpose high-level programming language with a large number of string-processing functions, developed by Ralph Griswold at the University of Arizona. Icon has a C-like syntax and is available both as a compiler and as an interpreter.

icon In a graphical user interface, a small screen image representing a specific element that the user can manipulate in some way,

selected by moving a mouse or another pointing device.

An icon can represent an application, a document, embedded and linked objects, a hard disk drive, or several programs collected in a group.

IDE *See* Integrated Drive Electronics.

identity In Microsoft Outlook Express, a feature that allows you to create multiple accounts so that you can keep business and personal e-mail associated with different identities.

idle cell In Asynchronous Transfer Mode, a cell transmitted purely to keep network traffic at a specific level.

See also Asynchronous Transfer Mode.

IDS *See* Intrusion Detection System.

IDT *See* Integrated Device Technology, Inc.

IE *See* Internet Explorer.

IEC *See* Interexchange Carrier.

IEEE *See* Institute of Electrical and Electronics Engineers.

IEEE standards The Institute of Electrical and Electronics Engineers (IEEE), acting as a coordinating body, has established a number of telecommunications standards, including Group 802 as follows:

- **802.1D** An access-control standard for bridges linking 802.3, 802.4, and 802.5 networks.
- **802.2** A standard that specifies the Data Link layer for use with 802.3, 802.4, and 802.5 networks.

- **802.3 1Base5** A standard matching the AT&T StarLAN product with a 1Mbps data transfer rate and a maximum cable-segment length of 500 meters (1640 feet).
 - **802.3 10Base2** An implementation of the Ethernet standard on thin Ethernet cable with a data transfer rate of 10Mbps, and a maximum cable-segment length of 185 meters (600 feet).
 - **802.3 10Base-T** A standard for Ethernet over unshielded twisted-pair wiring, the same wiring and RJ45 connectors used with telephone systems. The standard is based on a star topology, in which each node connects to a central wiring center, with a cable-length limitation of 100 meters (325 feet).
 - **802.3 10Broad36** A standard for long-distance Ethernet with a 10Mbps data rate and a maximum cable-segment length of 3600 meters (11,800 feet).
 - **802.4** A standard for bus topology networks that use token passing to control access and network traffic, running at 10Mbps. Not widely implemented.
 - **802.5** A standard for ring networks that use token passing to control access and network traffic, running at 4Mbps or 16Mbps. It is used by IBM's Token Ring network.
 - **802.6** An emerging standard for metropolitan area networks (MANs) transmitting voice, video, and data over two parallel fiber-optic cables, using signaling rates of up to 155Mbps.
 - **802.7** The Broadband Technical Advisory Committee provides advice on broadband techniques to other IEEE subcommittees.
 - **802.8** The Fiber-Optic Technical Advisory Committee provides advice on fiber-optic technology to other IEEE subcommittees.
 - **802.9** The Integrated Data and Voice (IDV) Networks group is currently working to integrate data, voice, and video to 802 LANs and ISDN. Now more commonly referred to as Iso-Ethernet.
 - **802.10** The Network Security Technical Advisory Group is developing a standard definition of a network security model.
 - **802.11** The Wireless Networking group is developing standards for wireless networks.
 - **802.12** The Demand Priority group is working on standards for the 100Mbps Ethernet standard.
 - **802.14** The Cable Modems group is defining standards for data transport over traditional cable TV networks.
- You will also see many of these standards referred to by their ISO reference numbers. IEEE standards 802.1 through 802.11 are also known as ISO standards 8802.1 through 8802.11.
- For more information, see the entries on the individual standards.
- IETF** See Internet Engineering Task Force.
- IFS** See installable file system.
- IGMP** See Internet Group Management Protocol.

IGP See Interior Gateway Protocol.

IGRP See Interior Gateway Routing Protocol.

IIOIP See Internet Inter-ORB Protocol.

IKE Abbreviation for Internet Key Exchange.

See ISAKMP/Oakley.

ILEC See Incumbent Local Exchange Carrier.

IMA See Inverse Multiplexing over ATM.

image map A graphical inline image on a Web page that contains more than one link. Each region of the image map is linked to a different Web resource; you can click a part of the image to retrieve the appropriate resource.

See also inline image.

imaging The process of capturing, storing, cataloging, displaying, and printing graphical information, as well as scanning paper documents for archival storage.

Network users can store and then retrieve imaged documents from large, centralized image-storage systems, using applications such as Lotus Notes or other groupware.

See also document management; high-capacity storage system; optical character recognition.

IMAP See Internet Mail Access Protocol.

impedance An electrical property of a cable that combines capacitance (the ability to store an electrical charge), inductance

(the ability to store energy in the form of a magnetic field), and resistance (the ability to impede or resist the flow of electric current), measured in ohms.

Impedance can be described as the apparent resistance to the flow of alternating current at a given frequency. Mismatches in impedance along a cable cause distortions and reflections. Each transmission protocol and network topology specifies its own standards for impedance.

impersonation attack An attack in which a hostile computer system masquerades as a trusted computer.

See also brute-force attack; dictionary attack; social engineering

implied security equivalence In Novell Directory Services, when an object receives the rights of the object's parent, it is said to be security equivalent. Also known as container security equivalence. The Inherited Rights Filter does not affect this.

See also explicit security equivalence; Inherited Rights Filter; Novell Directory Services; security equivalence.

I-Mux See inverse multiplexing.

incremental backup A backup of a hard disk that consists of only those files created or modified since the last backup.

See also differential backup; full backup.

Incumbent Local Exchange Carrier

Abbreviated ILEC. A term coined from the Telecommunications Act of 1996 to describe the incumbent local telephone company,

Independent Computing Architecture

providing local transmission and switching services.

See also Competitive Local Exchange Carrier.

Independent Computing Architecture

Abbreviated ICA. A presentation service protocol developed by Citrix that transports mouse clicks, keystrokes, and screen updates between a thin client and the server. ICA works with Microsoft Windows, Macintosh, and Unix clients and runs on top of TCP/IP, IPX/SPX, and NetBIOS.

See also thin client.

Independent Software Vendor Abbreviated ISV. A company that develops and sells computer software but is completely independent of the makers of the hardware upon which the software runs.

Industry Standard Architecture Abbreviated ISA. The 16-bit bus design first used in IBM's PC/AT computer in 1984. ISA has a bus speed of 8MHz and a maximum throughput of 8MBps. EISA (Extended Industry Standard Architecture) is a 32-bit extension to this bus.

INETCFG A Novell NetWare 4 NLM used to set up and configure AppleTalk, Internet Protocol (IP) and Internetworking Packet eXchange (IPX) protocols on the server.

infection The presence of a virus or a Trojan Horse within a computer system; the virus may be active in memory or present on the hard disk.

The infection may remain hidden from the user for a considerable length of time. Some

viruses are triggered by particular dates, others by specific events on the system.

See also antivirus program; boot sector virus; file-infecting virus; inoculate; macro virus; multipart virus; polymorphic virus; stealth virus; Trojan Horse; vaccine; virus.

information node Abbreviated i-node, sometimes written inode, pronounced "eye-node." In Unix, a data structure on disk that describes a file.

Each directory entry associates a filename with an i-node; although a single file may have several filenames, one for each link, a file has only one i-node. Within a filesystem, the number of i-nodes, and therefore the number of files, is defined when the system is first initialized.

An i-node contains all the information Unix needs to be able to access the file, including the file's length, the times that the file was last accessed or modified, owner and group ID information, access permissions, the number of links to the file, and the disk address of the data blocks that contain the file itself.

See also i-node table.

information warehouse A central repository containing a company's current and historical data in a form that can be accessed quickly and easily by users to aid in their business decision making. IBM has a large number of products for building fully automated enterprise-wide data warehouses.

Informix-4GL Certified Professional

A professional certification from Informix Software aimed at developers proficient in

creating custom database applications using Informix-4GL.

Informix Software, Inc. A major supplier of object-oriented and relational database products, based in Menlo Park, California.

For more information on Informix Software, Inc., see www.informix.com.

Infrared Data Association Abbreviated IrDA. A trade association of more than 150 computer and telecommunications hardware and software suppliers, including Hewlett-Packard, Apple Computer, AST, Compaq, Dell, IBM, Intel, Motorola, Novell, and others.

IrDA is concerned with defining standards for products that use wireless communications.

For more information on IrDA, see www.irda.org.

infrared transmission A method of wireless transmission that uses part of the infrared spectrum to transmit and receive signals.

Infrared transmissions take advantage of a frequency range just below that of visible light, and they usually require a line-of-sight connection between transmitter and receiver.

Infrared transmission can be used to send documents from portable computers to printers, to transmit data between portable computers, to exchange information between computers and cellular telephones and faxes, and to connect to home entertainment systems. Almost every manufacturer

of portable devices is implementing infrared communications at some level.

See also Infrared Data Association; mobile computing; wireless communications.

inherited rights Rights (in Novell NetWare) or permissions (in Microsoft Windows 2000) received by an object from its parent object. In Novell NetWare, inherited rights can be blocked by the Inherited Rights Filter or by an explicit assignment.

See also Inherited Rights Filter.

Inherited Rights Filter Abbreviated IRF. In Novell Directory Services (NDS), the mechanism that controls the rights a trustee can inherit from parent directories or container objects.

Inheritance allows an assignment applied at one point to apply to everything below that point in the file and directory structure. The IRF for any file, directory, or object is a part of NetWare's access control information.

In NetWare 3, the IRF was known as the Inherited Rights Mask and applied only to the file system.

See also Inherited Rights Mask; NETADMIN; trustee; trustee assignment.

Inherited Rights Mask Abbreviated IRM. In NetWare 3, the mechanism that controls the rights a trustee can inherit. By default, IRM allows all rights to be inherited. Both files and directories have individual IRM controls.

See also Inherited Rights Filter; trustee; trustee assignment.

initialize 1. To start up a computer.

2. In the Macintosh, the process of preparing a new, blank floppy or hard disk for use. Initializing completely obliterates any information previously stored on the disk.

3. To assign a beginning value to a variable.

See also formatting.

inline image On a Web page, an image displayed along with accompanying text. The process of placing the image on the page is known as inlining.

inoculate To protect a file against attack from a virus by recording characteristic information about it and then monitoring any changes.

See also antivirus program; boot sector virus; file-infecting virus; infection; macro virus; multipart virus; polymorphic virus; stealth virus; Trojan Horse; vaccine; virus.

i-node *See* information node.

i-node table In Unix, a list of all the i-nodes in a filesystem. Within the i-node table, each i-node is known by a number—the i-number, or index number. If a file is defined by i-node #300, it is said to have an i-number of 300.

See also information node.

in-place migration In Novell NetWare, a method of upgrading an existing NetWare server to NetWare 4, which includes converting the file system and updating the network operating system.

See also across-the-wire migration.

Inprise Corporation Originally known as Borland International, the company

changed its name in 1988. Well known for its dBASE database products, the company also markets Delphi Client/Server Suite, a visual rapid applications development environment, and JBuilder, a set of Java programming language development tools.

For more information on Inprise Corporation, see www.inprise.com.

input/output Abbreviated I/O. The transfer of data between the computer and its peripheral devices, disk drives, terminals, and printers.

input/output bound Abbreviated I/O bound. A condition in which the speed of operation of the input/output port limits the speed of program execution. Getting the data into and out of the computer is more time-consuming than actually processing that same data.

See also computation bound.

INSTALL A Novell NetWare server console NetWare Loadable Module (NLM) used for managing, maintaining, and updating NetWare servers. INSTALL can be used for the following tasks:

- Creating, deleting, and managing hard-disk partitions and NetWare volumes on the server
- Installing NetWare and other additional products and updating the license or registration disk
- Adding, removing, checking, and unmirroring hard disks
- Changing server startup and configuration files

install To configure and prepare hardware or software for operation.

Many application packages have their own installation programs, which copy all the required files from the original distribution disks into appropriate directories on your hard disk and then help to configure the program to your own operating requirements. Microsoft Windows programs are installed by a program called Setup.

installable file system Abbreviated IFS. A file system that is loaded dynamically by the operating system when it is needed. Different file systems can be installed to support specific needs, in just the same way as device drivers are loaded to support specific hardware.

installation program A program whose sole function is to install (and sometimes configure) another program.

The program guides the user through what might otherwise be a rather complex set of choices, copying the correct files into the right directories, decompressing them if necessary, and asking for the next disk when appropriate. An installation program may also ask for a person's name and a company name so that the startup screen can be customized. Microsoft Windows programs are installed by a program called Setup.

Installer service In Microsoft Windows 2000, a service that manages all aspects of application installation, removal, and the separate configuration of components of large packages such as Office 2000.

Institute of Electrical and Electronics Engineers Abbreviated IEEE, pronounced "eye-triple-ee." A membership organization, founded in 1963, including engineers, students, and scientists. IEEE also acts as a coordinating body for computing and communications standards.

For more information in IEEE, see www.ieee.org.

See also IEEE standards.

instruction set The set of machine-language instructions that a processor recognizes and can execute.

An instruction set for reduced instruction set computing (RISC) may only contain a few instructions; a computer that uses complex instruction set computing (CISC) may be able to recognize several hundred instructions.

INT 14 *See* Interrupt 14.

integrated circuit Abbreviated IC, also known as a chip. A small semiconductor circuit that contains many electronic components.

Integrated Device Technology, Inc.

Abbreviated IDT. An established manufacturer of SRAM, specialty memory, and embedded microprocessors, who has recently moved into producing clones of Intel microprocessors. IDT's first WinChip, originally known as the C6, shipped in limited quantities in 1997 and 1998, and the company recently announced the WinChip 2 and WinChip 2+.

For more information on Integrated Device Technology, see www.idt.com.

Integrated Drive Electronics Abbreviated IDE. A popular hard-disk interface standard, used for disks in the range of 40MB to 1.2GB, requiring medium to fast data-transfer rates. The electronic control circuitry is located on the drive itself, thus eliminating the need for a separate hard-disk controller card.

See also Enhanced Small Device Interface; Small Computer System Interface; ST506 Interface.

Integrated On-Demand Network

Abbreviated ION. A new high-capacity residential and business service from Sprint that allows customers to simultaneously make a phone call, send/receive a fax, and access the Internet using a single connection.

The service is based on a mixture of broadband technology, such as Asynchronous Transfer Mode (ATM) and Digital Subscriber Line (DSL). Deployment of the network is based on a partnership with three other companies—Cisco Systems, Bellcore, and Radio Shack.

See also Asynchronous Transfer Mode; Digital Subscriber Line.

Integrated Services Architecture Abbreviated ISA. A proposed extension to the Internet standards that would provide integrated services in support of real-time applications over the Internet.

Using current standards, real-time applications do not work well over the Internet because of variable and unpredictable queuing delays and other losses. To solve

this problem, ISA suggests that routers be able to reserve resources for specific data streams called flows. A flow might be video data from a source to a destination or from a source to multiple destinations.

See also IP over ATM; IP switching; quality of service.

Integrated Services Digital Network

Abbreviated ISDN. A standard for a worldwide digital communications network originally designed to replace all current systems with a completely digital, synchronous, full-duplex transmission system.

Computers and other devices connect to ISDN via simple, standardized interfaces. They can transmit voice, video, and data, all on the same line.

See also Basic Rate ISDN; Broadband Integrated Services Digital Network; Primary Rate ISDN.

integrated software Application software that combines the functions of several major applications, such as a spreadsheet, a database, a word processor, and a communications program, into a single package. Microsoft Works is an example of integrated software.

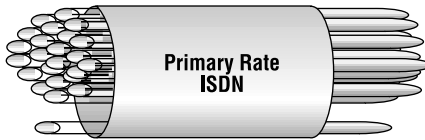
Integrated software provides a consistent user interface in all the modules and allows the user to transfer data from one part of the system to another quickly and easily. It is also usually inexpensive. Unfortunately, integrated software packages do not typically offer all the complex features available with their stand-alone counterparts.

See also software suite.

INTEGRATED SERVICES DIGITAL NETWORK

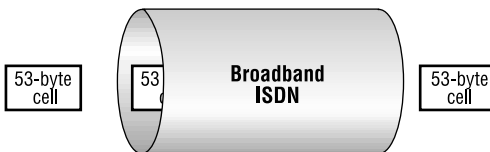
Two 64Kbps B channels

16Kbps D channel



Twenty-three 64Kbps B channels

64Kbps D channel



Integrated-Private Network-to-Network Interface

Abbreviated I-PNNI. A routing protocol based on the Asynchronous Transfer Mode (ATM) Forum's Private Network-to-Network Interface (PNNI) standard that allows ATM switches to communicate.

See also Private Network-to-Network Interface.

Intel Corporation The world's largest manufacturer of microprocessors, supplying the processors used in more than 80 percent of the world's personal computers.

Intel has developed a wide range of processors and board-level products that are used in applications as varied as personal computers, automobiles, robots, and supercomputers, as well as the multibus architecture that is used in many industrial and proprietary applications. Intel also manufactures modems, fax modems, memory chips, flash memory products, and other peripheral devices. The name Intel is a contraction of Integrated Electronics.

Intelligent Input/Output

For more information on Intel Corporation, see www.intel.com.

Intelligent Input/Output Abbreviated I2O; pronounced “eye-two-oh.” A specification from Intel Corporation that divides the traditional device driver into two parts;

- The part concerned with managing the device
- The part concerned with interfacing to the operating system

By making this distinction, that part of the driver concerned with managing the device now becomes portable across operating systems.

I2O is also designed to work with intelligent input/output subsystems, with support for message passing between multiple independent processors, so that the host system can be relieved of all the interrupt-intensive tasks associated with servicing device drivers.

Several operating system vendors, including SCO, Microsoft, and Novell, have announced support for I2O in future products.

intelligent hub *See* smart hub.

intelligent terminal A terminal connected to a large computer, often a mainframe, that has some level of local computing power and can perform certain operations independently from the remote computer, but does not usually have any local disk-storage capacity.

See also dumb terminal.

Interactive Unix A version of Unix from Sun Microsystems based on AT&T’s System V Release 3.2 kernel.

See also Solaris.

Inter-Application Communication

Abbreviated IAC. In the Macintosh operating system, a feature of the System software that allows independent applications to share and exchange information. IAC takes two main forms:

- **Publish-and-subscribe**, which allows users to create documents made of components created by multiple applications.
- **Apple events**, which let one application control another. For example, two programs can share common data, and one program can request that the other perform some action.

IAC is often referred to as program linking in the System manuals.

Interexchange Carrier Abbreviated IXC; sometimes abbreviated IEC. A term coined from the Telecommunications Act of 1996 to describe voice and data long-distance telephone companies, including AT&T, MCI, Sprint, and Worldcom.

See also Competitive Local Exchange Carrier; Incumbent Local Exchange Carrier; Local Exchange Carrier.

interface The point at which a connection is made between two hardware devices, between a user and a program or operating system, or between two applications.

In hardware, an interface describes the logical and physical connections, as in RS-232-C, and is often considered synonymous with the term *port*.

A user interface consists of the means by which a program communicates with the user.

er, including a command line, menus, dialog boxes, online help systems, and so on. User interfaces can be classified as character-based, menu-driven, or graphical.

Software interfaces are application programming interfaces (APIs) and consist of the codes and messages used by programs to communicate behind the scenes.

See also graphical user interface.

interface standard Any standard way of connecting two devices or elements that have different functions.

Personal computers use many different interface standards. These include Small Computer System Interface (SCSI), Integrated Drive Electronics (IDE), and the Enhanced Small Device Interface (ESDI) for hard disks; RS-232-C and the Centronics parallel interface for serial devices and parallel printers; and the OSI Reference Model for LAN communications over a network.

Interior Gateway Protocol Abbreviated IGP. The protocol used on the Internet to exchange routing information between routers within the same domain.

See also Border Gateway Protocol; External Gateway Protocol.

Interior Gateway Routing Protocol

Abbreviated IGRP. A distance-vector routing protocol from Cisco Systems for use in large heterogeneous networks.

See also heterogeneous network.

interleaved memory A method of speeding up access by dividing dynamic RAM (DRAM) into two (or more) separate banks.

DRAM requires that its contents be updated at least every thousandth of a second, and while this update is taking place, it cannot be read by the processor. Interleaved memory divides available memory into banks so that the processor can read from one bank while the other is cycling and so does not have to wait. Because interleaved memory does not require special hardware, it is one of the most cost-effective ways of speeding up system operation.

See also wait state.

internal command Any operating system command that is not a separate program and is always available to the user. In MS-DOS, DIR, COPY, and TYPE are examples of internal commands.

See also external command.

internal modem A modem that plugs into the expansion bus of a personal computer or into the PCMCIA connector of a laptop computer.

See also external modem.

internal security Security measures taken to prevent unauthorized computer access from within an organization.

See also Intrusion Detection System.

International Business Machines

Corporation Abbreviated IBM, also known as “Big Blue.” Known originally for its huge range of mainframe computers, IBM introduced the IBM PC, which quickly emerged as an industry standard. Since its introduction in 1981, the PC has seen many changes, and an enormous number of companies worldwide now manufacture or

market hardware, software, and peripheral devices for IBM-compatible computers.

IBM also has a huge research effort, owns thousands of patents, and has introduced many innovative products, including the small, touch-sensitive TrackPoint, which replaces the mouse on many portable computers, a 1000MHz chip, and a 3.5-inch hard disk drive capable of storing up to 17GB of information.

For more information on IBM, see www.ibm.com.

See also Advanced Program-to-Program Communications; Advanced Peer-to-Peer Networking; Advanced Interactive Executive; AS/400; Systems Network Architecture.

International Organization for Standardization Sometimes mistakenly referred to as the International Standards Organization, and commonly referred to as ISO, which is not an abbreviation but a derivation of the Greek word *isos*, which means equal and which is a term that was adopted by the International Organization for Standardization. An international standards-making body, based in Geneva, with representatives from more than 100 countries. The ISO establishes global standards for communications and information exchange. ANSI is the U.S. member of the ISO.

The seven-layer OSI Reference Model for computer-to-computer communications is one of the ISO's most widely accepted recommendations in the area of networking.

For more information on ISO, see www.iso.ch.

See also OSI Reference Model.

International Telecommunication

Union Abbreviated ITU. The U.N. umbrella organization that develops and standardizes telecommunications worldwide.

The ITU also contains the CCITT, the International Frequency Registration Board (IFRB), and the Consultative Committee on International Radio (CCIR). In popular usage, CCITT standards are being referred to as ITU standards.

For more information on ITU, see www.itu.ch.

Internet The world's largest computer network, consisting of millions of computers supporting tens of millions of users in hundreds of countries. The Internet is growing at such a phenomenal rate that any size estimates are quickly out of date.

The Internet was originally established to meet the research needs of the U.S. defense industry, but it has grown into a huge global network serving universities, academic researchers, commercial interests, government agencies, and private individuals, both in the United States and overseas.

The Internet uses TCP/IP protocols, and Internet computers run many different operating systems, including VMS, Microsoft Windows 2000, and many variations of Unix.

No government agency, single person, or corporate entity controls the Internet; there is no Internet Corporation working behind the scenes. All decisions on methods and standards are made by committees based on input from users.

Internet use falls into several major areas, including:

- **E-mail** Electronic mail. Well over 80 percent of the people who use the Internet regularly use it for e-mail. You can send e-mail to recipients in more than 150 countries, as well as to subscribers of commercial online services, such as America Online, CompuServe, Delphi, Genie, and Prodigy.
- **IRC chat** A service that connects large numbers of users in real-time group discussions.
- **Mailing lists** Private discussion groups accessed by e-mail.
- **Usenet newsgroups** Larger public discussion groups that focus on a specific subject. Posts and threads in newsgroups are accessed using a newsreader.
- **World Wide Web** Hypertext-based system for finding and accessing Internet resources; the World Wide Web is one of the fastest growing and most exciting of all Internet applications.

Other Internet applications such as Gopher, FTP and anonymous ftp, and Telnet have either been overshadowed by the growth of the World Wide Web or have

seen their function absorbed into the popular Web browsers.

The sheer volume of information available through the Internet is staggering; however, because the Internet is a casual grouping of many networks, there is often no easy way to determine the location of specific information. This has led to the emergence of several prominent portal sites and a number of popular search engines.

Internet access can be via a permanent network connection or by dial-up through one of the many Internet Service Providers (ISP).

See also Internet address; Internet Architecture Board; Internet Assigned Numbers Authority; Internet Engineering Task Force; Internet Research Task Force; Internet Society; portal; Request for Comment; search engine; Usenet; World Wide Web; World Wide Web Consortium.

internet *See* internetwork.

Internet abbreviations Like any culture, the Internet world has developed a whole language of abbreviations, acronyms, and slang expressions. The following list describes some of the common terms you are likely to encounter in Usenet newsgroups or in your e-mail.

See also emoticon; smiley.

Abbreviation	Description
aTdHvAaNnKcSe	Thanks in advance
AWTTW	A word to the wise
BRB	Be right back

Internet abbreviations

Abbreviation	Description
BTW	By the way
CU	See you
FAQ	Frequently asked question
FAQL	Frequently asked question list
FOAF	Friend of a friend
F2F	Face to face
FWIW	For what it's worth
GR&D	Grinning, running, and ducking
IMHO	In my (sometimes not very) humble opinion
IWBNI	It would be nice if
IYFEG	Insert your favorite ethnic group
LOL	Laughing out loud
MEGO	My eyes glaze over
MOTAS	Member of the appropriate sex
MOTOS	Member of the opposite sex
MOSS	Member of the same sex
Ob-	Obligatory, as in ob-joke
OTOH	On the other hand
PD	Public domain

Abbreviation	Description
PITA	Pain in the ass
PMFJI	Pardon me for jumping in
RL	Real life
ROTFL	Rolling on the floor laughing
RTFM	Read the (expletive deleted) manual
SO	Significant other
TIA	Thanks in advance
TTFN	Ta ta for now
WRT	With respect to
YMMV	Your mileage may vary
\$0.02	My two cents worth

Internet address A location on the Internet. An Internet address takes the form `someone@abc.def.xyz`, in which `someone` is a user's name or part of a user's name, `@abc` is the network computer of the user, and `def` is the name of the host organization. The last three letters denote the kind of institution the user belongs to:

- `edu` for educational
- `com` for commercial
- `gov` for government
- `mil` for the military
- `org` for non-profit organizations

`net` for Internet administrative organizations

See also bang path; IP address.

Internet Architecture Board Abbreviated IAB. A technical advisory group of the Internet Society (ISOC). The IAB manages the editing and publication of Request for Comments (RFCs), serves as an appeals board, and provides other services to the ISOC.

For more information on the IAB, see www.iab.org/iab.

Internet Assigned Numbers

Authority Abbreviated IANA. A central clearinghouse for the assignment and coordination of Internet protocol parameters, such as Internet addresses, protocol variables and numbers, and domain names. For more information on the IANA, see www.isi.edu/iana.

Internet Connection Sharing In Microsoft Windows 98 and Windows 2000, a mechanism that allows networked computers to share a single connection to the Internet.

Internet Connection Sharing works with any kind of Internet connection, including modems, Integrated Services Digital Network (ISDN), cable modem, or Digital Subscriber Line (DSL).

Internet Content Provider Abbreviated ICP. A company that will design and deliver content for your Web site. *See also* Internet Service Provider.

Internet Control Message Protocol Abbreviated ICMP. An error-reporting protocol that works with Internet Protocol (IP) and provides the functions used for network-layer management and control. Routers send ICMP messages to respond to undeliverable datagrams by placing an ICMP

message in an IP datagram and then sending the datagram back to the original source. ICMP is also used by the Ping command. *See also* Ping.

Internet Engineering Task Force Abbreviated IETF. Provides technical and development services for the Internet and creates, tests, and implements Internet standards that are then approved and published by the Internet Society (ISOC). The technical work is done within the IETF working groups.

For more information on IETF, see www.ietf.org.

Internet Explorer Abbreviated IE. In Microsoft Windows, a Web browser used to display Internet resources. *See also* security zone.

Internet file types The Internet offers many opportunities for downloading files from a huge number of Internet hosts. These files may have been generated on different computer systems, so before you spend time downloading a file, it is important to understand the type of file you are dealing with. Table I.1 lists many of the common file types you may encounter.

TABLE I.1 INTERNET FILE TYPES

Filename Extension	Description
tar	A tape archive created by the tar utility.
Z	A file created by the compress utility. You must use uncompress to restore the file before you can use it.

TABLE I.1 INTERNET FILE TYPES *(CONTINUED)*

tar.z	A compressed tape archive file.
z	A file created by the pack utility. You must use unpack to restore the file before you can use it.
zip	A file created by PKZIP or WinZip. You must unzip the file before you can use it.
gz	A file created by the GNU gzip utility. You must decompress the file before you can use it.
hqx	A compressed Macintosh file.
sit	A Macintosh file compressed by StuffIt.
tif/tiff	A graphics file in TIF format.
gif	A graphics file in GIF format.
htm/html	A Web page in HyperText Markup Language.
jpg/jpeg	A graphics file in JPEG format.
mpg/mpeg	A video file in MPEG format.
txt	A text file.
1	An nroff source file.
ps	A PostScript file ready for printing.
uue	A uuencoded file. You must use uudecode before you can use the file.
uue.z	A compressed uuencoded file.
shar	A Usenet newsgroup archive file created by the shar utility.
shar.z	A compressed shar file.

Internet Group Management Protocol Abbreviated IGMP. An Internet protocol used in multicasting.

IGMP allows hosts to add or remove themselves from a multicast group. A multicast group is a collection of computers receiving packets from a host that is transmitting multicast packets with IP Class D addresses. Group members can join the group and leave the group; when there are no more members, the group simply ceases to exist.

See also Class D network; IP Multicast; multicasting.

Internet Inter-ORB Protocol Abbreviated IIOP. That part of the Common Object Request Broker Architecture (CORBA) that allows CORBA-based interaction over TCP/IP networks (including the Internet) by replacing Hypertext Transfer Protocol (HTTP).

See also Common Object Request Broker Architecture.

Internet Key Exchange *See* ISAKMP/Oakley.

Internet Mail Access Protocol Abbreviated IMAP. A protocol that defines how users can access and store incoming e-mail messages.

Internet mail servers use Simple Mail Transfer Protocol (SMTP) to move e-mail from one server to another and then use either IMAP or Post Office Protocol (POP) to manage the e-mail and store messages in the appropriate mailboxes.

IMAP allows users to download mail selectively, to look at a message header, to download only part of a message, to store messages on the server in a hierarchical structure, and to link to documents and Usenet newsgroups. IMAP also has strong authentication features and supports Kerberos. Search commands are also available so that users can locate messages based on their subject or header or based on content.

See also Kerberos; Post Office Protocol; Simple Mail Transfer Protocol

Internet Network File System Abbreviated Internet NFS. A TCP/IP-based protocol from Sun Microsystems used for sharing and accessing files remotely over the Internet.

See also Network File System.

Internet Network Information Center

Abbreviated InterNIC. A cooperative venture between AT&T, the National Science Foundation, and Network Solutions, Inc., that provides domain name registration and assigns IP addresses for use on the Internet.

For more information on InterNIC, see www.internic.net.

Internet NFS *See* Internet Network File System.

Internet Protocol Abbreviated IP, IP version 4, and IPv4. The session-layer protocol that regulates packet forwarding by tracking addresses, routing outgoing messages, and recognizing incoming messages in TCP/IP networks and the Internet.

The IP packet header contains the following fields:

- **Version** Version number of the protocol.
- **IHL (Internet header length)** Length of the header information. The header length can vary; the default header is five 32-bit words, and a sixth word is optional.
- **TOS (Type of Service)** Various levels or types of service.
- **Total length** Length of the datagram in bytes, which can be a minimum of 576 bytes to a maximum of 65,535 bytes.
- **Identification** Information that the receiving system can use to reassemble fragmented datagrams.
- **Flags** The first flag bit (DF) specifies that the datagram should not be fragmented and must therefore travel over networks that can handle the size without fragmenting it. The second flag bit (MF) indicates whether this is the last fragment.
- **Fragment Offset** An indication of where this datagram belongs in the set of fragments; is used during reassembly.
- **Time-to-Live (TTL)** Originally, the time in seconds that the datagram could be in transit; if this time was exceeded, the datagram was considered lost. Now interpreted as a hop count and usually set to the default value 32 (for 32 hops), this value is decremented by each router through which the packet passes. Once it reaches zero, the datagram is discarded.
- **Protocol** Identifies the transport-layer process intended to receive the datagram.

A value of 6 indicates Transmission Control Protocol (TCP), and a value of 17 indicates User Datagram Protocol (UDP).

- **Header checksum** Checksum for the header.
- **Source address** IP address of the sender.
- **Destination address** IP address of the recipient.
- **Options/padding** Optional information and padding.

See also IP multicast; IPv6; Transmission Control Protocol; Transmission Control Protocol/Internet Protocol.

Internet Relay Chat Abbreviated IRC. An Internet client/server application that allows large groups of people to communicate interactively; developed by Jarkko Oikarinen in Finland.

Specific channels are devoted to a particular subject, from the sacred to the profane, and channels come and go regularly as interest levels change. Each channel has its own name, usually prefaced by the pound sign (#), as in #hottub.

When you join a channel, you can see what others have already typed; when you type a line and press Enter, everyone else sees your text. Most, but not all, of the conversations are in English. If someone asks you for a password during an IRC session, don't be tempted to give it; someone is trying to trick you into divulging important information about your system.

See also listserver; mailing list; newsgroup; Usenet.

INTERNET PROTOCOL FIELDS

Version	IHL	TOS	Total Length	
Identification			Flags	Fragment Offset
Time-to-Live	Protocol	Header Checksum		
Source Address				
Destination Address				
Options/Padding				

Internet Research Task Force Abbreviated IRTF. An Internet organization that creates long- and short-term research groups concentrating on protocols, architecture, and technology issues.

For more information on IRTF, see www.irtf.org.

Internet Service Provider Abbreviated ISP. A company that provides commercial or residential customers access to the Internet via dedicated or dial-up connections. An ISP will normally have several servers and a high-speed connection to an Internet backbone. Some ISPs also offer Web site hosting services and free e-mail to their subscribers.

See also Internet Content Provider

Internet Society Abbreviated ISOC. An international organization that promotes cooperation and coordination for the Inter-

net, Internet applications, and internet-working technologies.

ISOC also coordinates the activities of many other Internet groups, including the Internet Architecture Board (IAB), the Internet Engineering Task Force (IETF), the Internet Assigned Numbers Authority (IANA), and the Internet Research Task Force (IRTF).

For more information on ISOC, see www.isoc.org.

internetwork Abbreviated internet. Two or more networks using different networking protocols, connected by means of a router. Users on an internetwork can access the resources of all connected networks.

Internetwork Operating System *See* Internetworking Operating System.

Internetworking Operating System

Abbreviated IOS; sometimes referred to as Internetwork Operating System Proprietary operating system software that runs on the Cisco family of routers.

IOS is stored in flash memory or in RAM in the router and supports remote access and protocol translation services, with connectivity provided by terminals, modems, computers, printers, or workstations. LAN terminal support includes Transmission Control Protocol/Internet Protocol (TCP/IP) for telnet and rlogin connections to IP hosts, tn3270 for connections to IBM hosts, and LAT (Local Area Transport) for connections to DEC hosts.

For WANs, IOS supports connections over a dial-up line supporting AppleTalk Remote Access, Serial Line Internet protocol (SLIP), Compressed SLIP (CSLIP), and Point-to-Point protocol (PPP) and supports asynchronous terminal connections using terminal emulation software providing telnet, rlogin, or IBM 3270 protocols.

IOS also supports Simple Network Management Protocol (SNMP), Common Management Information Protocol/Common Management Information Services (CMIP/CMIS) and IBM's Network Management Vector Transport (NMVP) network management systems.

You can configure IOS to control which networking protocols are routed, which protocols are used to do the actual routing, which interfaces on the router are used for specific routed protocols, and which processes will execute on the router.

Internetwork Packet eXchange Abbreviated IPX. Part of Novell NetWare's native protocol stack, used to transfer data between the server and workstations on the network. IPX packets are encapsulated and carried by the packets used in Ethernet and the frames used in Token Ring networks.

IPX packets consist of a 30-byte header which includes the network, node, and socket addresses for the source and the destination, followed by the data area, which can be from 30 bytes (only the header) to 65,535 bytes in length. Most networks impose a more realistic maximum packet size of about 1500 bytes.

The IPX packet header contains the following fields:

- **Checksum** For data integrity checking..
- **Packet length** Length of the packet in bytes
- **Transport control** Number of routers a packet can cross before being discarded
- **Packet type** The service that created the packet
- **Destination network** Network address of the destination network
- **Destination node** Media access control (MAC) address of the destination node
- **Destination socket** Address of the process running on the destination node
- **Source network** Network address of the source network
- **Source node** Media access control (MAC) address of the source node
- **Source socket** Address of the process running on the source node

See also Sequenced Packet Exchange.

Internetwork Packet Exchange Open Data-Link Interface Abbreviated IPXODI. In Novell NetWare, the client software that accepts data from the DOS Requester, adds header and address information to each data packet, and transmits the data packet as a datagram.

InterNIC *See* Internet Network Information Center.

interoperability The ability to run application programs from different vendors across LANs, WANs, and MANs, giving users access to data and applications across heterogeneous networks. A network user need not know anything about the operating system or the configuration of the network hardware to access data from the file server.

Interoperability is boosted by the increasing availability of products that conform to open standards rather than to specific proprietary protocols. The products work in accordance with national and internationally accepted standards.

interpreter A programming language translator that converts high-level program source code into machine language statements one line at a time.

Unlike a compiler, which must translate the whole program before execution can begin, an interpreter translates and then executes each line one at a time; this usually means that an interpreted program runs more slowly than a compiled program.

Java is an interpreted language, BASIC was often interpreted, although recent releases

have used a compiler instead, and C and C++ are always compiled.

See also compiler; Java; just-in-time compiler.

interprocess communication Abbreviated IPC. A term that describes all the methods used to pass information between two programs running on the same computer in a multitasking operating system or between two programs running on a network, including pipes, shared memory, message queues, sockets, semaphores, and Object Linking and Embedding (OLE).

See also Inter-Application Communication.

Interrupt 14 Abbreviated INT 14. The PC interrupt used to reroute messages from the serial port to the network interface card (NIC); used by some terminal-emulation programs.

interrupt A signal to the processor generated by a device under its control, such as the system clock, that interrupts normal processing.

An interrupt indicates that an event requiring the processor's attention has occurred, causing the processor to suspend and save its current activity and then branch to an interrupt service routine (ISR).

In the PC, interrupts are often divided into three classes:

- Internal hardware
- External hardware
- Software

The Intel family of processors supports 256 prioritized interrupts, of which the first 64 are reserved for use by the system hardware or by the operating system.

See also interrupt request.

interrupt controller A chip used to process and prioritize hardware interrupts. In the PC, a programmable interrupt controller responds to each hardware interrupt, assigns a priority, and forwards it to the main processor.

See also interrupt request.

interrupt handler Special software located in the operating-system kernel that manages and processes system interrupts. Also known as an interrupt service routine (ISR).

When an interrupt occurs, the processor suspends and saves its current activity and then branches to the interrupt handler. This routine processes the interrupt, whether it was generated by the system clock, a key-stroke, or a mouse click. When the ISR is complete, it returns control to the suspended process.

Each type of interrupt is processed by its own specific interrupt handler. A table, called the interrupt vector table, maintains a list of addresses for these specific interrupt handlers.

See also hooked vector.

interrupt request Abbreviated IRQ. Hardware lines that carry a signal from a device to the processor.

A hardware interrupt signals that an event has taken place that requires the processor's

attention. The interrupt may come from the keyboard, the network interface card (NIC), or the system's disk drives.

See also interrupt controller.

interrupt service routine *See* interrupt handler.

interrupt vector table A list of addresses, maintained by the operating-system kernel, for specific software routines known as interrupt handlers.

See also interrupt handler.

intranet A private corporate network that uses Internet software and TCP/IP networking protocol standards.

Many companies use intranets for tasks as simple as distributing a company newsletter and for tasks as complex as posting and updating technical support bulletins to service personnel worldwide. An intranet does not always include a permanent connection to the Internet.

See also extranet; Internet; internet.

intruder An unauthorized user of a computer system, usually a person with malicious intent.

See also cracker; firewall; hacker; Intrusion Detection System.

Intrusion Detection System Abbreviated IDS. A software package designed to detect specific actions on a network that are typical of an intruder or that might indicate an act of corporate espionage.

An IDS package monitors the network or the server for specific "attack signatures" that might indicate an active intruder is

inverse multiplexing

attempting to gain access to the network; such actions are carefully documented by the system.

Specific actions, such as opening or renaming certain important files, opening specific applications, downloading large amounts of data from key documents, or sending classified documents out as e-mail attachments, are also monitored by the IDS software.

See also firewall.

inverse multiplexing Abbreviated I-Mux. In communications, a technique that splits a high-speed data stream into two or more parts for transmission over multiple lower-speed channels. Once at the receiving end of the circuit, the data streams are combined into a single stream.

Inverse multiplexing provides bandwidth on demand, to avoid the high costs associated

with a leased line that may not be fully utilized at all times and to take advantage of low-cost low-speed lines.

See also multiplexing.

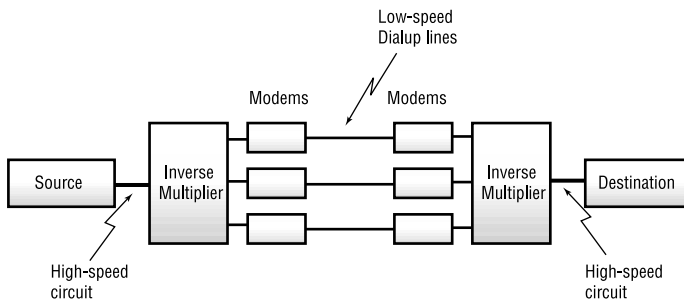
Inverse Multiplexing over ATM Abbreviated IMA. A specification defined by the Asynchronous Transfer Mode (ATM) Forum that defines a method of inverse multiplexing ATM over two or more T1 circuits.

See also Asynchronous Transfer Mode; inverse multiplexing.

inverted backbone A network architecture in which the wiring hub and routers become the center of the network; all the network segments attach to this hub.

See also backbone.

INVERSE MULTIPLEXING



I/O *See* input/output.

I/O bound *See* input/output bound.

Iomega Corporation A major manufacturer of removable storage media, best known for its popular Zip drive, which has

shipped more than 12 million units worldwide. Iomega also makes the higher-capacity Jaz drives, the Ditto line of tape-backup products, and Klik! storage devices for digital cameras and hand-held computers.

For more information on Iomega Corporation, see www.iomega.com.

ION *See* Integrated On-Demand Network.

IOS *See* Internetworking Operating System.

IP *See* Internet Protocol.

IP address The unique 32-bit number that identifies a computer on the Internet or some other Internet Protocol network.

An IP address is usually written (in decimal) as four numbers separated by dots or periods and can be divided into two parts. The network address is made up from the high-order bits in the address, and the host address comprises the rest. In addition, the host part of the address can be further subdivided to allow for a subnet address.

These numbers are very difficult for most people to remember, so humans tend to refer to computers by their domain names instead.

See also address classes; domain name; dotted decimal; Internet address; subnet.

IPC *See* interprocess communication.

ipconfig A utility program used to list various Internet Protocol configuration information, including host address, subnet mask, and gateway addresses.

See also IP address; netstat; Ping; subnet mask; tracer.

IP datagram *See* Internet Protocol.

IP Multicast An Internet standard that allows a single host to distribute data to multiple recipients.

IP multicasting can deliver audio and video content in real time so that the person using the system can interact with the data stream. A multicast group is created, and every member of the group receives every datagram. Membership is dynamic; when you join a group, you start to receive the datastream, and when you leave the group, you no longer receive the datastream.

See also Internet Group Management Protocol; multicast backbone; multicasting; streaming.

IPng *See* IPv6.

I-PNNI *See* Integrated-Private Network-to-Network Interface.

IP over ATM A set of proposals that define general methods of integrating Internet Protocol over an Asynchronous Transfer Mode (ATM) network, while preserving the best features of the connectionless IP and connection-oriented ATM.

These proposals have a variety of names, including layer 3 switching, high-speed routing, short-cut routing, and multilayer switching, and come from a variety of sources, including the ATM Forum, the Internet Engineering Task Force, and a vendor collective known as Layer 3 Switching/Routing Advocates.

See also IP switching.

IPSec A suite of protocols under development by the Internet Engineering Task Force (IETF) designed to add security provisions to the Internet Protocol (IP). Also known as IP Security.

The Authentication Header (AH) ensures that the datagram has not been tampered with during transmission, and the Encapsulating Security Payload (ESP) defines encryption methods for IP data.

IPSec operates in two modes:

- **Transport mode** AH or ESP is placed immediately after the original IP datagram header and provides security between two end systems such as a server and a workstation.
- **Tunnel mode** The original IP datagram is placed inside a new datagram, and AH and ESP are inserted between the IP header of the new packet and the original IP datagram. The new header points to the tunnel endpoint, and the original header points to the final destination of the datagram. Tunnel mode is best suited to Virtual Private Network (VPN) use, securing remote access to your corporate network through the Internet.

See also encapsulation; Internet Protocol; ISAKMP/Oakley; tunneling; Virtual Private Network.

IP security *See* IPSec.

IP switching **1.** A switch, developed by Ipsilon Networks, that combines intelligent Internet Protocol (IP) routing with high-speed Asynchronous Transfer Mode (ATM) switching hardware. The IP protocol stack is implemented on ATM hardware, allowing

the system to adapt dynamically to the flow requirements of the network traffic as defined in the packet header.

2. A technique that uses network-layer protocols which provide routing services to add capabilities to layer 2 switching. IP switching locates paths in a network by using routing protocols and then forwards packets along that route at layer 2. IP switching is designed for networks that use switches rather than networks built around repeater hubs and routers.

See also IP over ATM.

IPv4 *See* Internet Protocol.

IPv6 The next version of the Internet Protocol, also called IP version 6 or IPng for IP next generation.

The most striking feature of IPv6 is that it uses a 128-bit address space rather than the 32-bit system in use today. This provides for a truly astronomical number of possible addresses. The address format consists of 8 sections separated by colons. Each section contains 16 bits expressed as 4 hexadecimal numbers. An address might look like this:

1234:5678:9ABC:DEF0:1234:5678:9ABC:
:DEF0

In any address, one set of leading zeros can be replaced by two colons.

In addition to the 128-bit address space, IPv6 designates a 128-bit hierarchical address for point-to-point communication called an Aggregatable Global Unicast Address Format (AGUAF). In this format, a top level aggregator (TLA) is assigned a block of addresses by bodies such as the Internet Assigned Numbers Authority

(IANA). In turn, the TLA assigns addresses to a next level aggregator (NLA), which in turn assigns addresses to the site level aggregator (SLA). In turn, the SLA assigns blocks of contiguous addresses to its subscribers. The last level is the host interface ID, which identifies a single host interface. Companies assign host interface IDs by using a unique number on the subnet.

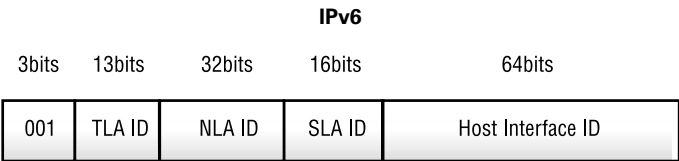
IPv6 assigns addresses to interfaces, and because a node can have multiple interfaces, it can also have multiple IP addresses. A single interface can also have multiple addresses.

An address can be multicast, unicast, or anycast, which is a special case of multicast. The IP datagram header has been simplified, by dropping some fields and making others optional. IPv6 also allows for several types of optional header extensions, some of which might be used for specialized handling instructions. In addition, IPv6 includes the IPSec security extensions.

See also Internet Protocol; IPSec.

IP version 4 See Internet Protocol.

IP version 6 See IPv6.



IPX See Internetwork Packet eXchange.

IPX address In Novell NetWare, a network address. Also known as an IPX internetwork address.

The IPX address includes a 4-byte network number assigned to every segment on a LAN (the IPX external network number), a 6-byte node number that identifies a specific system and is usually derived from the interface card address originally assigned by the manufacturer, and a 2-byte socket number. See also IPX external network number.

IPX external network number In Novell NetWare, a hexadecimal number that

identifies a network cable segment, assigned when the NetWare Internetwork Packet eXchange (IPX) protocol is bound to a network interface board in the server. An IPX external network number can have from one to eight digits (1 to FFFFFFFE).

IPX internal network number In Novell NetWare, a hexadecimal number that identifies a NetWare server. Each server on the network must have a unique Internetwork Packet Exchange (IPX) internal network number. The number can have from one to eight digits (1 to FFFFFFFE) and is assigned to the server during NetWare installation.

IPX internetwork address *See* IPX address.

IPXODI *See* Internetwork Packet Exchange Open Data-Link Interface.

IRC *See* Internet Relay Chat.

IrDA *See* Infrared Data Association.

IRF *See* Inherited Rights Filter.

IRM *See* Inherited Rights Mask.

IRQ *See* interrupt request.

IRTF *See* Internet Research Task Force.

ISA *See* Industry Standard Architecture; Integrated Services Architecture.

ISAKMP/Oakley Abbreviation for Internet Security Association and Key Management Protocol/Oakley security key management protocol. Also known as the Internet Key Exchange (IKE).

A security protocol that automatically manages the exchange of secret symmetric keys between sender and receiver.

See also IPSec.

ISDN *See* Integrated Services Digital Network.

ISO 10646 A 4-byte character encoding scheme that includes all the world's national standard character encodings, defined by the International Organization for Standardization (ISO). The 2-byte Unicode characters set maps into a part of ISO 10646.

See also American Standard Code for Information Interchange; double-byte character set; Unicode.

ISO *See* International Organization for Standardization.

ISOC *See* Internet Society.

isochronous service A method of transmitting real-time data using preallocated bandwidth on a communications link, allowing time-synchronized transmissions with very little delay. Isochronous service is required for real-time data such as synchronized voice and video, in which delays in packet delivery would be unacceptable.

Asynchronous Transfer Mode (ATM) can provide isochronous service because its cells are always the same size, so it is possible to guarantee accurate and timely delivery of packets. Other networks can provide isochronous service by using a priority scheme to dedicate bandwidth to video traffic.

Iso-Ethernet The IEEE standard 802.9a, which describes a combination of Ethernet and Integrated Services Digital Network (ISDN) on the same cable.

As well as providing standard 10Mbps Ethernet connections over Category 3 or Category 5 unshielded twisted-pair wiring, Iso-Ethernet also provides up to 96 ISDN Basic Rate B channels running at 64Kbps connected to internal or external users or systems. One D channel is used for control and signaling information.

The ISDN channels can be accessed over the network via a single point of connection, which in turn is connected to the ISDN service; this removes the need for each workstation to have its own ISDN hardware.

See also Integrated Services Digital Network.

ISO/OSI model *See* OSI Reference Model.

ISP *See* Internet Service Provider.

ISR Abbreviation for interrupt service routine.

See interrupt handler.

ISV *See* Independent Software Vendor.

ITU *See* International Telecommunication Union.

IXC *See* Interexchange Carrier.

J

jabber A continuous and meaningless transmission generated by a network device, usually the result of a user error or a hardware malfunction.

Java A programming language and development environment created by Sun Microsystems, designed to create distributed executable applications for use with a Web browser containing a Java runtime environment.

Java technology has been licensed by literally hundreds of companies, including IBM, Microsoft, Oracle, Hewlett-Packard, and others interested in developing Web-based and platform-independent applications.

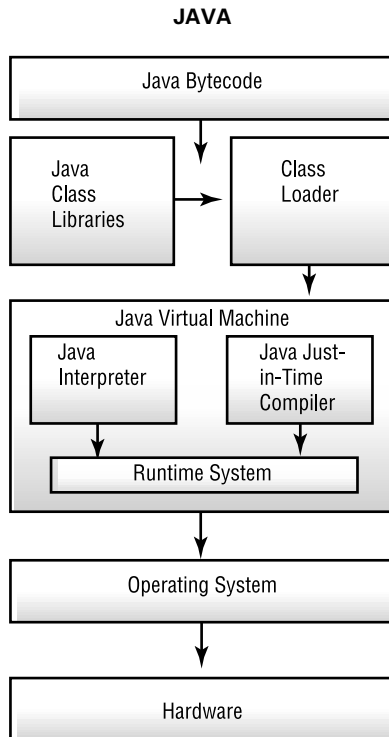
The Java programming language is a portable, object-oriented language, loosely modeled after C++, with some of the more troubling C++ constructs such as pointers removed. This similarity to C and C++ is no accident; it means that the huge population of professional programmers can quickly apply their previous C experience to writing code in Java.

Java is designed to support networking and networking operations right from the start and begins with the assumption that it can trust no one, implementing several important security mechanisms.

Java is architecturally neutral, does not care about the underlying operating system, and is portable because it makes no assumptions about the size of data types and explicitly defines arithmetic behavior. Java is also multithreaded to support different threads of execution and can adapt to a changing environment by loading classes as they are needed, even across a network.

Rather than writing code targeted at a specific hardware and operating-system platform, Java developers compile their source code into an intermediate form of bytecode that can be processed by any computer system with a Java runtime environment. The Java class loader transfers the bytecode to the Java Virtual Machine (JVM), which interprets the bytecode for that specific platform. Java class libraries, those files that make up the standard application programming interface (API), are also loaded dynamically. The runtime environment then executes the application, which can run within a Web browser or as a stand-alone application.

See also bytecode; interpreter; Java applet; Java Developer's Kit; Java Virtual Machine; just-in-time compiler; sandbox.



Java applet A small program written in the Java programming language designed to add a specific capability to a Web page. The applet is stored on the Web server and is downloaded to and executes within the browser.

Java applets are inserted into a Web page using the `<APPLET>` tag, which indicates the initial window size for the applet, various parameters for the applet, and the location from which the browser can download the bytecode file.

Stand-alone programs written in the Java programming language that can execute without needing the browser are known as Java applications.

See also bytecode; Java; Java Developer's Kit; Java Virtual Machine; just-in-time compiler; sandbox.

Java Database Connectivity Abbreviated JDBC. An application programming interface (API) that allows developers to write Java applications that can access a database.

See also Open Database Connectivity.

Java Developer's Kit Abbreviated JDK. A collection of software-development tools provided by Sun Microsystems for writing Java applications. This kit, distributed free, contains a Java compiler, an interpreter, a debugger, an applet viewer, and documentation.

JavaScript A simple scripting language created by Netscape Communications and Sun Microsystems that allows developers to add a specific capability to a Web page.

JavaScript is relatively easy to write when compared with the Java programming language, but it is slower in execution and has far fewer application programming interface (API) functions available. A JavaScript-compliant Web browser, such as Netscape Navigator, is necessary to run the JavaScript code.

See also Java; script.

JavaSoft A subsidiary of Sun Microsystems responsible for developing and promoting the Java programming language and related products.

For more information on JavaSoft, see www.javasoft.com.

Java Virtual Machine The runtime environment for Java applets and applications; sometimes abbreviated JVM.

The Java Virtual Machine creates a simulated environment that provides the same interface to applications, no matter what hardware and operating system combination is in use.

The Java Virtual Machine consists of two layers:

- The top layer is compatible with all Java applications.
- The bottom layer is compatible with a specific computing platform, such as the Intel family of processors.

See also Java.

JDBC *See* Java Database Connectivity.

JDK *See* Java Developer's Kit.

Jini A technology from Sun Microsystems that is based on Java concepts and that allows spontaneous networking of a large variety of network devices.

Jini creates a federation of Java Virtual Machines on a network and allows users, devices, data, and applications to connect dynamically to share information and to perform tasks without prior knowledge of one another's capabilities.

See also Java.

JIT compiler *See* just-in-time compiler.

jitter A type of distortion found on analog communications lines that results in data-transmission errors.

job A unit of work done by a computer, usually in a mainframe environment. A job can be the one-time execution of a single file or the execution of a whole series of programs to accomplish a complex task.

See also job control.

job control A mechanism used to manage jobs, allowing them to be started,

stopped, and moved between the background and the foreground.

Joint Photographic Experts Group

Abbreviated JPEG; also sometimes abbreviated JPG. An image-compression standard and file format that defines a set of compression methods for high-quality images such as photographs, single video frames, or scanned pictures. JPEG does not work very well when compressing text, line art, or vector graphics.

JPEG uses lossy compression methods that result in some loss of the original data.

When you decompress the original image, you don't get exactly the same image that you started with, although JPEG was specifically designed to discard information not easily detected by the human eye.

JPEG can store 24-bit color images in as many as 16 million colors; files in Graphics Interchange Format (GIF) form can only store 256 colors.

See also Graphics Interchange Format; lossy compression.

JPEG *See* Joint Photographic Experts Group.

JPG *See* Joint Photographic Experts Group.

jukebox A high-capacity storage device that uses an autochanger mechanism to mount or dismount disks automatically. A jukebox typically contains as many as 50 disks and a mechanism that picks up disks

from a bay and loads them into the drives as they are needed.

See also high-capacity storage system.

Julian date A method of representing the date that is often used in computer systems. The first digit represents the year, and the remaining digits represent the day of the year, counting from January 1.

See also Y2K problem.

jumper A small plastic and metal connector that completes a circuit, usually to select one option from a set of several user-definable options. Jumpers are often used to select one particular hardware configuration from a choice of configurations.

junction point In Microsoft Windows 2000, a drive or disk partition attached to an empty folder on an NTFS volume. Using junction points allows you to mount more than 26 (A–Z) storage volumes.

See also mounted drive.

just-in-time compiler Sometimes abbreviated JIT compiler. A compiler that receives the bytecode of a Java application and compiles the bytecode on the fly.

Java bytecode processed by a just-in-time compiler executes faster than the same bytecode processed by a Java Virtual Machine.

See also Java; Java Virtual Machine.

JVM *See* Java Virtual Machine.

K

K *See* kilo-.

K56Flex A modem technology from Rockwell Semiconductor Systems and Lucent Technologies that provides up to 56Kbps downstream and up to 40Kbps upstream. Replaced by the V.90 standard.

See also V.90; X2.

Kb *See* kilobit.

KB *See* kilobyte.

Kbit *See* kilobit.

Kbps *See* kilobits per second.

Kbyte *See* kilobyte.

KCC *See* Knowledge Consistency Checker.

keep-alive signal A signal transmitted to maintain a communications circuit during periods of idleness and to prevent the circuit from timing out and terminating the connection.

Kerberos A network security system developed as a part of Project Athena at MIT. Kerberos is used to authenticate a user who is asking for permission to use a particular network service.

Kerberos can be used to control the initial connection to a server or to authenticate every single request between a client and the server. It grants tickets to a client to allow the use of a specific service and is secure even on a nonsecure network.

Kerberos takes the following precautions:

- Passwords are never sent over the network unencrypted, making it impossible for network snoopers to capture passwords.
- All Kerberos messages are timestamped so that they cannot be captured and then replayed at a later time; Kerberos does not accept old messages.
- When you request access to a service, to access a file server, for example, Kerberos gives you a “ticket” that is valid for access to the file server but not valid for any other service. When you try to connect to the server, you send your ticket with the request. Once the server knows who you are, the server decides whether to grant you access. Tickets also expire, and if your session lasts longer than the predefined limit, you will have to reauthenticate yourself to Kerberos to get a new ticket.

Kerberos is named after the three-headed dog Cerberus, who guards the gates of the underworld in Greek mythology.

See also authentication; certificate; digital signature.

Kerberos Distribution Center Abbreviated KDC. In Microsoft Windows 2000 Server, a Kerberos function that runs on every domain controller and controls the distribution of Kerberos keys and tickets.

See also Kerberos.

KDC *See* Kerberos Distribution Center.

Kermit A file-transfer protocol developed at Columbia University and placed in the public domain that is used to transfer files between PCs and mainframe computers over standard telephone lines.

Data is transmitted in variable-length blocks up to 96 characters in length, and each block is checked for transmission errors. Kermit detects transmission errors and initiates repeat transmissions automatically.

See also Xmodem; Ymodem; Zmodem.

kernel The fundamental part of an operating system. The kernel stays resident in memory at all times, often hidden from the user, and manages system memory, the file system, and disk operations.

The kernel also runs processes and provides interprocess communications between those processes, including synchronizing events, scheduling, passing messages, managing input and output routines, and managing memory.

See also Linux; Mach, microkernel, shell.

key 1. An entry in the Microsoft Windows Registry that contains an element of configuration information. A key may also be empty and have no value set.

2. In encryption, a mechanism used to encode a message.

3. In a database system, a unique value used to identify data records. Also known as a primary key.

See also public key encryption; sub-key.

keyboard buffer A small amount of system memory used to store the most recently typed keys, also known as the type-ahead

buffer. Some utilities or shells let you collect a number of keystrokes or commands and edit or reissue them.

keyboard template A plastic card that fits over certain keys (usually the function keys) on the keyboard as a reminder of how to use them. These templates are specific to an application, and they can be a useful reminder for new or occasional users.

key combination In menu-driven and graphical user interfaces, some menu commands can be executed by certain combinations of keystrokes, also known as shortcut keystrokes. By using key combinations, users can bypass the menus and so speed up operations.

keypass attack *See* brute-force attack.

keyspace attack *See* brute-force attack.

key redefinition The ability of an application to assign different functions to specific keys.

keystroke The action of pressing and then releasing a key on the keyboard to initiate some action or enter a character.

keyword Any of the words, sometimes known as reserved words, that make up the vocabulary of a particular programming language or set of operating system commands and utilities.

kilo- A prefix indicating 1000 in the metric system. Because computing is based on powers of 2, kilo usually means 2^{10} , or 1024. To differentiate between these two uses, a lowercase k is used to indicate 1000

kilobaud

(as in kHz), and an uppercase K is used to indicate 1024 (as in KB).

See also mega-.

kilobaud One thousand baud. A unit of measurement of the transmission capacity of a communications channel.

See also baud.

kilobit Abbreviated Kb or Kbit. 1024 bits (binary digits).

See also gigabit; megabit.

kilobits per second Abbreviated Kbps. The number of bits, or binary digits, transmitted every second, measured in multiples of 1024 bits per second. Used as an indicator of communications transmission rates.

See also megabits per second.

kilobyte Abbreviated K, KB, or Kbyte. 1024 bytes.

See also exabyte; gigabyte; megabyte; petabyte; terabyte.

kluge Pronounced “klooj.” A program that doesn’t work as well as it should, is not carefully designed, and is not well written. A kluge may also be a program that works, but for all the wrong reasons, or only under very specific, highly unrealistic conditions.

See also hack.

Knowledge Consistency Checker

Abbreviated KCC. In Microsoft Windows 2000 Server, an Active Directory function that monitors and configures replication

connection objects between domain controllers.

Korn shell An upward-compatible extension to the original Unix shell, written by David Korn and released as part of System V.

The Korn shell is now the default shell on many Unix systems, particularly those based on System V, including UnixWare and many others.

Because the Korn shell is an extension of the Bourne shell, everything that works in the Bourne shell also works in the Korn shell. The Korn shell also adds the following:

- Interactive editing of the command line with either `vi` or `emacs`
- Better function definitions providing local variables and the ability to create recursive functions
- Extensive pattern matching for filenames, similar to regular expressions

Several features were adapted from the C shell, including:

- Command history allowing retrieval and reuse of previous commands
- Job control and the mechanism for moving jobs between the background and the foreground
- Aliases for abbreviated command names
- The tilde (~) used as a shorthand for the name of the home directory

See also Bash shell; Bourne shell; C shell; Unix shell.

L

L2 Cache *See* secondary cache.

L2TP *See* Layer 2 Tunneling Protocol.

LAN *See* local-area network.

LANalyzer for Windows A Microsoft Windows-based software product from Novell used to troubleshoot and monitor network performance and activity.

The LANalyzer main screen is a dashboard of gauges displaying real-time statistics, including LAN utilization, packets per second, and errors per second. User-specified alarms can also be set on these gauges, and detailed statistics screens plot frame rate, utilization, and error rate over time.

LAN-aware An application that contains mechanisms for file and record locking to prevent multiple simultaneous access when used on a network. The term is often applied in a broader sense to any application capable of running in a networked environment.

LANdesk Management Suite A package of network management utilities from Intel Corporation that includes hardware and software inventory, server monitoring, client monitoring and control, network traffic monitoring, virus protection, remote access, remote control, and print-queue management.

LANdesk also includes a Desktop Management Interface (DMI) remote management

console and DMI desktop agents or service layers for MS-DOS and Windows clients.

See also Desktop Management Interface.

LAN Distance A hardware/software combination product from Network Communications Corp. used to troubleshoot and monitor performance and activity on Ethernet and token-ring LANs.

The LAN Distance hardware includes an on-board processor and a large buffer for packet capturing. The package can decode most LAN protocols in common use, including NetWare, TCP/IP, DECnet, Banyan, Network File System (NFS), AppleTalk Filing Protocol (AFP), Open Systems Interconnect (OSI), Systems Network Architecture (SNA), NetBEUI, Xerox Network Services (XNS), and Server Message Block (SMB).

LANE *See* LAN Emulation.

LAN Emulation Abbreviated LANE. An Asynchronous Transfer Mode (ATM) specification that defines ways of connecting legacy local-area networks such as Ethernet and token ring to an ATM backbone, allowing ATM to replace older and slower backbone circuits.

LANE provides the translation services needed between the two types of networks and is totally transparent to the high-level protocols. LANE consists of four major components:

- Broadcast and unknown server (BUS), which manages address resolution.

- **LAN Emulation Client (LEC)**, which runs on all nodes connected to the ATM network, initiates data transfers between other clients, and makes mapping requests between MAC (media access control) addresses and ATM addresses.
- **LAN Emulation Configuration Server (LECS)**, which provides configuration information and tracks on which virtual LAN each LAN Emulation Client operates.
- **LAN Emulation Server (LES)**, which provides address mapping between MAC addresses and ATM addresses and usually runs on the same node as the BUS.

See also Asynchronous Transfer Mode; Integrated-Private Network-to-Network Interface; IP over ATM; IP switching.

LANRover A remote access package from Shiva Corporation that lets network users log in from their PC or portable computer when away from the office.

LAN Server An IBM network operating system, based on a version of OS/2, that runs on Intel processors as well as on the PowerPC.

LAN Server supports Microsoft Windows, MS-DOS, OS/2, and Macintosh clients as well as AppleTalk, IPX/SPX, TCP/IP, and NetBIOS protocols. LAN Server supports domain directory services and encrypted passwords, but does not include C-2 level certification.

LANtastic A popular peer-to-peer network operating system, from Artisoft, Inc., that runs with MS-DOS, OS/2, or Microsoft

Windows and supports Microsoft Windows, MS-DOS, OS/2, Macintosh, and Unix clients.

The network can be run with all stations sharing files with all other stations or, for enhanced performance, with one PC acting as a dedicated file server. LANtastic supports an unlimited number of users, includes built-in CD-ROM, network e-mail, and network fax support, scheduling and network management, file-level security, and Internet access and can connect easily into the NetWare environment.

LAN WorkGroup Novell's connectivity package that is located and managed centrally from the NetWare file server.

The system includes transparent file sharing, terminal emulation, remote command execution, and printer-redirection features for both MS-DOS and Windows clients. The LAN WorkGroup differs from its stand-alone counterpart, LAN WorkPlace, in that it can be managed from the file server.

See also LAN WorkPlace.

LAN WorkPlace Novell's connectivity software for MS-DOS and Windows clients. Features include terminal emulation, transparent Network File System (NFS) file sharing, and a Web browser.

See also LAN WorkGroup.

LAP *See* Link Access Procedure.

LAP-B *See* Link Access Procedure-Balanced.

LapLink A popular communications package from Traveling Software, used to

transfer and synchronize files between a laptop computer and a desktop or networked computer.

LAPM See Link Access Procedure for Modems.

laptop computer A small, portable computer that is light enough to carry comfortably, with a flat screen and keyboard that fold together.

Advances in battery technology allow laptop computers to run for many hours between charges. Laptop computers often have a thin, backlit or sidelit liquid-crystal display (LCD) or a plasma screen. Some models can mate with a docking station to perform as a full-sized desktop system back at the office, and many new laptop computers allow direct connection to the network with PCMCIA network interface cards. In some laptop computers, a set of business applications is built in to ROM.

See also hand-held computer; mobile computing; notebook computer; PC Memory Card International Association; port replicator; wireless communications.

Large Internet Packet Abbreviated LIP. A mechanism that allows the Novell NetWare internetwork packet size to be increased from the default 576 bytes, thus increasing throughput over bridges and routers.

LIP allows workstations to determine the packet size based on the largest packet supported by the router; the larger packet size is also supported by Ethernet and token-ring networks.

last known good configuration In Microsoft Windows NT, the last configuration that was used to boot the computer successfully. Windows NT saves this configuration and offers it as a startup option during the boot process.

last mile A term, not to be taken too literally, that describes the link between a customer site and the telephone company's central office. This link is often the most expensive and least efficient connection in the telephone company's system and is often a barrier to high-speed services.

latency The time delay involved in moving data traffic through a network. Latency can arise from several sources:

- Propagation delay, which is the time it takes the data to travel the length of the line.
- Transmission delay, which is the time it takes to move the data across the network media.
- Processing delay, which is the time it takes to establish the route, encapsulate the data, and take care of other switching tasks.

See also propagation delay.

Layer 2 Tunneling Protocol Abbreviated L2TP. A proposed standard for a secure, high-priority, temporary communications path through the Internet.

L2TP is based on Cisco's Layer 2 Forwarding (L2F) protocol and includes features from Microsoft's Point-to-Point Tunneling Protocol (PPTP). A tunnel is established from an Internet Service Provider (ISP) to a corporate site, and information is transmitted through the tunnel. Once L2TP is set up,

the user communicates with the corporate system through what seems to be a direct dial-up connection; the ISP is effectively removed from the picture.

See also tunneling.

LCD *See* liquid-crystal display.

LCD monitor A monitor that uses liquid-crystal display (LCD) technology. Many laptop and notebook computers use LCD monitors because of their low power requirements.

LDAP *See* Lightweight Directory Access Protocol.

leaf object In Novell NetWare Directory Services (NDS), an object that cannot contain any other objects; also known as a non-container object. Leaf objects can be printers, servers, print queues, or even users.

See also container object.

Learning Tree International A major provider of training classes and computer-technology certifications. Learning Tree International offers more than 150 networking and computer-related training classes in the United States, Europe, and Asia, as well as 27 professional certification programs.

For more information about Learning Tree International, see www.learningtree.com.

Learning Tree International Certified Professional A professional certification from Learning Tree International, available with a large number of subject specializations, including Local Area Networks,

Wide Area Networks, Internetworking, PC Service and Support, and Unix Systems.

See also Learning Tree International.

leased line A communications circuit or telephone line reserved for the permanent use of a specific customer; also called a private line.

See also dedicated line.

LEC *See* Local Exchange Carrier.

legacy application An application designed to run on what has now become a legacy system and that continues to be used.

A legacy application may perform admirably and provide little reason to upgrade to a more up-to-date system, or it may be inefficient, perform poorly, and continue in use because replacing it is simply too expensive.

See also legacy system; legacy wiring.

legacy system A computer system, developed to solve a particular business need, which, due to the passage of time, has become obsolete.

Legacy systems do not conform to the technical standards or performance standards of up-to-date systems. There is usually a requirement to maintain backward compatibility with or connections to legacy systems.

legacy wiring Preinstalled wiring that may or may not be suitable for use with a network.

See also legacy system.

Lempel-Ziv-Welch Abbreviated LZW, an algorithm used in data compression, developed by Abraham Lempel, Jacob Ziv, and Terry Welch.

LZW is a general-purpose compression algorithm suitable for use on almost any type of data. It is fast in both compressing and decompressing data and does not require the use of floating-point operations.

LZW is a lossless compression method and is used in several image formats such as Graphics Interchange Format (GIF) and Tag Image File Format (TIFF), as well as a part of the V.32 bis modem compression standard and PostScript Level 2.

The patent for LZW compression is held by Unisys, which levies a fee on any application that uses the LZW compression algorithm.

See also lossless compression; lossy compression.

level 2 cache A secondary static RAM cache located between the primary cache and the rest of the system. A level 2 cache is often larger than the primary cache, and it is usually slower. In Intel's Pentium II processor, the level 2 cache is implemented as a part of the package that holds the CPU.

See also cache; memory cache.

Lexmark International, Inc. A major manufacturer of printers, based in Lexington, Kentucky; spun off from IBM in 1991. Lexmark markets laser and ink jet printers for both home and office use, as well as laser-printer management software for servers.

For more information on Lexmark International, Inc., see www.lexmark.com

LF *See* line feed.

light-wave communications Usually refers to communications using fiber-optic cables and light generated by light-emitting diodes (LEDs) or lasers.

Lightweight Directory Access Protocol Abbreviated LDAP. A directory services specification that has recently gained wide acceptance.

A directory service provides "white page" service for an organization to help people find other people and network services and can also be used over the Internet. A directory service is a database that can be manipulated in a variety of ways to provide this information.

LDAP can be used as a directory service by itself, or it can be used for "lightweight" access to an X.500-compliant directory service. LDAP is a subset of X.500 that runs over TCP/IP; X.500 runs on OSI-compliant systems. Many companies have announced their intention to support LDAP in upcoming products.

See also Active Directory; Novell Directory Services; X.500.

LIM EMS *See* Expanded Memory Specification.

limited-distance modem *See* line driver.

line adapter In communications, a device that converts a signal into a form suitable for transmission over a communications channel.

A modem is a specific type of line adapter used to convert the computer's digital

signals into analog form so that they can be transmitted over a telephone line.

line analyzer Any device that monitors and displays information about a transmission on a communications channel. A line analyzer is used for troubleshooting and load monitoring.

line-of-business application *See* mission-critical application.

line driver In communications, a hardware device used to extend the transmission distance between computers that are connected using a serial interface based on the RS-232 standard. A line driver is required at each end of the line. Also known as a baseband modem, limited-distance modem, or short-haul modem.

See also short-haul modem.

line feed Abbreviated LF. A printer command that advances the paper in the printer by one line, leaving the print head in the same position. In the ASCII character set, a line feed has a decimal value of 10.

See also American Standard Code for Information Interchange; carriage return; Extended Binary Coded Decimal Interchange Code.

line printer Any high-volume printer that prints a complete line at a time, rather than printing one character at a time (as a dot-matrix printer or a daisy-wheel printer does) or one page at a time (as a laser printer does).

Line printers are very high-speed printers and are common in mainframe environments.

line-sharing device A small electronic mechanism that allows a fax machine and a telephone answering machine to share the same phone line. The device answers the call and listens for the characteristic high-pitched fax carrier signal. If this signal is detected, the call is routed to the fax machine; if it is not present, the call is sent to a telephone or answering machine.

line-of-sight An unobstructed path between the transmitter and receiver. Laser, microwave, and infrared transmissions usually require a clear line-of-sight.

See also wireless communications.

line speed In communications, the transmission speed that a line will reliably support for any given grade of service.

See also connection speed; data-transfer rate.

link On a Web page or a hypertext document, a connection between one element and another in the same or in a different document.

Link Access Procedure Abbreviated LAP. The link-level protocol specified by the CCITT X.1 recommendation used for communications between DCE (data communications equipment) and DTE (data terminal equipment).

See also Consultative Committee for International Telephony and Telegraphy; Link Access Procedure-Balanced.

Link Access Procedure-Balanced Abbreviated LAP-B. A common CCITT bit-oriented, data-link layer protocol used to

link terminals and computers to packet-switched networks. It is equivalent to the HDLC (High-level Data Link Control) asynchronous balanced mode, in which a station can start a transmission without receiving permission from a control station.

Link Access Procedure for Modems

Abbreviated LAPM. The data-link protocol used by V.32 error-correcting modems.

When two LAPM modems connect, they transmit data in frames using bit-oriented synchronous techniques, even though the attached computer communicates with the modem as a standard asynchronous device.

See also Link Access Procedure.

link level Part of the CCITT's X.25 standard that defines the link protocol. LAP (Link Access Procedure) and LAP-B (Link Access Procedure-Balanced) are the link access protocols recommended by CCITT.

See also Consultative Committee for International Telephony and Telegraphy.

link rot A slang expression used to describe an out-of-date URL on a Web page. Link rot occurs when the page indicated by the link is moved or erased.

See also link; Uniform Resource Locator.

link-state routing algorithm A routing algorithm in which each router broadcasts information about the state of the links to all other nodes on the internetwork. This algorithm reduces routing loops but has greater memory requirements than the distance vector algorithm.

See also distance vector algorithm; Open Shortest Path First.

link-support layer Abbreviated LSL. An implementation of the Open Datalink Interface (ODI) that works between the NetWare server's LAN drivers and communications protocols, such as IPX (Internet Packet eXchange) or TCP/IP, allowing network interface cards to service one or more protocol stacks. LSL is also used on workstations.

See also multiple-link interface driver; Open Data-link Interface/Network Driver Interface Specification Support.

Linux A free, Unix-compatible, 32-bit operating system developed in 1991 by Linus Torvalds while at the University of Helsinki in Finland.

Strictly speaking, Linux is the name of the operating system kernel, the central part of the operating system that manages system services, but many people use the name to refer to the complete operating system package, including utilities, editors and compilers, games, and networking components. Many of these important elements are actually part of the Free Software Foundation's GNU Project, and others have been written and released by volunteers.

Linux is supported and distributed by companies such as Red Hat Software, Caldera Software, Workgroup Solutions, Walnut Creek Software, and S.u.S.E. of Germany.

With the increasing use of Linux in the corporate world, several major companies have announced some level of support for the operating system, including Hewlett-Packard, Silicon Graphics, Sun Microsystems, and Intel, and several major applications packages have been ported to Linux,

including Oracle's Oracle8, Adaptive Server Enterprise from Sybase, and IBM's DB2.

See also Free Software Foundation; open source software.

LIP *See* Large Internet Packet.

liquid-crystal display Abbreviated LCD. A display technology common in portable computers that uses electric current to align crystals in a special liquid.

The rod-shaped crystals are contained between two parallel, transparent electrodes. When current is applied, the electrodes change their orientation, creating a dark area. Many LCD screens are also backlit or sidelit to increase visibility and reduce eyestrain.

LISTSERV A mailing list product available from L-Soft International, Inc. LISTSERV runs on Unix, Microsoft Windows, and several other operating systems.

See also listserver; mailing list.

listserver An automatic mailing system available on the Internet.

Rather than sending e-mail on a particular topic to a long list of people, you send it to a special e-mail address, where a program automatically distributes the e-mail to all the people who subscribe to the mailing list.

Several programs have been written to automate a mailing list; the most common is called LISTSERV, but you may also encounter mailserv, majordomo, or almanac.

Mailing lists are usually devoted to a specific subject, such as training dogs or playing

obscure types of bagpipes, rather than to general interest communications.

See also LISTSERV; mailing list; newsgroup; Usenet.

little endian A computer architecture in which the least significant byte has the lowest address and so is stored little end first.

Many processors, including those from Intel, the PDP-11, and the VAX family of computers, are all little endian. The term comes from Swift's *Gulliver's Travels*, in which wars were fought over whether boiled eggs should be opened at the big end or the little end.

See also big endian; holy wars.

liveware A slang term for the people who use computers, as distinct from hardware, firmware, or software.

LLC *See* logical link control.

loadable module In Novell NetWare, a program that can be loaded and unloaded from the server or workstation while the operating system is running.

Two common types of loadable modules are NetWare Loadable Modules (NLMs), which are run on the server, and Virtual Loadable Modules (VLMs), which run on a workstation.

See also NetWare Loadable Module; Virtual Loadable Module.

load average One measure of how much work the CPU is doing; defined as the average number of jobs in the run queue plus the number of jobs that are blocked

while waiting for a disk access. This measurement is usually taken at 1-, 5-, and 15-minute intervals during a 24-hour period to give useful information.

See also uptime.

load balancing A technique that distributes network traffic along parallel paths to make the most efficient use of the available bandwidth while also providing redundancy. Load balancing will automatically move a user's job from a heavily loaded network resource to a less-loaded resource.

See also clustering.

load sharing The ability of two or more remote bridges to share their load in a parallel configuration. If a bridge fails, traffic is routed to the next parallel bridge.

local-area network Abbreviated LAN. A group of computers and associated peripheral devices connected by a communications channel, capable of sharing files and other resources among several users.

See also file server; metropolitan-area network; peer-to-peer network; wide-area network; zero-slot LAN.

local bus A PC bus specification that allows peripheral devices to exchange data at a rate faster than the 8MBps allowed by the ISA (Industry Standard Architecture) definition and the 32MBps allowed by the EISA (Extended Industry Standard Architecture) definition.

Local bus capability must be built into the system's motherboard right from the start; it is not possible to convert an ISA-, EISA-,

or MCA (Microchannel Architecture)-based computer into a local-bus system.

local disk In networking, a disk drive on your workstation or PC, rather than a drive available to you over the network.

See also network drive.

local drive *See* local disk.

Local Exchange Carrier Abbreviated LEC. A term coined from the Telecommunications Act of 1996 to describe the local telephone company that provides local transmission and switching services.

See also Competitive Local Exchange Carrier; Incumbent Local Exchange Carrier.

local group In Microsoft Windows NT Server, a group granted rights and permissions to only the resources on the servers of its own domain.

See also global group.

local loop That part of a communications circuit that connects subscriber equipment to equipment in a local telephone exchange. This link is often the most expensive and least efficient connection in the telephone company's system and is often a barrier to high-speed services.

See also last mile.

local printer In networking, a printer attached to a workstation or a PC, rather than to the file server or a print server.

Local Procedure Call Abbreviated LPC. An interprocess communications method used in multitasking operating systems that allows tasks running concurrently to talk to

local security authority

each other. LPCs allow tasks to share memory space, synchronize tasks, and pass messages to one another.

See also interprocess communication; pipe; Remote Procedure Call; socket.

local security authority In Microsoft Windows NT, a security subsystem that manages the security policy on the local computer and provides user authentication services to other operating system components that need access to that information.

LocalTalk The shielded twisted-pair (STP) wiring and connectors available from Apple for connecting Macintosh computers using the built-in AppleTalk network hardware.

See also AppleTalk.

local user profile A user profile that is specific to the computer upon which it was created; a local user profile does not follow the user if he or she logs on to a different computer.

See also mandatory user profile; roaming user profile.

locational organization A way of organizing the Novell Directory Services (NDS) Directory tree that divides it into Organizational Units for each geographical location in a company.

See also Organizational Unit.

locked file A file that you can open and read, but not write to, delete, move, rename, or change in any way.

logical drive The internal division of a large hard disk into smaller units. One

single physical drive may be organized into several logical drives for convenience, with each one appearing to the user as a separate drive.

logical link control Abbreviated LLC. The upper component of the data-link layer that provides data repackaging functions for operations between different network types.

The media access control is the lower component that gives access to the transmission medium itself.

See also data-link layer; media access control.

logical unit Abbreviated LU. A suite of protocols developed by IBM to control communications in an SNA (Systems Network Architecture) network, as follows:

- **LU type 0** Uses SNA transmission control and flow-control layers.
- **LU type 1, LU type 2, and LU type 3** Control host sessions for IBM 3270 terminals and printers.
- **LU type 4** Supports peer-to-peer and host-to-device communications between peripheral nodes.
- **LU type 6.1** Supports a communications session for IBM databases and transaction management systems.
- **LU type 6.2** The peer-to-peer protocol of Advanced Program-to-Program Communications. It also features comprehensive end-to-end error processing and a generalized application program interface (API).

- **LU type 7** Data Stream terminals used on AS/400 systems.

See also Advanced Program-to-Program Communications; physical unit.

logical unit number Abbreviated LUN. The logical address of a Small Computer System Interface (SCSI) device when several devices are attached to a single SCSI device ID. The LUN is usually set to zero unless the SCSI adapter supports multiple LUNs on a single SCSI device ID.

See also Small Computer System Interface.

logic bomb A sabotage attack on a system timed to go off at some time in the future; essentially a Trojan Horse with a fuse.

A logic bomb goes off at a certain time or when triggered by a certain event and then performs some operation. It might release a virus, delete files, or send comments to a terminal. An unhappy programmer may plant a logic bomb on a system and time it to go off long after she has left the company so as to avoid suspicion.

See also boot sector virus; file-infecting virus; macro virus; multipart virus; polymorphic virus; stealth virus; Trojan Horse; vaccine.

log in *See* log on.

LOGIN directory In Novell NetWare, the SYS:LOGIN directory, created during the network installation, that contains the LOGIN and NLIST utilities used to support users who are not yet authenticated.

login script A small file or macro that executes the same set of instructions every time a user logs on to a computer system or network. Sometimes called a logon script.

Login scripts can map drives, display messages, set environment variables, and run programs and are critical for proper configuration of each user's network environment.

A communications script may send the user-identification information to an online information service each time a subscriber dials up the service.

See also script.

log off To terminate a session and sign off a computer system by sending an ending message. Also known as logoff and log out.

The computer may respond with its own message, indicating the resources consumed during the session or the period between log on and log off. Logging off is not the same as shutting down or turning off the computer.

See also log on.

log on To establish a connection to a computer system or online information service; also known as logon and log in. Many systems require the entry of an identification number or a password before the system can be accessed.

See also log off; password protection.

logon script *See* login script.

log out *See* log off.

long filename Any filename longer than the eight-character plus three-character filename extension allowed by MS-DOS.

long-haul modem

In many of today's operating systems, including Unix, Microsoft Windows, and the Macintosh, filenames can be more than 200 characters and can include upper- and lowercase letters and spaces.

long-haul modem A modem or other communications device that can transmit information over long distances.

See also line driver; short-haul modem.

loopback A troubleshooting test in which a signal is transmitted from a source to a destination and then back to the source again so that the signal can be measured and evaluated or so that the data contained in the signal can be examined for accuracy and completeness.

lossless compression Any data-compression method that compresses a file by rearranging or recoding the data that it contains in a more compact fashion.

With lossless compression, no original data is lost when the file is decompressed. Lossless compression methods are used on program files and on images such as medical X rays, when data loss cannot be tolerated, and can typically reduce a file to 40 percent of its original size.

Many lossless compression programs use a method known as the Lempel-Ziv-Welch (LZW) algorithm, which searches a file for redundant strings of data and converts them to smaller tokens. When the compressed file is decompressed, this process is reversed.

See also Lempel-Ziv-Welch; lossy compression.

lossy compression Any data-compression method that compresses a file by discarding any data that the compression mechanism decides is not needed.

Original data is lost when the file is decompressed. Lossy compression methods may be used for shrinking audio or image files when absolute accuracy is not required and the loss of data will not be noticed; however, this technique is unsuitable for more critical applications in which data loss cannot be tolerated, such as with medical images or program files. Lossy compression can typically reduce a file to as little as 5 percent of its original size.

See also Joint Photographic Experts Group; lossless compression; Moving Pictures Experts Group.

Lotus cc:Mail An e-mail and messaging package originally developed by Lotus Development Corporation and now supported by IBM.

cc:Mail works with MS-DOS, Macintosh, Microsoft Windows, Unix, and OS/2 networks and includes built-in discussion groups and connections to the Internet.

Lotus Domino A server technology from Lotus Development Corporation that turns the popular Lotus Notes groupware product into an Internet application server.

With Lotus Domino, users can access the Notes environment, including dynamic data and applications on Notes servers, and use a Web browser. Application developers can use the Notes environment to develop Web-based applications.

See also Lotus Notes.

Lotus Notes A popular groupware product originally developed by Lotus Development Corporation and now supported by IBM.

Lotus Notes is the defining force behind the entire groupware market and is the target toward which all other developers aim. Notes includes a flexible database that can contain a variety of data types with none of the restrictions that normally apply, such as fixed field lengths. A Notes field can contain text, scanned images, OLE embedded objects, and even hypertext links to other Notes documents.

This database is closely linked with an e-mail system that allows users to forward any document in any Notes database to any other Notes database; a user can send e-mail to the database, and an application can even send e-mail to an individual user.

Notes can also maintain multiple copies of the database, perhaps on the LAN or on remote workstations, and can synchronize these copies using background dial-up modem connections, IPX/SPX, or TCP/IP.

Notes is supported by a large number of networks, including AppleTalk, Banyan VINES, IBM APPC, Novell NetWare, TCP/IP, and X.25.

See also Lotus Domino; workflow software.

Lotus SmartSuite A popular software suite originally developed by Lotus Development Corporation and now supported by IBM.

SmartSuite consists of the 1-2-3 spreadsheet, WordPro word processor (previously known as Ami Pro), Approach database, Freelance Graphics, and the Organizer personal information manager.

SmartSuite is also available in a version called NotesSuite, which is integrated with Lotus Notes for groupware applications.

low-end Describes any inexpensive product, from the bottom of a company's product list, that includes a reduced set of capabilities.

See also high-end.

low-level language A hardware-specific programming language close to machine language. All assembly languages are considered low-level languages.

See also high-level language.

LPC *See* Local Procedure Call.

LPT ports In MS-DOS, the device name used to denote a parallel communications port, often used with a printer.

LSL *See* link-support layer.

LU6.2 *See* logical unit.

LU *See* logical unit.

Lucent Technologies, Inc. A major manufacturer of networking switches and routing products, spun off from AT&T in 1996.

Lucent Technologies provides products for the wired, wireless, and voice/data/video markets, as well as voice and fax messaging systems, Gigabit Ethernet, and Asynchronous Transfer Mode (ATM) products. In 1999, Lucent bought Ascend Communications, Inc.

For more information on Lucent Technologies Inc, see www.lucent.com

LUN *See* logical unit number.

lurking

lurking The practice of reading an Internet mailing list or Usenet newsgroup without posting anything yourself. In the online world, lurking is not considered particularly antisocial; in fact, it is a good idea to lurk

for a while when you first subscribe so that you can get a feel for the tone of the discussions in the group and come up to speed on recent history.

LZW See Lempel-Ziv-Welch.

M

M See mega-.

m See milli-.

MAC See media access control.

Mac See Macintosh.

MacBinary In the Macintosh, a file transfer protocol that ensures a proper transfer of Macintosh files and related data over a modem, including the file itself, the file's resource fork, data fork, and Finder information block.

Most Macintosh communications programs support sending and receiving files in MacBinary, but the protocol is not often supported in other environments.

Mach An operating system created from scratch at Carnegie-Mellon University, designed to support advanced features such as multiprocessing and multitasking.

Mach has its roots in the Unix world and was originally based on BSD 4.4; however, its most notable feature is that it employs a relatively small microkernel rather than a conventional monolithic kernel.

The microkernel is designed to manage only the most fundamental operations, including interrupts, task scheduling, messaging, and virtual memory; other modules can be added as necessary for file management, network support, and other tasks.

See also microkernel.

machine code See machine language.

machine collating sequence The sequence in which the computer orders characters. Because most systems use ASCII, except the large IBM systems that use EBCDIC, the machine collating sequence is usually based on the ordering of characters in the ASCII character set; see Appendix C for details.

See also American Standard Code for Information Interchange; Extended Binary Coded Decimal Interchange Code.

machine language The native binary language used internally by the computer; also known as machine code.

Machine language is difficult for humans to read and understand. Programmers create applications using high-level languages, which are translated into a form that the computer can understand by an assembler, a compiler, or an interpreter. Whichever method is used, the result is machine language.

See also assembly language; compiler; interpreter; microcode.

Macintosh Abbreviated Mac. A range of personal computers first introduced in 1984 by Apple Computer, Inc., featuring a popular and easy-to-use graphical user interface. The computer was based on the Motorola 68000 series of microprocessors and used a proprietary operating system to simulate the user's desktop on the screen.

Macintosh client

The original Mac was a portable, self-contained unit with a small monochrome screen, 128KB of memory, two serial ports, extended sound capabilities, and a single 400KB, 3.5-inch floppy disk. The computer was an instant success, and users quickly began to demand more power and additional features.

Apple released many new models over the years, expanding the range to include Macs based on the 68020, 68030, and 68040 processors and adding color, more memory, a built-in SCSI interface, built-in networking, a 32-bit bus, and larger and faster hard disks.

Introduced in 1991, the PowerBook series of notebook computers offered both power and convenience in a very small package, posing a real challenge to the MS-DOS-based laptops.

During the 1990s, the Mac lost overall market share, but remained popular in the desktop-publishing, music, and graphics-related fields.

The iMac, introduced in 1998, was an instant success with its sleek new shape and bold color schemes; the one criticism was that it lacked a floppy disk drive.

Macintosh client Any Macintosh computer attached to a network. A Macintosh client can store and retrieve information from a NetWare server running NetWare for Macintosh modules and can run executable Macintosh network files.

Macintosh File System Abbreviated MFS. In the Macintosh, an older system that stored files in a flat structure rather

than the hierarchical system used in more recent versions. All current Macintosh models can read disks created using MFS.

See also Hierarchical File System.

MacOS The operating system that runs on Macintosh computers. MacOS contains the famous Mac graphical user interface, it is multithreaded to allow execution of multiple concurrent tasks, and it includes Internet access in the form of TCP/IP and a Web browser.

MacOS 8.5 requires a PowerPC processor and so will not run on Macs based on earlier Motorola 680x0 processors; you will also need 16MB of physical RAM and virtual memory set to at least 24MB.

macro A stored group of keystrokes or instructions that can automate a complex or repetitive sequence of application commands.

Many of the major spreadsheet, word-processing, and database programs let users create and edit macros to speed up operations. Some macros can incorporate control structures, such as DO/WHILE loops and IF/THEN branching statements.

See also login script; script.

macro virus An executable program that attaches itself to a document created in Microsoft Word or Excel. When you open the document and execute the macro, the virus runs and does whatever damage it was programmed to do.

See also boot sector virus; file-infecting virus; multipart virus; polymorphic virus; stealth virus; vaccine; virus.

magic A substance that is sprinkled or a phrase that is chanted by a guru to make something work.

A magic character is one with special, often hard to understand capabilities, used at every opportunity by gurus wanting to demonstrate their powers.

A magic number is one that represents a particular condition, but whose value is unusual, given the conditions, and which would certainly never be guessed by anyone who is not a guru.

See also guru.

Magneto-optical storage Abbreviated MO. A high-capacity network storage device capable of storing 5.2GB on a 5.25-inch removable cartridge.

The data on the cartridge is highly stable, and so MO storage is suitable for use as backup and archival storage. MO offers access speeds of 35ms, compared with CD-RW and DVD-R, which offer speeds of between 100 and 200ms.

See also CD ReWritable; digital video disc-recordable.

mail In the networking world, e-mail, rather than the postal service.

See also electronic mail.

mail-aware application Any application with the ability to send and receive e-mail. Applications in the document management, groupware, and workflow cate-

gories all use e-mail to interconnect users and help with the flow of information. This integration of e-mail is made possible in part by APIs such as Microsoft's Messaging API and Novell's Message Handling Service.

Sometimes known as a message-enabled application.

See also groupware; Lotus Notes; workflow software.

mail bombing To send a huge number of long e-mail messages to the same e-mail address, effectively bringing the e-mail server to a halt. A variation on the denial of service attack, but one that can be aimed at a specific person.

This kind of attack is relatively easy to carry out and does not require actual access to the server. A malicious person can easily write a program to send a huge e-mail message to the same address a few hundred thousand times, overwhelming the server and annoying the recipient.

See also denial of service attack.

mailbox In e-mail systems, an area of hard-disk space used to store e-mail messages until users can access them. An on-screen or audio message often tells users that they have mail.

MAIL directory In Novell NetWare, a default directory created during installation for the use of e-mail applications.